

STEM versus Humanities: High School Curriculum and the Malleability of Beliefs, Preferences, and Values

Robert Ainsworth*
Rajeev Dehejia+
Andrei Munteanu†
Cristian Pop-Eleches±
Miguel Urquiola±

June 2026

Abstract

We estimate causal effects of assignment to STEM vs. humanities-focused high school curricula in Romania. We study impacts not only on educational/career trajectories, but also on beliefs, preferences, and values. STEM assignment strongly increases STEM college enrollment and career aspirations, with limited heterogeneity by gender, initial curricular preferences, or math vs. language ability, but important differences by baseline achievement. The large effects on trajectories arise because curricula shape students' interests and beliefs about their abilities. They also influence values, with STEM making boys more conservative. The breadth of impacts we document helps explain why curricular reforms are often controversial.

For helpful comments and conversations, we thank Joe Altonji, Zach Bleemer, José Montalban Castilla, Sarah Cohodes, David Gill, Richard Romano, David Sappington, and seminar participants at IIES, NBER, the World Bank, the University of Florida, ERMAS 2024, SCSE 2024, AEDE 2025, Concordia University, Hitotsubashi University, the University of Tokyo, Keio University, the National Taiwan University, Academia Sinica, CUNY, Purdue University, Simon Fraser University, and the Berlin Applied Micro Seminar. For outstanding research assistance, we thank Poorvi Goel. We thank Ioana Veghes for help in implementing the surveys. This research was supported by the Social Sciences and Humanities Research Council of Canada (SSHRC) [Grant number 430-2024-00780] and the Unicredit Foundation.

* University of Florida. + New York University. † Université du Québec à Montréal. ± Columbia University.

1 Introduction

In recent decades, policymakers have sought to steer young people toward STEM, based in part on the view that STEM generates economic returns. A common strategy has been to expand STEM exposure in high school, often at the expense of humanities and social studies (HSS). A growing literature suggests that additional high school STEM exposure can push students toward STEM college majors and careers and ultimately increase earnings.¹

While the literature has focused on effects on the STEM pipeline, high school education serves wider goals than just influencing the type of job that students pursue. Rather, it is one of the primary means by which societies attempt to shape who individuals are as people, including their beliefs, preferences, and values. Shifting curricular emphasis between STEM and HSS may therefore impact core aspects of students’ identities. Indeed, debates about STEM versus HSS often take as given that these fields impart different knowledge and forms of reasoning, which implies that they may influence how students see and engage with the world. Yet there is little causal evidence on the effects of STEM vs. HSS curricular exposure on outcomes beyond educational and career trajectories.²

This paper asks: to what extent and in what ways does a STEM- vs. HSS-focused high school curriculum shape who students are? Who students are encompasses their educational and career trajectories but also other, perhaps deeper traits. Effects on the latter outcomes are often of intrinsic interest; further, they help illuminate the mechanisms by which curriculum influences trajectories. Central to our analysis is the question of whether students’ core attributes are already fixed before high school or are instead malleable to curricular exposure. If students’ core attributes are already fixed, then curricular policies may generate mismatch, leading some students to struggle, disengage, and eventually sort away from fields that conflict with their comparative advantage. If students are malleable, they may not only adapt to their assigned curriculum but also experience changes in their sense of self, outlook, and conduct.

The setting for our analysis is Romania’s high school system. This system assigns students to nationally standardized curricula through a centralized mechanism that creates sharp assignment cutoffs. Cutoffs between STEM- and HSS-focused curricula occur in much of the achievement distribution, meaning that many types of students are at risk of assignment to either curriculum type. Combining administrative and survey data, we estimate impacts both on traditional educational/career outcomes and on beliefs, preferences, wellbeing/school satisfaction, and social/civic

1. E.g., Altonji (1995), Levine and Zimmerman (1995), Rose and Betts (2004), Wiswall et al. (2014), Joensen and Nielsen (2016), Goodman (2019), Darolia et al. (2020), De Philippis (2023), Cohodes, Ho, and Robles (2022), Dahl, Rooth, and Stenberg (2023), and Liu, Conrad, and Blazar (2024). These papers complement a large body of research on the pecuniary benefits of studying STEM at the college level.

2. Despite the lack of causal evidence, people seem to believe that STEM vs. HSS exposure affects non-economic outcomes, with curricular debates often centering on disagreement over the desirability of the effects. For instance, in the U.S., some voices attack HSS as left-wing indoctrination (e.g., Trump 2025 and Luscombe 2026), while others argue for it by claiming it has social and civic benefits (e.g., Nussbaum 2010).

outcomes. We do this for students with different initial curricular preferences, relative ability in math vs. language, gender, and baseline achievement. We also confirm that the effects we estimate reflect curricular exposure itself rather than correlated factors such as peer composition.

Our main finding is that high school curriculum exerts large effects on core aspects of who students are. This includes what students like and think they are good at, as well as their personalities, friendships, family expectations, and political views. High school students are sufficiently malleable to what they study that almost all types seem able to adapt to their assigned curriculum in terms of wellbeing and school satisfaction. Further, students’ malleability means that high school curricula strongly influence educational and career outcomes—though we find evidence of a leaky pipeline for some groups, especially students with low baseline achievement. The breadth of impacts we document suggests that fully evaluating a curriculum requires going beyond effects on educational and career trajectories and helps explain why debates about curricular reforms are often contentious.

Our empirical strategy is a regression discontinuity (RD) design that exploits the Romanian setting. Romanian students apply to high school by submitting a rank-ordered list of high school-curriculum combinations called *tracks*. They are assigned to tracks using a serial dictatorship that prioritizes students according to their academic achievement in middle school. Admission to an oversubscribed track depends only on whether a student’s achievement surpasses a cutoff—the score of the last student who fills the track. Thus, a student who ranks both a STEM and an HSS option may be assigned to either, depending on how the student’s achievement compares to the cutoffs for the tracks he/she ranks.

Cutoffs that determine assignment to STEM or HSS curricula come in two types: those where the track with the STEM curriculum is more selective and those where the opposite holds. The first step in our strategy is to calculate the effect of STEM (vs. HSS) assignment for each cutoff type. Importantly, these effects combine the impact of a change in curricular assignment with the impact of scoring above vs. below a cutoff. To remove the latter channel, we calculate the average effect of STEM assignment over both cutoff types. This effect continues to reflect a compound treatment: it captures both the effect of curriculum (i.e., exposure to STEM or HSS subjects and to the teachers who teach each subject type) and the fact that peers in STEM and HSS tracks have different horizontal attributes (including gender composition and relative ability in math vs. language). In contrast, it nets out differences in vertical dimensions of track quality, changes in class rank, and any psychological effects from just getting/missing a preferred option.³ Finally, to isolate the effect of curriculum, we exploit the large number of cutoffs in our setting and run specifications that control for track peer composition.

A benefit of our setting is that we can directly study the impact of scoring above vs. below a cutoff by using cutoffs between tracks with the same curricula. We find that this impact is

3. In the appendix, we scrutinize the averaging approach using a potential outcomes framework.

always small. In addition, we find that controlling for track peer composition hardly matters. Thus, it appears that curriculum is the key driver of the effects we document, even when we do not average over the two cutoff types.

Our administrative data covers students who entered high school in 2015-2017 and 2019. Our survey data comes from multiple instruments administered to the 2019 cohort, beginning with an in-person classroom survey just before the end of high school and continuing with phone surveys through the first year and a half after high school. We use the administrative data to construct the RD samples and the surveys to obtain outcomes not available in the administrative data.⁴

Our analysis produces five specific findings. The first two extend the literature that studies effects on educational and career trajectories. The remaining three are about how high school curriculum affects students' core traits, including (i) beliefs about abilities and preferences over academic subjects and job tasks, (ii) wellbeing and school satisfaction, and (iii) social and civic outcomes. The findings are as follows.

1. High school curriculum affects college and career outcomes, with limited heterogeneity.

On average, being assigned to STEM rather than HSS makes students 25 percentage points (pp) more likely to enroll in a STEM college program and 26 pp less likely to enroll in humanities, law, or social science. It also makes students 25 pp more likely to want a career in technology or engineering and 15 pp less likely to want one in art, education, law, or social services.

Effects on college enrollment exhibit little heterogeneity by initial curricular preferences, relative ability in math vs. language, or gender. Effects on wanting a technology/engineering career do not differ by math vs. language ability but are somewhat smaller for students who initially preferred HSS and for females. The main dimension along which we find heterogeneity is baseline achievement, to which we return below. Also, as mentioned, effects hardly change when we control for track peer composition.

These results contribute to the literature on the effects of high school STEM exposure on students' trajectories. They substantiate the finding that curricular exposure can shift trajectories using a credible research design in a contemporary setting with a diverse student sample. In addition, they add a systematic analysis of heterogeneity and show that effects reflect curriculum itself rather than correlated factors.⁵

2. STEM is more difficult than HSS, with negative consequences for low-achievers.

STEM assignment increases time spent on homework and raises values on a grit index, indi-

4. Bleemer and Mehta (2022) and Cohodes, Ho, and Robles (2022) also study curricular effects using a combination of administrative and survey data.

5. The closest paper is Dahl, Rooth, and Stenberg (2023). It also uses an RD design to estimate the effects of curricularly-differentiated high school tracks, but focuses on Swedish students who were enrolled in the 1970s-1990s. It finds that completing a STEM rather than HSS track boosted STEM college degrees, STEM occupations, and earnings. Due to data constraints, it does not explore heterogeneity by absolute or relative ability, nor does it isolate the effect of curriculum. The rest of the literature studies less intensive interventions, such as STEM summer programs and expansions of STEM course offerings; it finds mixed but mostly positive effects.

cating that it spurs greater effort. In addition, it reduces scores on Romania’s national high school exit exam (the “baccalaureate” exam). This binds for students with low baseline achievement, who become 11 pp less likely to pass the exam, and thus obtain a baccalaureate diploma, within the year after high school. Obtaining this diploma is an important labor-market credential and is required for college admission; meanwhile, a high score on the baccalaureate exam facilitates access to college scholarships and selective universities. Depressed baccalaureate performance contributes to a 9 pp reduction in college attendance for low-achievers and a 24 pp drop in their probability of receiving a scholarship. In contrast, the added rigor of STEM does not seem to affect college outcomes for high-achievers.

These results highlight that STEM is risky for low-achieving students. Further, they provide a causal foundation for a large body of descriptive research that finds that STEM is more demanding than HSS and that this difficulty may explain leaks in the STEM pipeline (e.g., Stinebrickner and Stinebrickner 2014 and Ahn et al. 2024 on U.S. college majors).

3. High school curriculum shapes students’ beliefs about their abilities and preferences over academic subjects and job tasks. Effects on beliefs and preferences exhibit almost no heterogeneity and seem to be the main mechanism behind the impacts on college and career outcomes.

At the end of high school, STEM assignment raises students’ self-assessed STEM abilities by about two-thirds of a standard deviation (sd) and lowers their self-perceived HSS abilities by 0.25-0.5 sd. A year later, STEM assignment increases reported enjoyment of STEM subjects by nearly 0.75 sd and reduces enjoyment of HSS subjects by 0.4-0.6 sd.⁶ In an O*NET questionnaire, STEM assignment raises (lowers) the share of STEM- (HSS-) related job tasks that students report enjoying by 10 (9) pp. In addition, it changes the composition of the jobs that O*NET flags as matching a student’s interests, raising (lowering) the share that are STEM- (HSS-) related by 11 (10) pp.

To gauge the role of alternative mechanisms, we elicited students’ beliefs about a representative set of jobs and asked students what attributes they consider when making college or career choices. The former data rule out a few plausible channels: While high school curriculum influences whether a student would like the work content in a job and feel prepared for the job, it does not affect beliefs about earnings, coworkers, or work conditions. The latter data reveal the elevated importance that students place on what matches their interests and abilities. Almost all students report caring about whether a college or career option is a good fit along these dimensions. Far fewer report being swayed by other factors that could differ by curriculum, such as switching costs or peer and teacher influences.

These results contribute to a large literature on how students make college and career choices

6. Students learn their baccalaureate performance in the summer after high school. At this time, the effects of STEM assignment on beliefs about abilities shift in a negative direction, consistent with STEM-assigned students receiving a negative update. However, the effect on confidence in STEM remains positive and large.

(see Patnaik, Wiswall, and Zafar 2020 for a review). The existing literature usually takes students’ interests and abilities as given. Our results show that these are not cemented in early childhood but can instead be molded by what students study in high school. The results also offer a hopeful message: the effects on students’ post-high-school trajectories are not simply a consequence of students’ inability to switch fields.

4. High school curriculum has nuanced effects on wellbeing and school satisfaction, but we find little evidence of acute psychic costs due to mismatch.

At the end of high school, STEM assignment improves these outcomes, seemingly via better social dynamics. In contrast, by 1-1.5 years after high school, STEM assignment has no impact on wellbeing and more variable effects on high school satisfaction; in addition, it reduces college enjoyment. The reduction in college enjoyment is driven by lower satisfaction with the college experience, not by whether students found the college curriculum to be a good fit for their abilities; this is consistent with Romanian STEM college programs having a reputation as competitive and unwelcoming. Despite the shifting effects on wellbeing and school satisfaction, an important finding is that students generally report strong contentment with their educational choices, regardless of their curricular assignment.

We believe we are the first to examine the causal effects of focusing on STEM vs. HSS in high school on wellbeing and school satisfaction. Our findings show that high school curriculum can impact these outcomes but also counter the worry that students may suffer if assigned to a curriculum for which they are an imperfect fit.

5. High school curriculum influences a range of social and civic outcomes.

First, STEM assignment causes boys—but not girls—to hold more traditionalist expectations about their future family structure. Second, STEM assignment pushes boys’ political views to the right. For girls, it has no effect on political views on average, but it increases agreement that climate change is a serious issue and disagreement that boys are naturally better at math. Third, STEM assignment does not affect trust, whether students turned out to vote in Romania’s Fall 2024 elections, or how many friends they have. However, it reduces the number of female vs. male friends, verbal development (including time spent reading), and empathy.

Exploring mechanisms behind the impacts on family expectations and political views, we do not find that the shifts operate through altered economic expectations: STEM assignment does not affect how rich students expect to be or how much education they expect to obtain. Nor, as always, do we find meaningful support for a peer effects channel. We conclude that impacts are likely due to the differing curricular content of STEM and HSS tracks and the effects on verbal development and empathy.

These results relate to research showing that HSS-related majors and occupations correlate with left-wing political views (Goldstein and Kolerman 2025; Hooghe, Marks, and Kamphorst 2025; Martin, Scott, and Kappe 2025), as do verbal development and empathy (Nussbaum 2010;

Wagaman and Segal 2014; Hasson et al. 2018; Morris 2020; Zebarjadi et al. 2023; Edwards et al. 2025; Martin, Scott, and Kappe 2025). Our results add causality to the first link and point to the second as a likely mechanism.⁷

In sum, we find that high school curriculum shapes students’ core traits, including their beliefs, preferences, and values. The effects on these traits underpin the large impacts that curricula have on educational and career outcomes and help explain why the educational and career impacts exhibit limited heterogeneity (with low-achievers being a key exception). Overall, our results offer both an optimistic and cautionary perspective on using high school curriculum as a policy lever: Curricula can shift students’ trajectories, and students seem able to adapt to what they study in terms of wellbeing and school satisfaction; however, people may disagree on the desirability of the many ways in which curricula influence who students are.

2 Setting

Romanian students first experience significant curricular differentiation during high school (grades 9 to 12). As mentioned, high school students are assigned to tracks—units within schools with set curricula. Each curriculum covers a variety of subjects but emphasizes a particular area. Students in a track share the same coursework and do not take classes with students from other tracks. In contrast, teachers specialize in particular subjects and may teach in multiple tracks, depending on which tracks include their subject(s).

In Romania, curricula are classified into categories. Two of these—STEM and HSS—are academically oriented and intended to lead to college. The others concentrate on vocational fields, technical subjects, the arts, religion, or preparation for the military. In this paper, we analyze the two academic categories, both of which group a few specific curricula—for STEM: math/computer science, natural science, and business/economics; and for HSS: humanities and social studies.⁸ The curricula within a category are similar in terms of their core coursework and differ mainly in the specialization classes that students take. Given the modest within-category differences, we distinguish curricula at the category level.

Table 1 lists the weekly hours of instruction in different subjects for STEM and HSS tracks. Compared to HSS, STEM tracks have about 6 more (fewer) hours per week devoted to STEM (HSS) subjects in grades 9 and 10, with an even greater difference in grades 11 and 12.

Romania’s high school application process begins at the end of 8th grade, when students take a national admission exam that we call the “transition” exam. They then receive a “transition

7. Other papers have studied how curricula affect social and civic outcomes, but none, to our knowledge, has examined the impacts of STEM vs. HSS assignment in high school. Notable in the prior literature, Clots-Figueras and Masella (2013) find that Catalan language instruction reinforces Catalan identity, while Cantoni et al. (2017) find that a pro-government curriculum led Chinese students to hold more pro-government views.

8. Business and economics are also often designated as STEM in classifications of U.S. college majors (see, e.g., Imberman et al. 2026). We find that excluding them from the STEM category yields similar results, as does broadening STEM to include the technical category (not reported).

Table 1: Weekly hours of instruction for STEM and HSS tracks

	Grades 9-10		Grades 11-12	
	STEM	HSS	STEM	HSS
STEM subjects	14-15	8	12-13	2-4
HSS subjects	11-12	17	11	18-20
Other subjects	3-4	4	4-6	6-7
Total	29-30	29	28-29	28-29

The table shows the mandated weekly hours of instruction in different types of subjects for STEM and HSS tracks, separately for grades 9-10 and 11-12. “Other subjects” include physical education and classes that are designed locally. The values come from the Romanian Ministry of Education, Research and Innovation (2009).

score”, which is an average of their exam score and their middle school GPA (grades 5-8). After learning their transition score, students submit a rank-ordered list (or “preference ranking”) of the tracks they wish to attend. Students are assigned to tracks via a nationwide serial dictatorship that considers preference rankings in the order of transition scores. Specifically, the mechanism starts by assigning the student with the highest transition score to his/her most-preferred track. It then proceeds down the transition score distribution, placing each student in their most-preferred track that is not yet at capacity.

Students can list an almost unlimited number of tracks. Thus, the assignment mechanism is incentive compatible: the dominant strategy is to rank according to one’s preferences. In addition, the mechanism creates an admissions cutoff for each oversubscribed track. This is equal to the transition score of the last student who fills the track (i.e., the minimum score among the track’s students). Our empirical strategy exploits the cutoffs in an RD design.

Once assigned to a track, students can transfer. However, this is not easy and is especially hard when switching to a track with a different curriculum.⁹

At the end of 12th grade, students may opt to take the national baccalaureate exam. The exam has three main parts. Two vary by curriculum, while the third is on Romanian language and literature and applies to all curricula. Students can retake the exam, which is offered twice per year. As mentioned, there are rewards both to passing the exam and to obtaining a high score. Passing yields a baccalaureate diploma. This is a more advanced credential than high school graduation and is necessary for college admission; for students who do not pursue higher education, it serves as a labor market signal. A high score helps students access college scholarships and prestigious universities.¹⁰

9. Students can transfer only to tracks with spare capacity. In 9th grade (later grades), a student’s transition score (high school GPA) must match that of the weakest student in the track that the student wishes to enter. When switching curricula, a student must also pass an exam related to the destination curriculum (Romanian Ministry of Education, Research and Innovation 2016).

10. Romanian students pay for college in one of three ways: the highest-achievers receive a scholarship, which covers tuition and living expenses; students with middling achievement receive only a tuition waiver; the lowest-

At the college level, students are admitted to a specific field of study (or “program”) within a school. Importantly, in determining admissions, programs usually do not consider the curriculum of a student’s high school track. Thus, with a few exceptions, students from any track can be admitted to any program. That said, some programs administer their own field-specific admission exams, which may be difficult for students who focused on unrelated areas in high school.

Finally, in Romania, STEM careers—and especially IT jobs—are widely encouraged. To spur the growth of the IT industry, until 2024 the government offered income tax exemptions to IT workers. In addition, like many former communist countries, Romania has a tradition of female participation in STEM. Nonetheless, as of 2016, only about 41 percent of tertiary graduates in STEM fields are women (Robayo-Abril et al. 2023). Similarly, Romania has recently ranked at the bottom of the EU in EIGE’s Gender Equality Index (EIGE 2024).

3 Data

Our first data source is administrative information from the Romanian Ministry of Education. For each high school student in the country, the administrative data lists gender, middle school GPA, scores on the components of the transition exam (math and language), overall transition score, and track assignment. For students who attended middle school in 16 of Romania’s 41 counties, it also provides track preference rankings, track enrollment histories, and performance on all attempts on the baccalaureate exam (including the overall score, the component scores, whether the student passed, and the curriculum connected to the exam that the student took).¹¹

The administrative data is for the 2015-2017 and 2019 admissions cohorts. The 2018 cohort is excluded because of a reporting issue, while enrollment histories are incomplete for the 2019 cohort, which had not graduated from high school when we obtained the enrollment data. Appendix A1 provides summary statistics. It shows that the students from the 16 counties with track preference rankings are reasonably similar to those in the entire country. For convenience, in the rest of the paper, we refer to the students from the 16 counties as the “administrative data”, though we sometimes make use of the full sample (such as in calculating track attributes).¹²

To complement the administrative data, we administered a series of surveys on students in the 2019 admissions cohort. Our first survey (the “end-of-high-school survey”) took place in May 2023, one month before graduation and students’ first baccalaureate attempt. We visited high schools and distributed a paper-and-pencil questionnaire in students’ homeroom classes. To capture transfer students, we surveyed all students in a high school, not just those in STEM or HSS tracks. We included questions on the high school experience, wellbeing, friendships, time

achievers pay out of pocket. There are also financial aid packages for low-income students.

11. The Ministry agreed to provide the additional information for only part of the country. Romania is organized into eight regions; we were given two counties per region. Switching counties (or even towns) to attend high school is uncommon (Ainsworth et al. 2023); it does not pose a problem for our empirical strategy.

12. A group that is potentially of interest in Romania is the Roma. Unfortunately, the Roma are hard to study, as they rarely self-identify on government forms and many drop out before high school.

use, beliefs about academic abilities, college plans, expectations for life at age 30, political views, grit, trust, and empathy. We also gathered students’ contact information to enable follow-ups. We obtained responses from 10,267 students in 292 schools across 94 towns.¹³

In Spring 2024, we surveyed students from the end-of-high-school survey by phone (the “first follow-up survey”). We repeated questions relating to the high school experience and beliefs about academic abilities. In addition, we asked about baccalaureate performance, college enrollment, the college experience, preferences for academic subjects, desired careers, and reasons for college choices. A total of 2,051 students responded, representing 257 schools in 87 towns.

In late spring/early summer of 2024, we phoned students from the first follow-up survey to administer an additional questionnaire (the “peer survey”). It asked students about the baccalaureate performance and college enrollment of their closest high school peers. The goal was to obtain information on these outcomes for students who did not complete the first follow-up.¹⁴ In the peer survey, 1,759 students answered questions about 6,359 of their peers. Together, the first follow-up and peer surveys concern 8,339 students from 289 high schools in 93 towns.

Finally, in Fall 2024, we again surveyed students from the end-of-high-school survey by phone (the “second follow-up survey”). This survey repeated a number of earlier questions, including those related to college enrollment, wellbeing, friendships, time use, expectations, grit, trust, and empathy. In addition, the survey asked further questions about political views and behaviors, included an abridged version of the O*NET Interest Profiler, elicited beliefs about a representative set of jobs, and asked what factors students expect to weigh in their future career choices.¹⁵ We obtained responses from 3,006 students in 285 schools in 93 towns. As we detail later, the survey samples are representative of the students in the administrative data.¹⁶

4 Empirical strategy

Our empirical strategy is an RD design that leverages students’ preference rankings over high school tracks. We exploit the fact that students who rank both STEM and HSS tracks may be assigned to either track type, depending on how their transition score compares to the cutoff scores for the tracks they ranked.

Since we are interested in comparing STEM and HSS, we restrict attention to students who are assigned to these curricula and who face cutoffs that involve a switch between them. Specifically, we consider two cutoff types: (i) cases where a student ranks a more selective STEM track above a less selective HSS track, which we call STEM-above vs. HSS-below, or “STEM-above” for

13. To select towns, we started by restricting to those in the 16 counties with track preference rankings. We then balanced geographic representativeness with convenience for the surveyors. Within a town, we visited all high schools that agreed to the survey.

14. In Romania, a student’s baccalaureate performance is generally known to their family and friends.

15. The O*NET Interest Profiler is an assessment tool that helps people identify careers that match their preferences over job tasks. It was developed by the U.S. Department of Labor based on Holland (1997).

16. The text of the surveys is available online at <https://www.andrei-munteanu.com/additional.html>.

short, and (ii) cases where a student ranks a more selective HSS track above a less selective STEM track, which we call HSS-above vs. STEM-below, or “HSS-above” for short. Given the connection between cutoff types and preference rankings, we often label cutoff types based on the preferences of the students to whom they apply: “Prefer STEM” for STEM-above and “Prefer HSS” for HSS-above.

To select cutoffs for a particular student, we focus on the lowest (highest) that the student is below (above) where crossing the cutoff would change the student’s curricular assignment between STEM and HSS. In addition, we require a student’s transition score to be close to the cutoff score for a relevant cutoff. In this setup, a student may be in the RD sample for zero, one, or two cutoffs. Importantly, when students are in-sample for two cutoffs, the cutoffs are necessarily of different types.¹⁷ Given the possibility of multiple cutoffs per student, our analysis is at the student-cutoff level. We let the letter i denote a student-cutoff combination, although for brevity we sometimes refer to i simply as a student. In the full administrative data, among students who are in the RD sample for at least one cutoff, 18% are in-sample for two cutoffs. Appendix A2 illustrates our selection of cutoffs via an example.

We estimate separate treatment effects for STEM-above and HSS-above cutoffs. Formally, let i ’s transition score be written t_i , let i be near cutoff j , let f_j denote j ’s cutoff type (with f_j equaling SH for STEM-above and HS for HSS-above), and let c_j be j ’s cutoff score. For STEM-above cutoffs, STEM assignment occurs when $t_i > c_j$; for HSS-above cutoffs, it is when $t_i < c_j$.¹⁸ To be consistent, we orient the RD running variable, r_i , such that it is always positive on the STEM side of a cutoff:

$$r_i = \begin{cases} t_i - c_j & \text{if STEM-above,} \\ c_j - t_i & \text{if HSS-above.} \end{cases}$$

We then fit the following model via OLS:

$$Y_i = \alpha_j + \beta_{f_j} \cdot \mathbb{1}\{r_i > 0\} + \gamma_{f_j} \cdot r_i \cdot \mathbb{1}\{r_i < 0\} + \delta_{f_j} \cdot r_i \cdot \mathbb{1}\{r_i > 0\} + \epsilon_i, \quad (1)$$

for $|r_i| < b$ and $r_i \neq 0$. In (1), α_j is a cutoff fixed effect, γ_{SH} , γ_{HS} , δ_{SH} , and δ_{HS} are coefficients on a linear spline in the running variable that varies by cutoff type, and b is a chosen bandwidth. β_{SH} and β_{HS} are the parameters of interest: they are cutoff-type-specific coefficients on an indicator

17. To see this, suppose a student is assigned to STEM: if the student is close to getting into a more selective HSS track, the student will be in the sample for an HSS-above cutoff; if the student is close to falling back to a less selective HSS track, the student will be in-sample for a STEM-above cutoff. The opposite holds for students assigned to HSS. Also, a student may not be in the sample for any cutoff if they are not at risk of switching between STEM and HSS. This occurs if the student does not rank both STEM and HSS tracks, if they do but in an irrelevant manner (such as ranking a less selective track of one type before a more selective track of the other type), or if their transition score is far from all applicable cutoff scores.

18. When $t_i = c_j$, additional tie-breaking rules apply. As such, we drop students who fall exactly on a cutoff.

for being on the STEM side of a cutoff, $r_i > 0$. Thus, they measure the effects of STEM (vs. HSS) assignment for STEM-above and HSS-above cutoffs, respectively.

To estimate (1), we define a cutoff as a combination of a cutoff type, cutoff score, and admission year. This definition reflects that our treatment of interest is assignment to a curricular category (i.e., STEM or HSS), not a specific track. Unless otherwise noted, we use a bandwidth of 1.25 transition score points. The transition score is on a scale of 1 to 10, and 1.25 is 77% of the standard deviation in the administrative data (Appendix Table A1). The appendix shows that our results are robust to a variety of alternative specifications.¹⁹ We cluster standard errors in two ways, by student and by cutoff.

In addition to estimating effects by cutoff type, we estimate an average effect over both cutoff types: $\beta = (\beta_{SH} + \beta_{HS})/2$. We call β the average treatment effect (ATE) of STEM assignment, as it is for both students who prefer STEM and those who prefer HSS. We estimate β by plugging in estimates of β_{SH} and β_{HS} , and we calculate standard errors using the Delta Method.²⁰

We underline that the effects we estimate (β_{SH} , β_{HS} , and β) are more general than those typically associated with RD designs. First, we have a large number of cutoffs—almost 2,000 in our largest sample—and these cover a wide achievement range (Appendix A1). In RD designs with multiple cutoffs, the overall effect is a weighted average of cutoff-specific effects (Cattaneo et al. 2016; Bertanha 2020). Thus, β_{SH} , β_{HS} , and β are not local to a single achievement level; instead, they are averages over our diverse set of cutoffs. Second, one might worry that students near STEM vs. HSS cutoffs are unrepresentative. Namely, their willingness to rank both STEM and HSS may be because they are somewhat balanced in their STEM vs. HSS abilities and preferences. If so, our results may hold little relevance for students who lean strongly toward one field. In contrast with this worry, we find that the RD samples are quite representative, although tilting high-achieving on average. Means and standard deviations in the RD samples are similar to those for all students in the top 70% of the national achievement distribution (Appendix Table A1).²¹ The reason for the representativeness is that ranking both STEM and HSS tracks is common: among the highest-achieving 70% of the administrative data, 80% of students do

19. These include: (i) adding controls for student characteristics, (ii) using alternative bandwidths, including MSE-optimal bandwidths from Calonico et al. (2017), (iii) fitting global quartic polynomials instead of local linear splines, and (iv) altering the fixed effects, α_j . Following Abdulkadiroglu et al. (2017) and Abdulkadiroglu et al. (2022), we calculate the optimal bandwidths by cutoff; these usually range between 1 and 1.25.

20. Since Equation (1) is fully interacted with cutoff type, it could be broken into separate regressions by cutoff type. The reason for a single regression is to permit using the Delta Method. Also, the approach of defining β as the average of β_{SH} and β_{HS} is desirable because it ensures that β does not depend on the shares of students at STEM- and HSS-above cutoffs. In our setting, these shares are almost even (51 vs. 49 percent, Appendix Table A2). Thus, we would get similar results if we simply fit a single coefficient on $\mathbb{1}\{r_i > 0\}$ in (1).

21. The high average achievement is because Romania’s lowest-achievers usually attend vocational/technical tracks, not STEM or HSS. Key evidence for the representativeness of the RD samples is that standard deviations of math and language transition exam scores in these samples match those for the administrative-data top 70%. If the RD samples were filled with students who are balanced in math vs. language, the math and language standard deviations would instead be smaller in the RD samples than in the relevant administrative data.

(Appendix Table A3). This likely reflects that students care about a variety of track attributes beyond curriculum (travel time, teaching quality, prestige, whether siblings/friends attend, etc.) and that students often face situations where they must give up their preferred curriculum in exchange for such attributes (Abdulkadiroglu et al. 2020; Ainsworth et al. 2023).

4.1 Characterizing the treatment

For a given cutoff type, assignment to STEM vs. HSS involves three categories of changes in the high school experience, which we call “sub-treatments”. First, a change in curriculum: i.e., more focus on STEM subjects and less on HSS subjects, together with more (less) time with the teachers who teach STEM (HSS) subjects. Second, a change in horizontal dimensions of track peer composition, such as share female or relative academic strength in math vs. language. Third, a change in whether a student scored above or below a cutoff (together with all attendant changes, such as in vertical dimensions of track quality).

All three sub-treatments are potentially of interest. The first is relevant to policymakers contemplating a curriculum reform, such as shifting emphasis between STEM and HSS subjects. The second, in combination with the first, is relevant both to students choosing among schooling options in a tracked school system and to policymakers considering changing the number of seats in different track types. The third has been explored in prior literature, both in Romania (Pop-Eleches and Urquiola 2013, Goff et al. 2023, and Malamud et al. 2025) and other settings (e.g., Abdulkadiroglu, Angrist, and Pathak 2014). Thus, our interest here is in the first two.

Our approach for eliminating the third sub-treatment is to estimate the average effect of STEM assignment over both STEM- and HSS-above cutoffs (i.e., we estimate β , not just β_{SH} and β_{HS}). For STEM-above cutoffs, STEM assignment coincides with scoring above a cutoff, while for HSS-above it occurs when scoring below. Thus, on average over the two cutoff types, STEM assignment is unrelated to scoring above or below, and β captures only the effects of the first two sub-treatments (we present estimates in Section 5.2). Appendix A4 probes possible limitations of our approach using a potential outcomes framework. In addition, it provides evidence that the third sub-treatment may not matter even for β_{SH} and β_{HS} : using cutoffs between tracks with the same curricula, we find that scoring above vs. below a cutoff hardly affects the outcomes we consider.²²

For simplicity, in the main text we do not disentangle the relative importance of the first two sub-treatments. Instead, we focus on the impact of curricular assignment, which incorporates both the curriculum and peer changes associated with placement in STEM vs. HSS.

Appendix A5 presents an attempt to separate the roles of curriculum and peers. We use two strategies, which we summarize here. First, we modify Equation (1) by adding controls for

22. The prior studies in Romania do find impacts from scoring above a cutoff. However, they calculate effects using all cutoff types, not just same-curriculum cutoffs.

attributes of the peer composition in students’ assigned tracks. To the extent that peers matter, the estimated effects of STEM assignment should diminish once we adjust for the fact that STEM and HSS tracks have different types of peers. Second, we conduct a cutoff-level analysis. In a first step, we run cutoff-specific RD regressions both for the paper’s outcomes of interest and for measures of track peer composition. Then, for each outcome of interest, we regress the cutoff-specific effects for the outcome on the cutoff-specific effects for the track peer attributes, weighting by the number of students in a cutoff’s RD sample. The intercept in this regression is the predicted effect of STEM assignment for cutoffs with no change in peer composition across the RD threshold. Thus, for this approach, we can gauge the relative importance of curriculum versus peers by comparing the intercept estimates with our main estimates.

Overall, we find that controlling for peer composition hardly matters. This suggests that the change in curriculum is the key sub-treatment for effects on the outcomes we study. (We discuss the one notable exception to this in Section 6.5.)

5 Initial results

We now present some initial results that shed light on our setup.

5.1 RD samples and validity tests

Since we use outcomes from different datasets, our analysis draws on multiple RD samples. We summarize the samples and conduct validity tests on them in Appendix A3. First, we rule out bunching of the running variable on the STEM or HSS sides of the RD threshold. Second, we show that the samples have good covariate balance across the threshold.²³ Third, we show that STEM vs. HSS assignment is not associated with differential selection into survey response. Fourth, we show that the RD samples for the survey data are representative of those for the administrative data. Namely, covariate means and standard deviations are similar, as are treatment effects on outcomes that can be observed for all samples. Finally, to test survey quality, we show that reported baccalaureate performance from the survey data closely matches actual performance from the administrative data.

5.2 Evidence on sub-treatments

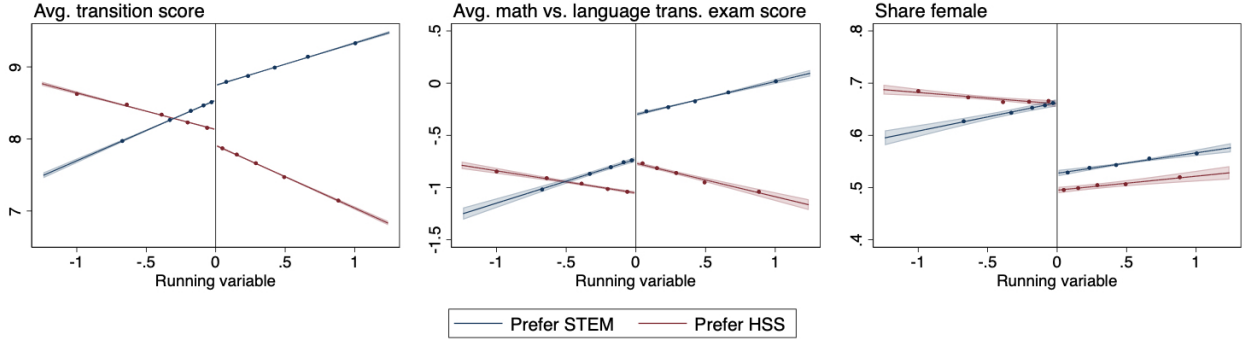
Figure 1 presents evidence on the sub-treatments associated with placement in STEM vs. HSS. It displays RD plots of the effects of STEM assignment on three track peer attributes: absolute ability (as measured by the average transition score), relative academic strength (as measured by the average difference in math vs. language transition exam scores), and share female.

The RD plots are obtained by fitting Equation (1) and provide a graphical illustration of the mechanics of this regression equation. The two colors in the plots differentiate cutoff types, with

23. The lack of evidence for running variable manipulation is expected—transition exams are graded non-locally, and cutoffs are endogenously determined by the assignment system after students submit their preference rankings.

blue for STEM-above (“Prefer STEM”) and red for HSS-above (“Prefer HSS”). For each cutoff type, the lines are the estimated spline in the running variable, and the vertical gap between the right and left lines at the RD threshold is the estimated effect of STEM assignment. In particular, the gap in the blue lines is the estimate of β_{SH} and that in the red lines is the estimate of β_{HS} . Our estimate of β is the simple average of the two vertical gaps.²⁴

Figure 1: RD plots of effects on peer composition in students’ assigned tracks



The figure displays RD plots for effects on three variables related to the peer composition in students’ assigned tracks. See the text of Section 5.2 for definitions of these variables. “Prefer STEM” (“Prefer HSS”) refers to observations for STEM-above (HSS-above) cutoffs. For Prefer STEM, the running variable is a student’s transition score minus their cutoff’s cutoff score; for Prefer HSS, it is the cutoff score minus the transition score. Thus, in both cases, students with positive (negative) running variable values are assigned to STEM (HSS). The blue and red lines are linear spline estimates from Equation (1), and the shaded areas are 95% confidence intervals. The dots are estimates from a version of Equation (1) that replaces the linear spline running variable controls with cutoff-type-specific indicators for bins in the running variable. The bins are constructed to have equal populations by cutoff type and curricular assignment, and the dots are plotted at the mean running variable value within a bin. The estimated effect of STEM assignment for Prefer STEM (Prefer HSS) is equal to the vertical difference between the right and left blue (red) lines at the RD threshold. The estimated ATE of STEM assignment is the simple average of the two vertical differences. The plots use the administrative data for the 2015-2017 and 2019 cohorts and a bandwidth of 1.25 transition score points.

Regression results corresponding to Figure 1 are in Table 2, with Panel A being for β and Panels B and C for β_{SH} and β_{HS} , respectively. In all panels, “STEM” is the coefficient estimate and standard error for the effect of STEM assignment, while “Intercept” is the predicted value of the outcome variable for students at the RD threshold who are assigned to HSS.

Figure 1 and Table 2 yield two takeaways. First, conditional on cutoff type, placement in STEM rather than HSS entails a change in both vertical and horizontal dimensions of track peer composition. For STEM-above cutoffs, it means gaining peers who have higher transition scores, are relatively more oriented toward math vs. language, and are less likely to be female. For HSS-above cutoffs, it again leads to peers who are more mathematically oriented and less female, but now with lower transition scores. The discrepancy in effects on peer absolute ability is because STEM assignment occurs when scoring above a cutoff for STEM-above cutoffs and when scoring below for HSS-above cutoffs.

24. Since Figure 1 is an exact representation of Equation (1), it estimates separate linear splines by cutoff type. In the appendix, we instead present RD plots that fit a single spline over both cutoff types. These plots do not give insight into β_{SH} or β_{HS} , but yield an estimate of an alternative version of β . As a robustness check, we provide the plots for all of the paper’s main outcomes.

Table 2: Regression results for effects on peer composition in students’ assigned tracks

	Average		Share female
	Transition score	Math vs. language transition exam score	
<i>Panel A: Average treatment effects</i>			
STEM	-0.007 (0.007)	0.357*** (0.010)	-0.150*** (0.004)
Intercept	8.33	-0.89	0.66
Std. dev.	0.92	0.75	0.15
Student-cutoffs	73,470	73,470	73,470
<i>Panel B: Effects for Prefer STEM</i>			
STEM	0.207*** (0.010)	0.429*** (0.015)	-0.135*** (0.006)
Intercept	8.54	-0.73	0.66
Std. dev.	0.75	0.66	0.15
Student-cutoffs	37,376	37,376	37,376
<i>Panel C: Effects for Prefer HSS</i>			
STEM	-0.221*** (0.010)	0.284*** (0.012)	-0.166*** (0.005)
Intercept	8.13	-1.05	0.66
Std. dev.	0.91	0.73	0.15
Student-cutoffs	36,094	36,094	36,094

The table presents results from estimating Equation (1) for three variables describing the peer composition in students’ assigned tracks; see Section 5.2 for variable definitions. Panel A is for the ATE of STEM assignment, β . Panel B (C) is for the effect for students at STEM-above (HSS-above) cutoffs, β_{SH} (β_{HS}). The sample information in Panels B and C is for the specified cutoff type. “STEM” is the coefficient estimate and standard error for β , β_{SH} , or β_{HS} . “Intercept” is the predicted value of the outcome for students at the RD threshold who are assigned to HSS. In Panels B and C, it is the mean of the cutoff fixed effects for the listed cutoff type; in Panel A, it is the simple average of the means for the two cutoff types. “Std. dev.” is the standard deviation of the outcome among students in a given RD sample. “Student-cutoffs” is the number of observations. The table uses the administrative data for the 2015-2017 and 2019 cohorts and a bandwidth of 1.25 transition score points. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Second, when we average over the two cutoff types, we eliminate the association between curricular assignment and peer absolute ability; however, we do not remove the link between curricular assignment and the horizontal peer attributes. Specifically, the ATE of STEM assignment on the average transition score in a student’s assigned track is essentially zero; this is because the positive effect for STEM-above cutoffs cancels with the negative effect for HSS-above cutoffs. In contrast, ATEs for relative academic strength and share female are non-zero; this is because effects on these outcomes go in the same direction for both cutoff types.

In sum, curricular assignment is a compound treatment. Conditional on cutoff type (β_{SH} and β_{HS}), it is the combination of a curriculum change, a change in horizontal dimensions of peer composition, and all changes associated with scoring above or below a cutoff (such as in peer absolute ability). When averaged over the two cutoff types (β), it is the combination of the first two sub-treatments. Despite this compound nature, we show in the appendix that the curriculum change is the main factor.

5.3 The first stage on curricular exposure

Next, we quantify the first-stage impact of curricular assignment on curricular exposure. Using the paper’s administrative data, we estimate how placement in STEM rather than HSS affects enrollment in and graduation from different curricula. Table 3 presents results for β for all students. Heterogeneity by initial curricular preferences (i.e., β_{SH} and β_{HS}), relative academic strength in math vs. language, gender, and baseline achievement is in Appendix A6.

Table 3: Effects on high school enrollment and graduation

	Years of enrollment				Graduate			
	All	STEM	HSS	Other	All	STEM	HSS	Other
STEM	0.002 (0.007)	3.08*** (0.023)	-3.09*** (0.023)	0.015** (0.006)	-0.003 (0.003)	0.694*** (0.007)	-0.703*** (0.008)	0.007*** (0.002)
Intercept	3.91	0.50	3.36	0.05	0.96	0.16	0.80	0.01
Std. dev.	0.41	1.89	1.88	0.36	0.18	0.50	0.49	0.12
Student-cutoffs	55,221	55,221	55,221	55,221	55,221	55,221	55,221	55,221

The table presents results for the ATE of STEM assignment, β , for outcomes related to high school enrollment and graduation. The table uses the administrative data for the 2015-2017 cohorts, as enrollment histories for the 2019 cohort are incomplete. “Years of enrollment” is the number of years that a student is enrolled in a track of a given type. “Graduate” is whether the student graduates from a track of the given type. Outcomes are measured within four years of track assignment; see Appendix Table A17 for results that do not restrict outcome timing. Other details are the same as in Panel A of Table 2.

Table 3 shows that curricular assignment has large effects on what students study in high school. Being placed in STEM rather than HSS induces students to have an average of 3.1 years of additional STEM enrollment and correspondingly fewer years of HSS enrollment, out of the 4 years of high school. It also makes students about 70 pp more (less) likely to graduate in STEM (HSS). In contrast, placement in STEM vs. HSS on average does not affect the total number of years a student is enrolled in high school or whether the student graduates (“All” columns). It also has little impact on enrollment in or graduation from tracks with non-STEM and non-HSS curricula (“Other” columns).²⁵

Appendix A6 shows that the effects on high school enrollment and graduation are similar across student types and are highly robust.

In short, in Romanian high schools, curricular assignment has a strong but imperfect first stage on curricular exposure. This is consistent with the fact that transferring to a track with a different curriculum is difficult. An implication of the size of the first stage is that the effects we report constitute the impact of about 3 years of STEM vs. HSS enrollment, not 4. To gauge the impact of spending all of high school in STEM, one can multiply the estimated effects by 4/3.²⁶

25. Making use of the intercept values, one can see that 96% of students at STEM vs. HSS cutoffs graduate. For HSS-assigned students at these cutoffs, 16% graduate in STEM, 80% graduate in HSS, and 1% graduate in other track types. For STEM-assigned students, the analogous values are 85%, 10%, and 2%.

26. In theory, one could instrument for enrollment using assignment. We do not do so because we cannot fully observe enrollment (or graduation) for the 2019 cohort, which is the basis for our surveys.

6 Main results

We now turn to impacts on the paper’s outcomes of interest. We divide our results into five subsections, which correspond to the five findings discussed in the introduction.

6.1 College and career outcomes

We begin by assessing how STEM vs. HSS curricular assignment affects college and career outcomes. Our analysis extends the existing literature in several ways: the average treatment effects that we estimate average over a wide variety of student types; we examine heterogeneity across these student types; we confirm that effects are due to curriculum and not correlated factors; and we consider a contemporary setting.

To start, Table 4 presents effects on college enrollment for all students. Panel A relates to students’ college intentions at the end of high school—which is before students learn where they are admitted—while Panel B concerns actual enrollment a year after high school. The outcomes in Panel A are from the end-of-high-school survey, while those in Panel B use data from both the first follow-up and peer surveys. In all of the surveys, we gave students a list of options and asked them to select the option that best represents their intentions or enrollment status.

Table 4: Effects on college plans and enrollment

	Any	STEM				Hum., law, or social science				Other/ unsure
		Any	Math or CS	Natural science	Business or econ.	Any	Humanities	Law	Social science	
Panel A: College plans at the end of high school										
STEM	0.010 (0.018)	0.233*** (0.035)	0.147*** (0.024)	0.041* (0.024)	0.046* (0.027)	-0.224*** (0.032)	-0.075*** (0.020)	-0.057*** (0.021)	-0.092*** (0.020)	0.001 (0.023)
Intercept	0.87	0.31	0.08	0.10	0.14	0.39	0.13	0.12	0.14	0.16
Student-cutoffs	3,987	3,987	3,987	3,987	3,987	3,987	3,987	3,987	3,987	3,987
Panel B: College enrollment one year after high school										
STEM	-0.037 (0.025)	0.246*** (0.036)	0.192*** (0.031)	0.015 (0.022)	0.039 (0.026)	-0.261*** (0.031)	-0.123*** (0.021)	-0.092*** (0.021)	-0.046*** (0.017)	-0.022 (0.027)
Intercept	0.82	0.28	0.11	0.08	0.10	0.38	0.17	0.13	0.08	0.16
Student-cutoffs	3,327	3,327	3,327	3,327	3,327	3,327	3,327	3,327	3,327	3,327

The table presents results for the ATE of STEM assignment, β , for outcomes related to college plans and enrollment. Outcomes are indicator variables. Those in Panel A are from the end-of-high-school survey; those in Panel B are from either the first follow-up survey (for students who responded to this survey) or the peer survey (for students who did not respond to the first follow-up). “Any” equals 1 if a student plans to attend college (Panel A) or is enrolled in college (Panel B). The remaining outcomes equal 1 if a student plans to attend or is enrolled in a college program of the listed type. Other details match those in Panel A of Table 2.

Table 4 offers two main findings. First, averaged over all students, curricular assignment has little influence on either planning to attend college or actually being enrolled in college (“Any” column). By contrast, it has large impacts on students’ field choices (remaining columns). Specifically, placement in STEM rather than HSS makes students 23 pp more likely to want a STEM college program when asked at the end of high school and 25 pp more likely to be enrolled

in a STEM program a year after high school. It reduces plans for (enrollment in) humanities, law, or social science programs by 22 (26) pp. The shift toward STEM is driven by a shift into math or computer science programs, although effect estimates are also positive for the natural sciences and for business or economics. The shift away from humanities, law, or social science is somewhat evenly distributed across program types. Finally, STEM vs. HSS assignment has no effect for an “Other/unsure” option that we included in our survey questions.

The results in Table 4 are robust to alternative specifications and hold when we use the same sample for the two panels (Appendix A7).²⁷ In addition, as always, the results change little when we implement strategies that control for track peer composition (Appendix A5).

Appendix Table A19 shows how the college effects vary by student type. It reveals that there is little heterogeneity by initial curricular preferences (β_{SH} and β_{HS}), relative academic strength, or gender, indicating that college field of study is malleable to high school curricular assignment for many types of students. (We defer discussion of heterogeneity by baseline achievement until the next section.) We explore the reasons behind this malleability in Section 6.3.

Table 5 presents effects on the careers that students want. Outcomes are based on free-response questions in the first follow-up survey; they thus reflect desires a year after high school. We manually coded students’ responses into the five career categories presented in the table.²⁸

Table 5: Effects on desired careers one year after high school

	Technology or engineering	Medicine	Business or economics	Art, education, law, or social services	Other/ unsure
STEM	0.254*** (0.051)	-0.041 (0.044)	-0.040 (0.063)	-0.145*** (0.046)	-0.028 (0.040)
Intercept	0.10	0.17	0.33	0.23	0.17
Student-cutoffs	1,159	1,159	1,159	1,159	1,159

The table presents results for the ATE of STEM assignment, β , on students’ desired careers. Outcomes are from the first follow-up survey and are indicators equal to 1 if a student’s desired career falls into the listed category. Other details follow Panel A of Table 2.

As before, Table 5 shows that high school curricular assignment is impactful. Being placed in STEM rather than HSS on average makes students 25 pp more likely to want a career in technology or engineering and 15 pp less likely to want one in art, education, law, or social services. Interestingly, STEM assignment has insignificant effects on wanting a career in medicine or in business or economics. It also has little effect on the probability of giving responses that we categorized as “Other/unsure”. These results are again robust (Appendix A7).

Appendix Table A25 displays heterogeneity in the career effects, and here we do find some. The boost in wanting a technology or engineering career is similar by relative academic strength;

27. We repeated our questions about college enrollment in the second follow-up survey, which occurred 1.5 years after high school. Effects remain similar (Appendix Table A22).

28. The questions asked students what industry and specific job they want to work in.

however, it is smaller, although still statistically significant and sizable, for students who initially preferred HSS (18 pp) vs. those who preferred STEM (33 pp) and for females (18 pp) vs. males (36 pp). In contrast, the reduction in wanting an art, education, law, or social services career is similar in size over each of these student groups. We again defer discussion of heterogeneity by baseline achievement until the next section.

Overall, our results show that changing what students study in high school can have large effects on their college and career outcomes—and that this is the case for many types of students. The finding of some heterogeneity for effects on career choices—but not college choices—is consistent with research showing that STEM pipelines can be leaky. Specifically, it aligns with evidence that career choices are subject to additional considerations beyond college choices, such as tastes for work hours and conditions, which may vary across student types (Patnaik, Wiswall, and Zafar 2020). As noted, our results add to an existing literature on the effects of high school STEM exposure on students’ trajectories. Unfortunately, we cannot precisely benchmark our magnitudes to this literature: most papers consider treatments that are less intensive than ours, while Dahl, Rooth, and Stenberg (2023) do not provide estimates for college or career effects.

6.2 Difficulty and risk

We next continue studying the effects of curricular assignment on students’ trajectories, but with a specific focus on the fact that STEM and HSS may differ in terms of difficulty and risk. Our analysis is motivated by descriptive research that suggests that STEM subjects are more strenuous and time-intensive than other fields and that the elevated difficulty of STEM may explain leaks in the STEM pipeline (e.g., Stinebrickner and Stinebrickner 2014 and Ahn et al. 2024). We contribute by studying this topic causally and for various types of students.

We begin by using our survey data to provide evidence that STEM is indeed harder than HSS. First, placement in STEM rather than HSS spurs students to report spending more time on homework on a typical weekday in high school (Appendix Table A27). Over all students, the estimated effect is 0.25 hours per day, or 15% of the sd of homework time in the RD sample. Effects are larger for students with low baseline achievement, but otherwise exhibit little heterogeneity. Next, STEM assignment also leads students to score higher on a grit index, implying again that it demands more effort (Appendix Table A31). As before, the all-student effect is equal to 15% of the sd of the outcome in the RD sample. Interestingly, effects are similar by baseline achievement but are larger for males than females. Third, when students were asked a year after high school, STEM assignment makes them more convinced that the baccalaureate exam for their curriculum is difficult in comparison to the exams for other curricula; impacts are pronounced for males and low-achievers (defined as students who are in-sample for cutoffs with cutoff scores of 8 or less). In contrast, STEM assignment does not affect whether students believe their track provides good preparation for the baccalaureate exam, suggesting that the

difficulty of STEM is not due to poor teaching quality in STEM tracks; here, effects are small for all student types.²⁹

We next examine whether the difficulty of STEM affects students’ trajectories. We begin by using the administrative data to see how STEM vs. HSS curricular assignment influences performance on the baccalaureate exam. As mentioned, this exam is the capstone to high school and is a key factor in college admissions.

Table 6: Effects on baccalaureate performance

	Take the exam	Pass the exam	Exam score	Pass in		
				STEM	HSS	Other
STEM	-0.011*** (0.003)	-0.036*** (0.004)	-0.351*** (0.015)	0.621*** (0.007)	-0.698*** (0.007)	0.040*** (0.004)
Intercept	0.95	0.92	8.08	0.13	0.78	0.01
Std. dev.	0.21	0.29	1.21	0.50	0.49	0.16
Student-cutoffs	73,470	73,470	69,938	73,470	73,470	73,470

The table presents results for the ATE of STEM assignment, β , on performance on the baccalaureate exam. The table uses the administrative data for the 2015-2017 and 2019 cohorts. “Exam score” is on a scale of 1-10 and is available only for students who took the exam. All other outcomes are indicators. The outcomes for the “Pass in” columns equal 1 if a student passed the version of the exam for the listed track type. Outcomes reflect a student’s highest-scoring exam attempt within four years of track assignment; see Appendix Table A35 for results that do not restrict outcome timing. Other details follow Panel A of Table 2.

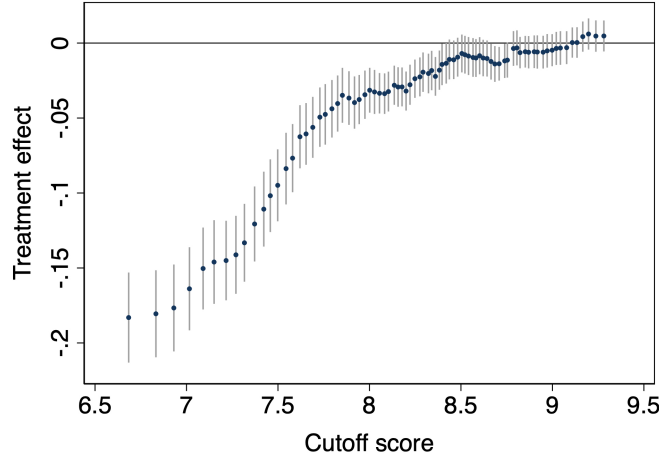
Table 6 presents effects on baccalaureate performance for all students. The table shows that STEM assignment, on average, leads to worse performance. First, it makes students slightly less likely to take and pass the exam—by 1 and 4 pp, respectively. Second, it also causes students to receive worse scores conditional on taking the exam—a 0.35 point reduction on a scale of 1-10, equal to 29% of the sd for students in the RD sample. On the other hand, STEM assignment substantially increases the chance of passing the version of the exam for STEM tracks; this is not surprising given that it strongly raises the probability of being in a STEM track at the end of high school (Table 3).

Appendix Table A34 displays heterogeneity for the effects on baccalaureate performance. It shows that effects are reasonably similar by initial curricular preferences, relative academic strength, and gender, which is consistent with the finding that there is little heterogeneity for college outcomes along these dimensions. By contrast, effects differ by baseline achievement. For low-achievers, effects are severe: STEM assignment makes these students 4 pp less likely to take the baccalaureate exam and 11 pp less likely to pass it; further, it reduces their scores conditional on taking the exam by 0.50 points and boosts their chance of passing in STEM by only 56 pp (while lowering their chance of passing in HSS by 71 pp). For high-achievers, STEM assignment hardly affects the probability of taking or passing the exam, although it continues to reduce exam scores (by 0.30 points).

Figure 2 elaborates on the heterogeneity by baseline achievement. It shows how effects on

29. Robustness checks for this section are in Appendices A8 and A9.

Figure 2: Heterogeneity in effects on passing the baccalaureate exam by cutoff score



The figure displays heterogeneity by cutoff score in the effect of STEM assignment on the probability of passing the baccalaureate exam. The figure uses the administrative data for the 2015-2017 and 2019 cohorts. Each dot and line is the point estimate and 95% confidence interval for β for 20% of the full RD sample. The left-most value is for the fifth of observations with the least selective cutoff scores. The remaining values are for fifths that are successively shifted up the cutoff score distribution by one percentile. Values are plotted at the mean cutoff score in each sub-sample. Other details are the same as in Table 6.

passing the exam vary across rolling windows of the distribution of cutoff scores. Beginning with the bottom fifth of observations by cutoff-score selectivity, the figure moves up the cutoff score distribution one percentile at a time, plotting effect estimates and 95% confidence intervals at the mean cutoff score within each window. The figure shows that effects on passing the exam vary widely by achievement. Effects are zero at the most selective cutoffs but become increasingly negative at less selective cutoffs. For the lowest-achieving students, STEM assignment reduces the probability of passing by almost 20 pp.³⁰

The drop in baccalaureate performance due to STEM assignment would be expected to affect college outcomes, given the importance of the baccalaureate exam in college admissions. This is especially the case for low-achievers, who become less likely to pass the exam, a requirement for progressing to college. Accordingly, our survey data shows that STEM assignment decreases low-achievers' chance of being enrolled in college a year after high school by 9 pp (Appendix Table A19). This is despite only modestly affecting their intention to attend college at the end of high school. The reduction in college enrollment is driven by reduced enrollment in public colleges, which in Romania are more prestigious than privates (Appendix Table A24). In

30. It's worth reiterating that STEM assignment reduces scores on the baccalaureate exam for all students, even those with high baseline achievement. This means that the difference in effects on exam passage for low- and high-achievers is mainly because the two groups differ in their distance to the passing threshold. That said, the effect on exam scores is larger for low- than high-achievers. This difference could arise if low-achievers are especially mismatched to STEM material or if less selective STEM tracks have unusually low value added. Our survey evidence favors the mismatch story. As mentioned, the effect of STEM assignment on thinking one's version of the baccalaureate exam is hard is bigger for low- than high-achievers; meanwhile, the effect on believing one's track provides good baccalaureate preparation is similar for the two groups (Appendix Table A31).

addition, STEM assignment makes low-achievers 24 pp less likely to receive a college scholarship. For high-achievers, STEM assignment does not affect overall college enrollment or enrollment in specific college types; however, it modestly influences the method of paying for college, reducing the chance of receiving a (full) scholarship while increasing the chance of getting a tuition waiver.

The heterogeneity in effects on baccalaureate and college outcomes for low- and high-achievers also appears in effects on desired careers (Appendix Table A25). For both groups, STEM assignment shifts students away from wanting a humanities-related career. However, for high-achievers the shift is entirely toward a technology or engineering career, while for low-achievers it is split between technology/engineering and business/economics.

The results in this subsection highlight that STEM assignment is risky, though mainly for students with low baseline achievement. These students are already marginal in terms of achieving key milestones, such as obtaining a baccalaureate diploma or attending college. The negative shock from the elevated difficulty of STEM lowers their chance of doing so. Thus, our results causally confirm and refine the findings from the existing descriptive literature: STEM is harder than other fields, and this rigor does contribute to leaks in STEM pipelines; however, the degree of risk differs markedly across the achievement distribution. Our results are somewhat dependent on the fact that Romania’s education system has a binding performance threshold; nonetheless, this caveat seems minor, as thresholds are common throughout the world.

6.3 Beliefs and Preferences

We next turn to effects on students’ core traits, starting with beliefs about abilities and preferences over academic subjects and job tasks. Effects on beliefs and preferences are important in their own right; in addition, they are potential mechanisms explaining how high school curricular assignment affects college and career outcomes. As part of our investigation into their role as mechanisms, we also evaluate a few alternative channels. We believe we are the first to study the causal effects of STEM vs. HSS curricular assignment on students’ core traits; thus, the effects in this subsection and in Sections 6.4-6.5 are novel contributions.

We begin with effects on students’ self-perceived strength in and enjoyment of academic subjects. We make use of survey questions where students were asked to rate subjects on these dimensions on a scale of 1 to 5. The questions dealt with subjects at both the high school and college levels.

Table 7 presents all-student effects on ratings of high school subjects. Panels A and B concern students’ beliefs about their abilities in the listed subjects, with Panel A reflecting beliefs at the end of high school and Panel B capturing beliefs one year after high school. Panel C concerns students’ preferences over the subjects; these are available only one year after high school, as we did not ask about preferences in the end-of-high-school survey.

Panels A and B of Table 7 show that STEM vs. HSS curricular assignment powerfully affects

Table 7: Effects on beliefs about own abilities and preferences: high school subjects

	STEM subjects	Humanities subjects	Social studies subjects
<i>Panel A: Beliefs at the end of high school</i>			
STEM	0.873*** (0.077)	-0.235*** (0.064)	-0.565*** (0.074)
Intercept	2.60	4.07	3.98
Std. dev.	1.29	0.90	1.15
Student-cutoffs	3,987	3,987	3,987
<i>Panel B: Beliefs one year after high school</i>			
STEM	0.744*** (0.124)	-0.267*** (0.087)	-0.745*** (0.139)
Intercept	2.96	4.20	4.06
Std. dev.	1.10	0.84	1.13
Student-cutoffs	1,159	1,159	1,159
<i>Panel C: Preferences one year after high school</i>			
STEM	1.01*** (0.151)	-0.405*** (0.102)	-0.697*** (0.135)
Intercept	2.72	4.13	4.08
Std. dev.	1.37	1.04	1.21
Student-cutoffs	1,159	1,159	1,159

The table presents results for the ATE of STEM assignment, β . Outcomes are on a scale of 1-5. Those in Panels A and B are students' ratings of their abilities in the listed high school subjects. Those in Panel C are students' ratings of how much they like the subjects. The Panel A outcomes are from the end-of-high-school survey. The remaining outcomes are from the first follow-up survey and thus are measured a year after high school. Other details are the same as in Panel A of Table 2.

how strong students feel in different subjects. By the end of high school, placement in STEM induces students to give substantially higher ratings to their abilities in STEM subjects (by 0.87 points on average, or about two-thirds of the sd in the RD sample) and lower ratings to their abilities in humanities subjects (0.26 sd) and social studies subjects (0.49 sd). In the year after high school, treatment effects shift in a negative direction, which is to be expected given that (a) students learn their baccalaureate performance in this period and (b) STEM assignment reduces baccalaureate performance. Nonetheless, STEM assignment continues to generate a positive and sizable boost in students' self-confidence in STEM (0.74 points, on average).³¹

Panel C of Table 7 shows that curricular assignment also affects how much students like different subjects. In particular, being assigned to STEM rather than HSS spurs students to rate their enjoyment of STEM subjects higher (by an average of 1 point, or 0.74 sd) and their enjoyment of humanities and social studies subjects lower (by 0.39 and 0.58 sd, respectively).

Table 8 is analogous to Table 7, but for college subjects. The outcomes are from the first follow-up survey (one year after high school) and are missing for students who had not attended college by the time of this survey—as we did not want to ask students about college if they had never been to college. We deal with the missing responses by imputing values of 1; results are

31. A complication with comparing Panels A and B of Table 7 is that they use different samples. Appendix Table A38 replicates the analysis using a consistent sample and confirms the negative shift in treatment effects.

similar if we instead exclude them (Appendix Table A41).

Table 8 shows that the effects on beliefs and preferences extend to the college level. STEM assignment makes students feel stronger in college STEM subjects and like them more (in both cases, by an average of about 0.5 sd). Conversely, it makes students feel weaker in college subjects that involve reading, writing, and memorization, and like these subjects less.

Table 8: Effects on beliefs about own abilities and preferences: college subjects

	STEM subjects	Subjects that involve reading and writing	Subjects that involve memorization
<i>Panel A: Beliefs one year after high school</i>			
STEM	0.659*** (0.151)	-0.396** (0.154)	-0.278* (0.153)
Intercept	2.57	3.23	3.17
Std. dev.	1.39	1.32	1.32
<i>Panel B: Preferences one year after high school</i>			
STEM	0.776*** (0.149)	-0.525*** (0.164)	-0.372** (0.157)
Intercept	2.48	3.24	3.06
Std. dev.	1.45	1.31	1.31
Student-cutoffs	1,159	1,159	1,159

The table is analogous to Table 7 but for college subjects, not high school subjects. Outcomes are from the first follow-up survey. The relevant questions were not asked to students who had never attended college by the time of this survey. We impute values of 1 for these students; see Appendix Table A41 for results without imputations. The number of observations is the same across panels, so it is displayed in a single row at the bottom of the table. Other details match those in Table 7.

Appendix A10 shows that the belief and preference effects differ little across student types and are highly robust. In total, the results indicate that what a student studies in high school shapes core features of the student—both how confident the student is in different areas and what the student likes.

An important question is whether the effects on beliefs reflect actual skill acquisition or just changes in students’ perceptions of their skills. To explore this, we exploit the fact that the Romanian language and literature component of the baccalaureate exam is common to all track types. Appendix Table A43 shows that STEM assignment reduces Romanian baccalaureate scores by 0.25 sd, on average. This is almost the exact magnitude of the effect on students’ beliefs about their humanities abilities in Panel A of Table 7, which suggests that impacts on beliefs may mostly reflect skill acquisition. In a similar vein, we find that STEM assignment lowers the amount of time that students spend reading on a typical weekday, both during high school and 1.5 years later (Appendix Tables A27 and A29). In contrast, it does not seem to affect the types of books or movies that students enjoy (Appendix Table A43). The reduction in Romanian baccalaureate scores is similar for all student types, while the effect on time spent reading differs by gender, being larger for males than females.³²

32. Appendix Table A43 also presents effects on performance on the non-Romanian parts of the baccalaureate

As an additional test of whether curricular assignment affects preferences, we included an abbreviated version of the O*NET Interest Profiler in the second follow-up survey (1.5 years after high school). The full version of the O*NET module provides a list of 60 job tasks—which fall into six groups—and asks a respondent to select the tasks that they would enjoy doing, regardless of remuneration. O*NET then recommends jobs that are a good fit for the respondent’s interests. In our shortened version, we kept three tasks per group and also added a task related to computer programming. We classify the tasks into three categories (STEM-related, HSS-related, or Other) and calculate the share of tasks in each category that a student reports they would enjoy. We also follow O*NET’s job recommendation algorithm to obtain a list of five jobs that O*NET recommends to a student, and we calculate the shares of the recommended jobs that are in each category.³³

Table 9: Effects on O*NET outcomes

	Share of X job tasks that student enjoys			Share of O*NET- recommended jobs that are:		
	STEM- related	HSS- related	Other	STEM- related	HSS- related	Other
STEM	0.096*** (0.024)	-0.091** (0.041)	0.010 (0.028)	0.105*** (0.032)	-0.097** (0.048)	-0.007 (0.044)
Intercept	0.48	0.67	0.55	0.14	0.43	0.43
Std. dev.	0.24	0.28	0.24	0.32	0.35	0.37
Student-cutoffs	1,199	1,199	1,199	1,199	1,199	1,199

The table presents results for the ATE of STEM assignment, β , for outcomes derived from the O*NET Interest Profiler included in the second follow-up survey. Outcomes in the first three columns are equal to the share of job tasks in the listed categories that a student reports they would enjoy; Appendix Table A45 presents effects on specific tasks within the categories. Outcomes in the fourth through sixth columns are equal to the share of the five jobs that O*NET recommends to a student that are in the given categories; see Appendix A10.1 for details on the construction of these outcomes. Other details are as in Panel A of Table 2.

Table 9 presents all-student effects on the O*NET outcomes. It shows that curricular assignment has a large impact on students’ preferences over job tasks. On average, assignment to STEM rather than HSS increases the share of STEM-related job tasks that a student enjoys by 10 pp and the share of O*NET-recommended jobs that are STEM-related by 11 pp. Conversely, STEM assignment reduces the share of HSS-related job tasks that a student enjoys by 9 pp and the share of O*NET-recommended jobs that are HSS-related by 10 pp. STEM assignment does not affect outcomes for the Other category. Overall, we interpret the O*NET evidence as underscoring the depth of the changes induced by STEM vs. HSS assignment: the effects on

exam. These effects are not informative about skill acquisition, as the exam content varies across the RD threshold. However, they help explain why STEM assignment reduces overall scores. Namely, the table shows that STEM assignment lowers both Romanian and non-Romanian scores, implying that the reduction in overall scores is partly due to harsher grading in STEM—and is not merely because STEM-assigned students are forced to take the Romanian component, for which they receive less preparation than HSS-assigned students.

33. The tasks that we asked about are displayed by category in Appendix Table A45. Appendix A10.1 details how we obtained and classified the recommended jobs.

preferences persist beyond an academic setting and into students' enjoyment of work content.

To assess some other potential mechanisms, in the second follow-up survey we also asked students about three representative jobs: a technology or engineering worker, a humanities teacher, and a person who runs a small business. We had students rate the jobs on a scale of 1 to 5 on various quality dimensions and tell us which job they would choose if they were required to do one at age 30. Effects on outcomes related to the representative jobs are in Table 10. As before, placement in STEM increases the probability of wanting to be a technology or engineering worker; in addition, it makes students like the work content in this job more and be more confident that they are prepared for the job. The opposite holds for the other jobs, though effects are not always statistically significant. Interestingly, curricular assignment does not influence students' beliefs about (i) how much the different jobs pay or (ii) how much they would like the jobs' coworkers and work conditions; thus, changes in job perceptions along these dimensions are unlikely to explain the effects of curricular assignment on college and career outcomes. Finally, assignment to STEM increases the extent to which students believe their friends and family would approve of them being a technology or engineering worker. This could be because Romanians tend to value STEM jobs and thus want students who have studied STEM to attain a STEM career.³⁴

To gain a broader understanding of students' decision-making, we asked students what attributes they consider when making college or job choices. Specifically, in the first follow-up survey, we provided a list of factors that might explain a student's choice of college field of study, and we asked students to select those that mattered in their choice. In the second follow-up survey, we did the same but for the factors that students expect to influence their job choice at age 30. In Appendix A11, we estimate the share of students at the RD threshold who select each factor, as well as the difference in shares across the threshold.³⁵

We find that over 95% of students care about whether a college or job option matches their interests and abilities. Much lower shares cite as an explanation that (i) an option aligns with what they studied in high school and they believe that changing would be difficult (38% for college choice and 43% for job choice), (ii) they were encouraged to choose the option by their high school teachers (46% and 42%), or (iii) the option mirrors what their high school friends chose (22% for college choice and unavailable for job choice). Intermediate shares of students are influenced by factors such as family approval, location, and prestige. Shares are generally similar on the STEM and HSS sides of the RD threshold, though STEM assignment somewhat raises the shares who expect that family approval and teacher encouragement will affect their job choice.³⁶

34. As with the effects on beliefs and preferences regarding academic subjects, Appendix A10 shows that effects on the O*NET and representative jobs outcomes differ little by student type and are robust.

35. As before, we did not ask the college questions to students who had never attended college by the time we surveyed them. We drop these students when calculating the college results; results are qualitatively unchanged if we instead impute values of 0.

36. STEM assignment also slightly raises the share of students who care about earnings. Thus, it does not

Table 10: Effects on choices and beliefs regarding representative jobs

	Would choose	Good pay	Enjoyable work content	Good coworkers & work conditions	Am well prepared	Friends & family would approve
<i>Panel A: A technology or engineering worker</i>						
STEM	0.146*** (0.047)	0.037 (0.077)	0.614*** (0.185)	0.185 (0.157)	0.543*** (0.199)	0.336** (0.135)
Intercept	0.20	4.50	2.58	3.20	2.60	4.05
Std. dev.	0.46	0.63	1.51	1.26	1.49	1.06
<i>Panel B: A humanities teacher</i>						
STEM	-0.054 (0.049)	0.019 (0.082)	-0.153 (0.151)	0.022 (0.139)	-0.371** (0.148)	0.149 (0.166)
Intercept	0.18	2.82	2.26	2.86	2.70	3.52
Std. dev.	0.37	0.77	1.26	1.21	1.36	1.33
<i>Panel C: A person who runs a small business</i>						
STEM	-0.092 (0.061)	0.066 (0.113)	-0.090 (0.142)	-0.152 (0.124)	-0.336*** (0.115)	-0.086 (0.108)
Intercept	0.62	4.19	4.20	4.18	3.96	4.49
Std. dev.	0.50	0.81	1.22	1.10	1.16	0.92
Student-cutoffs	1,199	1,199	1,199	1,199	1,199	1,199

The table presents results for the ATE of STEM assignment, β , for outcomes related to students’ choices and beliefs regarding representative jobs. The outcomes are from the second follow-up survey, and the jobs are displayed in the panels of the table. “Would choose” is a set of mutually exclusive indicator variables equal to one when a student would choose a given job if forced to do one of the three at age 30. The outcomes in the other columns are students’ ratings of the jobs on the listed dimensions on a scale of 1-5. The number of observations is the same across panels, so it is displayed in a single row at the bottom of the table. Other details are as in Panel A of Table 2.

In sum, our results reveal that high school curriculum shapes what students like and think they are good at—and that this holds for all student types. Moreover, nearly all students treat whether something matches their interests and abilities as important in their decision-making. These findings help explain why STEM vs. HSS assignment has such large effects on college and career outcomes for so many student types. Of course, other factors also contribute to the college and career effects, such as whether students can handle the rigor of STEM.

6.4 Wellbeing and school satisfaction

We now explore how curricular assignment affects outcomes related to wellbeing and school satisfaction. Beyond their intrinsic importance, effects on these outcomes provide additional insight into the effect of curricular assignment on post-high school trajectories. They also shed more light on how malleable high school students are: We have already seen that curricular assignment affects what students like and think they are good at. But do these shifts come with psychic costs for some students, especially those whose interests and abilities do not naturally align with their assigned curriculum?

influence students’ choices by changing their beliefs about earnings (Table 10), but it may influence their choices by affecting the amount of utility weight that they place on this attribute, albeit slightly.

Our analysis makes use of our survey data. In the surveys, we measured wellbeing and school satisfaction both at the end of high school and 1-1.5 years later, which is after students learned their baccalaureate performance and transitioned to college or the labor market. In addition, we asked about school satisfaction with respect to both high school and college.

The results for all students are in Tables 11-13. Table 11 presents effects on a wellbeing index and on measures of high school satisfaction. The wellbeing index is based on questions related to happiness, belonging, and depression; its construction is detailed in Appendix Table A52. The other outcomes in the table are students' ratings (on a scale of 1 to 5) of (a) how much they liked different aspects of high school and (b) how well the high school curriculum fit their abilities. Panel A is for outcomes derived from the end-of-high-school survey, while the outcomes in Panel B are from the follow-up surveys. Next, Table 12 provides effects on students' regret over their high school application choices. These effects are measured one year after high school, which is the only time we asked the associated questions. The first outcome in the table is a student's 1 to 5 rating of how happy the student is with their high school application choices. The other outcomes indicate the ways that students would change their choices if they could go back in time. Finally, Table 13 presents effects on college satisfaction; these are measured one year after high school, which is again the only time we asked the questions on which they are based.³⁷

Table 11: Effects on wellbeing and high school satisfaction

	Wellbeing	Liked the high school:				H.s. curriculum was a good fit for my abilities
		Experience	Curriculum	Peers	Teachers	
<i>Panel A: End of high school</i>						
STEM	0.133*** (0.051)	0.113* (0.068)	0.002 (0.066)	0.233*** (0.072)	-0.022 (0.068)	0.001 (0.053)
Intercept	-0.14	3.69	3.32	3.53	3.60	3.39
Std. dev.	0.74	0.99	1.04	1.12	1.02	0.86
Student-cutoffs	3,987	3,987	3,987	3,987	3,987	3,987
<i>Panel B: 1-1.5 years after high school</i>						
STEM	-0.033 (0.068)	0.217* (0.110)	-0.004 (0.137)	0.287** (0.138)	0.128 (0.130)	-0.123 (0.118)
Intercept	0.66	3.90	3.13	3.82	3.75	3.49
Std. dev.	0.56	0.89	1.06	1.08	0.93	1.02
Student-cutoffs	1,199	1,159	1,159	1,159	1,159	1,159

The table presents results for the ATE of STEM assignment, β , for outcomes related to wellbeing and high school satisfaction. In Panel A, outcomes are from the end-of-high-school survey. In Panel B, "Wellbeing" is from the second follow-up survey, while the other outcomes are from the first. "Wellbeing" is an index; details on its construction are in Appendix Table A52. The remaining outcomes are students' ratings on a scale of 1-5. Other details are the same as in Panel A of Table 2.

The results in Tables 11-13 offer a few findings. First, at the end of high school, assignment to STEM rather than HSS boosts wellbeing (Table 11). However, second, the boost fades after

37. As always, we did not ask the college questions to students who had not attended college by the time we surveyed them. For these students, we impute values of 1. We find that results without imputations are similar.

high school. Specifically, at the end of high school, STEM assignment causes an average increase in the wellbeing index of almost a fifth of the sd in the RD sample; by 1.5 years after high school, the effect is small, negative, and statistically insignificant.³⁸ Third, the boost in wellbeing at the end of high school coincides with an increase in satisfaction with social dimensions of high school. In particular, STEM assignment makes students give higher ratings to the high school experience and to their peers. In contrast, it does not change students’ ratings of the high school curriculum or teachers, nor does it affect their evaluation of how well the high school curriculum fit their abilities. Fourth, the increase in liking the social aspects of high school persists through the year after high school. However, fifth, by this time, STEM assignment has a negative (though statistically insignificant) effect on how well students think the high school curriculum fit their abilities; this is consistent with negative updating after seeing the baccalaureate results.³⁹

Next, by a year after high school, curricular assignment hardly affects students’ retrospective satisfaction with how they applied to high school (Table 12). In particular, at the RD threshold, STEM- and HSS-assigned students have similar contentment with their high school application choices and are similar in their likelihood of wishing they could change them.

Table 12: Effects on regret over high school application choices one year after high school

	Am happy w/ the appl. choices I made	If could do over, would:			
		Make no change	Rank STEM tracks higher	Rank HSS tracks higher	Rank Other tracks higher
STEM	0.049 (0.129)	0.045 (0.042)	-0.032 (0.035)	-0.018 (0.032)	0.005 (0.013)
Intercept	4.20	0.80	0.10	0.08	0.02
Std. dev.	0.94	0.38	0.28	0.27	0.14
Student-cutoffs	1,159	1,159	1,159	1,159	1,159

The table presents results for the ATE of STEM assignment, β , for outcomes related to regret over high school application choices. Outcomes are from the first follow-up survey. “Am happy with the application choices I made” is on a scale of 1-5. The remaining outcomes are indicator variables equal to one if a student agrees with the listed statement. Other details follow Panel A of Table 2.

Finally, assignment to STEM reduces how much students enjoy college, both socially and academically (Table 13). However, it does not affect perceptions of curricular fit in college, nor does it influence students’ happiness with their college application choices. The reduction in liking college is not surprising, as the conventional wisdom in Romania is that STEM college programs are stern and competitive.

Overall, the evidence reveals that, on average, there is a positive effect of STEM assignment on wellbeing during high school and a null effect after high school. There are a few channels that could explain this. First, as discussed, STEM assignment increases how much students like the social dimensions of high school, worsens baccalaureate performance, and reduces how much students like college. Second, in Appendix A8, we find that STEM assignment lowers time

38. Appendix Table A57 replicates Table 11 using a constant sample and confirms the difference in effects.

39. The effect on the change in ratings of curricular fit is statistically significant (Appendix Table A57).

spent on social media; research indicates that social media usage harms wellbeing (e.g., Allcott et al. 2025).⁴⁰ Third, as mentioned previously, STEM is widely valued in Romania. Thus, one could tell a story where improved high school social dynamics, lower social media use, and elevated parental and societal approval lead to a boost in wellbeing in high school; meanwhile, the latter two channels counteract worsened baccalaureate performance and reduced college enjoyment to yield no impact on wellbeing after high school. Of course, additional factors could be at play.

Table 13: Effects on college satisfaction one year after high school

	Like the college:				College curric. is a good fit for my abilities	Am happy w/ the appl. choices I made
	Experience	Curriculum	Peers	Instructors		
STEM	-0.443*** (0.138)	-0.227* (0.129)	-0.398** (0.166)	-0.352*** (0.134)	0.082 (0.156)	0.122 (0.179)
Intercept	2.20	2.25	2.18	2.31	3.48	3.72
Std. dev.	1.11	1.09	1.20	1.14	1.33	1.48
Student-cutoffs	1,159	1,159	1,159	1,159	1,159	1,159

The table is analogous to Table 11 and the first column of Table 12, but for college outcomes, rather than high school outcomes. Outcomes are from the first follow-up survey. The relevant questions were not asked to students who had never attended college by the time of this survey. We impute values of 1 for these students; see Appendix Table A61 for results without imputations. Other details are the same as in Panel A of Table 2.

Appendix A12 presents robustness and heterogeneity for the effects on wellbeing and school satisfaction. It shows that effects are robust and that there is little systematic heterogeneity across student groups. One exception is that by a year after high school, STEM assignment makes low-achievers less happy with their high school application choices (by 0.45 points out of 5, Appendix Table A58); this is not surprising given its negative impact on their chance of achieving key milestones. Interestingly, even for low-achievers, STEM assignment does not affect the likelihood of wishing one could change one’s choices.

Despite the shifting effects of curricular assignment on wellbeing and school satisfaction, a main takeaway from the results in this subsection is that students tend to be satisfied with their educational choices by a year after high school, regardless of their curricular assignment. At the RD threshold, the average happiness rating for a student’s high school application choices is 4.20 (4.25) out of 5 for students assigned to HSS (STEM) (Table 12). Similarly, 80% (85%) of HSS- (STEM-) assigned students would not change their choices if they could go back in time. Across student-group by curricular-assignment cells, the minimums of these values are 3.99 and 77%; meanwhile, the maximum share that wishes they had ranked STEM (HSS) tracks higher is 12% (11%) (Appendix Table A58). Happiness ratings tend to be lower for college choices, but this is distorted by the fact that we impute values of 1 for students who had never attended college. Without imputations, ratings of college choices are well above 4 (Appendix Table A61).

40. On average, the reduction is 26 minutes on a typical weekday in high school and 21 minutes on a typical weekday 1.5 years after high school.

To summarize, curricular assignment has a nuanced relationship with wellbeing and school satisfaction. This relationship changes in just the first year after high school and may continue to evolve as students age. Nonetheless, by a year after high school, most students are happy with their educational decisions, with no evidence of acute psychic costs for any group (except, perhaps, low-achievers). This broad contentment reinforces the earlier finding that high school students are malleable to what they study: Not only does curricular assignment influence what students like and think they are good at, but students also seem able to adapt to their assigned curriculum in terms of wellbeing and school satisfaction. The lack of mismatch-related psychic costs helps explain why curricular assignment has large college and career effects for most student types; meanwhile, the reduction in liking college when assigned to STEM may help explain why STEM pipelines can be leaky.

6.5 Social and civic outcomes

We finish by examining the effects of curricular assignment on social and civic outcomes. Historically, shifting these outcomes was seen as one of the main purposes of education, alongside pecuniary returns. Even today, large majorities of parents believe that schools should focus on influencing them.⁴¹ How curriculum affects these outcomes is of special interest, as there is a long tradition that claims they are best cultivated by a liberal arts education (rather than a technical/vocational one) and specifically by the study of humanities and social studies.⁴² Adjacent to this claim, research shows that people in HSS-related occupations and educational fields tend to be more left-leaning than those in STEM (Goldstein and Kolerman 2025; Hooghe, Marks, and Kamphorst 2025; Martin, Scott, and Kappe 2025).

We again make use of our surveys, which asked questions about family expectations, political views and behaviors, trust, friendships, and empathy. We combine all questions on a given topic into indices; however, we also estimate effects on the indices’ components.⁴³ We find considerable heterogeneity in effects by gender; thus, in the main text we provide results separately for boys and girls. Remaining heterogeneity (of which we find little) and robustness are in Appendix A13. We start by presenting effects on key outcomes, such as family expectations and political views. We then assess possible mechanisms.

Table 14 displays effects on students’ expectations about their future family structure. In the table, the columns under the “Components” heading correspond to different dimensions of

41. A recent survey of parents of U.S. K-12 students finds that most want schools to help their children navigate social situations (86%), respond ethically (85%), prepare to be an active, informed citizen (83%), and empathize with the feelings of others (81%) (Tyner 2021).

42. See Nussbaum (2010) and Roth (2014) for contemporary reviews.

43. When applicable, the indices combine data across surveys. Such indices can be interpreted as averages of an outcome during the end of high school through 1 or 1.5 years after. Results are similar if we instead restrict to responses from the end-of-high-school survey (Appendix Table A69). We also find limited impacts on changes in outcomes over the analysis period, although estimates are noisy (Appendix Table A72). Details on indices are provided in the notes to the tables and figures.

expectations, while the first column is the index. The index is constructed such that higher values indicate greater traditionalism.

Table 14: Effects on traditionalist expectations

	Index	Components				
		Work amount	Bread- winner	Wealth decile	Number of children	Smaller locale
<i>Panel A: All students</i>						
STEM	0.096*** (0.034)	-0.006 (0.028)	0.097*** (0.033)	0.133 (0.086)	0.067 (0.057)	0.049 (0.055)
Intercept	-0.10	2.85	2.13	7.40	1.42	2.21
Std. dev.	0.49	0.44	0.48	1.33	0.87	0.86
Student-cutoffs	3,987	3,987	3,987	3,987	3,987	3,987
<i>Panel B: Male students</i>						
STEM	0.116** (0.048)	0.045 (0.060)	0.133** (0.059)	0.030 (0.147)	0.164* (0.094)	0.076 (0.110)
Intercept	0.22	2.77	2.33	7.51	1.40	2.21
Std. dev.	0.41	0.53	0.51	1.34	0.84	0.89
Student-cutoffs	1,396	1,396	1,396	1,396	1,396	1,396
<i>Panel C: Female students</i>						
STEM	0.034 (0.033)	-0.030 (0.031)	0.030 (0.032)	0.172 (0.114)	0.035 (0.069)	0.039 (0.064)
Intercept	-0.29	2.90	2.01	7.33	1.41	2.17
Std. dev.	0.41	0.37	0.38	1.32	0.88	0.85
Student-cutoffs	2,510	2,510	2,510	2,510	2,510	2,510

The table presents results for the ATE of STEM assignment, β , on students' expectations regarding their family at age 30. "Index" combines the expectations in a traditionalist direction. The components relate to specific types of expectations. The questions from which the components are derived were asked in both the end-of-high-school and second follow-up surveys. The components use data from both surveys. They are equal to the student fixed effects from a panel regression of a question's responses on student and survey fixed effects, centered at the mean response for students with non-missing responses for the question in both surveys. The index is the average of standardized versions of all of a student's responses. The average ignores any responses that are missing for the student, and the standardization uses survey-specific means and standard deviations that are calculated over the students with no missing responses. The procedure for obtaining the components and index is described in more detail in the notes to Table A32. "Work amount" is how much a student expects to work on a scale of 1-3, with 1 being "not at all", 2 "part time", and 3 "full time". "Breadwinner" is also on a scale of 1-3, with 1 being if the student's spouse will contribute most of the household's money, 2 if the partners will contribute equally, and 3 if the student will contribute the most. "Wealth decile" is for a student's household and is on a scale of 1-10. "Number of children" is capped at 3. "Smaller locale" concerns the type of community in which the student expects to live, with 1 being outside Romania, 2 a big city, 3 a medium town, and 4 a rural area or small town. When we construct the traditionalist expectations index, we reverse the order of the responses for "Work amount" and "Breadwinner" for female students, as the traditionalist expectation for women is to work less and not be the breadwinner. Other details follow Panel A of Table 2.

Table 14 shows that assignment to STEM rather than HSS causes male students to hold expectations that are more traditionalist. The estimated effect on the index is equal to 0.28 of the sd for male students in the RD sample. The effect is driven by increases in expectations about being the family "breadwinner" and in the expected number of children. For the other dimensions, effects are statistically insignificant, but point estimates continue to be in a more traditionalist direction. In contrast, curricular assignment does not measurably affect family expectations for female students.

Table 15 is analogous to Table 14, but for effects on political views. Here, the index is constructed such that higher values indicate views that are more right-wing. The component

questions concern a variety of economic and social issues and mostly come from the European Values Study, the General Social Survey, and the World Values Survey.

Table 15: Effects on right-wing political views

	Index	Components							
		The poor are lazy	Redistri- bution is bad	Ability- tracking is desirable	Tradition is good	Wife earning more is a problem	Divorce is immoral	Climate change is not so serious	Boys are better in math
Panel A: All students									
STEM	0.042 (0.031)	0.073 (0.087)	-0.022 (0.089)	0.030 (0.024)	0.065 (0.068)	0.029 (0.080)	0.149** (0.073)	-0.147 (0.154)	-0.212 (0.161)
Intercept	-0.06	2.42	3.13	0.54	3.75	1.84	1.72	2.77	2.81
Std. dev.	0.44	1.19	1.26	0.33	1.12	1.21	1.16	1.20	1.22
Student-cutoffs	3,987	3,987	3,987	3,987	3,987	3,987	3,987	1,199	1,199
Panel B: Male students									
STEM	0.145*** (0.052)	0.298** (0.138)	0.077 (0.161)	0.118*** (0.038)	0.062 (0.130)	0.114 (0.140)	0.062 (0.119)	0.011 (0.294)	0.034 (0.287)
Intercept	-0.05	2.55	3.05	0.48	3.75	1.86	1.97	2.61	2.76
Std. dev.	0.45	1.20	1.27	0.33	1.16	1.24	1.21	1.27	1.18
Student-cutoffs	1,396	1,396	1,396	1,396	1,396	1,396	1,396	373	373
Panel C: Female students									
STEM	-0.012 (0.036)	-0.096 (0.108)	-0.062 (0.108)	-0.012 (0.029)	0.060 (0.089)	0.014 (0.097)	0.196** (0.098)	-0.371** (0.176)	-0.415** (0.205)
Intercept	-0.08	2.34	3.15	0.57	3.75	1.80	1.53	2.86	2.83
Std. dev.	0.42	1.16	1.25	0.34	1.10	1.18	1.09	1.16	1.23
Student-cutoffs	2,510	2,510	2,510	2,510	2,510	2,510	2,510	728	728

The table presents results for the ATE of STEM assignment, β , for students' political views. "Index" combines the views, with higher values indicating views that are more right-wing. The components capture students' views on particular issues. All components except "Ability-tracking is desirable" are the responses to questions where students were asked how much they agree with a statement on a scale of 1-5. The respective statements are: (a) "When a person is poor, it's usually because the person made bad choices and didn't work hard", (b) "The government should do more to make sure that everyone is cared for, even if it means that some people can get by without working", (c) "Things are good if they follow traditions and customs", (d) "If a woman earns more money than her husband, it's almost certain to cause problems", (e) "Divorce is always immoral", (f) "Climate change is one of the most serious issues that we face today", and (g) "Boys' natural ability in mathematics is greater than that of girls". To ensure that outcomes are always in a right-wing direction, we reverse the order of the responses for (b) and (f). "Ability-tracking is desirable" is the fraction of "yes" answers for three yes or no questions about splitting students between classes based on academic ability. The questions are whether a student agrees that this method is fair, that it does not stigmatize students, and that it improves learning. The first six components are from the end-of-high-school survey; the last two are from the second follow-up survey. The index is constructed using the same strategy as in Table 14. Other details follow Panel A of Table 2.

Similar to before, Table 15 shows that STEM assignment pushes males' political views to the right. The estimated effect on the index is equal to 0.32 of the sd for males in the RD sample. Effects on views over specific issues are often statistically insignificant, but point estimates are always in a right-wing direction. Interestingly, for girls, effects on political views are mixed. The effect on the index is small and statistically insignificant, while effects on views over specific issues have contrasting signs. Notably, STEM assignment makes female students more strongly agree with the statement that divorce is immoral (by 0.18 of the sd for females in the RD sample), but it also makes them more concerned about climate change (by 0.32 sd), and it increases disagreement with the claim that boys are naturally better than girls in math (by 0.34 sd).

Next, the first four columns of Table 16 show that curricular assignment does not affect students’ values of a trust index, whether students turned out to vote in Romania’s Fall 2024 elections, or how many friends students have. On the other hand, curricular assignment does affect the gender composition of students’ friendships (fifth column): Placement in STEM causes boys (girls) to report having about 0.7 (0.5) fewer good female versus male friends. In this way, curricular assignment seems to influence the nature of students’ family expectations, political views, and friendships, but not students’ degree of political engagement and social connection.⁴⁴

Table 16: Effects on additional social and civic outcomes

	Trust	Turn out in Fall 2024 elections	Friends			Verbal development	Empathy
			Good	Very close	Good female vs. male		
<i>Panel A: All students</i>							
STEM	0.016 (0.050)	0.027 (0.036)	-0.005 (0.185)	-0.041 (0.096)	-0.764*** (0.191)	-0.279*** (0.033)	-0.116** (0.054)
Intercept	-0.03	0.87	6.07	3.22	0.62	0.18	0.06
Std. dev.	0.78	0.26	2.76	1.50	2.76	0.49	0.89
Student-cutoffs	3,987	1,199	3,987	3,987	3,987	3,987	3,987
<i>Panel B: Male students</i>							
STEM	-0.012 (0.085)	-0.005 (0.077)	-0.066 (0.321)	0.004 (0.171)	-0.706** (0.281)	-0.260*** (0.052)	-0.145 (0.093)
Intercept	0.05	0.89	6.61	3.42	-1.03	-0.01	-0.04
Std. dev.	0.77	0.28	2.69	1.51	2.48	0.48	0.92
Student-cutoffs	1,396	373	1,396	1,396	1,396	1,396	1,396
<i>Panel C: Female students</i>							
STEM	0.028 (0.068)	0.049 (0.043)	-0.041 (0.236)	-0.063 (0.114)	-0.491*** (0.188)	-0.273*** (0.043)	-0.082 (0.069)
Intercept	-0.10	0.87	5.78	3.07	1.50	0.29	0.10
Std. dev.	0.78	0.24	2.78	1.48	2.30	0.47	0.86
Student-cutoffs	2,510	728	2,510	2,510	2,510	2,510	2,510

The table presents results for the ATE of STEM assignment, β , for additional social and civic outcomes. All outcomes except “Turn out in Fall 2024 elections” use data from both the end-of-high-school and second follow-up surveys. “Trust”, “Verbal development”, and “Empathy” are indices; see Appendix Tables A66, A67, and A68 for details on their construction. The “Friends” outcomes are based on questions where students were asked how many good male, good female, and very close friends they have; “Good female vs. male” is the difference in the number of good female and male friends. As in Table 14, the “Friends” outcomes equal the student fixed effects from a panel regression of a given variable on student and survey fixed effects, centered at the mean for students with non-missing values in both surveys. “Turn out in Fall 2024 elections” is from the second follow-up survey. It is the share of times that a student turned out to vote in Romania’s two Fall 2024 elections (presidential and parliamentary). Other details match those in Panel A of Table 2.

The last two columns of Table 16 show that STEM assignment reduces students’ scores on an index of verbal development and an index of empathy. The estimated impact on verbal development is similar in size across genders; that on empathy is slightly larger for boys.⁴⁵ The

44. We also asked students about who they supported in the Fall 2024 elections. Unfortunately, most refused to answer, likely because the elections were subject to foreign interference and were highly controversial.

45. The verbal development index combines all available variables related to verbal skills and interests; its construction is detailed in Appendix Table A67. The empathy index combines a question about whether students can notice others’ emotions with one about whether they enjoy helping people with their personal and emotional problems; it is detailed in Appendix Table A68. Empathy has cognitive (“theory of mind”) and affective aspects (Baron-Cohen 2011; Dodell-Feder and Tamir 2018; Morris 2020; Fé, Gill, and Prowse 2022; Zebarjadi et al. 2023);

fact that curricular assignment affects verbal development is not surprising; nonetheless, it is worth noting, since verbal skills play an important role in socialization. The fact that curricular assignment affects empathy is consistent with findings from psychology. Namely, placement in HSS means that students spend more time reading literature, and psychological research shows that reading literature boosts empathy. The hypothesized mechanism is that reading involves inhabiting external minds, which makes people more attuned to others’ emotions and mental states (Mar and Oatley 2008; Nussbaum 2010; Koopman and Hakemulder 2015; Mumper and Gerrig 2017; Dodell-Feder and Tamir 2018; Lecce, Bianco, and Hughes 2021).

We next explore the mechanisms that could underlie the shifts in social and civic outcomes, paying particular attention to those that explain the impacts on family expectations and political views. A first candidate is that STEM assignment may make students expect to be richer. Among other consequences, this could make them expect to have a higher relative position in society, be less dependent on government assistance, and have less need for a co-equal partner, all of which could influence what they expect for their family structure and how they view the world politically. Given our results, a requirement for this channel to be operative is that STEM assignment would specifically need to raise wealth expectations for boys. Interestingly, Appendix Table A64 shows that curricular assignment hardly affects wealth expectations, with the estimate for boys being a relatively precise zero. Thus, we find little support for this channel.

A second potential mechanism relates to years of schooling. Research finds that gaining more education tends to cause people’s political views to become more left-wing (Apfeld et al. 2023; Firoozi 2026). This channel is unlikely to be relevant in our setting, as STEM- and HSS-assigned students had similar amounts of education when we surveyed them. Nonetheless, curricular assignment could conceivably have impacts by influencing how much education students expect to obtain. Appendix Table A65 shows that assignment to STEM slightly reduces boys’ expectations about their final amount of education, but in a way that seems unlikely to matter. Namely, it shifts boys away from expecting a doctoral degree and toward expecting a master’s or professional degree. For girls, curricular assignment does not affect expected years of schooling.

A third potential mechanism is peer effects. As we have noted, assignment to STEM rather than HSS causes a student’s classmates to be less female and more oriented toward math vs. language. Different types of peers likely differ in their family expectations and political views. For instance, girls are more left-leaning than boys on average, both in our data and across countries (e.g., Burn-Murdoch 2024). Thus, curricular assignment could affect a student’s social and civic outcomes via peer spillovers. Interestingly, we find that the effects on family expectations and political views persist when we control for track peer attributes, as do those on verbal development and empathy (Appendix A5). In contrast, the effects on the gender split of students’ friends grow smaller. Thus, peer exposure may partly explain why STEM assignment causes

our measure is meant to capture both.

students to have more male and fewer female friends, but it does not explain our other results.

Another set of stories centers on the direct impacts of the curricular differences between STEM and HSS tracks. Spending more time on STEM or HSS subjects could matter for family expectations and political views in a few ways. First, the subjects impart different types of knowledge, which may lead students to have different views. For instance, learning about science could cause STEM students to become more worried about climate change; meanwhile, learning about social issues could make HSS students more concerned about poverty. Second, HSS subjects often contain an explicit focus on moral reasoning; in addition, the process of analyzing a text and crafting and communicating an argument may spur forms of critical thinking and perspective-taking that are distinct from those developed in STEM subjects (Nussbaum 2010). Third, we have shown that curricular assignment affects verbal development and empathy; studies find that both correlate with having views that are more left-wing and theorize that the correlations may be causal (Wagaman and Segal 2014; Hasson et al. 2018; Morris 2020; Zebarjadi et al. 2023; Edwards et al. 2025).⁴⁶

While we cannot prove any of the stories described in the previous paragraph, the evidence generally supports them. First, the effects on political views are roughly consistent with what one might expect given the different types of knowledge supplied by STEM and HSS. For instance, assignment to STEM increases concern about climate change and reduces gender math stereotypes, though only for girls; meanwhile, assignment to HSS lowers agreement with the claim that the poor are lazy, though only for boys.⁴⁷ Second, our results also mostly align with the verbal development and empathy channels identified by the prior literature. Namely, placement in STEM rather than HSS reduces these; further, for boys, it shifts political views to the right and makes family expectations more traditionalist. Thus, for these channels, curricular assignment has the predicted effects for both the intermediate and final outcomes, though only for boys. Finally, we do not have concrete evidence on moral reasoning, critical thinking, or perspective-taking. However, we note that questions related to these factors are included in the language and literature component of the baccalaureate exam, performance on which is affected by STEM vs. HSS assignment.⁴⁸

46. For instance, having more empathy may make someone more concerned about the plight of disadvantaged groups, which may make them more supportive of redistribution.

47. Our data lends insight into the information about male vs. female math abilities provided by STEM vs. HSS assignment. Appendix Tables A73 and A74 present estimates of the average math gender gap in students' tracks for students on the STEM and HSS sides of the RD threshold, both at the beginning and end of high school. A first finding is that the gap at the beginning of high school (i.e., in terms of math transition exam scores) is the same on either side of the threshold. Thus, girls in STEM tracks are not positively selected relative to boys on math abilities. A second finding is that the sign of the gap reverses during high school, with girls performing worse than boys on the transition exam but better on the baccalaureate exam. The math component of the baccalaureate exam is not taken by HSS students. Thus, students in STEM can observe this reversal, while students in HSS likely cannot. This suggests that, by the end of high school, students in STEM receive a signal that girls are at least as proficient in math as boys, while students in HSS do not.

48. An additional story is that the teachers who teach HSS subjects may be left-leaning and may thus choose

Despite the general consistency of the evidence, some questions remain. First is why family expectations and political views do not change, on net, for girls. A noteworthy fact is that, at our cutoffs, girls display higher levels of verbal development and empathy than boys. In fact, values of these indices are higher for girls assigned to STEM than for boys assigned to HSS (Table 16). Thus, one possibility is that girls face diminishing returns, where additional increases in verbal development and empathy do not translate into further changes in outcomes. Another question is why curricular assignment does not influence views about climate change and gender math stereotypes for boys. A potential explanation is that the information provided in these domains from study of STEM subjects is counterbalanced by the general rightward shift that boys receive from STEM assignment. Of course, these answers are only speculative; we leave further investigation to future work.

Overall, the results in this subsection again highlight the power of high school curricula in shaping who students are: What students study impacts a number of social and civic outcomes, and the effects seem to be at least partially due to changes in students' knowledge and personalities, not merely in their sense of their future economic situation. The gender heterogeneity in effects on family expectations and political views is particularly interesting. Recent years have seen growing gender gaps among young people in these areas (e.g., Burn-Murdoch 2024 and Kamarck and Muchnick 2024), coincident with strong societal emphasis on STEM education; our results suggest that these phenomena may be connected.

7 Conclusion

The last decades have seen an increasing emphasis on STEM education, motivated by evidence that STEM produces not only private labor market returns (e.g., Arcidiacono, Hotz, and Kang 2012) but also innovation and macroeconomic growth (Aghion et al. 2009; Peri, Shih, and Sparber 2015). Yet shifting curricular emphasis toward STEM necessarily displaces humanities and social studies. This paper examines this tradeoff, asking how intensive STEM rather than HSS curricular exposure shapes not only students' educational and career trajectories but also deeper aspects of who they are.

Our key finding is that high school curriculum shapes students' beliefs, preferences, and values and that this malleability not only matters in its own right but also helps to explain the broad and largely uniform impact of STEM exposure on educational and career trajectories. Specifically, intensive exposure to a STEM curriculum reshapes what students think they are good at and what they enjoy, redirecting their pursuit of education and careers, with only modest effects on wellbeing and school satisfaction. This malleability is consistent with the mindset strand of the psychology literature (e.g., Blackwell, Trzesniewski, and Dweck 2007 and Yeager et al. 2019); it

left-wing texts and topics. We believe this story does not apply in the Romanian setting, as the HSS curriculum is both nationally standardized and relatively classical in nature. Nonetheless, research indicates that ideologically motivated curricular content can be successful in indoctrinating students (e.g., Cantoni et al. 2017).

suggests that high school is a promising venue for STEM-focused interventions. In short, while research rightly emphasizes early childhood interventions (Heckman 2006; Heckman et al. 2010; Del Boca, Flinn, and Wiswall 2016; Akabayashi and Tanaka 2024), adolescence also offers a window in which to shift trajectories and, more broadly, to shape who students become.

The malleability we document has implications for increasing women’s representation in STEM. Notably, the effects on preferences, beliefs about abilities, and college enrollment are similarly large for both female and male students, indicating that high school STEM exposure can do much to push women toward STEM. That said, the effect on wanting a technology or engineering career is about half as large for females as for males, which is consistent with studies finding gender gaps in career outcomes conditional on college field of study (e.g., Wiswall and Zafar 2017 and Aguirre, Matta, and Montoya 2024). More research on this post-college divergence—and on the extent to which it is due to barriers—would be valuable.

Another important dimension of heterogeneity concerns students with low baseline achievement, who become more likely to fail a high school exit exam and less likely to attend college when assigned to STEM. This finding adds causality to an established descriptive result that coursework and assessments are harder in STEM than in other fields (e.g., Ahn et al. 2024).

Our results also highlight the impact of what students study in high school on social and civic outcomes, notably with boys exposed to a STEM rather than HSS curriculum becoming more socially and politically conservative. This result lends causal credence to a long tradition of claims summarized in Nussbaum (2010) and Roth (2014) and accords with research finding social and civic effects from other dimensions of education. It underlines that an assessment of education policy must contend with the impacts on these types of outcomes, in addition to test scores and labor market returns.

Overall, our results strike both an optimistic and a cautionary note on emphasizing STEM in high school education. Doing so can be effective at shifting students toward STEM education and careers, driven by the surprisingly deep and comprehensive impact of high school curriculum on students’ beliefs and preferences. But this same malleability means that curriculum shapes students in ways that go well beyond educational and career trajectories, with consequences for social and civic outcomes that reasonable people may evaluate differently.

References

- Abdulkadiroglu, Atila, Joshua Angrist, Yusuke Narita, and Parag Pathak. 2022. “Breaking Ties: Regression Discontinuity Design Meets Market Design.” *Econometrica* 90 (1): 117–151.
- Abdulkadiroglu, Atila, Joshua Angrist, Yusuke Narita, Parag Pathak, and Roman Zarate. 2017. “Regression Discontinuity in Serial Dictatorship: Achievement Effects at Chicago’s Exam Schools.” *American Economic Review* 107, no. 5 (May): 240–45.
- Abdulkadiroglu, Atila, Joshua Angrist, and Parag Pathak. 2014. “The Elite Illusion: Achievement Effects at Boston and New York Exam Schools.” *Econometrica* 82 (1): 137–196.
- Abdulkadiroglu, Atila, Parag Pathak, Jonathan Schellenberg, and Christopher Walters. 2020. “Do Parents Value School Effectiveness?” *American Economic Review* 110, no. 5 (May): 1502–39.
- Aghion, Philippe, Leah Boustan, Caroline Hoxby, and Jérôme Vandenbussche. 2009. “The Causal Impact of Education on Economic Growth: Evidence from U.S.” *Brookings Papers on Economic Activity* 2009 (1): 1–73.
- Aguirre, Josefa, Juan Matta, and Ana Maria Montoya. 2024. “Joining the Old Boys’ Club: Women’s Returns to Majoring in Technology and Engineering.” *Unpublished manuscript*.
- Ahn, Tom, Peter Arcidiacono, Amy Hopson, and James Thomas. 2024. “Equilibrium Grading Policies With Implications for Female Interest in STEM Courses.” *Econometrica* 92 (3): 849–880.
- Ainsworth, Robert, Rajeev Dehejia, Cristian Pop-Eleches, and Miguel Urquiola. 2023. “Why Do Households Leave School Value Added on the Table? The Roles of Information and Preferences.” *American Economic Review* 113, no. 4 (April): 1049–82.
- Akabayashi, H., and R. Tanaka. 2024. “The Rate of Return to Early Childhood Education in Japan: Estimates from the Nationwide Expansion.” *Education Economics* 32 (5): 581–598.
- Allcott, Hunt, Matthew Gentzkow, Benjamin Wittenbrink, Juan Carlos Cisneros, Adriana Crespo-Tenorio, Drew Dimmery, Deen Freelon, et al. 2025. *The Effect of Deactivating Facebook and Instagram on Users’ Emotional State*. Working Paper, Working Paper Series 33697. National Bureau of Economic Research, April.
- Altonji, Joseph. 1995. “The Effects of High School Curriculum on Education and Labor Market Outcomes.” *Journal of Human Resources* 30, no. 3 (July): 410–438.
- Apfeld, Brendan, Emanuel Coman, John Gerring, and Stephen Jessee. 2023. “Higher Education and Cultural Liberalism: Regression Discontinuity Evidence from Romania.” *The Journal of Politics* 85 (1): 34–48.
- Arcidiacono, Peter, Esteban Aucejo, and Ken Spenner. 2012. “What Happens After Enrollment? An Analysis of the Time Path of Racial Differences in GPA and Major Choice.” *IZA Journal of Labor Economics* 1 (1): 5.
- Arcidiacono, Peter, Joseph Hotz, and Songman Kang. 2012. “Modeling College Major Choices Using Elicited Measures of Expectations and Counterfactuals.” *Journal of Econometrics* 166, no. 1 (January): 3–16.

- Baron-Cohen, Simon. 2011. *The Science of Evil: On Empathy and the Origins of Cruelty*. Basic Books.
- Bertanha, Marinho. 2020. “Regression Discontinuity Design with Many Thresholds.” *Journal of Econometrics* 218 (1): 216–241.
- Blackwell, Lisa, Kali Trzesniewski, and Carol Dweck. 2007. “Implicit Theories of Intelligence Predict Achievement Across an Adolescent Transition: A Longitudinal Study and an Intervention.” *Child Development* 78 (1): 246–263.
- Bleemer, Zachary, and Aashish Mehta. 2022. “Will Studying Economics Make You Rich? A Regression Discontinuity Analysis of the Returns to College Major.” *American Economic Journal: Applied Economics* 14, no. 2 (April): 1–22.
- Burn-Murdoch, John. 2024. “A New Global Gender Divide is Emerging.” *Financial Times*.
- Calonico, Sebastian, Matias Cattaneo, Max Farrell, and Rocío Titiunik. 2017. “rdrobust: Software for regression-discontinuity designs.” *The Stata Journal* 17 (2): 372–404.
- Cantoni, Davide, Yuyu Chen, David Yang, Noam Yuchtman, and Jane Zhang. 2017. “Curriculum and Ideology.” *Journal of Political Economy* 125 (2): 338–392.
- Cattaneo, Matias, Michael Jansson, and Xinwei Ma. 2020. “Simple Local Polynomial Density Estimators.” *Journal of the American Statistical Association* 115 (531): 1449–1455.
- Cattaneo, Matias, Luke Keele, Rocío Titiunik, and Gonzalo Vazquez-Bare. 2016. “Interpreting Regression Discontinuity Designs with Multiple Cutoffs.” *The Journal of Politics* 78 (4): 1229–1248.
- Clots-Figueras, Irma, and Paolo Masella. 2013. “Education, Language and Identity.” *The Economic Journal* 123 (570).
- Cohodes, Sarah, Helen Ho, and Silvia Robles. 2022. *STEM Summer Programs for Underrepresented Youth Increase STEM Degrees*. Working Paper 30227. National Bureau of Economic Research, July.
- Dahl, Gordon, Dan-Olof Rooth, and Anders Stenberg. 2023. “High School Majors and Future Earnings.” *American Economic Journal: Applied Economics* 15, no. 1 (January): 351–82.
- Darolia, Rajeev, Cory Koedel, Joyce Main, Felix Ndashimye, and Junpeng Yan. 2020. “High School Course Access and Postsecondary STEM enrollment and Attainment.” *Educational Evaluation and Policy Analysis* 42 (1): 22–45.
- De Philippis, Marta. 2023. “STEM Graduates and Secondary School Curriculum: Does Early Exposure to Science Matter?” *Journal of Human Resources* 58 (6): 1914–1947.
- Del Boca, Daniela, Christopher Flinn, and Matthew Wiswall. 2016. “Transfers to Households with Children and Child Development.” *The Economic Journal* 126 (596): F136–F183.
- Dodell-Feder, David, and Diana Tamir. 2018. “Fiction Reading Has a Small Positive Impact on Social Cognition: A Meta-Analysis.” *Journal of Experimental Psychology: General* 147 (11): 1713.

- Edwards, Tobias, Christopher Dawes, Emily Willoughby, Matt McGue, and James Lee. 2025. "More than G: Verbal and Performance IQ as Predictors of Socio-Political Attitudes." *Intelligence* 108:101876.
- EIGE. 2024. *Gender Equality Index 2024: Sustaining Momentum on a Fragile Path*. EU report. Publications Office of the European Union.
- Fé, Eduardo, David Gill, and Victoria Prowse. 2022. "Cognitive Skills, Strategic Sophistication, and Life Outcomes." *Journal of Political Economy* 130 (10): 2643–2704.
- Firoozi, Daniel. 2026. *Education and Partisanship*. Working Paper. Claremont McKenna College.
- Goff, Leonard, Ofer Malamud, Cristian Pop-Eleches, and Miguel Urquiola. 2023. "Interactions Between Family and School Environments." *Journal of Human Resources*.
- Goldstein, Yoav, and Matan Kolerman. 2025. *Changing Minds: How Academic Fields Shape Political Attitudes*. Working Paper. Tel Aviv University.
- Goodman, Joshua. 2019. "The Labor of Division: Returns to Compulsory High School Math Coursework." *Journal of Labor Economics* 37 (4): 1141–1182.
- Hasson, Yossi, Maya Tamir, Kea Brahms, Christopher Cohrs, and Eran Halperin. 2018. "Are liberals and conservatives equally motivated to feel empathy toward others?" *Personality and Social Psychology Bulletin* 44 (10): 1449–1459.
- Heckman, James. 2006. "Skill Formation and the Economics of Investing in Disadvantaged Children." *Science* 312 (5782): 1900–1902.
- Heckman, James, Seong Hyeok Moon, Rodrigo Pinto, Peter Savelyev, and Adam Yavitz. 2010. "Analyzing Social Experiments as Implemented: A Reexamination of the Evidence from the HighScope Perry Preschool Program." *Quantitative Economics* 1 (1): 1–46.
- Holland, John. 1997. *Making Vocational Choices: A Theory of Vocational Personalities and Work Environments*. 3rd. Psychological Assessment Resources.
- Hooghe, Liesbet, Gary Marks, and Jonne Kamphorst. 2025. "Field of education and political behavior: predicting GAL/TAN voting." *American Political Science Review* 119 (2): 794–811.
- Imberman, Scott, Michael Lovenheim, Patrick Massey, Kevin Stange, and Rodney Andrews. 2026. *The Contribution of College Majors to Gender and Racial Earnings Differences*. Working Paper 34726. National Bureau of Economic Research, January.
- Joensen, Juanna Schrøter, and Helena Skyt Nielsen. 2016. "Mathematics and Gender: Heterogeneity in Causes and Consequences." *The Economic Journal* 126 (593): 1129–1163.
- Kamarck, Elaine, and Jordan Muchnick. 2024. *The growing gender gap among young people*. Technical report. The Brookings Institution.
- Koopman, Eva Maria, and Frank Hakemulder. 2015. "Effects of Literature on Empathy and Self-Reflection: A Theoretical-Empirical Framework." *Journal of Literary Theory* 9 (1): 79–111.

- Kroenke, Kurt, Robert Spitzer, Janet Williams, and Bernd Löwe. 2009. “An Ultra-Brief Screening Scale for Anxiety and Depression: The PHQ-4.” *Psychosomatics* 50 (6): 613–621.
- Lecce, Serena, Federica Bianco, and Claire Hughes. 2021. “Reading Minds and Reading Texts: Evidence for Independent and Specific Associations.” *Cognitive Development* 57:101010.
- Levine, Phillip, and David Zimmerman. 1995. “The Benefit of Additional High-School Math and Science Classes for Young Men and Women.” *Journal of Business & Economic Statistics* 13 (2): 137–149.
- Liu, Jing, Cameron Conrad, and David Blazar. 2024. *Computer Science for All? The Impact of High School Computer Science Courses on College Majors and Earnings*. Technical report. IZA Discussion Paper No. 16758.
- Luscombe, Richard. 2026. “Florida axes sociology as required class at state universities in latest attack on ‘woke’.” *The Guardian* (March).
- Malamud, Ofer, Andreea Mitrut, Cristian Pop-Eleches, and Miguel Urquiola. 2025. “Self-, Peer-, and Teacher Perceptions Under School Tracking.” *The Review of Economics and Statistics* August:1–41.
- Mar, Raymond, and Keith Oatley. 2008. “The Function of Fiction is the Abstraction and Simulation of Social Experience.” *Perspectives on Psychological Science* 3 (3): 173–192.
- Martin, Nicole, Ralph Scott, and Roland Kappe. 2025. “School subject choices in adolescence affect political party support.” *West European Politics*.
- Morris, Stephen. 2020. “Empathy and the Liberal-Conservative Political Divide in the US.” *Journal of Social and Political Psychology* 8 (1): 8–24.
- Mumper, Micah, and Richard Gerrig. 2017. “Leisure Reading and Social Cognition: A Meta-Analysis.” *Psychology of Aesthetics, Creativity, and the Arts* 11 (1): 109.
- Nussbaum, Martha. 2010. *Not for Profit: Why Democracy Needs the Humanities*. Princeton University Press.
- Patnaik, Arpita, Matthew Wiswall, and Basit Zafar. 2020. *College Majors*. Working Paper 27645. National Bureau of Economic Research, August.
- Peri, Giovanni, Kevin Shih, and Chad Sparber. 2015. “STEM Workers, H-1B Visas, and Productivity in US Cities.” *Journal of Labor Economics* 33 (S1): S225–S255.
- Pop-Eleches, Cristian, and Miguel Urquiola. 2013. “Going to a Better School: Effects and Behavioral Responses.” *American Economic Review* 103 (4): 1289–1324.
- Robayo-Abril, Monica, Chifundo Patience Chilera, Britta Rude, and Irina Costache. 2023. *Gender Equality in Romania: Where Do We Stand?* Country Gender Assessment. World Bank.
- Romanian Ministry of Education, Research and Innovation. 2009. *Annexes 1 and 2 to the Order of the Minister of Education, Research and Innovation No. 3411 of March 16, 2009*. Gazette of Romania, Part I.

- Romanian Ministry of Education, Research and Innovation. 2016. *Regulament-cadru de organizare și funcționare a unităților de învățământ preuniversitar*. Aprobă prin Ordinul nr. 5.079 din 31 august 2016, Monitorul Oficial al României, Partea I, nr. 720 din 19 septembrie 2016. Available at <https://legislatie.just.ro/Public/DetaliuDocument/181773>.
- Rose, Heather, and Julian Betts. 2004. "The Effect of High School Courses on Earnings." *The Review of Economics and Statistics* 86 (2): 497–513.
- Roth, Michael. 2014. *Beyond the University: Why Liberal Education Matters*. Yale University Press.
- Stinebrickner, Ralph, and Todd Stinebrickner. 2014. "A Major in Science? Initial Beliefs and Final Outcomes for College Major and Dropout." *Review of Economic Studies* 81 (1): 426–472.
- Trump, Donald. 2025. *Ending Radical Indoctrination in K-12 Schooling*. Executive Order 14190. The White House, January.
- Tyner, Adam. 2021. *How to Sell SEL: Parents and the Politics of Social-Emotional Learning*. White Paper. Thomas B. Fordham Institute.
- Wagaman, Alex, and Elizabeth Segal. 2014. "The relationship between empathy and attitudes toward government intervention." *Journal of Sociology & Social Welfare* 41.
- Wiswall, Matthew, Leanna Stiefel, Amy Ellen Schwartz, and Jessica Boccoardo. 2014. "Does attending a STEM high school improve student performance? Evidence from New York City." *Economics of Education Review* 40:93–105.
- Wiswall, Matthew, and Basit Zafar. 2017. "Preference for the Workplace, Investment in Human Capital, and Gender." *The Quarterly Journal of Economics* 133, no. 1 (August): 457–507.
- Yeager, David, Paul Hanselman, Gregory Walton, Joseph Murray, Robert Crosnoe, Chandra Muller, Elizabeth Tipton, et al. 2019. "A National Experiment Reveals Where a Growth Mindset Improves Achievement." *Nature* 573:364–369.
- Zebarjadi, Niloufar, Eliyahu Adler, Annika Kluge, Mikko Sams, and Jonathan Levy. 2023. "Ideological Values are Parametrically Associated with Empathy Neural Response to Vicarious Suffering." *Social Cognitive and Affective Neuroscience* 18 (1).

A1 STEM vs. HSS cutoffs

This appendix characterizes STEM vs. HSS cutoffs. Notably, it describes their frequency and achievement range, compares students near them with all high school students, and examines students' choice behavior with respect to STEM and HSS tracks.

To start, Table A1 provides summary statistics for the raw administrative data and the administrative-data RD sample. In the table, “Administrative data-All counties” is the administrative data for all of Romania, “Administrative data-16 counties” restricts to students from the 16 counties with track preference rankings, and “Administrative data-Top 70%” further restricts to students with transition scores in the top 70% of the yearly national distribution. “RD sample-All students” is the administrative-data RD sample (i.e., student-cutoff observations for students from the 16 counties who are near STEM vs. HSS cutoffs). “Prefer STEM” and “Prefer HSS” are the parts of the administrative-data RD sample for STEM- and HSS-above cutoffs.

Table A1 reveals that the administrative data and the administrative-data RD sample are both large. Across all of Romania, more than 470,000 students entered high school during 2015-2017 and 2019. In the 16 counties with track preference rankings, this number was almost 210,000. In comparison, the administrative-data RD sample has 62,301 students, 1,904 cutoffs, and 73,470 student-cutoff observations. These observations are split almost evenly between STEM- and HSS-above cutoffs. In particular, the “Prefer STEM” sample has 37,376 students and 973 cutoffs, while “Prefer HSS” has 36,094 students and 931 cutoffs.⁴⁹

Table A1 also gives insight into the achievement levels and ranges of the different samples. It shows first that the students from the 16 counties with track preference rankings are slightly higher achieving on average than the full set of students in the country. In results not reported, we find that this is because the 16 counties include Bucharest and Cluj, which contain Romania's two largest cities and have its highest-achieving students. Second, the RD sample is also high-achieving on average—though it covers a wide achievement range. In fact, the RD sample's covariate means and standard deviations are similar to those for “Administrative data-Top 70%”, the subset of the 16-county version of the administrative data that limits to students in the top 70% of the national achievement distribution.⁵⁰

Figure A1 offers additional detail on the achievement range covered by the administrative-data RD sample. The figure presents histograms of cutoff and transition scores for this sample, separately for the full sample and by cutoff type. The histograms reveal substantial variation in achievement, in a manner that accords with “Administrative data-Top 70%”. For the full RD sample, the 10th and 90th percentiles of the cutoff (transition) score distribution are, respectively, 6.92 and 9.23 (7.02 and 9.51) on a scale of 1 to 10. Meanwhile, the 10-90 range of transition scores for “Administrative data-Top 70%” is 6.98 to 9.61. Next, when conditioning on cutoff type, Figure A1 shows that the sample for STEM-above cutoffs has higher average achievement than that for HSS-above cutoffs, yet both have wide dispersion.

Table A2 provides additional information on the frequency of cutoff types within the RD sample. For various student groups, it lists the percent of RD sample observations that are for a STEM- rather than HSS-above cutoff. The table shows that the two cutoff types remain balanced in frequency for male and female students. In contrast, STEM-above cutoffs are more common for students who are stronger in math, while HSS-above cutoffs are more common for students

49. In our data construction, students can be in-sample for only one cutoff of a given type. Thus, conditional on cutoff type, the number of student-cutoffs and students is the same.

50. The high average achievement of the RD sample is because Romania's lowest-achieving students usually attend vocational/technical tracks, not STEM or HSS.

who are stronger in language. Also, the split in cutoff types tilts toward HSS-above for students with low baseline achievement and toward STEM-above for high-achievers. Regardless, all student groups have a substantial fraction of observations for each cutoff type. Further, by calculating simple averages of effects over both cutoff types, we ensure that we treat the types equally.

To better characterize the RD sample, we next investigate how students rank STEM and HSS tracks. Table A3 displays the percent of students in the administrative data who rank both track types, while Table A4 summarizes ranking behavior for each track type separately. Results are provided for all students, for students in the top 70% of the national achievement distribution, and for subsets of the top-70% sample by student attributes.⁵¹

Tables A3 and A4 show that ranking STEM and HSS tracks is common. Among all students, 95% rank a STEM track and 77% rank an HSS track. Also, 75% rank both track types, 64% rank both in their top 10 choices, and 50% rank both in their top 5 choices. For the top-70% sample, the shares are even higher, with 98% ranking a STEM track, 83% ranking an HSS track, 80% ranking both, 69% ranking both in their top 10 choices, and 54% ranking both in their top 5 choices. As expected, there is some variation in ranking behavior by student characteristics. However, the variation is relatively limited. For instance, 77% of top-70% male students rank both a STEM and HSS track versus 84% of top-70% female students.

In sum, our setting offers a large number of STEM vs. HSS cutoffs, and the students who are in-sample for them have characteristics that are similar to the top 70% of high school students. The representativeness arises because students commonly rank both STEM and HSS tracks. For some students, this is because they have weak curricular preferences. For others, it is because they face situations where they must trade off their preferred curriculum for other track attributes that they care about (Ainsworth et al. 2023).

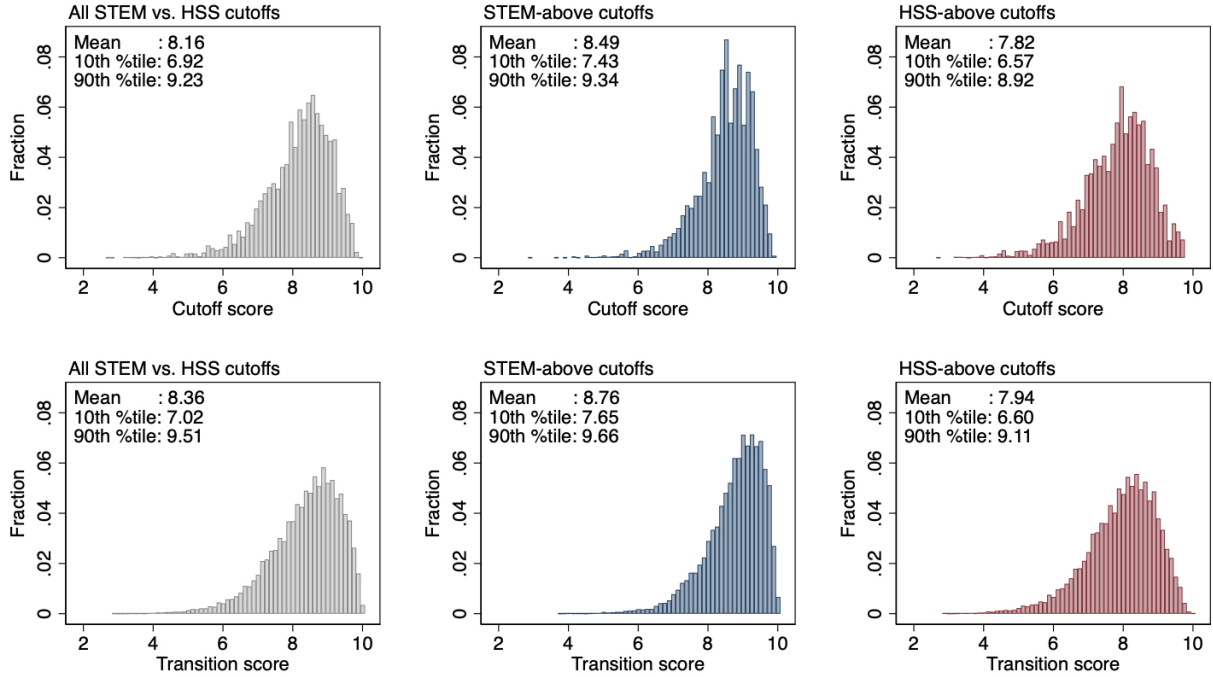
51. These results are necessarily for students from the 16 counties with track preference rankings.

Table A1: Comparing the administrative data
with the administrative-data RD sample

	Transition score								Female		N
	Overall score		Math exam score		Lang. exam score		Grades 5-8 GPA				
	Mean	Std. dev.	Mean	Std. dev.	Mean	Std. dev.	Mean	Std. dev.	Mean	Std. dev.	
Administrative data											
All counties	7.42	1.62	6.52	2.20	7.42	1.85	8.87	0.86	0.53	0.50	472,777
16 counties	7.57	1.61	6.79	2.19	7.57	1.80	8.90	0.86	0.50	0.50	207,632
Top 70%	8.36	0.96	7.78	1.52	8.40	1.04	9.27	0.58	0.53	0.50	152,636
RD sample											
All students	8.36	1.01	7.76	1.57	8.42	1.10	9.27	0.57	0.56	0.50	73,470
Prefer STEM	8.76	0.82	8.39	1.29	8.74	0.93	9.44	0.49	0.55	0.50	37,376
Prefer HSS	7.94	1.01	7.11	1.56	8.10	1.16	9.09	0.60	0.58	0.49	36,094

The table presents summary statistics for different samples. “Administrative data-All counties” is for all of Romania. “Administrative data-16 counties” is for the 16 counties with data on track preference rankings. “Administrative data-Top 70%” is for students in the 16 counties who have transition scores in the top 70% of the yearly national distribution. “RD sample-All students” is the RD sample for the full administrative data. “Prefer STEM” and “Prefer HSS” are, respectively, the portions of “RD sample-All students” for STEM- and HSS-above cutoffs. Transition score variables are on a scale of 1 to 10. In the “Administrative data” rows, N is the number of students; in the other rows, it is the number of student-cutoff combinations. “RD sample-All students” has 62,301 students and 1,904 cutoffs; “Prefer STEM” (“Prefer HSS”) has 37,376 (36,094) students and 973 (931) cutoffs.

Figure A1: Histograms of cutoff scores and transition scores for observations
in the administrative-data RD sample



The figure displays histograms of cutoff scores and transition scores for student-cutoff observations in the administrative-data RD sample. “All STEM vs. HSS cutoffs”, “STEM-above cutoffs”, and “HSS-above cutoffs” correspond to “RD sample-All students”, “Prefer STEM”, and “Prefer HSS” in Table A1, respectively. “Mean”, “10th %tile”, and “90th %tile” are the mean, 10th, and 90th percentiles of the listed variables in the given samples. See Table A1 for information on sample sizes.

Table A2: The percent of observations in the administrative-data RD sample that are for STEM-above cutoffs

	STEM-above	Student-cutoffs
All students	51	73,470
Stronger in math	61	45,077
Stronger in language	35	28,060
Male	53	31,900
Female	49	41,295
Low-achiever	30	27,204
High-achiever	63	46,266

The table lists the percent of student-cutoff observations in the administrative-data RD sample that are in-sample for a STEM-above (rather than HSS-above) cutoff. “Stronger in math” (“language”) is whether a student’s math transition exam score is $\geq$ ($<$) their language score after demeaning scores by cohort. “Low-achiever” (“High-achiever”) is whether the cutoff score for an observation’s cutoff is $\leq$ ($>$) 8. “Student-cutoffs” is the total number of observations, over both STEM- and HSS-above cutoffs.

Table A3: The percent of students in the administrative data who rank both STEM and HSS

	At all	In top 10	In top 5	Students
All students	75	64	50	207,632
Top 70%	80	69	54	152,636
Stronger in math	77	64	49	95,755
Stronger in language	87	78	63	56,881
Male	77	64	50	72,181
Female	84	73	58	80,455
Low-achiever	84	75	61	56,618
High-achiever	79	65	50	96,018

The table displays the percent of students in the administrative data who rank both a STEM and an HSS track. “At all” is whether both track types are ranked among any of a student’s choices. “In top 10” and “In top 5” are whether both are ranked in a student’s 10 or 5 most-preferred choices. “Top 70%” is students with transition scores in the top 70% of the yearly national distribution. “Stronger in math”, “Stronger in language”, “Male”, “Female”, “Low-achiever”, and “High-achiever” are subsets of “Top 70%”. “Stronger in math” (“language”) is whether a student’s math transition exam score is $\geq$ ($<$) their language score after demeaning scores by cohort. “Low-achiever” (“High-achiever”) is whether a student’s transition score is $\leq$ ($>$) 8.

Table A4: Summary statistics on ranking STEM and HSS for students in the administrative data

	STEM				HSS				Students
	% rank at all	% rank in top 10	% rank in top 5	Mean min. rank	% rank at all	% rank in top 10	% rank in top 5	Mean min. rank	
All students	95	92	86	3.1	77	68	58	6.8	207,632
Top 70%	98	95	90	2.6	83	73	63	6.2	152,636
Stronger in math	98	97	94	2.1	78	67	54	7.4	95,755
Stronger in language	96	92	85	3.5	90	85	77	4.3	56,881
Male	98	96	93	2.2	78	67	55	7.3	72,181
Female	97	93	88	3.0	87	80	69	5.3	80,455
Low-achiever	97	94	88	3.1	86	80	71	5.0	56,618
High-achiever	98	95	92	2.3	81	70	58	7.0	96,018

The table provides summary statistics on students’ ranking behavior with respect to STEM and HSS tracks. “Mean min. rank” is the across-student mean of the lowest (most favorable) rank that a student gives to a track of a given type; if a student does not rank any track of the type, we impute a value of 14, which is the median total number of tracks ranked. Other details match those in Table A3.

A2 An example of our selection of cutoffs

This appendix provides an example of how we select cutoffs.

Consider a student with a transition score of 7.6 who ranks seven tracks. Let these tracks be labeled alphabetically in order of the student’s preferences over them (i.e., the student prefers A to B to C, etc.). Also, suppose the curricula and cutoff scores for the tracks are:

A: STEM, 9.5; B: STEM, 9.6; C: HSS, 8.5; D: HSS, 7.5; E: STEM, 7.0; F: HSS, 6.5; G: Other, 0.

Here, the 0 for G indicates that this track was not over-subscribed and thus accepts all students regardless of their transition score.

We start by identifying the mapping from potential transition scores, t_i , to curricular assignment for the student. This is:

$$\begin{cases} \text{STEM} & \text{if } t_i > 9.5 \\ \text{HSS} & \text{if } t_i \in (7.5, 9.5) \\ \text{STEM} & \text{if } t_i \in (7.0, 7.5) \\ \text{HSS} & \text{if } t_i \in (6.5, 7.0) \\ \text{Other} & \text{if } t_i < 6.5. \end{cases}$$

To select cutoffs, we focus on the lowest (highest) cutoff that the student is below (above) where scoring above (below) the cutoff would change the student’s curricular assignment from STEM to HSS or HSS to STEM. Since the student’s transition score in this example is 7.6, the student is assigned to HSS, and the relevant cutoffs are those with cutoff scores of 9.5 and 7.5. 9.5 is for a “STEM-above” cutoff, while 7.5 is for an “HSS-above” cutoff. Finally, we impose a bandwidth restriction, requiring the gap between a cutoff score and the student’s transition score to be less than 1.25 points. As a result, we drop the STEM-above cutoff. Thus, the student is in-sample for an HSS-above cutoff with a cutoff score of 7.5.

Suppose instead that the student’s transition score were 7.4. Then the student would be assigned to STEM and would be in-sample for two cutoffs: an HSS-above cutoff with a cutoff score of 7.5 and a STEM-above cutoff with a cutoff score of 7.0. Suppose next that the student’s transition score were 6.6. Then the student would be assigned to HSS and would be in-sample for a STEM-above cutoff with a cutoff score of 7.0; the 6.5 cutoff score would be irrelevant because it is for a cutoff that involves a switch between HSS and Other. Finally, suppose the student’s transition score were 6.4. Then the student would be in-sample for no cutoffs because the student’s curricular assignment would be Other.

One decision is what to do if a student ranks an Other track between STEM and HSS tracks. For instance, consider a scenario that is the same as above except that C is Other rather than HSS. The new mapping from potential transition scores to curricular assignment is:

$$\begin{cases} \text{STEM} & \text{if } t_i > 9.5 \\ \text{Other} & \text{if } t_i \in (8.5, 9.5) \\ \text{HSS} & \text{if } t_i \in (7.5, 8.5) \\ \text{STEM} & \text{if } t_i \in (7.0, 7.5) \\ \text{HSS} & \text{if } t_i \in (6.5, 7.0) \\ \text{Other} & \text{if } t_i < 6.5. \end{cases}$$

Suppose a student has a transition score of 8.4. This student is assigned to HSS and is in-sample for an HSS-above cutoff with a cutoff score of 7.5. The question is whether the student should also be in-sample for a STEM-above cutoff with a cutoff score of 9.5. 9.5 is the lowest score such that if the student scored above that score, the student's curricular assignment would switch from HSS to STEM. However, in the process, the student's assignment would first switch from HSS to Other. We decided to include the student as in-sample for the STEM-above cutoff. However, we find that not doing so does not change our results. This is because such scenarios are rare.

A3 RD samples and validity tests

This appendix summarizes the various RD samples that we use in obtaining results and conducts a series of validity tests on them.

Depending on the outcome under consideration, we use six different RD samples. These are named according to the data source of the outcome.⁵² The “2015-2017 and 2019 cohorts” sample is for outcomes that come from the administrative data (the “16 county” version) and can be observed for all cohorts. “2015-2017 cohorts” is also from the administrative data but for outcomes based on high school enrollment histories, which are not available for the 2019 cohort. “End-of-high-school survey”, “First follow-up survey”, and “Second follow-up survey” are for outcomes obtained from the listed surveys. Finally, “First follow-up or peer surveys” is for outcomes based on questions asked in both of these surveys; here, we first use the first follow-up survey and then impute missing values using the peer survey.⁵³

In this appendix, we also study a seventh RD sample (the “2019 cohort”), which employs the administrative data for the 2019 cohort. In the main paper, we never use this sample to calculate results. However, the surveys apply to the 2019 cohort; thus, we include the sample as a benchmark.

Figure A2 provides histograms of the running variable for each RD sample. The histograms show no evidence of bunching on either side of the RD thresholds. To rule out bunching more formally, we run the Cattaneo, Jansson, and Ma (2020) discontinuity-in-density test. For all samples, we fail to reject that the running variable density is continuous across the threshold. Moreover, none of the p-values are close to standard levels of significance.

Table A5 presents summary statistics and balance tests for the RD samples. The columns are for different covariates, and “Mean” and “Std. dev.” are the mean and standard deviation of the listed covariate in the given RD sample. “STEM” and “Intercept” are the same as in Panel A of Table 2—namely, the estimate of the ATE of STEM assignment, β , and the predicted value on the HSS side of the RD threshold. Finally, “Student-cutoffs” is the number of observations; the numbers of students and cutoffs are provided in the table notes.

Table A5 shows that covariates are well-balanced across the RD threshold for all samples. This is confirmed by tests of joint significance, which reveal that in no sample can we reject that all across-threshold covariate differences are zero.⁵⁴ When providing results in the rest of the paper, we always run a robustness specification that controls for the covariates in Table A5. As expected given the observed balance, we find that adding the covariates has little impact.

Table A6 explores whether STEM assignment affects the probability that a student appears in our surveys. In each panel, the sample is the “2019 cohort” RD sample, and the outcome is an indicator for being in the listed survey(s). “Mean” is the mean of the outcome among the sample; as always, “STEM” and “Intercept” are, respectively, the estimated effect of STEM assignment and the predicted value for students at the RD threshold who are assigned to HSS. Results are provided for all students and by initial curricular preferences, relative strength in math vs. language, gender, and baseline achievement.

Table A6 has a few takeaways. First, 22% of the observations in the “2019 cohort” RD sample appear in the end-of-high-school survey. The corresponding shares in the first follow-up survey,

52. The running variable is always from the administrative data.

53. Recall that first follow-up and peer surveys occurred at a similar time.

54. A complication is that β is mechanically zero for some covariates (scoring above a cutoff, being in-sample for a STEM- rather than HSS-above cutoff, and the overall transition score). We re-run the joint tests excluding these variables. P-values remain insignificant for all samples except “First follow-up or peer surveys”, where we reject the null at a $p = 0.045$ significance level.

the combination of the first follow-up and peer surveys, and the second follow-up survey are 7%, 18%, and 7%, respectively. Second, in no survey does STEM assignment significantly affect the overall response rate; however, effects on response rates can sometimes be statistically significant for particular student types. Third, the significance seems to be due to multiple hypothesis testing: When we run joint tests, we always fail to reject that effects are zero for all student types.

An important question is whether the survey-based RD samples are representative of those for the administrative data. Some insight can be gained from Table A5 by comparing covariate means and standard deviations for the different samples. These are usually quite similar, regardless of whether we compare the survey samples with the “2019 cohort” sample or with the entire “2015-2017 and 2019 cohorts” sample. An exception is that females represent a larger share of observations in the survey samples (64-67% versus 57% for “2019 cohort” and 56% for “2015-2017 and 2019 cohorts”), which suggests that females were more willing than males to respond to our surveys.⁵⁵

We gain additional evidence on representativeness by comparing treatment effects for different samples. Specifically, on each sample, we estimate the ATE of STEM assignment, β , for the outcomes that can be observed in all samples: attributes of students’ assigned tracks (Table A7) and students’ baccalaureate performance (Table A8). In the tables, the first panel is for the full data (the “2015-2017 and 2019 cohorts” sample) and matches the values presented elsewhere in the paper. The other panels restrict to observations from the given sample. In terms of results, the tables show that effect estimates for the survey samples are close to those for the administrative-data samples; this is especially the case when comparing to the “2019 cohort” sample.

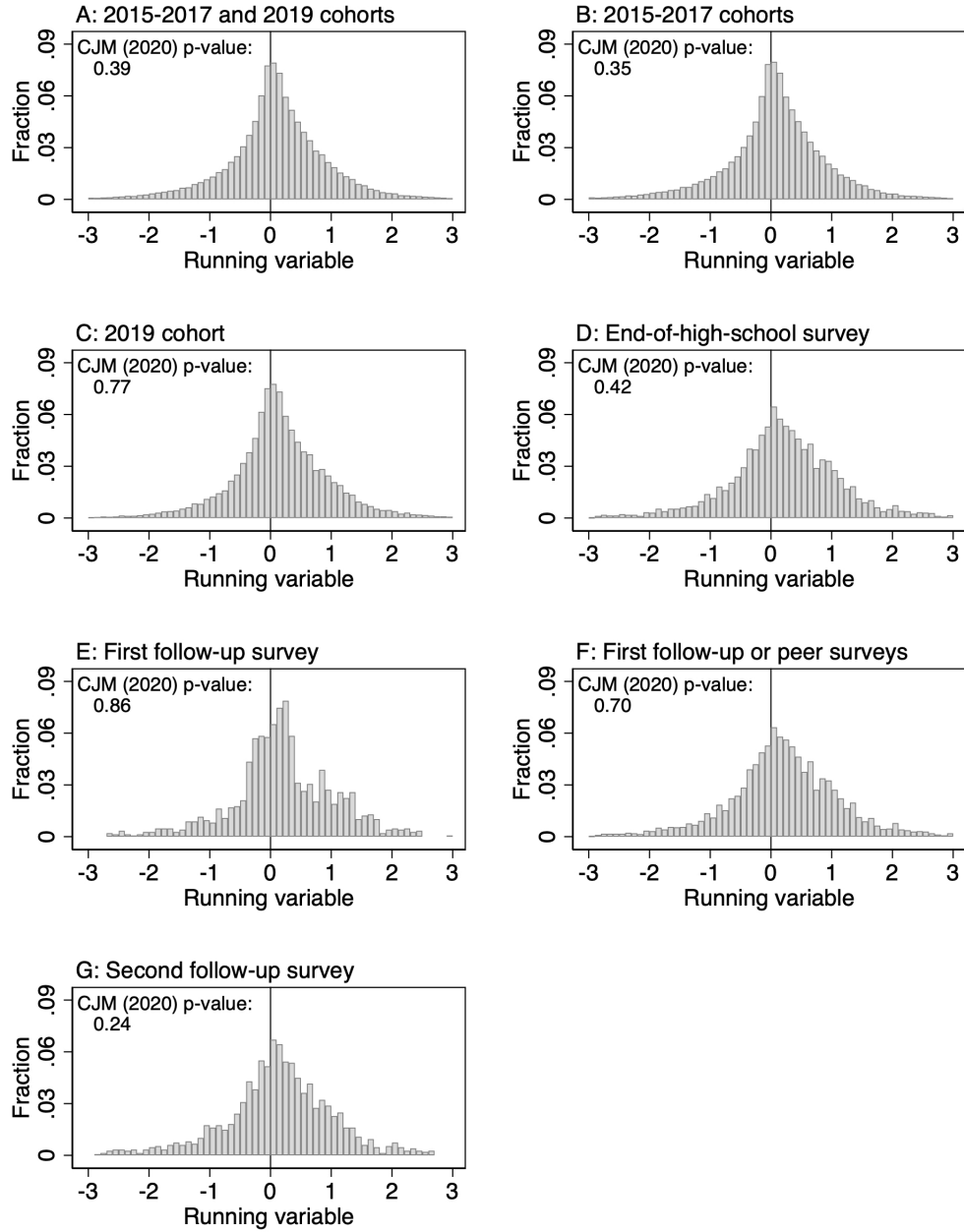
As a last exercise, we test survey quality. We exploit the fact that we have measures of baccalaureate performance in both the administrative data and in some of the surveys. In particular, in the administrative data, we observe actual performance for all students. Meanwhile, in the first follow-up survey we asked students to report their performance, and in the peer survey we asked students to report the performance of their peers.

Table A9 provides two sets of results related to the accuracy of reported baccalaureate performance. First, it displays mean absolute differences (MAD) between reported and actual performance. Second, it presents estimates of the effects of STEM assignment on reported performance, which can be compared with the estimates for effects on actual performance in Table A8. Results are presented separately for the “First follow-up” and the “First follow-up or peer” RD samples, so as to illuminate whether response quality differs for self- and peer-reports. The results suggest that the survey data are high quality. Mean absolute differences between reported and actual performance are small, and the effect estimates in Table A9 are similar to those in Table A8.

Overall, the results lend confidence to our empirical strategy and surveys. There is little evidence of manipulation of the running variable or selection into survey response. The RD samples for the surveys appear representative of those for the administrative data. Further, the quality of the survey data seems high.

55. Another possibility is that more females were in class when we conducted the end-of-high-school survey. This is somewhat contradicted by the fact that high school graduation rates are similar by gender (Table A16).

Figure A2: Histograms of the running variable for the RD samples



The figure displays histograms of the running variable for the paper's various RD samples. See the text of Appendix A3 for definitions of the samples. The "CJM (2020) p-value" is the p-value from the Cattaneo, Jansson, and Ma (2020) discontinuity-in-density test. A small p-value would indicate rejection of the null hypothesis that the running variable density is continuous at the RD threshold.

Table A5: Summary statistics and balance tests for the RD samples

	Above the cutoff	Prefer STEM	Transition score				Female	Years of schooling	
			Overall score	Math exam score	Language exam score	Grades 5-8 GPA		Father	Mother
Panel A: 2015-2017 and 2019 cohorts									
STEM	0.000 (0.000)	0.000 (0.000)	0.000 (0.000)	-0.005 (0.008)	0.000 (0.008)	0.005 (0.005)	-0.010 (0.007)	-	-
Intercept	0.50	0.50	8.16	7.49	8.23	9.16	0.52	-	-
Mean	0.66	0.51	8.36	7.76	8.42	9.27	0.56	-	-
Std. dev.	0.47	0.50	1.01	1.57	1.10	0.57	0.50	-	-
Student-cutoffs	73,470	73,470	73,470	73,470	73,470	73,470	73,470	-	-
Panel B: 2015-2017 cohorts									
STEM	0.000 (0.000)	0.000 (0.000)	0.000 (0.000)	-0.000 (0.009)	-0.002 (0.009)	0.003 (0.006)	-0.007 (0.009)	-	-
Intercept	0.50	0.50	8.21	7.67	8.19	9.13	0.52	-	-
Mean	0.66	0.52	8.42	7.94	8.40	9.24	0.56	-	-
Std. dev.	0.47	0.50	1.01	1.54	1.09	0.59	0.50	-	-
Student-cutoffs	55,221	55,221	55,221	55,221	55,221	55,221	55,221	-	-
Panel C: 2019 cohort									
STEM	0.000 (0.000)	0.000 (0.000)	0.000 (0.000)	-0.011 (0.016)	0.002 (0.016)	0.006 (0.010)	-0.021 (0.015)	-	-
Intercept	0.50	0.50	7.98	6.95	8.37	9.26	0.55	-	-
Mean	0.64	0.48	8.16	7.22	8.49	9.33	0.57	-	-
Std. dev.	0.48	0.50	0.99	1.50	1.11	0.52	0.49	-	-
Student-cutoffs	18,249	18,249	18,249	18,249	18,249	18,249	18,249	-	-
Panel D: End-of-high-school survey									
STEM	0.000 (0.000)	0.000 (0.000)	0.000 (0.000)	-0.016 (0.039)	0.007 (0.038)	-0.007 (0.021)	-0.067* (0.034)	-0.089 (0.116)	-0.088 (0.117)
Intercept	0.50	0.50	7.88	6.76	8.31	9.27	0.64	13.0	13.5
Mean	0.65	0.48	8.08	7.09	8.45	9.34	0.64	13.2	13.5
Std. dev.	0.48	0.50	1.00	1.52	1.11	0.49	0.48	1.94	1.73
Student-cutoffs	3,987	3,987	3,987	3,987	3,987	3,987	3,987	3,987	3,987
Panel E: First follow-up survey									
STEM	0.000 (0.000)	0.000 (0.000)	0.000 (0.000)	0.021 (0.067)	0.001 (0.061)	-0.043 (0.039)	-0.026 (0.061)	-0.198 (0.270)	-0.116 (0.213)
Intercept	0.50	0.50	7.98	6.88	8.41	9.32	0.66	13.2	13.5
Mean	0.66	0.51	8.19	7.25	8.54	9.38	0.67	13.1	13.5
Std. dev.	0.47	0.50	0.93	1.47	1.00	0.46	0.47	1.95	1.75
Student-cutoffs	1,159	1,159	1,159	1,159	1,159	1,159	1,159	1,159	1,159
Panel F: First follow-up or peer surveys									
STEM	0.000 (0.000)	0.000 (0.000)	0.000 (0.000)	-0.021 (0.039)	0.009 (0.036)	-0.014 (0.023)	-0.063* (0.037)	-0.082 (0.135)	-0.095 (0.130)
Intercept	0.50	0.50	7.90	6.78	8.34	9.28	0.64	13.1	13.5
Mean	0.65	0.48	8.10	7.10	8.47	9.34	0.64	13.2	13.5
Std. dev.	0.48	0.50	1.00	1.52	1.09	0.50	0.48	1.92	1.73
Student-cutoffs	3,327	3,327	3,327	3,327	3,327	3,327	3,327	3,327	3,327
Panel G: Second follow-up survey									
STEM	0.000 (0.000)	0.000 (0.000)	0.000 (0.000)	0.030 (0.067)	-0.012 (0.056)	0.001 (0.042)	-0.061 (0.063)	0.206 (0.258)	-0.071 (0.216)
Intercept	0.50	0.50	7.93	6.81	8.35	9.30	0.63	12.7	13.2
Mean	0.66	0.48	8.14	7.17	8.51	9.36	0.64	13.1	13.4
Std. dev.	0.47	0.50	0.95	1.49	1.02	0.48	0.48	1.96	1.75
Student-cutoffs	1,199	1,199	1,199	1,199	1,199	1,199	1,199	1,199	1,199

The table presents summary statistics and balance tests for the RD samples. The columns list the covariates that are used as outcomes in the balance tests. The panels are for the different samples; see the text of Appendix A3 for definitions of the samples. “Mean” and “Std. dev.” are the mean and standard deviation of a listed covariate in a given sample. “STEM” and “Intercept” have the same interpretation as in Panel A of Table 2. “Above the cutoff” and “Prefer STEM” are indicators for scoring above a cutoff and being in-sample for a STEM- rather than HSS-above cutoff. “Years of schooling” is from the end-of-high-school survey. Sample-specific p-values for tests that effects on the listed covariates are all zero are 0.353, 0.737, 0.585, 0.379, 0.959, 0.156, and 0.968, respectively. The effect of STEM assignment is mechanically zero for “Above the cutoff”, “Prefer STEM”, and the overall transition score. Excluding these variables, p-values become 0.362, 0.846, 0.333, 0.146, 0.888, 0.045, and 0.900. The numbers of students (cutoffs) in the samples are 62,301 (1,904), 46,789 (1,411), 15,512 (493), 3,562 (293), 1,046 (190), 2,978 (280), and 1,089 (198), respectively. Other details match those in Panel A of Table 2.

Table A6: Selection into the surveys

	All	Prefer		Stronger in		Gender		Achievement	
		STEM	HSS	Math	Language	Male	Female	Low	High
<i>Panel A: End-of-high-school survey</i>									
STEM	0.010 (0.015)	0.005 (0.015)	0.015 (0.024)	0.007 (0.016)	0.009 (0.021)	0.039** (0.018)	-0.007 (0.019)	0.020 (0.025)	0.006 (0.016)
Intercept	0.19	0.18	0.21	0.18	0.20	0.15	0.22	0.20	0.17
Mean	0.22	0.22	0.22	0.21	0.23	0.19	0.25	0.24	0.20
<i>Panel B: First follow-up survey</i>									
STEM	0.002 (0.008)	0.006 (0.009)	-0.003 (0.011)	0.007 (0.010)	-0.003 (0.012)	0.010 (0.010)	-0.006 (0.012)	0.010 (0.012)	-0.003 (0.010)
Intercept	0.07	0.07	0.08	0.06	0.08	0.05	0.09	0.07	0.07
Mean	0.07	0.07	0.06	0.07	0.07	0.05	0.08	0.07	0.07
<i>Panel C: First follow-up or peer surveys</i>									
STEM	0.008 (0.012)	0.005 (0.013)	0.010 (0.019)	0.002 (0.015)	0.009 (0.018)	0.030* (0.016)	-0.005 (0.017)	0.022 (0.020)	0.001 (0.014)
Intercept	0.16	0.15	0.18	0.15	0.17	0.13	0.19	0.17	0.15
Mean	0.18	0.18	0.19	0.18	0.19	0.15	0.21	0.20	0.17
<i>Panel D: Second follow-up survey</i>									
STEM	0.008 (0.007)	0.007 (0.008)	0.008 (0.010)	0.006 (0.010)	0.010 (0.011)	0.014 (0.010)	0.006 (0.011)	0.021* (0.011)	-0.004 (0.008)
Intercept	0.06	0.05	0.06	0.06	0.06	0.05	0.07	0.06	0.06
Mean	0.07	0.07	0.07	0.07	0.07	0.06	0.08	0.08	0.06
Student-cutoffs	18,249	8,850	9,399	9,320	8,837	7,773	10,399	8,703	9,546

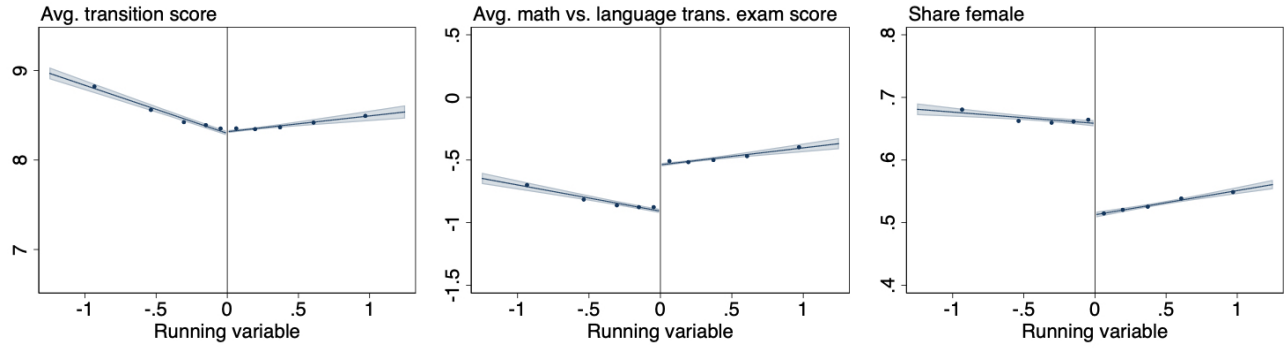
The table presents results on selection into the surveys for observations from the “2019 cohort” RD sample. In each panel, the outcome is an indicator equal to 1 if an observation appears in the listed survey(s). The columns are for sub-samples representing different student types. “Prefer STEM” (“Prefer HSS”) is students near STEM-above (HSS-above) cutoffs, “Stronger in Math” (“Language”) is whether a student’s math transition exam score is $\geq$ ($<$) their language score after demeaning scores by cohort, “Gender” is a student’s gender, and “Achievement-Low” (“High”) is whether a cutoff score is $\leq$ ($>$) 8. “Mean” is the mean of the outcome among a given sample. “STEM” and “Intercept” have the same interpretation as in Table 2. For “Prefer STEM” (“Prefer HSS”), the effect of STEM assignment is β_{SH} (β_{HS}). For the other columns, it is β . Panel-specific p-values for tests that the effects by student type are all zero are 0.397, 0.924, 0.449, and 0.337, respectively. The number of observations is the same for all outcomes, so it is displayed in a single row at the bottom of the table.

Table A7: Effects on attributes of students' assigned tracks by RD sample

	Average		Share female
	Transition score	Math vs. language transition exam score	
Panel A: 2015-2017 and 2019 cohorts			
STEM	-0.007 (0.007)	0.357*** (0.010)	-0.150*** (0.004)
Intercept	8.33	-0.89	0.66
Student-cutoffs	73,470	73,470	73,470
Panel B: 2015-2017 cohorts			
STEM	-0.011 (0.008)	0.351*** (0.011)	-0.151*** (0.005)
Intercept	8.40	-0.67	0.66
Student-cutoffs	55,221	55,221	55,221
Panel C: 2019 cohort			
STEM	0.009 (0.015)	0.387*** (0.019)	-0.149*** (0.008)
Intercept	8.14	-1.56	0.66
Student-cutoffs	18,249	18,249	18,249
Panel D: End-of-high-school survey			
STEM	0.012 (0.027)	0.405*** (0.030)	-0.132*** (0.013)
Intercept	8.08	-1.65	0.65
Student-cutoffs	3,987	3,987	3,987
Panel E: First follow-up survey			
STEM	0.013 (0.038)	0.437*** (0.038)	-0.121*** (0.018)
Intercept	8.16	-1.65	0.65
Student-cutoffs	1,159	1,159	1,159
Panel F: First follow-up or peer surveys			
STEM	0.014 (0.028)	0.410*** (0.031)	-0.129*** (0.014)
Intercept	8.10	-1.65	0.65
Student-cutoffs	3,327	3,327	3,327
Panel G: Second follow-up survey			
STEM	-0.017 (0.034)	0.415*** (0.045)	-0.134*** (0.021)
Intercept	8.14	-1.67	0.65
Student-cutoffs	1,199	1,199	1,199

The table is the same as Panel A of Table 2 but for the listed RD samples.

Figure A3: RD plots of effects on attributes of students' assigned tracks by sample



The figure is an alternative version of Figure 1. It displays RD plots that fit a single linear spline over the entire data (i.e., for both STEM- and HSS-above cutoffs). These plots correspond to a version of Equation (1) that does not allow β_{f_j} , γ_{f_j} , or δ_{f_j} to vary by cutoff type. As in Figure 1, the lines are linear spline estimates, and the shaded areas are 95% confidence intervals. The dots are estimates from a version of the regression equation that replaces the linear spline running variable controls with indicators for bins in the running variable. The bins are constructed to have equal populations by curricular assignment, and the dots are plotted at the mean running variable value within a bin. The estimated ATE of STEM assignment is equal to the vertical difference between the right and left lines at the RD threshold. All other details follow Figure 1.

Table A8: Effects on baccalaureate performance by RD sample

	All students			Low-achievers			High-achievers		
	Take the exam	Pass the exam	Exam score	Take the exam	Pass the exam	Exam score	Take the exam	Pass the exam	Exam score
<i>Panel A: 2015-2017 and 2019 cohorts</i>									
STEM	-0.011*** (0.003)	-0.036*** (0.004)	-0.351*** (0.015)	-0.035*** (0.007)	-0.111*** (0.010)	-0.502*** (0.032)	-0.001 (0.003)	-0.008** (0.003)	-0.302*** (0.016)
Intercept	0.95	0.92	8.08	0.92	0.86	7.45	0.98	0.97	8.44
Student-cutoffs	73,470	73,470	69,938	27,204	27,204	24,751	46,266	46,266	45,187
<i>Panel B: 2015-2017 cohorts</i>									
STEM	-0.012*** (0.003)	-0.039*** (0.005)	-0.341*** (0.017)	-0.047*** (0.010)	-0.135*** (0.013)	-0.507*** (0.040)	-0.001 (0.003)	-0.008** (0.004)	-0.298*** (0.018)
Intercept	0.95	0.91	8.06	0.91	0.84	7.34	0.97	0.97	8.41
Student-cutoffs	55,221	55,221	52,369	18,501	18,501	16,561	36,720	36,720	35,808
<i>Panel C: 2019 cohort</i>									
STEM	-0.007 (0.005)	-0.030*** (0.007)	-0.390*** (0.028)	-0.013 (0.010)	-0.068*** (0.015)	-0.503*** (0.051)	-0.002 (0.005)	-0.005 (0.005)	-0.318*** (0.036)
Intercept	0.97	0.94	8.14	0.95	0.92	7.69	0.99	0.98	8.54
Student-cutoffs	18,249	18,249	17,569	8,703	8,703	8,190	9,546	9,546	9,379
<i>Panel D: End-of-high-school survey</i>									
STEM	0.001 (0.006)	-0.023** (0.010)	-0.455*** (0.057)	-0.003 (0.015)	-0.050** (0.020)	-0.479*** (0.107)	0.006 (0.005)	0.006 (0.009)	-0.376*** (0.083)
Intercept	0.99	0.96	8.27	0.98	0.95	7.81	0.99	0.99	8.71
Student-cutoffs	3,987	3,987	3,929	2,093	2,093	2,038	1,894	1,894	1,891
<i>Panel E: First follow-up survey</i>									
STEM	-0.004 (0.009)	-0.015 (0.019)	-0.551*** (0.096)	-0.007 (0.015)	-0.061* (0.033)	-0.678*** (0.172)	0.000 (0.000)	0.023 (0.019)	-0.499*** (0.148)
Intercept	0.99	0.99	8.45	0.99	0.99	8.04	1.00	0.99	8.86
Student-cutoffs	1,159	1,159	1,149	547	547	537	612	612	612
<i>Panel F: First follow-up or peer surveys</i>									
STEM	-0.009 (0.006)	-0.027** (0.012)	-0.438*** (0.061)	-0.023 (0.014)	-0.065*** (0.023)	-0.435*** (0.113)	0.002 (0.002)	0.008 (0.008)	-0.353*** (0.086)
Intercept	0.99	0.97	8.28	1.00	0.97	7.80	1.00	0.99	8.71
Student-cutoffs	3,327	3,327	3,282	1,703	1,703	1,659	1,624	1,624	1,623
<i>Panel G: Second follow-up survey</i>									
STEM	0.010 (0.008)	-0.014 (0.021)	-0.358*** (0.107)	0.005 (0.007)	-0.075* (0.039)	-0.134 (0.165)	0.013 (0.012)	0.022 (0.014)	-0.391*** (0.143)
Intercept	0.99	0.97	8.25	0.99	0.98	7.66	0.98	0.98	8.75
Student-cutoffs	1,199	1,199	1,185	618	618	606	581	581	579

The table presents results for the ATE of STEM assignment, β , on baccalaureate performance for the listed RD samples. The values in Panel A under the “All students” heading match those in the first three columns of Table 6. The Panel A values under “Low-achievers” (“High-achievers”) match the values for “Achievement-Low” (“High”) in the first three panels of Table A34. “Exam score” is not available for students who did not take the exam. Other details are as in Panel A of Table 2.

Table A9: Effects on reported baccalaureate performance in the survey RD samples

	First follow-up			First follow-up or peer	
	Take the exam	Pass the exam	Exam score	Take the exam	Pass the exam
<i>Panel A: All students</i>					
STEM	-0.004 (0.009)	-0.011 (0.018)	-0.469*** (0.126)	-0.004 (0.003)	-0.030*** (0.011)
Intercept	1.00	0.99	8.55	1.00	0.98
Std. dev.	0.08	0.14	1.09	0.08	0.20
MAD: survey vs. admin.	0.01	0.01	0.24	0.01	0.02
Student-cutoffs	1,159	1,159	1,130	3,327	3,327
<i>Panel B: Low-achievers</i>					
STEM	-0.018 (0.015)	-0.045 (0.033)	-0.783*** (0.187)	-0.008 (0.006)	-0.055*** (0.021)
Intercept	1.00	0.98	8.37	0.99	0.96
Std. dev.	0.10	0.19	1.12	0.10	0.26
MAD: survey vs. admin.	0.01	0.02	0.31	0.02	0.04
Student-cutoffs	547	547	527	1,703	1,703
<i>Panel C: High-achievers</i>					
STEM	0.000 (0.000)	0.008 (0.012)	-0.313 (0.194)	0.000 (0.000)	-0.006 (0.006)
Intercept	1.00	1.01	8.83	1.00	1.00
Std. dev.	0.04	0.07	0.94	0.02	0.07
MAD: survey vs. admin.	0.00	0.00	0.19	0.00	0.00
Student-cutoffs	612	612	603	1,624	1,624

The table presents results for the ATE of STEM assignment, β , on reported baccalaureate performance from the surveys. “First follow-up” uses self-reports from the first follow-up survey. “First follow-up or peer” fills in peer-reports for students who did not respond to the first follow-up survey. “MAD: survey vs. admin.” is the mean absolute difference between the given survey-based outcome and the corresponding outcome from the administrative data. “Low-achievers” and “High-achievers” use the same sample split as in the “Achievement-Low” and “High” columns in Table A6. Other details match those in Panel A of Table 2.

A4 Adjusting for the effect of scoring above a cutoff

As discussed in Section 4.1, the RDs for STEM- and HSS-above cutoffs combine three sub-treatments, with the third being whether a student scores above or below a cutoff. Since the third sub-treatment has been studied in prior work, we hope to eliminate it and thereby recover an effect that reflects only the first two. This effect is the impact of a change in curriculum together with exposure to the types of peers who congregate in STEM vs. HSS. Alternatively, it can be understood as the impact of being assigned to a STEM track when choosing among STEM and HSS tracks that are equal on vertical quality dimensions.

In the main paper, our approach for eliminating the third sub-treatment is to calculate an average over STEM- and HSS-above cutoffs. In this appendix, we delve into the validity of our approach. We first use potential outcomes notation to derive what the approach recovers. We then show that results are robust to including an additional adjustment.

A4.1 Theory

To understand the treatment effect that our approach recovers, denote i 's treatment status as T_i , and let T_i take on four values: SA, SB, HA, and HB, where the first letter indicates whether i is assigned to STEM (S) or HSS (H), and the second letter indicates whether i scores above (A) or below (B) a cutoff. Also, let $Y_i(\cdot)$ be i 's potential outcome function, with $Y_i = Y_i(T_i)$ being i 's realized outcome.

We observe four types of cutoffs: STEM-above vs. HSS-below (SH), HSS-above vs. STEM-below (HS), STEM-above vs. STEM-below (SS), and HSS-above vs. HSS-below (HH). Importantly, we observe the four cutoff types for different sets of students. Since membership in a set depends on a student's preference ranking over tracks, we name the sets Prefer STEM (PS), Prefer HSS (PH), Strongly Prefer STEM (SPS), and Strongly Prefer HSS (SPH), respectively.⁵⁶

The RD design lets us estimate treatment effects for each cutoff type. The treatment effects are always “at the RD threshold”—i.e., conditioned on the running variable being equal to 0. To simplify notation, we omit the conditioning.

For STEM-above vs. HSS-below and HSS-above vs. STEM-below cutoffs, we estimate the effect of being on the STEM side of a cutoff. The specific estimands are:

- STEM-above vs. HSS-below: $E[Y_i(\text{SA}) - Y_i(\text{HB}) | i \in \text{PS}] =$

$$E[Y_i(\text{SB}) - Y_i(\text{HB}) | i \in \text{PS}] + E[Y_i(\text{SA}) - Y_i(\text{SB}) | i \in \text{PS}]. \quad (2)$$

- HSS-above vs. STEM-below: $E[Y_i(\text{SB}) - Y_i(\text{HA}) | i \in \text{PH}] =$

$$E[Y_i(\text{SB}) - Y_i(\text{HB}) | i \in \text{PH}] - E[Y_i(\text{HA}) - Y_i(\text{HB}) | i \in \text{PH}]. \quad (3)$$

These estimands are, respectively: (i) the average effect of being assigned to STEM vs. HSS while scoring below a cutoff plus the average effect of scoring above vs. below a cutoff while being assigned to STEM for students in Prefer STEM, and (ii) the average effect of being assigned to STEM vs. HSS while scoring below a cutoff minus the average effect of scoring above vs. below a cutoff while being assigned to HSS for students in Prefer HSS.

56. The “strongly” modifier for SPS and SPH is because students likely have a strong preference for a curriculum if they rank it multiple times in succession.

We also estimate the average of (2) and (3). Here, the estimand is:

$$\begin{aligned}
& \frac{E[Y_i(\text{SA}) - Y_i(\text{HB})|i \in \text{PS}] + E[Y_i(\text{SB}) - Y_i(\text{HA})|i \in \text{PH}]}{2} \\
&= \frac{E[Y_i(\text{SB}) - Y_i(\text{HB})|i \in \text{PS}] + E[Y_i(\text{SB}) - Y_i(\text{HB})|i \in \text{PH}]}{2} \\
&+ \frac{E[Y_i(\text{SA}) - Y_i(\text{SB})|i \in \text{PS}] - E[Y_i(\text{HA}) - Y_i(\text{HB})|i \in \text{PH}]}{2}.
\end{aligned} \tag{4}$$

The first term after the equality is the average effect of being assigned to STEM vs. HSS while scoring below a cutoff, averaged over students in Prefer STEM and Prefer HSS. The second term is half the difference in (a) the average effect of scoring above vs. below a cutoff while being assigned to STEM for students in Prefer STEM and (b) the average effect of scoring above vs. below a cutoff while being assigned to HSS for students in Prefer HSS.

Finally, for STEM-above vs. STEM-below and HSS-above vs. HSS-below cutoffs, we can estimate the effect of being on the higher-scoring side of a cutoff. The specific estimands are:

- STEM-above vs. STEM-below:

$$E[Y_i(\text{SA}) - Y_i(\text{SB})|i \in \text{SPS}]. \tag{5}$$

- HSS-above vs. HSS-below:

$$E[Y_i(\text{HA}) - Y_i(\text{HB})|i \in \text{SPH}]. \tag{6}$$

The interpretations are: (i) the average effect of scoring above vs. below a cutoff while being assigned to STEM for students in Strongly Prefer STEM, and (ii) the average effect of scoring above vs. below a cutoff while being assigned to HSS for students in Strongly Prefer HSS.

In the formulas for (2)-(4), the first terms are of interest, while the second terms are not. We cannot estimate the second terms, as we do not observe same-curriculum cutoffs for the Prefer STEM or Prefer HSS sets. As such, in the main paper, we do not attempt to isolate the first terms and instead simply estimate (2)-(4). That said, we do observe same-curriculum cutoffs for groups that are similar to Prefer STEM and Prefer HSS, namely Strongly Prefer STEM and Strongly Prefer HSS. In the next subsection, we estimate (5) and (6) in order to gain insight into the magnitudes of the second terms. In addition, we adjust our main approach to estimate an approximate version of the first term in (4). Specifically, we estimate the left-hand side of (4) minus half the difference in (5) and (6):

$$\begin{aligned}
& \frac{E[Y_i(\text{SA}) - Y_i(\text{HB})|i \in \text{PS}] + E[Y_i(\text{SB}) - Y_i(\text{HA})|i \in \text{PH}]}{2} \\
&- \frac{E[Y_i(\text{SA}) - Y_i(\text{SB})|i \in \text{SPS}] - E[Y_i(\text{HA}) - Y_i(\text{HB})|i \in \text{SPH}]}{2} \\
&\approx \frac{E[Y_i(\text{SB}) - Y_i(\text{HB})|i \in \text{PS}] + E[Y_i(\text{SB}) - Y_i(\text{HB})|i \in \text{PH}]}{2}.
\end{aligned} \tag{7}$$

The amount of insight provided by estimating (5) and (6) and the quality of the approximation in (7) depend on the extent to which effects for Strongly Prefer STEM and Strongly Prefer HSS are a good stand-in for those for Prefer STEM and Prefer HSS.

A4.2 Methods and results

As with the across-curriculum cutoffs analyzed in the main paper, we consider up to two same-curriculum cutoffs per student. These are: the lowest (highest) cutoff such that if the student had scored higher (lower), the student would switch into a different track with the same curriculum as the student’s assigned track. For these cutoffs, we define the running variable as simply the centered transition score: $r_i = t_i - c_j$. We then stack the same-curriculum cutoffs on top of the across-curriculum cutoffs and estimate an extended version of (1). The extension is that f_j now takes on four values: $f_j \in \{SH, HS, SS, HH\}$. Since the equation is fully interacted with cutoff type, β_{SH} and β_{HS} are unchanged. Meanwhile, since $r_i = t_i - c_j$ for SS and HH , β_{SS} and β_{HH} are, respectively, the effects of scoring above same-curriculum STEM and HSS cutoffs (i.e., regression-equation versions of (5) and (6)). For (7), we estimate:

$$\beta_{\text{adj}} = \frac{\beta_{SH} + \beta_{HS}}{2} - \frac{\beta_{SS} - \beta_{HH}}{2}.$$

As with β in the main text, we estimate β_{adj} by plugging in estimates of β_{f_j} , and we calculate standard errors using the Delta Method.

Results are in Tables A10-A12. In the tables, each row corresponds to a different outcome, and rows are included for all of the paper’s main outcomes. The columns under the “STEM” and “HSS” headings are for the estimates of β_{SS} and β_{HH} , those under “Main approach” are for β and match the values from elsewhere in the paper, and those under “Adjusted approach” are for β_{adj} . The column titled “In C.I.” under “Adjusted approach” indicates whether the estimate of β_{adj} falls within the 95% confidence interval for β for the given outcome.

In terms of takeaways, the tables show that the effect of scoring above vs. below a cutoff likely exerts little distortion on our findings. The estimates of β_{SS} and β_{HH} are generally small, and the estimates of β_{adj} tend to be similar to those of β . For only two of the 89 outcomes is the estimate of β_{adj} outside the confidence interval for β , and even for those outcomes the differences in coefficient estimates are modest.

As mentioned, the analysis in this section faces a limitation in that same- and across-curriculum cutoffs are observed for different sets of students. Nonetheless, the fact that the effect of scoring above a cutoff is usually small for both observable groups (SPS and SPH) suggests that it is probably also small for the groups of interest (PS and PH). In turn, it seems reasonable to conclude that the second terms in (2)-(4) are usually small and that what we estimate elsewhere in the paper is close to the first terms. Moreover, when scoring above a cutoff has no impact, the first terms have a broader interpretation. They are not just the average effect of STEM vs. HSS assignment while scoring below a cutoff (i.e., at the cutoff, but on the lower-scoring side); instead, they are the average effect while scoring either above or below.

Table A10: Adjusting for the effect of scoring above a cutoff

Outcome	The effect of scoring above a same-curriculum cutoff						The effect of being on the STEM side of an across-curriculum cutoff					
	STEM			HSS			Main approach			Adjusted approach		
	Coef. est.	Std. error	N	Coef. est.	Std. error	N	Coef. est.	Std. error	N	Coef. est.	Std. error	N
	In C.I.											
Panel A: High school enrollment and graduation												
Years of enrollment	0.00	0.00	75,321	0.01	0.01	25,969	0.00	0.01	55,221	0.00	0.01	156,511
Years of STEM enrollment	0.00	0.01	75,321	-0.01	0.01	25,969	3.08	0.02	55,221	3.07	0.02	156,511
Years of HSS enrollment	0.00	0.00	75,321	0.02	0.02	25,969	-3.09	0.02	55,221	-3.07	0.02	156,511
Years of Other enrollment	0.01	0.00	75,321	-0.01	0.01	25,969	0.01	0.01	55,221	0.01	0.01	156,511
Graduate	0.00	0.00	75,321	0.00	0.00	25,969	0.00	0.00	55,221	0.00	0.00	156,511
Graduate in STEM	0.00	0.00	75,321	0.00	0.00	25,969	0.69	0.01	55,221	0.69	0.01	156,511
Graduate in HSS	0.00	0.00	75,321	0.00	0.01	25,969	-0.70	0.01	55,221	-0.70	0.01	156,511
Graduate in Other	0.00	0.00	75,321	0.00	0.00	25,969	0.01	0.00	55,221	0.01	0.00	156,511
Panel B: College outcomes												
College plans at the end of high school: any college	-0.03	0.01	4,570	0.05	0.02	1,558	0.01	0.02	3,987	0.05	0.02	10,115
College plans at the end of high school: STEM	-0.01	0.02	4,570	-0.01	0.04	1,558	0.23	0.04	3,987	0.23	0.04	10,115
College plans at the end of high school: HLSS	0.01	0.02	4,570	0.01	0.04	1,558	-0.22	0.03	3,987	-0.22	0.04	10,115
College enrollment one year after high school: any college	0.00	0.01	3,763	0.00	0.03	1,280	-0.04	0.03	3,327	-0.03	0.03	8,370
College enrollment one year after high school: STEM	0.00	0.02	3,763	0.02	0.03	1,280	0.25	0.04	3,327	0.25	0.04	8,370
College enrollment one year after high school: HLSS	0.00	0.01	3,763	0.05	0.04	1,280	-0.26	0.03	3,327	-0.24	0.04	8,370
Panel C: Desired careers one year after high school												
Technology or engineering	0.04	0.04	1,098	-0.10	0.05	381	0.25	0.05	1,159	0.18	0.06	2,638
Medicine	-0.05	0.03	1,098	-0.04	0.05	381	-0.04	0.04	1,159	-0.04	0.06	2,638
Business or economics	0.05	0.04	1,098	0.00	0.08	381	-0.04	0.06	1,159	-0.07	0.08	2,638
Art, education, law, or social services	0.00	0.02	1,098	0.10	0.07	381	-0.14	0.05	1,159	-0.10	0.06	2,638
Other/unsure	-0.05	0.03	1,098	0.04	0.05	381	-0.03	0.04	1,159	0.02	0.05	2,638
Panel D: Baccalaureate performance												
Take the exam	-0.01	0.00	98,802	0.00	0.00	34,736	-0.01	0.00	73,470	-0.01	0.00	207,008
Pass the exam	-0.02	0.00	98,802	0.00	0.00	34,736	-0.04	0.00	73,470	-0.03	0.00	207,008
Exam score	0.01	0.01	94,078	0.02	0.01	33,679	-0.35	0.01	69,938	-0.35	0.02	197,695
Pass in STEM	-0.02	0.01	98,802	0.00	0.00	34,736	0.62	0.01	73,470	0.63	0.01	207,008
Pass in HSS	0.00	0.00	98,802	0.00	0.00	34,736	-0.70	0.01	73,470	-0.70	0.01	207,008
Pass in Other	0.00	0.00	98,802	0.00	0.00	34,736	0.04	0.00	73,470	0.04	0.00	207,008
Panel E: Confidence in and preferences for high school subjects												
Beliefs about own high school STEM abilities at the end of high school	-0.11	0.05	4,570	0.00	0.08	1,558	0.87	0.08	3,987	0.93	0.09	10,115
Beliefs about own high school HSS abilities at the end of high school	0.01	0.05	4,570	-0.11	0.05	1,558	-0.40	0.06	3,987	-0.46	0.07	10,115
Beliefs about own high school STEM abilities one year after high school	0.01	0.06	1,098	0.06	0.17	381	0.74	0.12	1,159	0.77	0.15	2,638
Beliefs about own high school HSS abilities one year after high school	0.02	0.07	1,098	-0.05	0.09	381	-0.51	0.09	1,159	-0.54	0.11	2,638
Preferences for STEM high school subjects one year after high school	-0.03	0.09	1,098	0.18	0.20	381	1.01	0.15	1,159	1.11	0.19	2,638
Preferences for HSS high school subjects one year after high school	-0.02	0.09	1,098	-0.03	0.10	381	-0.55	0.08	1,159	-0.56	0.11	2,638

The table presents results for β_{SS} ("STEM" columns), β_{HH} ("HSS" columns), β ("Main approach" columns), and β_{adj} ("Adjusted approach" columns). Rows correspond to different outcomes. The outcomes included are all those from the main paper for which we show robustness and heterogeneity. For each outcome, results are obtained from a single regression that stacks same- and across-curriculum cutoffs, uses a 1.25 point bandwidth, and clusters standard errors by student and by cutoff. The number of student-cutoff observations in these regressions is displayed in the column titled "N" under the "Adjusted approach" heading. The other "N" columns reveal the number of student-cutoffs for the cutoffs that contribute to the estimation of the parameter associated with the given heading. "In C.I." indicates whether the estimate of β_{adj} is in the 95% confidence interval for β . See the text of Appendix A4 for additional methodological details.

Table A11: Adjusting for the effect of scoring above a cutoff, cont.

Outcome	The effect of scoring above a same-curriculum cutoff						The effect of being on the STEM side of an across-curriculum cutoff					
	STEM			HSS			Main approach			Adjusted approach		
	Coef. est.	Std. error	N	Coef. est.	Std. error	N	Coef. est.	Std. error	N	Coef. est.	Std. error	In C.I.
Panel F: Confidence in and preferences for college subjects												
Beliefs about own college STEM abilities one year after high school	-0.01	0.09	1,098	0.13	0.17	381	0.66	0.15	1,159	0.73	0.18	2,638 Y
Beliefs about own college non-STEM abilities one year after high school	-0.01	0.11	1,098	0.22	0.15	381	-0.34	0.14	1,159	-0.22	0.17	2,638 Y
Preferences for STEM college subjects one year after high school	0.02	0.09	1,098	0.13	0.20	381	0.78	0.15	1,159	0.83	0.19	2,638 Y
Preferences for non-STEM college subjects one year after high school	-0.03	0.10	1,098	0.23	0.16	381	-0.45	0.15	1,159	-0.32	0.18	2,638 Y
Panel G: O*NET outcomes 1.5 years after high school												
Share of STEM-related job tasks that enjoy	-0.03	0.02	1,326	-0.02	0.03	452	0.10	0.02	1,199	0.10	0.03	2,977 Y
Share of HSS-related job tasks that enjoy	0.02	0.02	1,326	0.02	0.04	452	-0.09	0.04	1,199	-0.09	0.05	2,977 Y
Share of Other job tasks that enjoy	-0.01	0.02	1,326	0.03	0.04	452	0.01	0.03	1,199	0.03	0.04	2,977 Y
Dif. in the share of STEM- vs. HSS-related job tasks that enjoy	-0.04	0.03	1,326	-0.04	0.05	452	0.19	0.05	1,199	0.19	0.05	2,977 Y
Share of O*NET-recommended jobs that are STEM-related	-0.02	0.03	1,326	0.02	0.04	452	0.10	0.03	1,199	0.12	0.04	2,977 Y
Share of O*NET-recommended jobs that are HSS-related	0.01	0.03	1,326	-0.06	0.05	452	-0.10	0.05	1,199	-0.14	0.06	2,977 Y
Share of O*NET-recommended jobs that are Other	-0.01	0.03	1,326	0.03	0.05	452	-0.01	0.04	1,199	0.01	0.05	2,977 Y
Panel H: Choices and beliefs regarding representative jobs 1.5 years after high school												
Would choose to be a technology or engineering worker	0.00	0.04	1,326	-0.05	0.07	452	0.15	0.05	1,199	0.12	0.06	2,977 Y
Difference in beliefs about pay	0.05	0.07	1,326	0.05	0.11	452	-0.01	0.10	1,199	-0.01	0.12	2,977 Y
Difference in beliefs about enjoying the work content	-0.01	0.13	1,326	-0.31	0.25	452	0.74	0.21	1,199	0.59	0.26	2,977 Y
Difference in beliefs about coworkers and work conditions	-0.01	0.12	1,326	-0.21	0.21	452	0.25	0.19	1,199	0.15	0.22	2,977 Y
Difference in beliefs about own preparation	0.04	0.14	1,326	-0.39	0.23	452	0.90	0.23	1,199	0.68	0.28	2,977 Y
Difference in beliefs about approval of friends and family	-0.05	0.10	1,326	-0.13	0.20	452	0.30	0.12	1,199	0.26	0.17	2,977 Y
Panel I: Wellbeing, high school satisfaction, and high school curricular fit at the end of high school												
Wellbeing	0.02	0.04	4,570	0.01	0.05	1,558	0.13	0.05	3,987	0.13	0.06	10,115 Y
Liked the high school experience	0.03	0.05	4,570	0.04	0.08	1,558	0.11	0.07	3,987	0.12	0.08	10,115 Y
Liked the high school curriculum	0.00	0.06	4,570	0.03	0.07	1,558	0.00	0.07	3,987	0.02	0.08	10,115 Y
Liked the high school peers	0.04	0.06	4,570	0.08	0.10	1,558	0.23	0.07	3,987	0.25	0.09	10,115 Y
Liked the high school teachers	-0.02	0.05	4,570	0.01	0.06	1,558	-0.02	0.07	3,987	-0.01	0.08	10,115 Y
The high school curriculum was a good fit for my abilities	0.00	0.04	4,570	-0.07	0.06	1,558	0.00	0.05	3,987	-0.03	0.06	10,115 Y
Panel J: Wellbeing, high school satisfaction, and high school curricular fit 1-1.5 years after high school												
Wellbeing	-0.01	0.05	1,326	-0.04	0.09	452	-0.03	0.07	1,199	-0.05	0.08	2,977 Y
Liked the high school experience	0.13	0.09	1,098	-0.27	0.14	381	0.22	0.11	1,159	0.02	0.14	2,638 Y
Liked the high school curriculum	0.09	0.08	1,098	-0.23	0.15	381	0.00	0.14	1,159	-0.16	0.16	2,638 Y
Liked the high school peers	0.12	0.12	1,098	-0.27	0.15	381	0.29	0.14	1,159	0.09	0.17	2,638 Y
Liked the high school teachers	0.11	0.08	1,098	-0.34	0.11	381	0.13	0.13	1,159	-0.10	0.15	2,638 Y
The high school curriculum was a good fit for my abilities	0.19	0.08	1,098	-0.06	0.21	381	-0.12	0.12	1,159	-0.25	0.16	2,638 Y

The table is a continuation of Table A10.

Table A12: Adjusting for the effect of scoring above a cutoff, cont.

Outcome	The effect of scoring above a same-curriculum cutoff					The effect of being on the STEM side of an across-curriculum cutoff							
	STEM			HSS		Main approach			Adjusted approach				
	Coef. est.	Std. error	N	Coef. est.	Std. error	N	Coef. est.	Std. error	N	In C.I.			
Panel K: Regret over high school application choices one year after high school													
Am happy with the high school application choices that I made	-0.09	0.09	1,098	0.34	0.16	381	0.05	0.13	1,159	0.26	0.16	2,638	Y
If could do over, would make no change	0.03	0.04	1,098	-0.01	0.07	381	0.05	0.04	1,159	0.03	0.06	2,638	Y
If could do over, would rank STEM tracks higher	-0.02	0.03	1,098	0.05	0.04	381	-0.03	0.03	1,159	0.00	0.04	2,638	Y
If could do over, would rank HSS tracks higher	-0.02	0.03	1,098	-0.02	0.05	381	-0.02	0.03	1,159	-0.02	0.04	2,638	Y
If could do over, would rank Other tracks higher	0.00	0.01	1,098	-0.02	0.02	381	0.00	0.01	1,159	-0.01	0.02	2,638	Y
Panel L: College satisfaction, curricular fit, and regret one year after high school													
Like the college experience	-0.01	0.10	1,098	0.02	0.19	381	-0.44	0.14	1,159	-0.43	0.18	2,638	Y
Like the college curriculum	-0.07	0.12	1,098	0.13	0.14	381	-0.23	0.13	1,159	-0.13	0.16	2,638	Y
Like the college peers	0.12	0.10	1,098	0.26	0.17	381	-0.40	0.17	1,159	-0.33	0.19	2,638	Y
Like the college instructors	0.02	0.10	1,098	0.26	0.15	381	-0.35	0.13	1,159	-0.23	0.16	2,638	Y
The college curriculum is a good fit for my abilities	0.06	0.11	1,098	0.14	0.19	381	0.08	0.16	1,159	0.12	0.19	2,638	Y
Am happy with the college application choices that I made	0.03	0.11	1,098	0.40	0.22	381	0.12	0.18	1,159	0.31	0.22	2,638	Y
Panel M: Social and civic outcomes: male students													
Traditionalist expectations	0.00	0.03	2,044	0.14	0.07	401	0.12	0.05	1,396	0.19	0.06	3,841	Y
Right-wing political views	-0.07	0.03	2,044	0.04	0.08	401	0.14	0.05	1,396	0.20	0.07	3,841	Y
Trust	-0.02	0.05	2,044	-0.12	0.14	401	-0.01	0.08	1,396	-0.06	0.11	3,841	Y
Turn out to vote in the Fall 2024 elections	0.06	0.02	582	0.20	0.11	67	0.00	0.08	373	0.07	0.10	1,022	Y
Good friends	0.02	0.18	2,044	-0.12	0.55	401	-0.07	0.32	1,396	-0.14	0.44	3,841	Y
Very close friends	0.09	0.10	2,044	0.20	0.26	401	0.00	0.17	1,396	0.06	0.23	3,841	Y
Good female vs. male friends	0.16	0.14	2,044	0.80	0.40	401	-0.71	0.28	1,396	-0.39	0.36	3,841	Y
Verbal development	0.04	0.03	2,044	-0.06	0.07	401	-0.25	0.06	1,396	-0.30	0.07	3,841	Y
Empathy	0.01	0.07	2,044	0.02	0.12	401	-0.15	0.09	1,396	-0.14	0.12	3,841	Y
Panel N: Social and civic outcomes: female students													
Traditionalist expectations	-0.01	0.03	2,477	0.02	0.03	1,126	0.03	0.03	2,510	0.05	0.04	6,113	Y
Right-wing political views	0.00	0.03	2,477	-0.04	0.03	1,126	-0.01	0.04	2,510	-0.03	0.04	6,113	Y
Trust	-0.01	0.05	2,477	-0.04	0.05	1,126	0.03	0.07	2,510	0.01	0.08	6,113	Y
Turn out to vote in the Fall 2024 elections	0.01	0.03	665	-0.03	0.04	339	0.05	0.04	728	0.03	0.05	1,732	Y
Good friends	-0.05	0.17	2,477	-0.32	0.19	1,126	-0.04	0.24	2,510	-0.18	0.27	6,113	Y
Very close friends	-0.02	0.09	2,477	-0.12	0.11	1,126	-0.06	0.11	2,510	-0.11	0.13	6,113	Y
Good female vs. male friends	0.27	0.15	2,477	0.14	0.21	1,126	-0.49	0.19	2,510	-0.56	0.23	6,113	Y
Verbal development	-0.03	0.03	2,477	0.02	0.04	1,126	-0.26	0.05	2,510	-0.24	0.05	6,113	Y
Empathy	0.07	0.06	2,477	0.00	0.06	1,126	-0.08	0.07	2,510	-0.12	0.08	6,113	Y

The table is a continuation of Table A10.

A5 Adjusting for the influence of peer composition

Appendix A4 showed that the effects reported in this paper are not due to the third sub-treatment discussed in Section 4.1: whether a student scores above or below a cutoff. Thus, the effects reflect the first two sub-treatments: the difference in curriculum between STEM and HSS tracks and/or the difference in horizontal dimensions of peer composition, such as relative academic strength and share female.

This appendix attempts to disentangle the influence of the curriculum and peer composition sub-treatments. It uses two strategies. First, we fit a modified version of Equation (1) that adds controls for three variables related to the peer composition in students’ assigned tracks: average transition score, the average difference in math vs. language transition exam scores, and share female.⁵⁷ Intuitively, to the extent that peer composition matters, the measured effects of STEM assignment should diminish once we adjust for the fact that STEM and HSS tracks differ in peer composition. Thus, we can assess the importance of the curriculum sub-treatment by seeing how estimates of β from the modified version of (1) compare with those from the main version.

The second approach is similar, but conducts analysis at the cutoff level. In a first step, we run cutoff-specific RD regressions that use as left-hand-side variables both the paper’s outcomes of interest and the three track attributes mentioned in the previous paragraph.⁵⁸ Then, for each outcome, we regress the cutoff-specific effects for the outcome on the cutoff-specific effects for the three track attributes, weighting by the number of students in a cutoff’s RD sample. The intercept in this regression is the predicted effect of STEM assignment for cutoffs with no change in peer composition across the RD threshold. Thus, for this approach, we can gauge the importance of the curriculum sub-treatment by comparing the intercept estimates with the estimates of β from (1).⁵⁹ A noteworthy feature of the second approach is that it does not use all the observations in our main RD sample. In particular, when estimating the cutoff-specific effects, we restrict attention to cutoffs with at least three students on both the STEM and HSS sides. For consistency, we re-run our main approach on this restricted sample.

Tables A13-A15 contain the results. Each row represents a different outcome, and rows are included for all the outcomes in Tables A10-A12. The columns under the heading “Full sample–Main approach” relate to β from (1) and match the values from elsewhere in the paper. The columns under “Full sample–Track-level covariates” are for β in the modified version of (1) that adds controls for attributes of students’ assigned tracks. The columns under “Restricted sample–Main approach” are for β from (1) estimated on the restricted sample. Meanwhile, “Restricted sample–Cutoff-level variation” is for the cutoff-level approach. In order to have a large enough number of cutoffs to confidently fit a cutoff-level regression, we implement the cutoff-level approach only for outcomes that are derived from the administrative data or the end-of-high-school survey. We do not implement it for outcomes that come from just a follow-up survey. Finally, the columns titled “Full sample–In C.I.” and “Restricted sample–In C.I.” indicate whether the coefficient estimates

57. These are the attributes examined in Section 5.2. As seen there, average transition score is related to STEM assignment conditional on cutoff type (though not when averaging over STEM- and HSS-above cutoffs). Thus, we include it even though it is a vertical dimension of peer quality; results are similar if we drop it.

58. For convenience, we run these regressions separately by cutoff. However, they can be represented in a single equation as a version of Equation (1) that is fully interacted with specific cutoffs, not just cutoff types.

59. A key difference between the two approaches is that the first uses all the available variation to measure the relationship between an outcome and peer composition, while the second uses only the jumps at the cutoffs; i.e., the second compares the size of the change in an outcome across an RD threshold with the size of the change in peer composition, while the first simply traces how students’ values of the outcome vary with their values of peer composition. Much of the variation in the first approach is due to the jumps at the cutoffs, but not all.

from “Track-level covariates” and “Cutoff-level variation” fall within the 95% confidence intervals for β in our main approach (under the full and restricted samples, respectively).

The tables reveal that adjusting for peer composition does not strongly alter our results. For only 5 of the 89 outcomes is the coefficient estimate from “Track-level covariates” outside the “Full sample–Main approach” confidence interval for β . For only 18 of the 44 outcomes available for “Cutoff-level variation” is the estimate from this strategy outside the “Restricted sample–Main approach” confidence interval. Further, the differences in estimates tend to be modest. The main exception is that the estimate for “Good female vs. male friends” is smaller under “Track-level covariates” than under “Full sample–Main approach”, for both boys and girls.

In short, the results in this appendix complement the finding that a limited share of students cite peers as an influence on their decision-making. Overall, the evidence suggests that the curriculum channel—and not peer exposure—drives the effects of assignment to STEM vs. HSS for the outcomes that we study.

Table A13: Adjusting for the influence of peer composition

Outcome	Full sample						Restricted sample					
	Main approach			Track-level covariates			Student-cutoffs			Main approach		
	Coef. est.	Std. error		Coef. est.	Std. error	In C.I.	Cutoffs			Coef. est.	Std. error	Cutoffs
Panel A: High school enrollment and graduation												
Years of enrollment	0.00	0.01		0.01	0.01	Y	1,411	55,221		0.00	0.01	954
Years of HSS enrollment	3.08	0.02		3.06	0.02	Y	1,411	55,221		3.07	0.02	954
Years of STEM enrollment	-3.09	0.02		-3.06	0.02	Y	1,411	55,221		-3.08	0.02	954
Years of Other enrollment	0.01	0.01		0.01	0.01	Y	1,411	55,221		0.01	0.01	954
Graduate	0.00	0.00		0.00	0.00	N	1,411	55,221		0.00	0.00	954
Graduate in STEM	0.69	0.01		0.69	0.01	Y	1,411	55,221		0.69	0.01	954
Graduate in HSS	-0.70	0.01		-0.69	0.01	Y	1,411	55,221		-0.70	0.01	954
Graduate in Other	0.01	0.00		0.01	0.00	Y	1,411	55,221		0.00	0.00	954
Panel B: College outcomes												
College plans at the end of high school: any college	0.01	0.02		0.00	0.02	Y	293	3,987		0.02	0.02	113
College plans at the end of high school: STEM	0.23	0.04		0.19	0.04	Y	293	3,987		0.26	0.04	113
College plans at the end of high school: HLSS	-0.22	0.03		-0.20	0.03	Y	293	3,987		-0.24	0.04	113
College enrollment one year after high school: any college	-0.04	0.03		-0.05	0.03	Y	280	3,327		-0.01	0.03	95
College enrollment one year after high school: STEM	0.25	0.04		0.19	0.04	Y	280	3,327		0.28	0.04	95
College enrollment one year after high school: HLSS	-0.26	0.03		-0.22	0.04	Y	280	3,327		-0.28	0.04	95
Panel C: Desired careers one year after high school												
Technology or engineering	0.25	0.05		0.13	0.06	N	190	1,159		-	-	-
Medicine	-0.04	0.04		0.02	0.05	Y	190	1,159		-	-	-
Business or economics	-0.04	0.06		-0.02	0.06	Y	190	1,159		-	-	-
Art, education, law, or social services	-0.14	0.05		-0.09	0.05	Y	190	1,159		-	-	-
Other/unsure	-0.03	0.04		-0.04	0.05	Y	190	1,159		-	-	-
Panel D: Baccalaureate performance												
Take the exam	-0.01	0.00		0.00	0.00	N	1,904	73,470		-0.01	0.00	1,284
Pass the exam	-0.04	0.00		-0.02	0.00	N	1,904	73,470		-0.03	0.00	1,284
Exam score	-0.35	0.01		-0.24	0.02	N	1,855	69,938		-0.34	0.01	1,225
Pass in STEM	0.62	0.01		0.62	0.01	Y	1,904	73,470		0.62	0.01	1,284
Pass in HSS	-0.70	0.01		-0.69	0.01	Y	1,904	73,470		-0.70	0.01	1,284
Pass in Other	0.04	0.00		0.05	0.00	Y	1,904	73,470		0.04	0.00	1,284
Panel E: Confidence in and preferences for high school subjects												
Beliefs about own high school STEM abilities at the end of high school	0.87	0.08		0.81	0.08	Y	293	3,987		0.88	0.09	113
Beliefs about own high school HSS abilities at the end of high school	-0.40	0.06		-0.34	0.06	Y	293	3,987		-0.42	0.06	113
Beliefs about own high school STEM abilities one year after high school	0.74	0.12		0.75	0.13	Y	190	1,159		-	-	-
Beliefs about own high school HSS abilities one year after high school	-0.51	0.09		-0.41	0.10	Y	190	1,159		-	-	-
Preferences for STEM high school subjects one year after high school	1.01	0.15		1.01	0.17	Y	190	1,159		-	-	-
Preferences for HSS high school subjects one year after high school	-0.55	0.08		-0.44	0.09	Y	190	1,159		-	-	-

The table presents results for two strategies that adjust for the influence of peer composition: “Track-level covariates” and “Cutoff-level variation”. See the text of Appendix A5 for an explanation of these strategies. The cutoff-specific RDs for “Cutoff-level variation” use a 1.25 point bandwidth. “Main approach” is Equation (1), fit on either the paper’s full sample or the sample used for “Cutoff-level variation”. “Cutoffs” and “Student-cutoffs” are the number of cutoffs and observations for the listed samples. “In C.I.” indicates whether the coefficient estimate from “Track-level covariates” (“Cutoff-level variation”) is in the main-approach 95% confidence interval for β from the full (restricted) sample. Rows correspond to different outcomes and are included for all the outcomes in Tables A10-A12.

Table A14: Adjusting for the influence of peer composition, cont.

Outcome	Full sample					Restricted sample					
	Main approach		Track-level covariates			Cutoffs	Student-cutoffs	Main approach		Cutoff-level variation	
	Coef. est.	Std. error	Coef. est.	Std. error	In C.I.			Coef. est.	Std. error	Coef. est.	In C.I.
Panel F: Confidence in and preferences for college subjects											
Beliefs about own college STEM abilities one year after high school	0.66	0.15	0.60	0.17	Y	190	1,159	-	-	-	-
Beliefs about own college non-STEM abilities one year after high school	-0.34	0.14	-0.24	0.16	Y	190	1,159	-	-	-	-
Preferences for STEM college subjects one year after high school	0.78	0.15	0.74	0.17	Y	190	1,159	-	-	-	-
Preferences for non-STEM college subjects one year after high school	-0.45	0.15	-0.31	0.17	Y	190	1,159	-	-	-	-
Panel G: O*NET outcomes 1.5 years after high school											
Share of STEM-related job tasks that enjoy	0.10	0.02	0.08	0.03	Y	198	1,199	-	-	-	-
Share of HSS-related job tasks that enjoy	-0.09	0.04	-0.09	0.04	Y	198	1,199	-	-	-	-
Share of Other job tasks that enjoy	0.01	0.03	0.01	0.03	Y	198	1,199	-	-	-	-
Dif. in the share of STEM- vs. HSS-related job tasks that enjoy	0.19	0.05	0.17	0.05	Y	198	1,199	-	-	-	-
Share of O*NET-recommended jobs that are STEM-related	0.10	0.03	0.13	0.04	Y	198	1,199	-	-	-	-
Share of O*NET-recommended jobs that are HSS-related	-0.10	0.05	-0.09	0.05	Y	198	1,199	-	-	-	-
Share of O*NET-recommended jobs that are Other	-0.01	0.04	-0.04	0.05	Y	198	1,199	-	-	-	-
Panel H: Choices and beliefs regarding representative jobs 1.5 years after high school											
Would choose to be a technology or engineering worker	0.15	0.05	0.08	0.06	Y	198	1,199	-	-	-	-
Difference in beliefs about pay	-0.01	0.10	0.03	0.11	Y	198	1,199	-	-	-	-
Difference in beliefs about enjoying the work content	0.74	0.21	0.48	0.24	Y	198	1,199	-	-	-	-
Difference in beliefs about coworkers and work conditions	0.25	0.19	0.14	0.20	Y	198	1,199	-	-	-	-
Difference in beliefs about own preparation	0.90	0.23	0.63	0.27	Y	198	1,199	-	-	-	-
Difference in beliefs about approval of friends and family	0.30	0.12	0.17	0.13	Y	198	1,199	-	-	-	-
Panel I: Wellbeing, high school satisfaction, and high school curricular fit at the end of high school											
Wellbeing	0.13	0.05	0.15	0.05	Y	293	3,987	0.13	0.06	0.20	Y
Liked the high school experience	0.11	0.07	0.16	0.07	Y	293	3,987	0.12	0.08	0.19	Y
Liked the high school curriculum	0.00	0.07	0.07	0.07	Y	293	3,987	-0.06	0.07	0.02	Y
Liked the high school peers	0.23	0.07	0.19	0.08	Y	293	3,987	0.28	0.08	0.12	N
Liked the high school teachers	-0.02	0.07	0.04	0.07	Y	293	3,987	-0.03	0.07	0.11	Y
The high school curriculum was a good fit for my abilities	0.00	0.05	0.06	0.06	Y	293	3,987	-0.01	0.06	-0.09	Y
Panel J: Wellbeing, high school satisfaction, and high school curricular fit 1-1.5 years after high school											
Wellbeing	-0.03	0.07	-0.09	0.07	Y	198	1,199	-	-	-	-
Liked the high school experience	0.22	0.11	0.14	0.13	Y	190	1,159	-	-	-	-
Liked the high school curriculum	0.00	0.14	-0.03	0.15	Y	190	1,159	-	-	-	-
Liked the high school peers	0.29	0.14	0.25	0.15	Y	190	1,159	-	-	-	-
Liked the high school teachers	0.13	0.13	0.21	0.13	Y	190	1,159	-	-	-	-
The high school curriculum was a good fit for my abilities	-0.12	0.12	-0.28	0.13	Y	190	1,159	-	-	-	-

The table is a continuation of Table A13.

Table A15: Adjusting for the influence of peer composition, cont.

Outcome	Full sample						Restricted sample					
	Main approach			Track-level covariates			Main approach			Cutoff-level variation		
	Coef. est.	Std. error		Coef. est.	Std. error	In C.I.	Coef. est.	Std. error		Coef. est.	In C.I.	Student-cutoffs
Panel K: Regret over high school application choices one year after high school												
Am happy with the high school application choices that I made	0.05	0.13		0.09	0.13	Y	-	-		-	-	-
If could do over, would make no change	0.05	0.04		0.03	0.04	Y	-	-		-	-	-
If could do over, would rank STEM tracks higher	-0.03	0.03		-0.04	0.04	Y	-	-		-	-	-
If could do over, would rank HSS tracks higher	-0.02	0.03		0.01	0.03	Y	-	-		-	-	-
If could do over, would rank Other tracks higher	0.00	0.01		0.00	0.01	Y	-	-		-	-	-
Panel L: College satisfaction, curricular fit, and regret one year after high school												
Like the college experience	-0.44	0.14		-0.37	0.16	Y	-	-		-	-	-
Like the college curriculum	-0.23	0.13		-0.20	0.15	Y	-	-		-	-	-
Like the college peers	-0.40	0.17		-0.36	0.19	Y	-	-		-	-	-
Like the college instructors	-0.35	0.13		-0.34	0.17	Y	-	-		-	-	-
The college curriculum is a good fit for my abilities	0.08	0.16		0.13	0.17	Y	-	-		-	-	-
Am happy with the college application choices that I made	0.12	0.18		0.15	0.21	Y	-	-		-	-	-
Panel M: Social and civic outcomes: male students												
Traditionalist expectations	0.12	0.05		0.12	0.05	Y	0.13	0.06		0.40	N	45
Right-wing political views	0.14	0.05		0.18	0.06	Y	0.08	0.06		0.27	N	45
Trust	-0.01	0.08		0.07	0.10	Y	-0.09	0.12		0.02	Y	45
Turn out to vote in the Fall 2024 elections	0.00	0.08		-0.03	0.09	Y	-	-		-	-	-
Good friends	-0.07	0.32		-0.09	0.36	Y	-0.18	0.42		0.80	N	45
Very close friends	0.00	0.17		0.16	0.19	Y	-0.04	0.24		0.35	Y	45
Good female vs. male friends	-0.71	0.28		-0.21	0.32	Y	-0.59	0.36		-1.27	Y	45
Verbal development	-0.25	0.06		-0.17	0.06	Y	-0.25	0.08		-0.39	Y	45
Empathy	-0.15	0.09		-0.15	0.11	Y	-0.22	0.12		-0.12	Y	45
Panel N: Social and civic outcomes: female students												
Traditionalist expectations	0.03	0.03		0.05	0.03	Y	0.03	0.04		0.03	Y	70
Right-wing political views	-0.01	0.04		0.00	0.04	Y	-0.03	0.04		0.01	Y	70
Trust	0.03	0.07		0.02	0.07	Y	0.00	0.08		-0.27	N	70
Turn out to vote in the Fall 2024 elections	0.05	0.04		0.03	0.04	Y	-	-		-	-	-
Good friends	-0.04	0.24		-0.05	0.24	Y	0.09	0.28		-0.90	N	70
Very close friends	-0.06	0.11		0.00	0.12	Y	0.04	0.13		-0.87	N	70
Good female vs. male friends	-0.49	0.19		-0.27	0.20	Y	-0.54	0.23		-1.72	N	70
Verbal development	-0.26	0.05		-0.25	0.05	Y	-0.25	0.05		-0.14	N	70
Empathy	-0.08	0.07		-0.04	0.07	Y	-0.09	0.08		-0.09	Y	70

The table is a continuation of Table A13.

A6 Additional results for high school enrollment and graduation

Table A16: Heterogeneity for effects on high school enrollment and graduation

	All	Prefer		Stronger in		Gender		Achievement	
		STEM	HSS	Math	Language	Male	Female	Low	High
<i>Panel A: Years of enrollment</i>									
STEM	0.002 (0.007)	0.000 (0.008)	0.003 (0.008)	0.010 (0.009)	-0.017 (0.011)	0.008 (0.010)	0.000 (0.009)	0.007 (0.016)	-0.000 (0.008)
Intercept	3.91	3.92	3.90	3.91	3.91	3.90	3.91	3.87	3.93
<i>Panel B: Years of STEM enrollment</i>									
STEM	3.08*** (0.023)	3.01*** (0.030)	3.14*** (0.027)	3.00*** (0.028)	3.22*** (0.031)	2.87*** (0.033)	3.26*** (0.025)	3.25*** (0.038)	3.02*** (0.028)
Intercept	0.50	0.68	0.32	0.66	0.22	0.68	0.34	0.13	0.66
<i>Panel C: Years of HSS enrollment</i>									
STEM	-3.09*** (0.023)	-3.02*** (0.031)	-3.16*** (0.027)	-3.00*** (0.028)	-3.27*** (0.031)	-2.86*** (0.034)	-3.28*** (0.025)	-3.27*** (0.040)	-3.03*** (0.028)
Intercept	3.36	3.21	3.52	3.21	3.63	3.17	3.53	3.65	3.24
<i>Panel D: Years of Other enrollment</i>									
STEM	0.015** (0.006)	0.006 (0.007)	0.024*** (0.009)	0.008 (0.007)	0.030** (0.011)	0.004 (0.009)	0.022*** (0.008)	0.035* (0.019)	0.004 (0.006)
Intercept	0.05	0.03	0.06	0.04	0.05	0.06	0.04	0.08	0.03
<i>Panel E: Graduate</i>									
STEM	-0.003 (0.003)	-0.006* (0.003)	0.001 (0.004)	-0.001 (0.004)	-0.007 (0.005)	-0.002 (0.004)	-0.001 (0.004)	-0.011 (0.008)	-0.001 (0.003)
Intercept	0.96	0.97	0.95	0.97	0.96	0.96	0.96	0.94	0.98
<i>Panel F: Graduate in STEM</i>									
STEM	0.694*** (0.007)	0.673*** (0.010)	0.714*** (0.009)	0.674*** (0.009)	0.732*** (0.011)	0.630*** (0.011)	0.751*** (0.009)	0.732*** (0.013)	0.683*** (0.009)
Intercept	0.16	0.22	0.10	0.21	0.07	0.21	0.11	0.04	0.21
<i>Panel G: Graduate in HSS</i>									
STEM	-0.703*** (0.008)	-0.682*** (0.010)	-0.724*** (0.009)	-0.679*** (0.009)	-0.751*** (0.010)	-0.637*** (0.011)	-0.760*** (0.008)	-0.755*** (0.013)	-0.687*** (0.009)
Intercept	0.80	0.75	0.84	0.75	0.88	0.74	0.85	0.88	0.76
<i>Panel H: Graduate in Other</i>									
STEM	0.007*** (0.002)	0.003 (0.002)	0.011*** (0.003)	0.005** (0.002)	0.011*** (0.004)	0.006* (0.003)	0.008*** (0.003)	0.012** (0.006)	0.004** (0.002)
Intercept	0.01	0.01	0.01	0.01	0.01	0.01	0.01	0.02	0.01
Student-cutoffs	55,221	28,526	26,695	35,757	19,223	24,127	30,896	18,501	36,720

The table shows heterogeneity for the effects in Table 3. The sample is the same for all outcomes, so it is presented in a single row at the bottom of the table. See the notes to Table A6 for definitions of the column headings.

Table A17: Effects on high school enrollment and graduation
with no restriction on outcome timing

	Years of enrollment				Graduate			
	All	STEM	HSS	Other	All	STEM	HSS	Other
STEM	0.004 (0.007)	3.08*** (0.023)	-3.09*** (0.023)	0.014** (0.006)	-0.001 (0.003)	0.697*** (0.007)	-0.704*** (0.008)	0.006*** (0.002)
Intercept	3.91	0.50	3.37	0.05	0.97	0.16	0.80	0.01
Student-cutoffs	55,221	55,221	55,221	55,221	55,221	55,221	55,221	55,221

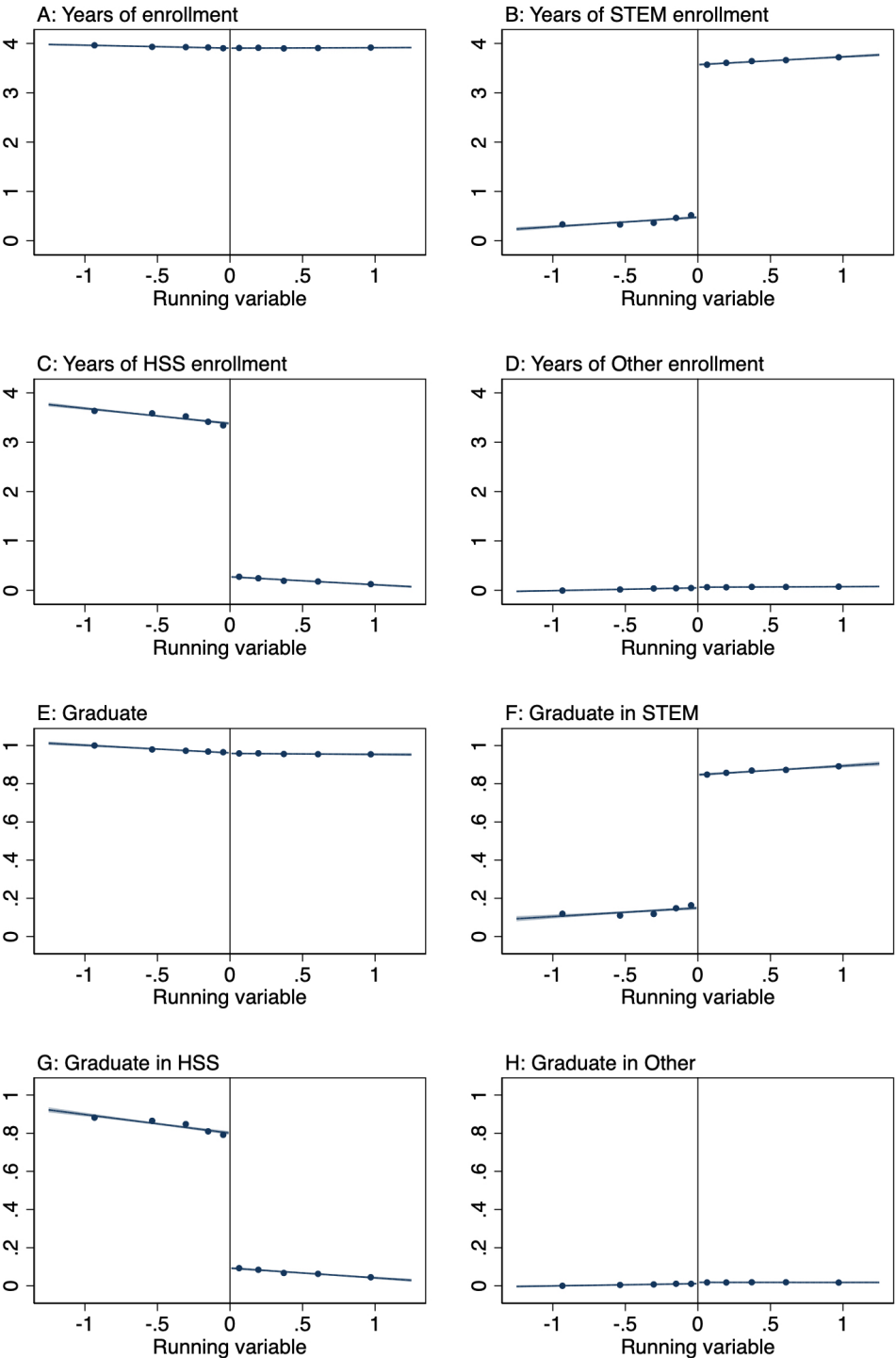
The table is analogous to Table 3; however, outcomes are measured over all years after track assignment, not just the first four.

Table A18: Robustness for effects on high school enrollment and graduation

	Main	Adding covariates	Optimal bandwidths		Alternative fixed bandwidths				Global quartic polynomial
			By cutoff	Mean	0.75	1.00	1.50	1.75	
Panel A: Years of enrollment									
STEM	0.002 (0.007)	0.002 (0.007)	0.000 (0.007)	0.001 (0.007)	0.001 (0.008)	0.001 (0.007)	0.001 (0.006)	-0.000 (0.006)	-0.002 (0.007)
Panel B: Years of STEM enrollment									
STEM	3.08*** (0.023)	3.08*** (0.023)	3.09*** (0.023)	3.06*** (0.023)	3.04*** (0.025)	3.06*** (0.024)	3.09*** (0.022)	3.10*** (0.021)	3.11*** (0.021)
Panel C: Years of HSS enrollment									
STEM	-3.09*** (0.023)	-3.09*** (0.023)	-3.11*** (0.023)	-3.07*** (0.024)	-3.05*** (0.026)	-3.07*** (0.024)	-3.11*** (0.022)	-3.12*** (0.022)	-3.12*** (0.022)
Panel D: Years of Other enrollment									
STEM	0.015** (0.006)	0.015** (0.006)	0.016** (0.006)	0.013** (0.006)	0.010 (0.006)	0.009 (0.006)	0.018*** (0.006)	0.018*** (0.006)	0.013** (0.006)
Panel E: Graduate									
STEM	-0.003 (0.003)	-0.003 (0.003)	-0.005 (0.003)	-0.003 (0.003)	-0.005 (0.003)	-0.004 (0.003)	-0.004 (0.003)	-0.004 (0.003)	-0.004 (0.003)
Panel F: Graduate in STEM									
STEM	0.694*** (0.007)	0.693*** (0.007)	0.699*** (0.007)	0.690*** (0.008)	0.683*** (0.008)	0.688*** (0.008)	0.697*** (0.007)	0.701*** (0.007)	0.703*** (0.007)
Panel G: Graduate in HSS									
STEM	-0.703*** (0.008)	-0.703*** (0.008)	-0.709*** (0.008)	-0.698*** (0.008)	-0.693*** (0.009)	-0.697*** (0.008)	-0.709*** (0.007)	-0.713*** (0.007)	-0.714*** (0.007)
Panel H: Graduate in Other									
STEM	0.007*** (0.002)	0.007*** (0.002)	0.007*** (0.002)	0.007*** (0.002)	0.005*** (0.002)	0.005*** (0.002)	0.008*** (0.002)	0.007*** (0.002)	0.007*** (0.002)
Student-cutoffs	55,221	55,221	48,648	52,892	44,749	51,050	58,122	60,052	64,289

The table shows robustness for the effects in Table 3. “Main” is the main specification, which uses a 1.25 point bandwidth. The added covariates in “Adding covariates” are the components of the transition score and an indicator for being female. “Optimal bandwidths” are calculated using the MSE-optimal bandwidth procedure from Calonico et al. (2017). “By cutoff” calculates these separately for each cutoff, following Abdulkadiroglu et al. (2017) and Abdulkadiroglu et al. (2022); “Mean” uses a fixed bandwidth equal to the mean of the cutoff-specific bandwidths. The columns under “Alternative fixed bandwidths” use fixed bandwidths of the listed lengths. “Global quartic polynomial” fits cutoff-type-specific quartic polynomials in the running variable and does not impose a bandwidth restriction. The sample is the same for all outcomes, so it is presented in a single row at the bottom of the table. Results for specifications that use alternative fixed effects are in Appendix A14.

Figure A4: RD plots of effects on high school enrollment and graduation



The figure displays RD plots for effects on the outcomes in Table 3. See the notes to Figure A3 for details on RD plots.

A7 Additional results for college and career outcomes

Table A19: Heterogeneity for effects on college enrollment

	All	Prefer		Stronger in		Gender		Achievement	
		STEM	HSS	Math	Language	Male	Female	Low	High
<i>Panel A: College plans at the end of high school: any college</i>									
STEM	0.010 (0.018)	0.009 (0.023)	0.010 (0.026)	-0.003 (0.029)	-0.007 (0.025)	0.110*** (0.042)	-0.030 (0.020)	-0.043 (0.030)	0.044* (0.026)
Intercept	0.87	0.92	0.83	0.88	0.88	0.76	0.93	0.86	0.92
Student-cutoffs	3,987	1,925	2,062	1,921	1,985	1,396	2,510	2,093	1,894
<i>Panel B: College plans at the end of high school: STEM</i>									
STEM	0.233*** (0.035)	0.270*** (0.048)	0.196*** (0.045)	0.267*** (0.050)	0.199*** (0.052)	0.293*** (0.058)	0.207*** (0.041)	0.208*** (0.062)	0.226*** (0.046)
Intercept	0.31	0.36	0.27	0.36	0.29	0.31	0.31	0.25	0.41
Student-cutoffs	3,987	1,925	2,062	1,921	1,985	1,396	2,510	2,093	1,894
<i>Panel C: College plans at the end of high school: humanities, law, or social science</i>									
STEM	-0.224*** (0.032)	-0.247*** (0.048)	-0.202*** (0.036)	-0.230*** (0.045)	-0.238*** (0.048)	-0.176*** (0.049)	-0.251*** (0.040)	-0.184*** (0.049)	-0.232*** (0.045)
Intercept	0.39	0.42	0.37	0.35	0.43	0.26	0.48	0.35	0.41
Student-cutoffs	3,987	1,925	2,062	1,921	1,985	1,396	2,510	2,093	1,894
<i>Panel D: College enrollment one year after high school: any college</i>									
STEM	-0.037 (0.025)	-0.065** (0.032)	-0.009 (0.035)	-0.055 (0.041)	-0.058 (0.038)	0.019 (0.050)	-0.054* (0.032)	-0.089* (0.046)	-0.002 (0.029)
Intercept	0.82	0.90	0.74	0.85	0.79	0.76	0.84	0.77	0.91
Student-cutoffs	3,327	1,600	1,727	1,591	1,658	1,146	2,098	1,703	1,624
<i>Panel E: College enrollment one year after high school: STEM</i>									
STEM	0.246*** (0.036)	0.259*** (0.052)	0.233*** (0.040)	0.221*** (0.065)	0.287*** (0.046)	0.276*** (0.060)	0.210*** (0.044)	0.219*** (0.052)	0.297*** (0.050)
Intercept	0.28	0.32	0.25	0.39	0.21	0.30	0.28	0.22	0.34
Student-cutoffs	3,327	1,600	1,727	1,591	1,658	1,146	2,098	1,703	1,624
<i>Panel F: College enrollment one year after high school: humanities, law, or social science</i>									
STEM	-0.261*** (0.031)	-0.308*** (0.037)	-0.215*** (0.043)	-0.285*** (0.044)	-0.271*** (0.046)	-0.207*** (0.051)	-0.280*** (0.045)	-0.219*** (0.042)	-0.313*** (0.054)
Intercept	0.38	0.42	0.34	0.34	0.41	0.26	0.46	0.34	0.44
Student-cutoffs	3,327	1,600	1,727	1,591	1,658	1,146	2,098	1,703	1,624

The table shows heterogeneity for effects on some of the outcomes in Table 4. See the notes to Table A6 for definitions of the column headings.

Table A20: Effects on college enrollment: students from the first follow-up or peer surveys

	Any	STEM				Hum., law, or social science				Other/ unsure
		Any	Math or CS	Natural science	Business or econ.	Any	Humanities	Law	Social science	
Panel A: College plans at the end of high school										
STEM	0.018 (0.020)	0.250*** (0.036)	0.159*** (0.026)	0.035 (0.025)	0.056** (0.026)	-0.232*** (0.034)	-0.073*** (0.021)	-0.056** (0.023)	-0.103*** (0.022)	0.000 (0.024)
Intercept	0.88	0.31	0.08	0.10	0.13	0.41	0.13	0.12	0.15	0.16
Panel B: College enrollment one year after high school										
STEM	-0.037 (0.025)	0.246*** (0.036)	0.192*** (0.031)	0.015 (0.022)	0.039 (0.026)	-0.261*** (0.031)	-0.123*** (0.021)	-0.092*** (0.021)	-0.046*** (0.017)	-0.022 (0.027)
Intercept	0.82	0.28	0.11	0.08	0.10	0.38	0.17	0.13	0.08	0.16
Student-cutoffs	3,327	3,327	3,327	3,327	3,327	3,327	3,327	3,327	3,327	3,327

The table is analogous to Table 4, but restricted to students from the first follow-up or peer surveys. The sample is the same across panels, so it is presented in a single row at the bottom of the table.

Table A21: Effects on college enrollment: students from the first follow-up survey

	Any	STEM				Hum., law, or social science				Other/ unsure
		Any	Math or CS	Natural science	Business or econ.	Any	Humanities	Law	Social science	
<i>Panel A: College plans at the end of high school</i>										
STEM	0.018 (0.029)	0.178*** (0.061)	0.184*** (0.048)	0.000 (0.037)	-0.006 (0.041)	-0.224*** (0.056)	-0.061 (0.038)	-0.053 (0.035)	-0.110*** (0.037)	0.065 (0.045)
Intercept	0.88	0.39	0.10	0.13	0.16	0.40	0.13	0.12	0.16	0.10
<i>Panel B: College enrollment one year after high school</i>										
STEM	-0.021 (0.045)	0.223*** (0.057)	0.213*** (0.059)	-0.005 (0.039)	0.015 (0.048)	-0.225*** (0.052)	-0.081** (0.040)	-0.103*** (0.033)	-0.041 (0.033)	-0.019 (0.038)
Intercept	0.82	0.36	0.11	0.09	0.16	0.35	0.12	0.13	0.09	0.12
Student-cutoffs	1,159	1,159	1,159	1,159	1,159	1,159	1,159	1,159	1,159	1,159

The table is analogous to Table 4, but restricted to students from the first follow-up survey. The sample is the same across panels, so it is presented in a single row at the bottom of the table.

Table A22: Effects on college enrollment: students from the second follow-up survey

	Any	STEM				Hum., law, or social science				Other/ unsure
		Any	Math or CS	Natural science	Business or econ.	Any	Humanities	Law	Social science	
Panel A: College plans at the end of high school										
STEM	0.040 (0.031)	0.207*** (0.058)	0.196*** (0.043)	0.034 (0.039)	-0.023 (0.045)	-0.173*** (0.059)	-0.016 (0.037)	-0.050 (0.040)	-0.107*** (0.039)	0.006 (0.038)
Intercept	0.85	0.35	0.09	0.11	0.16	0.37	0.09	0.12	0.16	0.13
Panel B: College enrollment a year and a half after high school										
STEM	0.001 (0.031)	0.242*** (0.053)	0.204*** (0.054)	0.001 (0.037)	0.038 (0.043)	-0.170*** (0.051)	0.040 (0.036)	-0.123*** (0.039)	-0.087*** (0.029)	-0.071* (0.039)
Intercept	0.87	0.35	0.13	0.10	0.12	0.32	0.04	0.16	0.12	0.19
Student-cutoffs	1,199	1,199	1,199	1,199	1,199	1,199	1,199	1,199	1,199	1,199

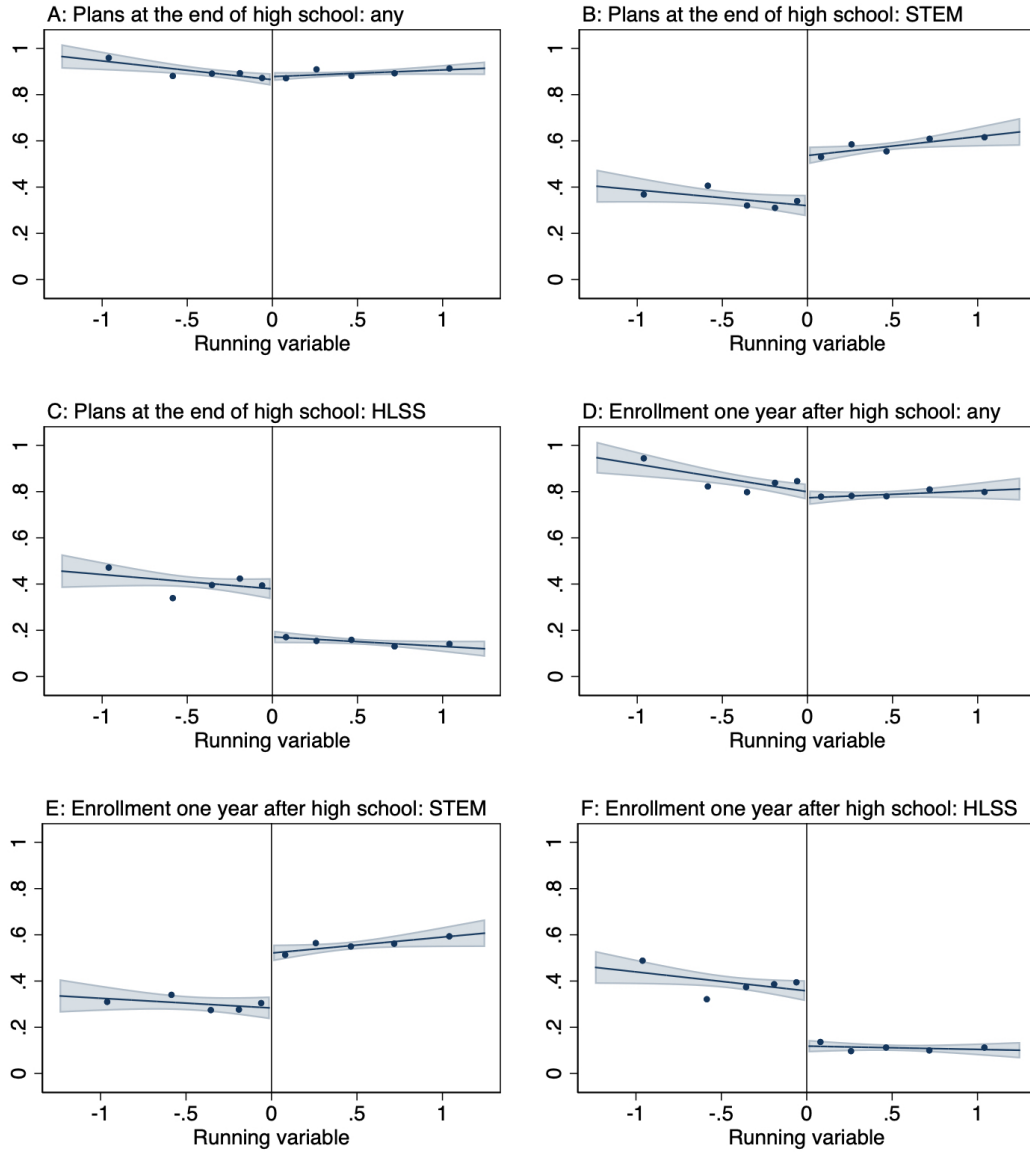
The table is analogous to Table 4, but restricted to students from the second follow-up survey, which occurred one and a half years after high school. The sample is the same across panels, so it is presented in a single row at the bottom of the table.

Table A23: Robustness for effects on college enrollment

	Main	Adding covariates	Optimal bandwidths		Alternative fixed bandwidths				Global quartic polynomial
			By cutoff	Mean	0.75	1.00	1.50	1.75	
Panel A: College plans at the end of high school: any college									
STEM	0.010 (0.018)	0.015 (0.018)	0.013 (0.018)	0.015 (0.019)	0.007 (0.021)	0.021 (0.019)	0.019 (0.017)	0.022 (0.017)	0.015 (0.018)
Student-cutoffs	3,987	3,987	3,629	3,783	2,993	3,573	4,249	4,415	4,822
Panel B: College plans at the end of high school: STEM									
STEM	0.233*** (0.035)	0.230*** (0.035)	0.235*** (0.034)	0.235*** (0.035)	0.242*** (0.039)	0.242*** (0.036)	0.229*** (0.034)	0.230*** (0.032)	0.224*** (0.034)
Student-cutoffs	3,987	3,987	3,395	3,675	2,993	3,573	4,249	4,415	4,822
Panel C: College plans at the end of high school: humanities, law, or social science									
STEM	-0.224*** (0.032)	-0.216*** (0.031)	-0.243*** (0.032)	-0.223*** (0.033)	-0.239*** (0.038)	-0.224*** (0.034)	-0.218*** (0.031)	-0.213*** (0.030)	-0.221*** (0.031)
Student-cutoffs	3,987	3,987	3,377	3,654	2,993	3,573	4,249	4,415	4,822
Panel D: College enrollment one year after high school: any college									
STEM	-0.037 (0.025)	-0.034 (0.025)	-0.057** (0.025)	-0.051** (0.025)	-0.058** (0.027)	-0.047* (0.025)	-0.031 (0.025)	-0.024 (0.025)	-0.027 (0.026)
Student-cutoffs	3,327	3,327	2,885	3,086	2,520	2,991	3,555	3,692	4,033
Panel E: College enrollment one year after high school: STEM									
STEM	0.246*** (0.036)	0.243*** (0.035)	0.241*** (0.035)	0.232*** (0.035)	0.243*** (0.040)	0.235*** (0.035)	0.251*** (0.033)	0.248*** (0.032)	0.234*** (0.033)
Student-cutoffs	3,327	3,327	2,783	3,019	2,520	2,991	3,555	3,692	4,033
Panel F: College enrollment one year after high school: humanities, law, or social science									
STEM	-0.261*** (0.031)	-0.255*** (0.030)	-0.288*** (0.032)	-0.252*** (0.032)	-0.277*** (0.036)	-0.257*** (0.032)	-0.264*** (0.030)	-0.260*** (0.030)	-0.248*** (0.031)
Student-cutoffs	3,327	3,327	2,854	3,054	2,520	2,991	3,555	3,692	4,033

The table shows robustness for the effects in Table A19. Specifications match those in Table A18. The one exception is that the added covariates in “Adding covariates” now also include mother’s and father’s years of schooling, which are available in the survey data but not the administrative data. Results for specifications that use alternative fixed effects are in Appendix A14.

Figure A5: RD plots of effects on college enrollment



The figure displays RD plots for effects on the outcomes in Table A19. See the notes to Figure A3 for details on RD plots.

Table A24: Effects on additional college outcomes

	Type of college				Paying for college			Pass winter exams	Expect to finish
	Any	Public	Private	Outside Romania	Scholar- ship	Tuition waiver	Out of pocket		
<i>Panel A: All students</i>									
STEM	-0.021 (0.045)	-0.018 (0.048)	-0.019 (0.016)	0.015 (0.013)	-0.125*** (0.046)	0.084 (0.060)	0.002 (0.052)	-0.020 (0.132)	-0.025 (0.134)
Intercept	0.82	0.78	0.04	0.00	0.27	0.35	0.27	3.24	3.28
Std. dev.	0.38	0.42	0.13	0.17	0.43	0.49	0.45	1.11	1.19
Student-cutoffs	1,159	1,159	1,159	1,159	1,159	1,159	1,159	1,159	1,159
<i>Panel B: Low-achievers</i>									
STEM	-0.124 (0.087)	-0.133 (0.083)	0.009 (0.019)	-0.001 (0.006)	-0.235*** (0.076)	0.013 (0.091)	-0.026 (0.107)	-0.385 (0.241)	-0.288 (0.252)
Intercept	0.82	0.79	0.01	0.02	0.32	0.31	0.29	3.34	3.21
Std. dev.	0.44	0.45	0.14	0.09	0.39	0.46	0.45	1.25	1.32
Student-cutoffs	547	547	547	547	547	547	547	547	547
<i>Panel C: High-achievers</i>									
STEM	-0.004 (0.046)	0.007 (0.058)	-0.022 (0.025)	0.011 (0.025)	-0.105 (0.073)	0.205** (0.090)	-0.049 (0.070)	0.151 (0.152)	-0.068 (0.139)
Intercept	0.91	0.86	0.04	0.01	0.27	0.36	0.31	3.33	3.62
Std. dev.	0.31	0.38	0.13	0.22	0.46	0.50	0.44	0.93	1.02
Student-cutoffs	612	612	612	612	612	612	612	612	612

The table presents results for the ATE of STEM assignment, β , for additional college outcomes. Outcomes are from the first follow-up survey and thus are measured a year after high school. The variables under the “Type of college” and “Paying for college” headings are indicators. For “Type of college”, “Any” equals 1 if a student is enrolled in college, while the other variables equal 1 if a student is enrolled in a college of the listed type. For “Paying for college”, the variables equal 1 if a student pays for college using the given method. The college types are mutually exclusive; the ways of paying for college are not. See Footnote 10 for details on paying for college. “Pass winter exams” and “Expect to finish” are on a scale of 1-4. For “Pass winter exams”, 1 is if a student passed none of the first-year winter exams for their college program or was never enrolled in college, 2 is if they passed some of the exams, 3 is if they passed most, and 4 is if they passed all. For “Expect to finish”, 1 is if a student does not think they will finish their college program or was not enrolled in college at the time of the survey, 2 is if finishing is somewhat likely, 3 is if it is likely, and 4 is if finishing is very likely. “Low-achievers” and “High-achievers” reflect the same sample split as “Achievement-Low” and “Achievement-High” in Table A6. Other details are as in Panel A of Table 2.

Table A25: Heterogeneity for effects on desired careers one year after high school

	All	Prefer		Stronger in		Gender		Achievement	
		STEM	HSS	Math	Language	Male	Female	Low	High
<i>Panel A: Technology or engineering</i>									
STEM	0.254*** (0.051)	0.333*** (0.072)	0.175** (0.067)	0.245*** (0.084)	0.257*** (0.063)	0.360*** (0.119)	0.183*** (0.051)	0.129 (0.105)	0.339*** (0.066)
Intercept	0.10	0.10	0.10	0.17	0.07	0.21	0.09	0.19	0.06
<i>Panel B: Medicine</i>									
STEM	-0.041 (0.044)	-0.033 (0.056)	-0.049 (0.059)	-0.051 (0.070)	-0.035 (0.059)	-0.084* (0.045)	-0.014 (0.068)	-0.050 (0.065)	0.002 (0.070)
Intercept	0.17	0.17	0.17	0.14	0.16	0.09	0.18	0.11	0.18
<i>Panel C: Business or economics</i>									
STEM	-0.040 (0.063)	-0.090 (0.072)	0.009 (0.093)	0.030 (0.094)	-0.035 (0.088)	-0.034 (0.117)	-0.033 (0.077)	0.157* (0.080)	-0.161 (0.102)
Intercept	0.33	0.35	0.31	0.37	0.25	0.29	0.33	0.12	0.44
<i>Panel D: Art, education, law, or social services</i>									
STEM	-0.145*** (0.046)	-0.169** (0.072)	-0.121** (0.049)	-0.166** (0.067)	-0.202*** (0.070)	-0.169** (0.077)	-0.140** (0.062)	-0.228** (0.112)	-0.173*** (0.056)
Intercept	0.23	0.22	0.24	0.18	0.32	0.18	0.26	0.36	0.22
<i>Panel E: Other/unsure</i>									
STEM	-0.028 (0.040)	-0.041 (0.056)	-0.014 (0.050)	-0.058 (0.065)	0.015 (0.051)	-0.074 (0.101)	0.004 (0.047)	-0.007 (0.090)	-0.008 (0.044)
Intercept	0.17	0.16	0.18	0.14	0.19	0.23	0.13	0.21	0.10
Student-cutoffs	1,159	589	570	546	520	325	746	547	612

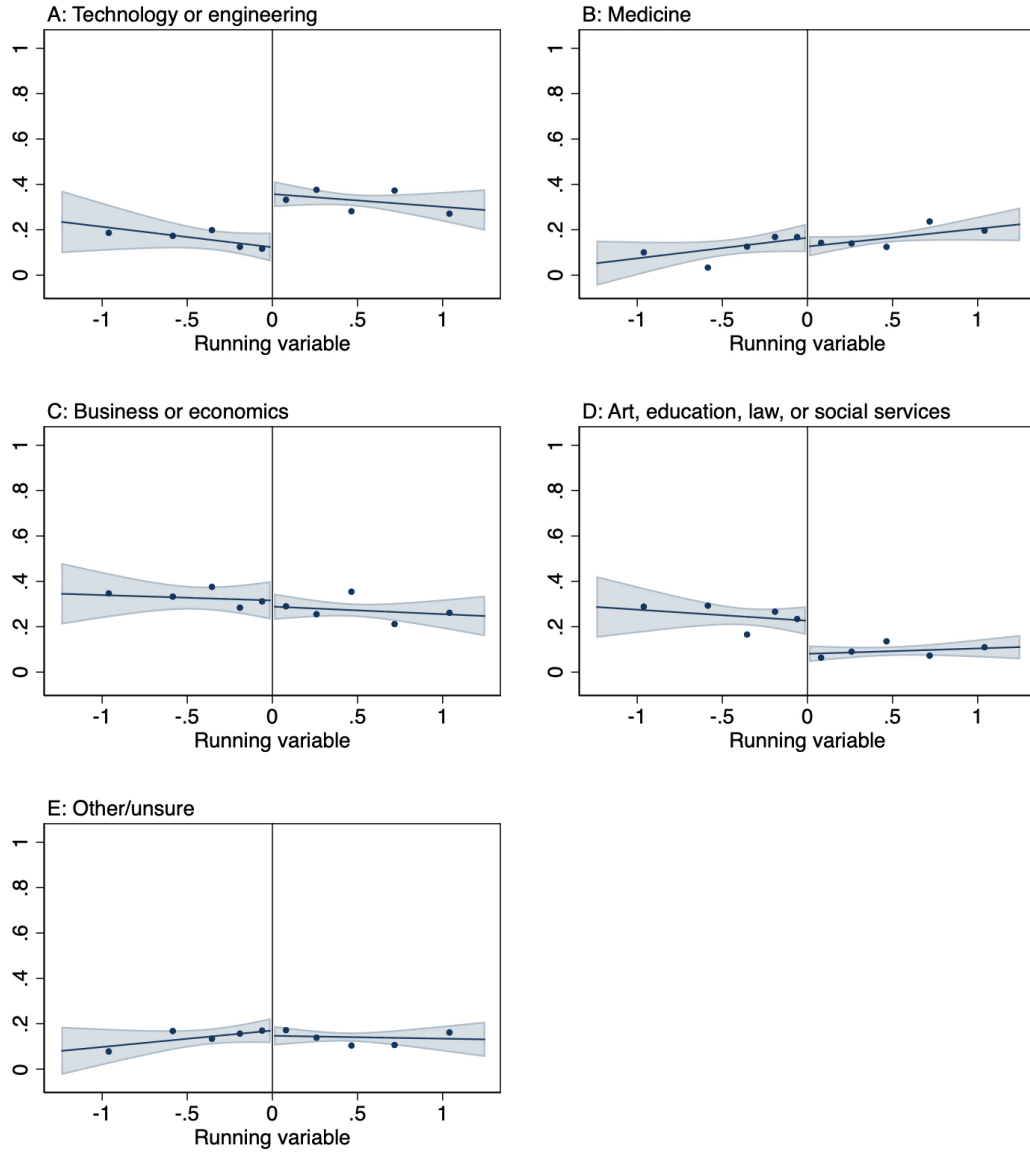
The table shows heterogeneity for the effects in Table 5. The sample is the same for all outcomes, so it is presented in a single row at the bottom of the table. See the notes to Table A6 for definitions of the column headings.

Table A26: Robustness for effects on desired careers one year after high school

	Main	Adding covariates	Optimal bandwidths		Alternative fixed bandwidths				Global quartic polynomial
			By cutoff	Mean	0.75	1.00	1.50	1.75	
<i>Panel A: Technology or engineering</i>									
STEM	0.254*** (0.051)	0.246*** (0.048)	0.230*** (0.053)	0.244*** (0.051)	0.245*** (0.059)	0.251*** (0.052)	0.242*** (0.049)	0.225*** (0.046)	0.226*** (0.052)
<i>Panel B: Medicine</i>									
STEM	-0.041 (0.044)	-0.035 (0.045)	-0.040 (0.045)	-0.040 (0.044)	-0.063 (0.052)	-0.063 (0.047)	-0.039 (0.041)	-0.026 (0.039)	-0.029 (0.046)
<i>Panel C: Business or economics</i>									
STEM	-0.040 (0.063)	-0.043 (0.063)	-0.029 (0.064)	-0.037 (0.063)	-0.040 (0.071)	-0.018 (0.066)	-0.040 (0.062)	-0.043 (0.059)	-0.045 (0.065)
<i>Panel D: Art, education, law, or social services</i>									
STEM	-0.145*** (0.046)	-0.141*** (0.044)	-0.146*** (0.048)	-0.144*** (0.046)	-0.155*** (0.055)	-0.150*** (0.050)	-0.141*** (0.046)	-0.141*** (0.043)	-0.155*** (0.045)
<i>Panel E: Other/unsure</i>									
STEM	-0.028 (0.040)	-0.027 (0.040)	-0.026 (0.040)	-0.028 (0.040)	0.013 (0.042)	-0.020 (0.040)	-0.021 (0.038)	-0.016 (0.036)	0.003 (0.040)
Student-cutoffs	1,159	1,159	1,118	1,140	886	1,041	1,263	1,319	1,389

The table shows robustness for the effects in Table 5. Specifications match those in Table A18. The one exception is that the added covariates in “Adding covariates” now also include mother’s and father’s years of schooling. The sample is the same for all outcomes, so it is presented in a single row at the bottom of the table. Specifications that use alternative fixed effects are in Appendix A14.

Figure A6: RD plots of effects on desired careers one year after high school



The figure displays RD plots for effects on the outcomes in Table 5. See the notes to Figure A3 for details on RD plots.

A8 Effects on time use

We included questions about time use in the end-of-high-school and second follow-up surveys. The questions asked students how many hours they spend on different activities on a typical weekday. Those in the end-of-high-school survey concerned a weekday during high school; those in the second follow-up survey were about a weekday at the time of the survey (approximately 1.5 years after high school).⁶⁰

Estimates for the effects of STEM assignment on time use are in Tables A27 (during high school) and A29 (1.5 years after high school). The tables provide results for all students and by initial curricular preferences, relative academic strength, gender, and baseline achievement. Tables A28 and A30 show robustness, and Figures A7 and A8 provide associated RD plots.

The tables yield a few findings. Starting with effects during high school, STEM assignment causes students to report spending more time on homework (Table A27). For all students, the estimated effect is 0.25 hours per day. For students with low baseline achievement, it is 0.5 hours; this is a 23% increase over the mean for low-achievers at STEM vs. HSS cutoffs who are assigned to HSS (2.1 hours), and it amounts to about 30% of the standard deviation for all students in the RD sample (1.6). In contrast, the effect for high-achievers is small and statistically insignificant. Next, STEM assignment does not influence how much time high school students spend working for pay, caring for others or volunteering, doing extracurricular activities, hanging out with friends, or watching TV. On the other hand, STEM assignment reduces the amount of time that high school students spend reading, especially for boys (0.5 hours, equal to 35% of the sd for all students in the RD sample). In addition, it reduces time spent on social media (0.4 hours, or 25% of the all-student sd). Finally, STEM assignment seems to lower the amount of time that boys spend playing video games, although the estimate is significant at only a 10% significance level.

By 1.5 years after high school, STEM assignment continues to reduce time spent reading and time on social media (Table A29). Surprisingly, it lowers the amount of time that students spend on homework, which could reflect impacts on whether students are enrolled in college together with differences in the structure of STEM and non-STEM college programs. Effects for the other activities are generally small and statistically insignificant.

Our finding that being placed in STEM causes high school students to spend more time on homework aligns with descriptive research that U.S. college students who major in STEM study more than those who major in non-STEM fields (Arcidiacono, Aucejo, and Spenner 2012; Ahn et al. 2024). We are unaware of prior research that examines the effects of curricular assignment on the other time use categories.

60. The activities asked about are not exhaustive; as a result, we did not require answers to sum to 24. We capped responses at a maximum of 6 hours.

Table A27: Effects on time use on a typical weekday in high school

	All	Prefer		Stronger in		Gender		Achievement	
		STEM	HSS	Math	Language	Male	Female	Low	High
<i>Panel A: Completing homework</i>									
STEM	0.248** (0.109)	0.331** (0.157)	0.165 (0.123)	0.308* (0.175)	0.172 (0.150)	0.342* (0.185)	0.404*** (0.124)	0.490** (0.192)	0.058 (0.153)
Intercept	2.44	2.48	2.41	2.46	2.45	1.86	2.66	2.13	2.75
Std. dev.	1.61	1.67	1.52	1.67	1.53	1.53	1.59	1.50	1.68
<i>Panel B: Working for pay</i>									
STEM	0.047 (0.114)	0.111 (0.166)	-0.017 (0.123)	0.278 (0.176)	-0.172 (0.172)	0.114 (0.205)	-0.061 (0.158)	0.121 (0.158)	0.040 (0.176)
Intercept	1.63	1.46	1.79	1.48	1.81	1.67	1.62	1.76	1.42
Std. dev.	1.85	1.83	1.85	1.85	1.84	1.88	1.82	1.84	1.83
<i>Panel C: Caring for others or volunteering</i>									
STEM	-0.036 (0.058)	-0.032 (0.076)	-0.040 (0.073)	0.082 (0.086)	-0.139 (0.090)	0.013 (0.087)	-0.058 (0.085)	0.074 (0.082)	-0.075 (0.087)
Intercept	0.53	0.56	0.49	0.41	0.59	0.41	0.58	0.43	0.56
Std. dev.	1.09	1.09	1.09	1.06	1.12	0.96	1.16	1.10	1.09
<i>Panel D: Doing extracurricular activities</i>									
STEM	0.032 (0.086)	0.080 (0.111)	-0.016 (0.108)	0.082 (0.139)	0.091 (0.113)	0.080 (0.185)	0.007 (0.100)	0.029 (0.133)	0.065 (0.138)
Intercept	0.88	0.92	0.83	0.96	0.71	1.04	0.78	0.79	0.97
Std. dev.	1.36	1.37	1.34	1.41	1.30	1.47	1.27	1.31	1.40
<i>Panel E: Hanging out with friends</i>									
STEM	-0.127 (0.112)	-0.198 (0.139)	-0.055 (0.150)	-0.013 (0.174)	-0.040 (0.167)	-0.029 (0.194)	-0.235 (0.148)	0.064 (0.184)	-0.180 (0.143)
Intercept	2.68	2.64	2.71	2.70	2.61	2.73	2.70	2.64	2.62
Std. dev.	1.76	1.70	1.79	1.72	1.78	1.86	1.70	1.78	1.72
<i>Panel F: Reading</i>									
STEM	-0.251*** (0.091)	-0.267* (0.139)	-0.235*** (0.089)	-0.264** (0.131)	-0.280** (0.141)	-0.472*** (0.153)	-0.103 (0.126)	-0.345** (0.163)	-0.287** (0.128)
Intercept	1.42	1.38	1.46	1.29	1.57	1.17	1.59	1.60	1.35
Std. dev.	1.36	1.31	1.39	1.31	1.39	1.26	1.37	1.36	1.34
<i>Panel G: Watching TV</i>									
STEM	-0.004 (0.074)	-0.046 (0.101)	0.038 (0.097)	0.106 (0.108)	-0.020 (0.117)	-0.050 (0.125)	0.037 (0.097)	0.094 (0.121)	-0.031 (0.095)
Intercept	0.86	0.76	0.96	0.71	0.97	0.75	0.93	0.84	0.81
Std. dev.	1.26	1.18	1.32	1.16	1.32	1.19	1.29	1.31	1.19
<i>Panel H: Playing video games</i>									
STEM	-0.096 (0.107)	-0.204 (0.147)	0.012 (0.126)	0.050 (0.174)	-0.076 (0.158)	-0.362* (0.191)	-0.137 (0.106)	0.050 (0.164)	-0.189 (0.157)
Intercept	1.41	1.52	1.30	1.49	1.37	2.57	0.78	1.45	1.37
Std. dev.	1.65	1.62	1.68	1.66	1.63	1.87	1.20	1.69	1.61
<i>Panel I: On social media</i>									
STEM	-0.428*** (0.111)	-0.462*** (0.146)	-0.394*** (0.138)	-0.453*** (0.163)	-0.325* (0.175)	-0.215 (0.189)	-0.437*** (0.147)	-0.064 (0.185)	-0.560*** (0.148)
Intercept	3.42	3.32	3.53	3.26	3.50	2.87	3.68	3.19	3.48
Std. dev.	1.69	1.67	1.70	1.67	1.68	1.69	1.64	1.70	1.68
Student-cutoffs	3,987	1,925	2,062	1,921	1,985	1,396	2,510	2,093	1,894

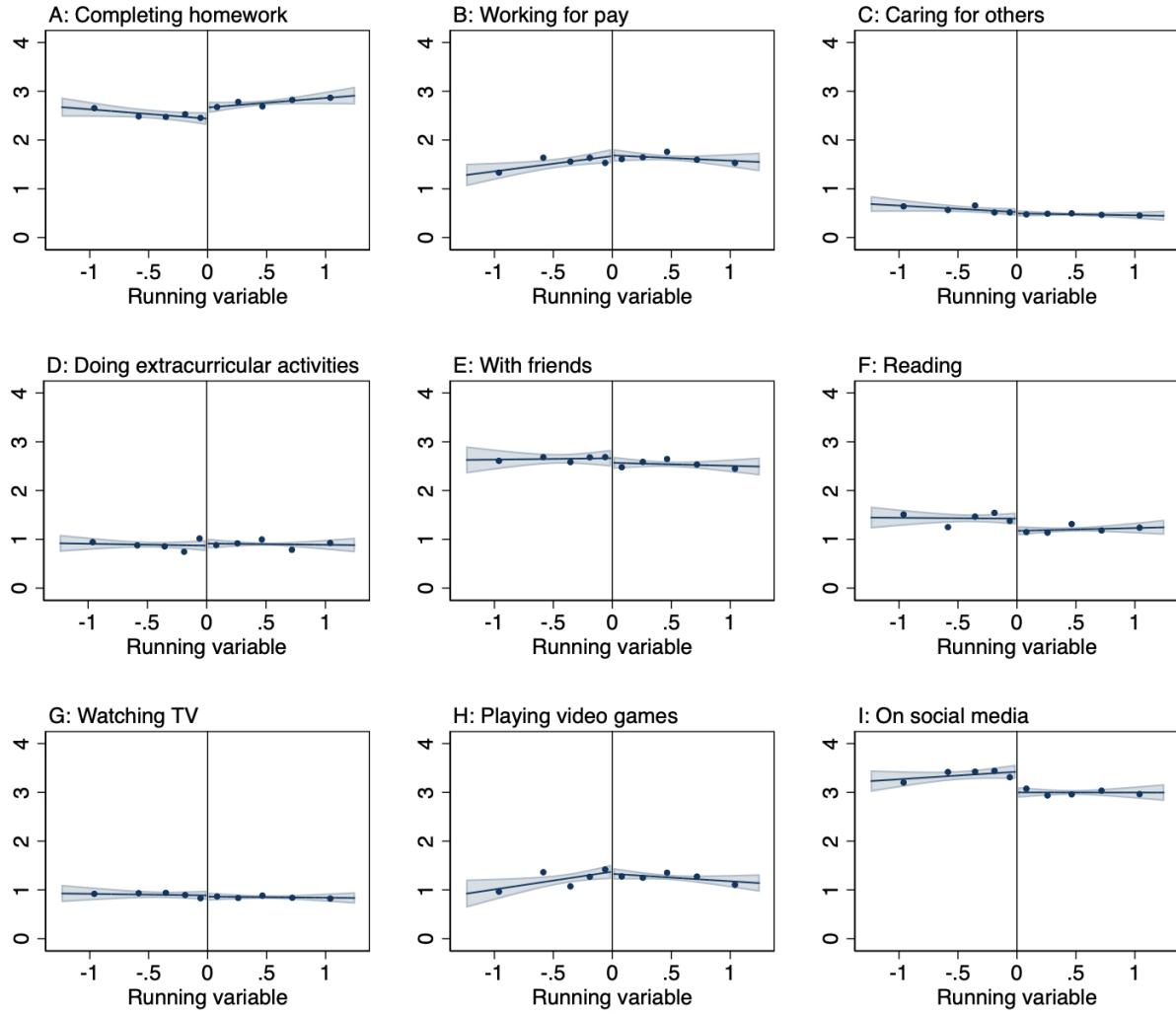
The table presents results for the effects of STEM assignment on the number of hours spent on the listed activities on a typical weekday in high school. Outcome variables are from the end-of-high-school survey. The sample is the same for all outcomes, so it is presented in a single row at the bottom of the table. See the notes to Table A6 for definitions of the column headings. Other details match those in Table 2.

Table A28: Robustness for effects on time use on a typical weekday in high school

	Main	Adding covariates	Optimal bandwidths		Alternative fixed bandwidths				Global quartic polynomial
			By cutoff	Mean	0.75	1.00	1.50	1.75	
Panel A: Completing homework									
STEM	0.248** (0.109)	0.297*** (0.103)	0.197* (0.114)	0.206* (0.118)	0.193 (0.126)	0.203* (0.118)	0.276*** (0.105)	0.305*** (0.103)	0.271** (0.106)
Panel B: Working for pay									
STEM	0.047 (0.114)	0.040 (0.114)	0.082 (0.116)	0.092 (0.118)	0.063 (0.134)	0.098 (0.119)	0.070 (0.108)	0.113 (0.106)	0.065 (0.112)
Panel C: Caring for others or volunteering									
STEM	-0.036 (0.058)	-0.022 (0.058)	-0.034 (0.061)	-0.022 (0.061)	-0.021 (0.067)	0.006 (0.060)	-0.048 (0.059)	-0.047 (0.058)	-0.052 (0.058)
Panel D: Doing extracurricular activities									
STEM	0.032 (0.086)	0.032 (0.083)	-0.043 (0.086)	0.020 (0.091)	0.025 (0.096)	0.048 (0.092)	0.046 (0.082)	0.028 (0.076)	0.048 (0.082)
Panel E: Hanging out with friends									
STEM	-0.127 (0.112)	-0.131 (0.113)	-0.147 (0.116)	-0.144 (0.118)	-0.136 (0.129)	-0.144 (0.119)	-0.128 (0.106)	-0.143 (0.102)	-0.178 (0.108)
Panel F: Reading									
STEM	-0.251*** (0.091)	-0.218** (0.089)	-0.272*** (0.101)	-0.303*** (0.096)	-0.386*** (0.114)	-0.294*** (0.098)	-0.247*** (0.086)	-0.235*** (0.083)	-0.267*** (0.085)
Panel G: Watching TV									
STEM	-0.004 (0.074)	0.008 (0.073)	-0.005 (0.080)	0.053 (0.078)	-0.009 (0.077)	0.044 (0.079)	-0.018 (0.071)	-0.061 (0.069)	-0.053 (0.071)
Panel H: Playing video games									
STEM	-0.096 (0.107)	-0.195** (0.097)	-0.050 (0.118)	-0.076 (0.117)	-0.058 (0.134)	-0.084 (0.119)	-0.064 (0.102)	-0.071 (0.098)	-0.064 (0.105)
Panel I: On social media									
STEM	-0.428*** (0.111)	-0.387*** (0.113)	-0.383*** (0.115)	-0.411*** (0.117)	-0.290** (0.136)	-0.403*** (0.118)	-0.426*** (0.104)	-0.406*** (0.102)	-0.414*** (0.107)
Student-cutoffs	3,987	3,987	3,322	3,609	2,993	3,573	4,249	4,415	4,822

The table shows robustness for the effects in Table A27. Specifications match those in Table A18. The one exception is that the added covariates in “Adding covariates” now also include mother’s and father’s years of schooling. The sample is the same for all outcomes, so it is presented in a single row at the bottom of the table.

Figure A7: RD plots of effects on time use on a typical weekday in high school



The figure displays RD plots for effects on the outcomes in Table A27. See the notes to Figure A3 for details on RD plots.

Table A29: Effects on time use on a typical weekday 1.5 years after high school

	All	Prefer		Stronger in		Gender		Achievement	
		STEM	HSS	Math	Language	Male	Female	Low	High
<i>Panel A: Completing homework</i>									
STEM	-0.383** (0.185)	-0.347 (0.284)	-0.419** (0.187)	-0.433 (0.399)	-0.398 (0.313)	-0.466 (0.341)	-0.233 (0.239)	0.163 (0.346)	-0.798*** (0.249)
Intercept	2.87	3.08	2.66	2.63	2.87	2.73	2.84	2.42	3.34
Std. dev.	1.70	1.63	1.72	1.68	1.70	1.69	1.68	1.70	1.64
<i>Panel B: Working for pay</i>									
STEM	-0.372 (0.285)	-0.636 (0.434)	-0.108 (0.322)	-0.885** (0.444)	-0.367 (0.397)	-0.580 (0.444)	-0.633* (0.381)	-0.933* (0.509)	-0.259 (0.388)
Intercept	1.82	1.75	1.88	2.00	2.09	2.36	1.84	2.85	1.26
Std. dev.	2.33	2.07	2.51	2.28	2.33	2.41	2.24	2.55	1.99
<i>Panel C: Caring for others or volunteering</i>									
STEM	0.115 (0.125)	0.126 (0.185)	0.103 (0.157)	-0.205 (0.237)	0.223 (0.219)	-0.238 (0.245)	0.235 (0.173)	0.560* (0.306)	0.013 (0.124)
Intercept	0.26	0.25	0.26	0.37	0.28	0.48	0.17	-0.12	0.32
Std. dev.	0.97	0.96	0.98	0.94	1.03	0.93	1.01	1.08	0.82
<i>Panel D: Doing hobbies</i>									
STEM	0.154 (0.111)	0.211 (0.158)	0.097 (0.140)	0.201 (0.199)	0.164 (0.166)	0.087 (0.230)	0.193 (0.147)	-0.013 (0.225)	0.362** (0.146)
Intercept	0.54	0.60	0.48	0.43	0.60	0.74	0.42	0.60	0.51
Std. dev.	0.91	0.92	0.89	0.88	0.95	0.96	0.85	0.91	0.90
<i>Panel E: Hanging out with friends</i>									
STEM	0.026 (0.163)	0.019 (0.219)	0.034 (0.215)	0.364 (0.295)	0.101 (0.268)	0.254 (0.314)	-0.140 (0.243)	0.048 (0.340)	0.170 (0.272)
Intercept	1.91	1.95	1.86	2.04	1.80	1.96	1.88	1.98	1.80
Std. dev.	1.35	1.32	1.39	1.33	1.40	1.38	1.31	1.42	1.28
<i>Panel F: Reading</i>									
STEM	-0.404* (0.225)	-0.096 (0.297)	-0.713** (0.316)	-0.896*** (0.334)	-0.217 (0.336)	-0.686** (0.331)	-0.267 (0.300)	-0.593 (0.471)	-0.528** (0.246)
Intercept	2.71	2.59	2.84	2.53	3.00	2.32	2.89	2.90	2.87
Std. dev.	1.67	1.60	1.73	1.59	1.72	1.63	1.65	1.73	1.60
<i>Panel G: Watching TV</i>									
STEM	0.076 (0.087)	0.017 (0.100)	0.134 (0.130)	0.062 (0.120)	0.273** (0.128)	0.147 (0.144)	0.083 (0.123)	0.256* (0.137)	0.031 (0.113)
Intercept	0.39	0.30	0.48	0.36	0.35	0.29	0.45	0.24	0.36
Std. dev.	0.76	0.68	0.83	0.72	0.80	0.73	0.79	0.81	0.71
<i>Panel H: Playing video games</i>									
STEM	0.072 (0.125)	0.152 (0.183)	-0.009 (0.147)	0.133 (0.185)	-0.013 (0.174)	-0.032 (0.318)	0.114 (0.125)	-0.169 (0.206)	0.176 (0.163)
Intercept	0.48	0.52	0.44	0.51	0.47	0.95	0.19	0.58	0.40
Std. dev.	1.08	1.05	1.11	1.08	1.05	1.34	0.78	1.11	1.05
<i>Panel I: On social media</i>									
STEM	-0.356** (0.170)	-0.370 (0.229)	-0.343 (0.224)	0.016 (0.321)	-0.336 (0.297)	-0.213 (0.313)	-0.430** (0.209)	0.019 (0.317)	-0.659*** (0.222)
Intercept	3.01	2.99	3.03	2.70	3.23	2.76	3.16	2.87	3.17
Std. dev.	1.42	1.40	1.44	1.39	1.45	1.30	1.46	1.46	1.38
Student-cutoffs	1,199	575	624	540	550	373	728	618	581

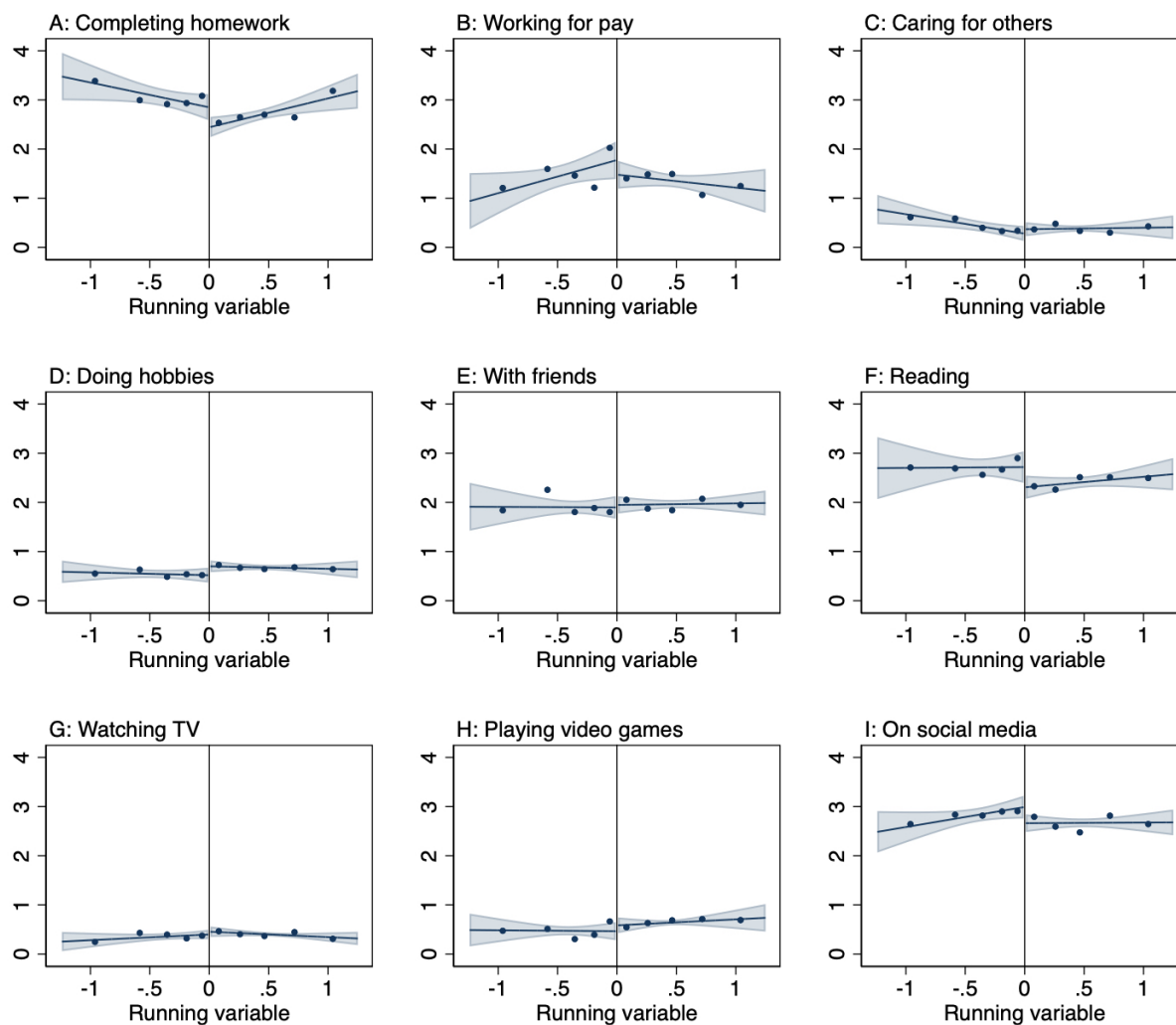
The table presents results for the effects of STEM assignment on the number of hours spent on the listed activities on a typical weekday a year and a half after high school. Outcome variables are from the second follow-up survey. The sample is the same for all outcomes, so it is presented in a single row at the bottom of the table. See the notes to Table A6 for definitions of the column headings. Other details match those in Table 2.

Table A30: Robustness for effects on time use on a typical weekday 1.5 years after high school

	Main	Adding covariates	Optimal bandwidths		Alternative fixed bandwidths				Global quartic polynomial
			By cutoff	Mean	0.75	1.00	1.50	1.75	
<i>Panel A: Completing homework</i>									
STEM	-0.383** (0.185)	-0.381** (0.184)	-0.370* (0.190)	-0.399** (0.188)	-0.469** (0.212)	-0.560*** (0.195)	-0.317* (0.181)	-0.218 (0.178)	-0.373** (0.189)
<i>Panel B: Working for pay</i>									
STEM	-0.372 (0.285)	-0.396 (0.283)	-0.449 (0.303)	-0.342 (0.286)	-0.403 (0.325)	-0.252 (0.296)	-0.416 (0.279)	-0.511* (0.273)	-0.284 (0.289)
<i>Panel C: Caring for others or volunteering</i>									
STEM	0.115 (0.125)	0.116 (0.127)	0.121 (0.125)	0.124 (0.125)	0.210 (0.144)	0.133 (0.123)	0.090 (0.120)	0.057 (0.114)	0.123 (0.123)
<i>Panel D: Doing hobbies</i>									
STEM	0.154 (0.111)	0.143 (0.111)	0.155 (0.114)	0.182 (0.111)	0.208 (0.130)	0.191 (0.119)	0.152 (0.107)	0.160 (0.106)	0.198* (0.115)
<i>Panel E: Hanging out with friends</i>									
STEM	0.026 (0.163)	0.016 (0.163)	0.041 (0.160)	0.059 (0.170)	0.235 (0.200)	0.122 (0.179)	0.004 (0.154)	0.002 (0.146)	0.060 (0.173)
<i>Panel F: Reading</i>									
STEM	-0.404* (0.225)	-0.364 (0.221)	-0.366 (0.224)	-0.445** (0.224)	-0.429* (0.230)	-0.435* (0.231)	-0.379* (0.214)	-0.306 (0.213)	-0.347 (0.223)
<i>Panel G: Watching TV</i>									
STEM	0.076 (0.087)	0.070 (0.086)	0.073 (0.088)	0.072 (0.087)	0.074 (0.097)	0.060 (0.088)	0.057 (0.085)	0.048 (0.081)	0.068 (0.089)
<i>Panel H: Playing video games</i>									
STEM	0.072 (0.125)	0.024 (0.124)	0.074 (0.125)	0.026 (0.126)	-0.015 (0.139)	-0.028 (0.130)	0.078 (0.117)	0.062 (0.110)	0.009 (0.126)
<i>Panel I: On social media</i>									
STEM	-0.356** (0.170)	-0.323* (0.164)	-0.323* (0.179)	-0.342** (0.167)	-0.219 (0.190)	-0.250 (0.171)	-0.330** (0.163)	-0.291* (0.155)	-0.324* (0.171)
Student-cutoffs	1,199	1,199	1,134	1,163	883	1,068	1,271	1,318	1,432

The table shows robustness for the effects in Table A29. Specifications match those in Table A18. The one exception is that the added covariates in “Adding covariates” now also include mother’s and father’s years of schooling. The sample is the same for all outcomes, so it is presented in a single row at the bottom of the table.

Figure A8: RD plots of effects on time use on a typical weekday 1.5 years after high school



The figure displays RD plots for effects on the outcomes in Table A29. See the notes to Figure A3 for details on RD plots.

A9 Additional results for difficulty and risk

Table A31: Effects on outcomes related to difficulty

	All	Prefer		Stronger in		Gender		Achievement	
		STEM	HSS	Math	Language	Male	Female	Low	High
<i>Panel A: Grit</i>									
STEM	0.083** (0.037)	0.079* (0.045)	0.087* (0.049)	0.090 (0.064)	0.044 (0.052)	0.157** (0.061)	0.037 (0.051)	0.068 (0.050)	0.057 (0.056)
Intercept	-0.10	-0.11	-0.09	-0.12	-0.05	-0.16	-0.06	-0.10	-0.06
Std. dev.	0.57	0.57	0.57	0.57	0.57	0.56	0.58	0.57	0.57
Student-cutoffs	3,987	1,925	2,062	1,921	1,985	1,396	2,510	2,093	1,894
<i>Panel B: The baccalaureate exam for my track type was particularly hard</i>									
STEM	0.173* (0.088)	0.228** (0.114)	0.118 (0.122)	0.192* (0.115)	0.075 (0.125)	0.580*** (0.193)	0.090 (0.118)	0.297** (0.135)	0.172 (0.127)
Intercept	1.88	1.85	1.92	1.77	1.95	1.61	1.97	1.74	1.89
Std. dev.	0.73	0.71	0.71	0.74	0.72	0.75	0.72	0.71	0.74
Student-cutoffs	1,159	589	570	546	520	325	746	547	612
<i>Panel C: My track prepares students well for the baccalaureate exam</i>									
STEM	0.039 (0.038)	0.036 (0.049)	0.041 (0.053)	0.089 (0.076)	0.003 (0.056)	0.071 (0.075)	0.021 (0.049)	-0.006 (0.074)	0.057 (0.056)
Intercept	0.84	0.84	0.84	0.87	0.84	0.79	0.85	0.87	0.84
Std. dev.	0.34	0.34	0.35	0.34	0.34	0.34	0.35	0.35	0.34
Student-cutoffs	1,159	589	570	546	520	325	746	547	612

The table presents results for the effects of STEM assignment on three outcomes related to the difficulty of a student's high school studies. "Grit" is an index and is defined in the notes to Table A32. The outcomes in Panels B and C are from the first follow-up survey. The Panel B outcome is on a scale of 1-3. 1 (3) is that a student believes the exam for their track type is easier (harder) than the exams for other track types; 2 is the same/unsure. The Panel C outcome is an indicator equal to 1 if a student agrees with the listed statement. See Table A6 for definitions of the column headings. Other details match those in Table 2.

Table A32: Effects on components of the grit index

	Index	Components				
		Give up now for later	Finish what I begin	Work hard	Don't get distracted	Follow through with goals
STEM	0.083** (0.037)	-0.009 (0.070)	0.151** (0.060)	0.121** (0.060)	0.175** (0.076)	0.011 (0.087)
Intercept	-0.10	3.81	3.68	3.85	2.57	3.23
Std. dev.	0.57	1.06	1.02	0.95	1.12	1.21
Student-cutoffs	3,987	3,987	3,987	3,987	3,987	3,987

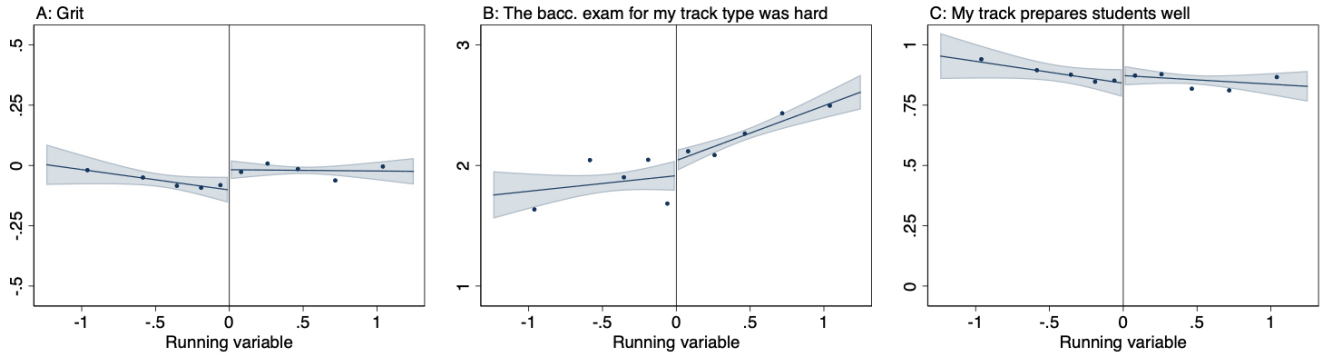
The table shows effects on components of the grit index used in Table A31. The components are based on questions where students were asked how much they agree with a statement on a scale of 1-5. The statements are: "I am willing to give up something that is beneficial for me today in order to benefit more in the future", "I finish whatever I begin", "I am a hard worker", "New ideas and projects sometimes distract me from previous ones", and "I often set a goal but later choose to pursue a different one". In calculating "Don't get distracted" and "Follow through with goals", we reverse the order of the responses so that higher values indicate higher grit. The questions were asked in both the end-of-high-school and second follow-up surveys. The components use data from both surveys. They are calculated in three steps. First, we demean a student's responses for a given question using survey-specific means that are calculated over the students with non-missing responses in both surveys (the "common" sample). Second, we average the demeaned responses by student, ignoring missing values. Third, for interpretability, we add back the overall mean response from the common sample. Steps 1 and 2 are equivalent to obtaining the student fixed effects from a panel regression of a question's responses on student and survey fixed effects. Finally, the grit index is the average of standardized versions of all of a student's responses. The average again uses only the responses that are non-missing for a student. The standardization uses survey-specific means and standard deviations that are calculated over the students without any missing responses. Other details follow Panel A of Table 2.

Table A33: Robustness for effects on outcomes related to difficulty

	Main	Adding covariates	Optimal bandwidths		Alternative fixed bandwidths				Global quartic polynomial
			By cutoff	Mean	0.75	1.00	1.50	1.75	
<i>Panel A: Grit</i>									
STEM	0.083** (0.037)	0.088** (0.037)	0.074* (0.039)	0.080** (0.038)	0.099** (0.043)	0.084** (0.039)	0.087** (0.035)	0.086** (0.034)	0.076** (0.036)
Student-cutoffs	3,987	3,987	3,344	3,609	2,993	3,573	4,249	4,415	4,822
<i>Panel B: The baccalaureate exam for my track type was particularly hard</i>									
STEM	0.173* (0.088)	0.169* (0.088)	0.174* (0.092)	0.162* (0.089)	0.241** (0.102)	0.181* (0.094)	0.210** (0.087)	0.246*** (0.086)	0.138 (0.095)
Student-cutoffs	1,159	1,159	1,115	1,140	886	1,041	1,263	1,319	1,389
<i>Panel C: My track prepares students well for the baccalaureate exam</i>									
STEM	0.039 (0.038)	0.039 (0.038)	0.022 (0.038)	0.045 (0.039)	0.069 (0.048)	0.057 (0.043)	0.032 (0.037)	0.036 (0.038)	0.049 (0.039)
Student-cutoffs	1,159	1,159	1,111	1,140	886	1,041	1,263	1,319	1,389

The table shows robustness for the effects in Table A31. Specifications match those in Table A18. The one exception is that the added covariates in “Adding covariates” now also include mother’s and father’s years of schooling.

Figure A9: RD plots of effects on outcomes related to difficulty



The figure displays RD plots for effects on the outcomes in Table A31. See the notes to Figure A3 for details on RD plots.

Table A34: Heterogeneity for effects on baccalaureate performance

	All	Prefer		Stronger in		Gender		Achievement	
		STEM	HSS	Math	Language	Male	Female	Low	High
<i>Panel A: Take the exam</i>									
STEM	-0.011*** (0.003)	-0.010*** (0.003)	-0.012*** (0.004)	-0.001 (0.003)	-0.025*** (0.005)	-0.004 (0.004)	-0.014*** (0.004)	-0.035*** (0.007)	-0.001 (0.003)
Intercept	0.95	0.97	0.94	0.96	0.95	0.95	0.96	0.92	0.98
Student-cutoffs	73,470	37,376	36,094	45,077	28,060	31,900	41,295	27,204	46,266
<i>Panel B: Pass the exam</i>									
STEM	-0.036*** (0.004)	-0.030*** (0.004)	-0.042*** (0.007)	-0.016*** (0.004)	-0.071*** (0.007)	-0.033*** (0.006)	-0.036*** (0.005)	-0.111*** (0.010)	-0.008** (0.003)
Intercept	0.92	0.95	0.89	0.93	0.90	0.91	0.93	0.86	0.97
Student-cutoffs	73,470	37,376	36,094	45,077	28,060	31,900	41,295	27,204	46,266
<i>Panel C: Exam score</i>									
STEM	-0.351*** (0.015)	-0.312*** (0.018)	-0.391*** (0.020)	-0.312*** (0.018)	-0.434*** (0.023)	-0.359*** (0.021)	-0.340*** (0.019)	-0.502*** (0.032)	-0.302*** (0.016)
Intercept	8.08	8.28	7.88	8.17	7.92	7.90	8.24	7.45	8.44
Student-cutoffs	69,938	36,282	33,656	43,454	26,154	30,364	39,321	24,751	45,187
<i>Panel D: Pass in STEM</i>									
STEM	0.621*** (0.007)	0.646*** (0.009)	0.597*** (0.010)	0.621*** (0.009)	0.621*** (0.010)	0.558*** (0.010)	0.678*** (0.008)	0.556*** (0.014)	0.649*** (0.008)
Intercept	0.13	0.19	0.07	0.19	0.05	0.18	0.09	0.03	0.20
Student-cutoffs	73,470	37,376	36,094	45,077	28,060	31,900	41,295	27,204	46,266
<i>Panel E: Pass in HSS</i>									
STEM	-0.698*** (0.007)	-0.688*** (0.009)	-0.708*** (0.008)	-0.674*** (0.008)	-0.738*** (0.009)	-0.631*** (0.010)	-0.755*** (0.007)	-0.712*** (0.012)	-0.692*** (0.008)
Intercept	0.78	0.75	0.81	0.74	0.85	0.72	0.84	0.82	0.76
Student-cutoffs	73,470	37,376	36,094	45,077	28,060	31,900	41,295	27,204	46,266
<i>Panel F: Pass in Other</i>									
STEM	0.040*** (0.004)	0.011*** (0.003)	0.070*** (0.007)	0.037*** (0.004)	0.046*** (0.005)	0.040*** (0.005)	0.042*** (0.004)	0.046*** (0.007)	0.036*** (0.004)
Intercept	0.01	0.01	0.01	0.01	0.01	0.01	0.01	0.01	0.01
Student-cutoffs	73,470	37,376	36,094	45,077	28,060	31,900	41,295	27,204	46,266

The table shows heterogeneity for the effects in Table 6. See the notes to Table A6 for definitions of the column headings.

Table A35: Effects on baccalaureate performance with no restriction on outcome timing

	Take the exam	Pass the exam	Exam score	Pass in		
				STEM	HSS	Other
STEM	-0.010*** (0.003)	-0.028*** (0.004)	-0.345*** (0.015)	0.633*** (0.007)	-0.701*** (0.007)	0.040*** (0.004)
Intercept	0.96	0.93	8.09	0.13	0.79	0.01
Student-cutoffs	73,470	73,470	70,141	73,470	73,470	73,470

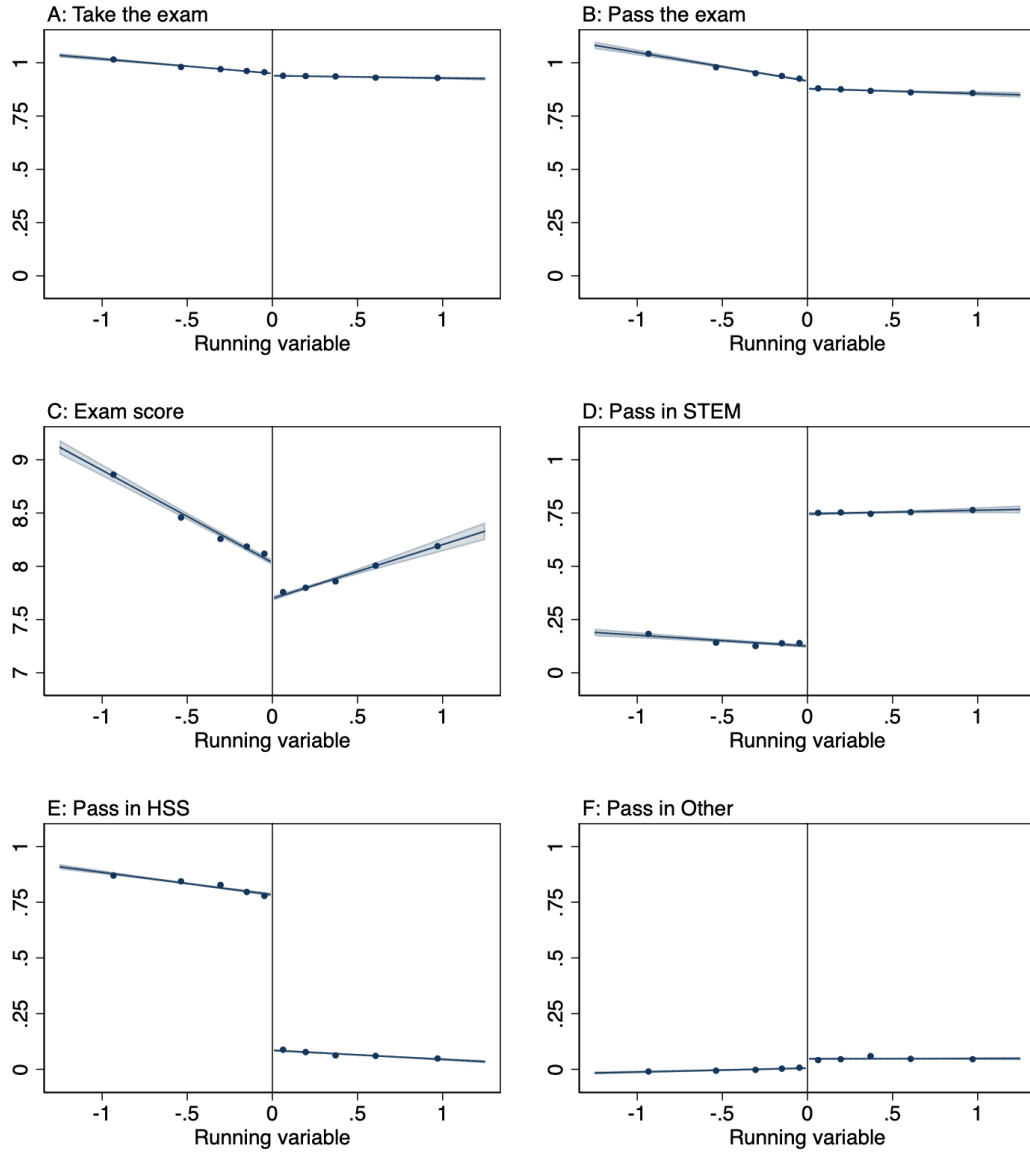
The table is analogous to Table 6; however, outcomes are measured over all years after track assignment, not just the first four.

Table A36: Robustness for effects on baccalaureate performance

	Main	Adding covariates	Optimal bandwidths		Alternative fixed bandwidths				Global quartic polynomial
			By cutoff	Mean	0.75	1.00	1.50	1.75	
Panel A: Take the exam									
STEM	-0.011*** (0.003)	-0.011*** (0.003)	-0.011*** (0.003)	-0.011*** (0.003)	-0.013*** (0.003)	-0.011*** (0.003)	-0.012*** (0.003)	-0.011*** (0.003)	-0.012*** (0.003)
Student-cutoffs	73,470	73,470	64,214	68,243	59,375	67,804	77,330	79,842	85,312
Panel B: Pass the exam									
STEM	-0.036*** (0.004)	-0.036*** (0.004)	-0.040*** (0.004)	-0.038*** (0.004)	-0.038*** (0.004)	-0.037*** (0.004)	-0.039*** (0.004)	-0.040*** (0.004)	-0.037*** (0.004)
Student-cutoffs	73,470	73,470	64,214	68,243	59,375	67,804	77,330	79,842	85,312
Panel C: Exam score									
STEM	-0.351*** (0.015)	-0.351*** (0.014)	-0.352*** (0.015)	-0.359*** (0.015)	-0.355*** (0.016)	-0.351*** (0.015)	-0.355*** (0.015)	-0.360*** (0.014)	-0.370*** (0.015)
Student-cutoffs	69,938	69,938	57,409	61,631	56,635	64,577	73,545	75,887	80,948
Panel D: Pass in STEM									
STEM	0.621*** (0.007)	0.621*** (0.007)	0.624*** (0.007)	0.618*** (0.007)	0.615*** (0.008)	0.619*** (0.007)	0.622*** (0.007)	0.623*** (0.007)	0.627*** (0.007)
Student-cutoffs	73,470	73,470	64,214	68,243	59,375	67,804	77,330	79,842	85,312
Panel E: Pass in HSS									
STEM	-0.698*** (0.007)	-0.698*** (0.007)	-0.704*** (0.007)	-0.692*** (0.007)	-0.688*** (0.008)	-0.694*** (0.007)	-0.704*** (0.007)	-0.707*** (0.006)	-0.708*** (0.006)
Student-cutoffs	73,470	73,470	64,214	68,243	59,375	67,804	77,330	79,842	85,312
Panel F: Pass in Other									
STEM	0.040*** (0.004)	0.040*** (0.004)	0.043*** (0.004)	0.040*** (0.004)	0.035*** (0.004)	0.038*** (0.004)	0.043*** (0.004)	0.044*** (0.004)	0.044*** (0.004)
Student-cutoffs	73,470	73,470	64,214	68,243	59,375	67,804	77,330	79,842	85,312

The table shows robustness for the effects in Table 6. Specifications match those in Table A18. Results for specifications that use alternative fixed effects are in Appendix A14.

Figure A10: RD plots of effects on baccalaureate performance



The figure displays RD plots for effects on the outcomes in Table 6. See the notes to Figure A3 for details on RD plots.

A10 Additional results for mechanisms

A10.1 Obtaining and classifying jobs recommended by O*NET

In the second follow-up survey, we administered an abridged O*NET Interest Profiler. We use each student’s O*NET responses to generate a list of five jobs recommended by O*NET for the student. In addition, we calculate the share of the recommended jobs that are STEM-related, HSS-related, or Other.

To generate the recommended jobs, we follow O*NET’s algorithm for job recommendations. When taking the Interest Profiler, a student indicates whether they would enjoy a given job task. O*NET groups job tasks as Realistic, Investigative, Artistic, Social, Enterprising, or Conventional. Based on a student’s responses, we calculate the share of tasks in each group that the student enjoys; these shares are called the student’s “RIASEC scores”. Similarly, for each job, O*NET provides RIASEC scores. Then, for each student, we identify the five jobs whose RIASEC scores are most strongly correlated with the student’s scores. These jobs are the best fit for the student in terms of the alignment between the types of tasks required by a job and the types of tasks that the student enjoys.

To classify which jobs are STEM-related, we rely on O*NET’s All STEM Occupations list.⁶¹ This list is expansive and includes some jobs that we do not think qualify as STEM. For instance, it categorizes all higher education teaching jobs as STEM, even those in non-STEM fields. Consequently, we modify the list by excluding all jobs with any of the following keywords in their title: Religion, History, Sociology, Social Sciences, Literature, Geography, Political Science, Psychology, Cultural Studies, Archaeology, Anthropology, Art Therapy, Music Therapist, or Sales Representative.

O*NET does not provide a list of HSS-related occupations; thus, we code these manually. In particular, we include all non-STEM jobs that (a) are in the Arts, Entertainment, & Design or Education O*NET Career Clusters, (b) are in the Behavioral & Mental Health, Community & Social Services, or Judicial Systems sub-clusters, or (c) contain any of the keywords mentioned in the previous paragraph except for Sales Representative.⁶² In addition, we exclude jobs in the Fashion & Interiors sub-cluster as well as those with the keywords Attendant, Data Entry, Occupational Therapist, Fundraising Manager, Printing Press Operator, Technician, Design, Sales, Fundraising, Business Manager, Athletics, Sports, or Scouts.

All jobs that do not fall into the STEM- or HSS-related categories get classified as Other.

61. <https://www.onetonline.org/find/stem?t=0>.

62. <https://www.onetonline.org/find/career?c=0>.

A10.2 Tables and figures

Table A37: Heterogeneity for effects on beliefs about own abilities and preferences:
high school subjects

	All	Prefer		Stronger in		Gender		Achievement	
		STEM	HSS	Math	Language	Male	Female	Low	High
<i>Panel A: Beliefs about own high school STEM abilities at the end of high school</i>									
STEM	0.873*** (0.077)	0.789*** (0.106)	0.956*** (0.092)	0.853*** (0.123)	0.916*** (0.105)	0.765*** (0.127)	0.912*** (0.093)	0.876*** (0.125)	0.854*** (0.110)
Intercept	2.60	2.85	2.36	2.85	2.41	2.75	2.51	2.38	2.87
Student-cutoffs	3,987	1,925	2,062	1,921	1,985	1,396	2,510	2,093	1,894
<i>Panel B: Beliefs about own high school HSS abilities at the end of high school</i>									
STEM	-0.400*** (0.056)	-0.439*** (0.073)	-0.360*** (0.073)	-0.505*** (0.078)	-0.322*** (0.077)	-0.395*** (0.103)	-0.382*** (0.069)	-0.238*** (0.079)	-0.531*** (0.081)
Intercept	4.03	4.02	4.03	3.96	4.07	3.93	4.08	3.99	4.05
Student-cutoffs	3,987	1,925	2,062	1,921	1,985	1,396	2,510	2,093	1,894
<i>Panel C: Beliefs about own high school STEM abilities one year after high school</i>									
STEM	0.744*** (0.124)	0.718*** (0.160)	0.770*** (0.159)	0.797*** (0.212)	0.737*** (0.171)	0.823*** (0.213)	0.656*** (0.161)	0.668** (0.259)	0.693*** (0.168)
Intercept	2.96	3.16	2.76	3.17	2.72	3.01	2.97	2.77	3.17
Student-cutoffs	1,159	589	570	546	520	325	746	547	612
<i>Panel D: Beliefs about own high school HSS abilities one year after high school</i>									
STEM	-0.506*** (0.090)	-0.494*** (0.123)	-0.518*** (0.117)	-0.574*** (0.167)	-0.602*** (0.120)	-0.287 (0.207)	-0.548*** (0.108)	-0.378** (0.176)	-0.575*** (0.129)
Intercept	4.13	4.02	4.24	4.00	4.33	3.90	4.18	4.14	4.08
Student-cutoffs	1,159	589	570	546	520	325	746	547	612
<i>Panel E: Preferences for STEM high school subjects one year after high school</i>									
STEM	1.01*** (0.151)	0.850*** (0.184)	1.17*** (0.208)	1.03*** (0.275)	1.14*** (0.192)	1.14*** (0.290)	0.952*** (0.191)	1.07*** (0.287)	1.04*** (0.208)
Intercept	2.72	3.00	2.43	3.01	2.39	2.78	2.69	2.41	2.83
Student-cutoffs	1,159	589	570	546	520	325	746	547	612
<i>Panel F: Preferences for HSS high school subjects one year after high school</i>									
STEM	-0.551*** (0.085)	-0.525*** (0.119)	-0.576*** (0.108)	-0.772*** (0.153)	-0.537*** (0.129)	-0.422** (0.177)	-0.539*** (0.113)	-0.459*** (0.145)	-0.660*** (0.113)
Intercept	4.11	3.91	4.31	4.03	4.29	3.84	4.16	4.20	4.04
Student-cutoffs	1,159	589	570	546	520	325	746	547	612

The table shows heterogeneity for effects on some of the outcomes in Table 7. The “HSS” outcomes are the average of the outcomes in the second and third columns of Table 7. See the notes to Table A6 for definitions of the column headings.

Table A38: Effects on beliefs about own abilities in high school subjects:
students from the first follow-up survey

	STEM subjects	Humanities subjects	Social studies subjects
<i>Panel A: Beliefs at the end of high school</i>			
STEM	0.966*** (0.139)	-0.038 (0.116)	-0.409*** (0.140)
Intercept	2.65	4.00	3.84
Std. dev.	1.28	0.84	1.16
<i>Panel B: Beliefs one year after high school</i>			
STEM	0.744*** (0.124)	-0.267*** (0.087)	-0.745*** (0.139)
Intercept	2.96	4.20	4.06
Std. dev.	1.10	0.84	1.13
<i>Panel C: Change in beliefs in the year after high school</i>			
STEM	-0.222 (0.140)	-0.229** (0.105)	-0.336** (0.141)
Intercept	0.31	0.20	0.22
Std. dev.	1.10	0.94	1.13
Student-cutoffs	1,159	1,159	1,159

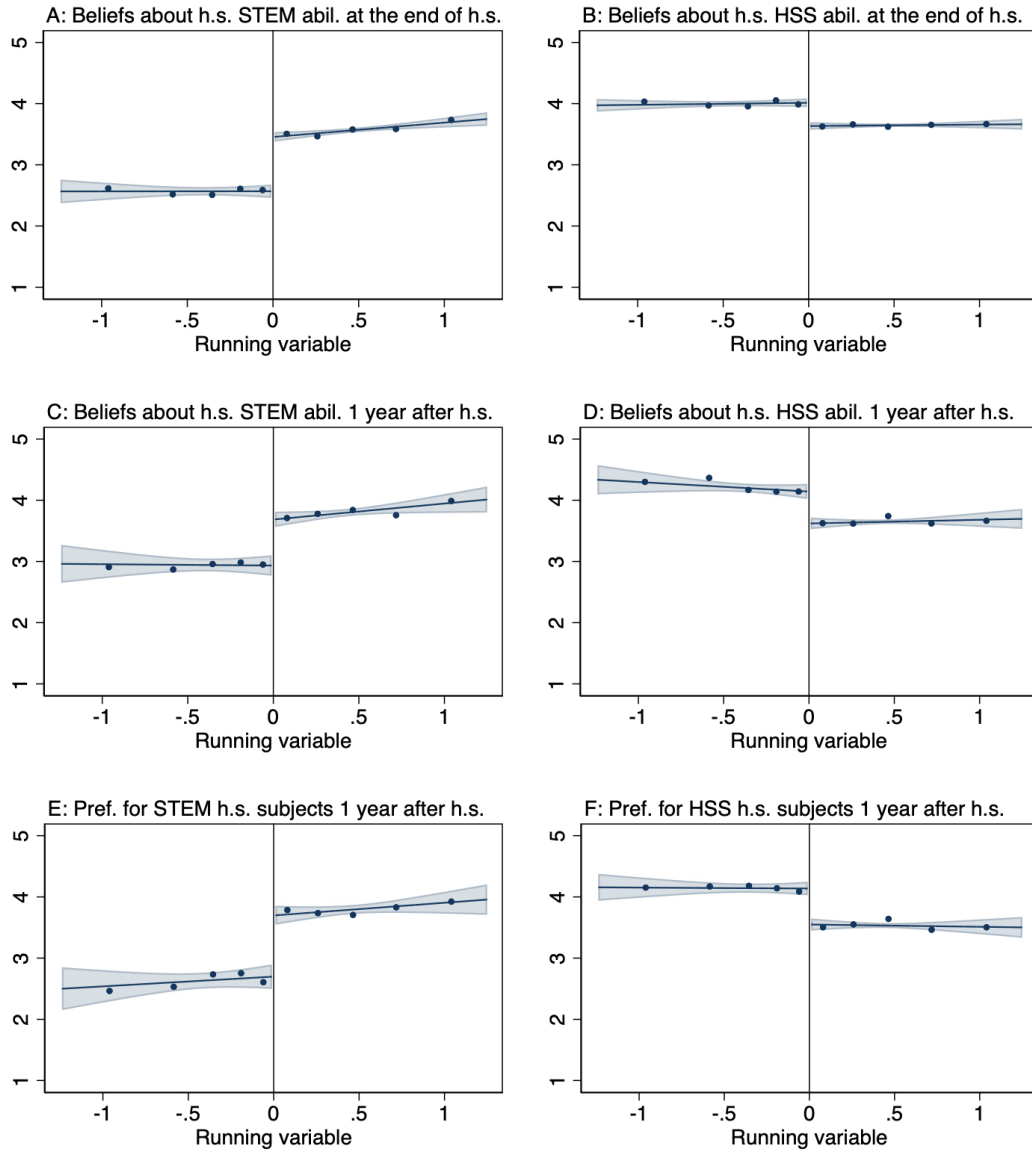
The table is analogous to Panels A and B of Table 7, but restricted to students from the first follow-up survey. The outcomes in Panel C are the difference between those in Panels B and A. The sample is the same across panels, so it is presented in a single row at the bottom of the table.

Table A39: Robustness for effects on beliefs about own abilities and preferences:
high school subjects

	Main	Adding covariates	Optimal bandwidths		Alternative fixed bandwidths				Global quartic polynomial
			By cutoff	Mean	0.75	1.00	1.50	1.75	
<i>Panel A: Beliefs about own high school STEM abilities at the end of high school</i>									
STEM	0.873*** (0.077)	0.870*** (0.075)	0.864*** (0.077)	0.864*** (0.079)	0.832*** (0.091)	0.868*** (0.079)	0.866*** (0.073)	0.878*** (0.071)	0.866*** (0.073)
Student-cutoffs	3,987	3,987	3,363	3,631	2,993	3,573	4,249	4,415	4,822
<i>Panel B: Beliefs about own high school HSS abilities at the end of high school</i>									
STEM	-0.400*** (0.056)	-0.391*** (0.056)	-0.414*** (0.057)	-0.388*** (0.058)	-0.397*** (0.063)	-0.381*** (0.058)	-0.388*** (0.053)	-0.374*** (0.051)	-0.383*** (0.054)
Student-cutoffs	3,987	3,987	3,349	3,609	2,993	3,573	4,249	4,415	4,822
<i>Panel C: Beliefs about own high school STEM abilities one year after high school</i>									
STEM	0.744*** (0.124)	0.745*** (0.122)	0.696*** (0.119)	0.739*** (0.124)	0.793*** (0.138)	0.701*** (0.130)	0.725*** (0.117)	0.736*** (0.111)	0.700*** (0.123)
Student-cutoffs	1,159	1,159	1,122	1,147	886	1,041	1,263	1,319	1,389
<i>Panel D: Beliefs about own high school HSS abilities one year after high school</i>									
STEM	-0.506*** (0.090)	-0.495*** (0.094)	-0.525*** (0.090)	-0.501*** (0.090)	-0.454*** (0.116)	-0.407*** (0.102)	-0.500*** (0.086)	-0.488*** (0.081)	-0.511*** (0.097)
Student-cutoffs	1,159	1,159	1,122	1,140	886	1,041	1,263	1,319	1,389
<i>Panel E: Preferences for STEM high school subjects one year after high school</i>									
STEM	1.01*** (0.151)	1.01*** (0.150)	0.973*** (0.144)	1.00*** (0.154)	1.13*** (0.169)	0.949*** (0.164)	1.01*** (0.144)	1.01*** (0.139)	1.00*** (0.151)
Student-cutoffs	1,159	1,159	1,118	1,140	886	1,041	1,263	1,319	1,389
<i>Panel F: Preferences for HSS high school subjects one year after high school</i>									
STEM	-0.551*** (0.085)	-0.537*** (0.088)	-0.580*** (0.086)	-0.553*** (0.086)	-0.553*** (0.104)	-0.454*** (0.094)	-0.530*** (0.083)	-0.524*** (0.079)	-0.535*** (0.094)
Student-cutoffs	1,159	1,159	1,112	1,140	886	1,041	1,263	1,319	1,389

The table shows robustness for the effects in Table A37. Specifications match those in Table A18. The one exception is that the added covariates in “Adding covariates” now also include mother’s and father’s years of schooling. Results for specifications that use alternative fixed effects are in Appendix A14.

Figure A11: RD plots of effects on beliefs about own abilities and preferences:
high school subjects



The figure displays RD plots for effects on the outcomes in Table A37. See the notes to Figure A3 for details on RD plots.

Table A40: Heterogeneity for effects on beliefs about own abilities and preferences:
college subjects

	All	Prefer		Stronger in		Gender		Achievement	
		STEM	HSS	Math	Language	Male	Female	Low	High
<i>Panel A: Beliefs about own college STEM abilities one year after high school</i>									
STEM	0.659*** (0.151)	0.622*** (0.197)	0.696*** (0.203)	0.773*** (0.211)	0.554** (0.223)	0.725** (0.315)	0.579*** (0.208)	0.263 (0.269)	0.895*** (0.194)
Intercept	2.57	2.80	2.34	2.74	2.36	2.53	2.63	2.52	2.69
<i>Panel B: Beliefs about own college non-STEM abilities one year after high school</i>									
STEM	-0.337** (0.143)	-0.409** (0.201)	-0.265 (0.184)	-0.329 (0.212)	-0.477** (0.209)	-0.276 (0.302)	-0.500*** (0.186)	-0.578** (0.250)	-0.316* (0.180)
Intercept	3.20	3.35	3.05	3.21	3.22	2.94	3.40	3.30	3.36
<i>Panel C: Preferences for STEM college subjects one year after high school</i>									
STEM	0.776*** (0.149)	0.723*** (0.194)	0.829*** (0.196)	0.856*** (0.220)	0.748*** (0.226)	1.05*** (0.330)	0.614*** (0.203)	0.396 (0.268)	0.914*** (0.182)
Intercept	2.48	2.72	2.25	2.73	2.22	2.25	2.60	2.44	2.66
<i>Panel D: Preferences for non-STEM college subjects one year after high school</i>									
STEM	-0.449*** (0.150)	-0.495** (0.221)	-0.403** (0.178)	-0.433* (0.227)	-0.484** (0.209)	-0.292 (0.308)	-0.620*** (0.176)	-0.711*** (0.248)	-0.395* (0.208)
Intercept	3.15	3.28	3.02	3.10	3.11	2.79	3.37	3.29	3.27
Student-cutoffs	1,159	589	570	546	520	325	746	547	612

The table shows heterogeneity for effects on some of the outcomes in Table 8. The “non-STEM” outcomes are the average of the outcomes in the second and third columns of Table 8. The sample is the same for all outcomes, so it is presented in a single row at the bottom of the table. See the notes to Table A6 for definitions of the column headings.

Table A41: Effects on beliefs about own abilities in and preferences for college subjects
with no imputations for missing responses

	STEM subjects	Subjects that involve reading and writing	Subjects that involve memorization
<i>Panel A: Beliefs one year after high school</i>			
STEM	0.901*** (0.152)	-0.409*** (0.124)	-0.280* (0.156)
Intercept	2.85	3.64	3.59
<i>Panel B: Preferences one year after high school</i>			
STEM	1.04*** (0.136)	-0.584*** (0.145)	-0.408** (0.157)
Intercept	2.75	3.66	3.47
Student-cutoffs	965	965	965

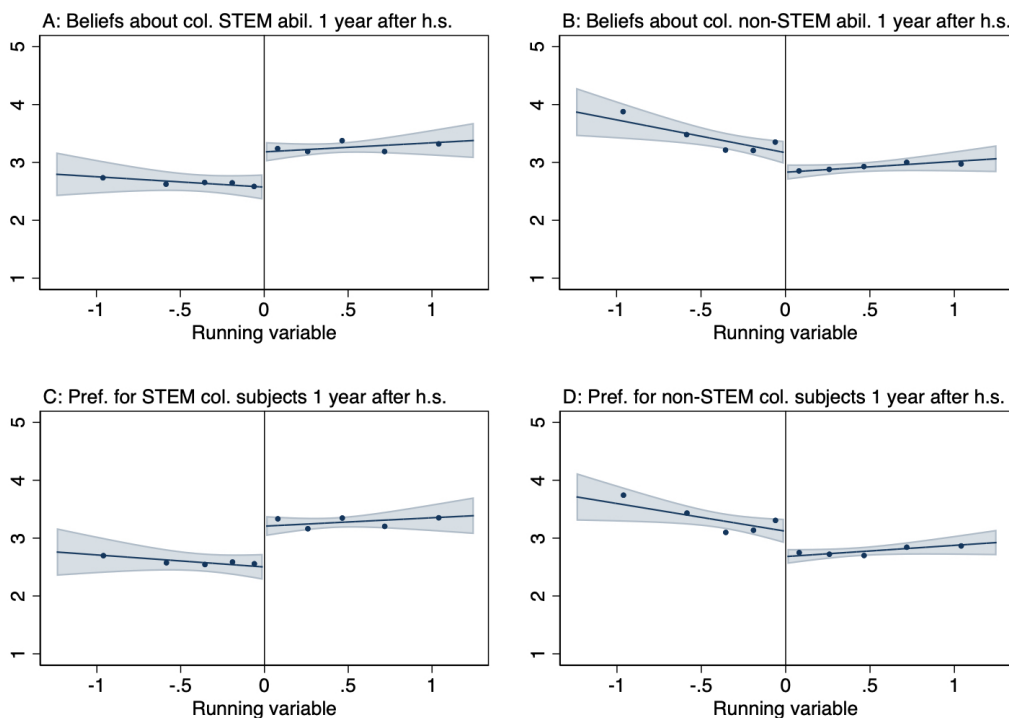
The table is analogous to Table 8, but it does not impute values of 1 for students who had never attended college by the time of the first follow-up survey. Instead, it drops these students.

Table A42: Robustness for effects on beliefs about own abilities and preferences: college subjects

	Main	Adding covariates	Optimal bandwidths		Alternative fixed bandwidths				Global quartic polynomial
			By cutoff	Mean	0.75	1.00	1.50	1.75	
<i>Panel A: Beliefs about own college STEM abilities one year after high school</i>									
STEM	0.659*** (0.151)	0.654*** (0.148)	0.618*** (0.152)	0.648*** (0.155)	0.580*** (0.183)	0.542*** (0.157)	0.631*** (0.144)	0.640*** (0.136)	0.587*** (0.153)
<i>Panel B: Beliefs about own college non-STEM abilities one year after high school</i>									
STEM	-0.337** (0.143)	-0.326** (0.144)	-0.374** (0.159)	-0.309** (0.143)	-0.399** (0.155)	-0.276* (0.145)	-0.367*** (0.139)	-0.339** (0.138)	-0.355** (0.141)
<i>Panel C: Preferences for STEM college subjects one year after high school</i>									
STEM	0.776*** (0.149)	0.769*** (0.148)	0.769*** (0.152)	0.778*** (0.151)	0.736*** (0.179)	0.679*** (0.152)	0.765*** (0.144)	0.760*** (0.136)	0.723*** (0.151)
<i>Panel D: Preferences for non-STEM college subjects one year after high school</i>									
STEM	-0.449*** (0.150)	-0.439*** (0.148)	-0.491*** (0.168)	-0.409*** (0.149)	-0.455*** (0.164)	-0.335** (0.155)	-0.454*** (0.145)	-0.426*** (0.142)	-0.459*** (0.146)
Student-cutoffs	1,159	1,159	1,098	1,134	886	1,041	1,263	1,319	1,389

The table shows robustness for the effects in Table A40. Specifications match those in Table A18. The one exception is that the added covariates in “Adding covariates” now also include mother’s and father’s years of schooling. The sample is the same for all outcomes, so it is presented in a single row at the bottom of the table. Results for specifications that use alternative fixed effects are in Appendix A14.

Figure A12: RD plots of effects on beliefs about own abilities and preferences: college subjects



The figure displays RD plots for effects on the outcomes in Table A40. See the notes to Figure A3 for details on RD plots.

Table A43: Effects on outcomes related to skill acquisition

	All	Prefer		Stronger in		Gender		Achievement	
		STEM	HSS	Math	Language	Male	Female	Low	High
<i>Panel A: Score on the Romanian language and literature part of the baccalaureate exam</i>									
STEM	-0.315*** (0.073)	-0.260*** (0.098)	-0.370*** (0.097)	-0.303*** (0.106)	-0.377*** (0.104)	-0.338*** (0.111)	-0.276*** (0.082)	-0.358*** (0.113)	-0.217* (0.116)
Intercept	7.92	8.12	7.73	7.76	8.01	7.54	8.18	7.50	8.34
Std. dev.	1.28	1.14	1.34	1.30	1.25	1.30	1.21	1.33	1.03
Student-cutoffs	3,951	1,920	2,031	1,905	1,965	1,384	2,488	2,060	1,891
<i>Panel B: Average score on the non-Romanian parts of the baccalaureate exam</i>									
STEM	-0.528*** (0.066)	-0.559*** (0.084)	-0.496*** (0.088)	-0.555*** (0.099)	-0.515*** (0.095)	-0.558*** (0.100)	-0.495*** (0.084)	-0.545*** (0.128)	-0.456*** (0.085)
Intercept	8.43	8.64	8.23	8.53	8.29	8.31	8.52	7.95	8.89
Std. dev.	1.28	1.09	1.38	1.18	1.33	1.28	1.27	1.38	0.96
Student-cutoffs	3,930	1,915	2,015	1,897	1,951	1,377	2,474	2,039	1,891
<i>Panel C: Prefer books and movies with exciting plots to those with rich characters</i>									
STEM	-0.071 (0.075)	-0.128 (0.106)	-0.014 (0.083)	-0.131 (0.121)	0.043 (0.116)	-0.036 (0.133)	-0.056 (0.103)	-0.121 (0.121)	-0.107 (0.111)
Intercept	3.37	3.38	3.37	3.45	3.26	3.39	3.32	3.36	3.45
Std. dev.	1.18	1.16	1.21	1.17	1.20	1.16	1.19	1.21	1.15
Student-cutoffs	3,987	1,925	2,062	1,921	1,985	1,396	2,510	2,093	1,894

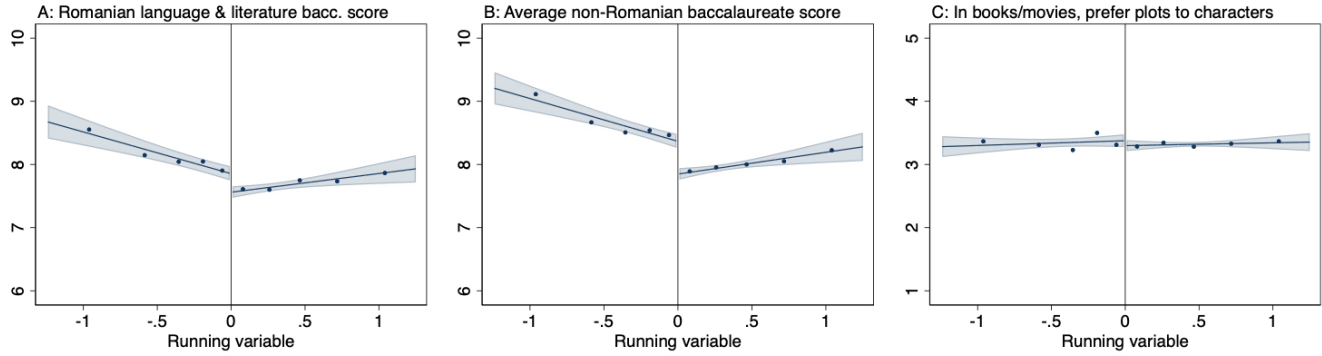
The table presents results for the effects of STEM assignment on three outcomes related to skill acquisition: a student's score on the Romanian language and literature part of the baccalaureate exam, the student's average score on the other parts of the baccalaureate exam, and the student's preferences over books and movies. The first two outcomes are from the administrative data, but we limit the sample to students from the end-of-high-school survey in order to allow comparability with the results on students' beliefs about their academic abilities; these outcomes are missing for students who did not complete the relevant exam components. The third outcome is based on a question where students were asked how much they agree (on a scale of 1-5) with the statement: "I prefer books and movies with exciting plots to those that focus on the feelings and motivations of the characters". The question was asked in both the end-of-high-school and second follow-up surveys. The outcome uses data from both surveys. Specifically, as in Table A32, it is equal to the student fixed effects from a panel regression of the responses on student and survey fixed effects, centered at the mean response for students with non-missing responses in both surveys. See the notes to Table A6 for definitions of the column headings.

Table A44: Robustness for effects on outcomes related to skill acquisition

	Main	Adding covariates	Optimal bandwidths		Alternative fixed bandwidths				Global quartic polynomial
			By cutoff	Mean	0.75	1.00	1.50	1.75	
<i>Panel A: Score on the Romanian language and literature part of the baccalaureate exam</i>									
STEM	-0.315*** (0.073)	-0.275*** (0.066)	-0.331*** (0.075)	-0.328*** (0.077)	-0.372*** (0.086)	-0.317*** (0.079)	-0.314*** (0.071)	-0.307*** (0.069)	-0.317*** (0.070)
Student-cutoffs	3,951	3,951	3,345	3,599	2,969	3,542	4,213	4,374	4,773
<i>Panel B: Average score on the non-Romanian parts of the baccalaureate exam</i>									
STEM	-0.528*** (0.066)	-0.523*** (0.065)	-0.536*** (0.069)	-0.524*** (0.068)	-0.583*** (0.079)	-0.504*** (0.068)	-0.513*** (0.064)	-0.519*** (0.062)	-0.551*** (0.064)
Student-cutoffs	3,930	3,930	3,273	3,581	2,956	3,524	4,191	4,350	4,744
<i>Panel C: Prefer books and movies with exciting plots to those with rich characters</i>									
STEM	-0.071 (0.075)	-0.089 (0.075)	-0.059 (0.081)	-0.048 (0.079)	-0.075 (0.090)	-0.050 (0.079)	-0.048 (0.073)	-0.059 (0.071)	-0.085 (0.075)
Student-cutoffs	3,987	3,987	3,380	3,631	2,993	3,573	4,249	4,415	4,822

The table shows robustness for the effects in Table A43. Specifications match those in Table A18. The one exception is that the added covariates in "Adding covariates" now also include mother's and father's years of schooling.

Figure A13: RD plots of effects on outcomes related to skill acquisition



The figure displays RD plots for effects on the outcomes in Table A43. See the notes to Figure A3 for details on RD plots.

Table A45: Effects on enjoyment of specific O*NET job tasks

Outcome	Coefficient estimate	Standard error	Intercept	Student-cutoffs
<i>Panel A: STEM-related job tasks</i>				
Develop or code a computer/software program	0.141**	0.062	0.376	1,199
Install software across computers on a large network	0.138**	0.063	0.383	1,199
Operate a calculator	0.063	0.051	0.820	1,199
Develop a new medicine	0.053	0.052	0.483	1,199
Conduct chemical experiments	0.042	0.056	0.496	1,199
Examine blood samples using a microscope	0.090	0.059	0.416	1,199
Assemble electronic parts	0.153***	0.056	0.319	1,199
Manage a department within a large company	0.124***	0.047	0.691	1,199
Keep shipping and receiving records	0.058	0.058	0.341	1,199
<i>Panel B: HSS-related job tasks</i>				
Write books or plays	-0.191***	0.073	0.558	1,199
Write scripts for movies or television shows	-0.084	0.068	0.526	1,199
Help people with personal or emotional problems	-0.108**	0.044	0.901	1,199
Do volunteer work at a non-profit organization	-0.067	0.048	0.837	1,199
Teach a high school class	-0.005	0.062	0.525	1,199
<i>Panel C: Other job tasks</i>				
Repair household appliances	0.019	0.055	0.311	1,199
Drive a truck to deliver packages to offices and homes	-0.031	0.053	0.294	1,199
Start your own business	-0.001	0.040	0.889	1,199
Negotiate business contracts	-0.021	0.050	0.782	1,199
Compose or arrange music	0.083	0.058	0.452	1,199

The table expands on Table 9 by showing effects on whether a student enjoys each job task in the O*NET questionnaire.

Table A46: Heterogeneity for effects on O*NET outcomes

	All	Prefer		Stronger in		Gender		Achievement	
		STEM	HSS	Math	Language	Male	Female	Low	High
<i>Panel A: Share of STEM-related job tasks that enjoy</i>									
STEM	0.096*** (0.024)	0.102*** (0.033)	0.090*** (0.032)	0.173*** (0.041)	0.052 (0.037)	0.098* (0.050)	0.085** (0.033)	0.079** (0.036)	0.093** (0.037)
Intercept	0.48	0.48	0.48	0.45	0.51	0.55	0.46	0.50	0.48
<i>Panel B: Share of HSS-related job tasks that enjoy</i>									
STEM	-0.091** (0.041)	-0.090* (0.052)	-0.092 (0.059)	-0.030 (0.066)	-0.156*** (0.051)	-0.030 (0.061)	-0.087* (0.045)	-0.158** (0.065)	-0.101* (0.053)
Intercept	0.67	0.65	0.69	0.59	0.72	0.53	0.73	0.74	0.67
<i>Panel C: Share of Other job tasks that enjoy</i>									
STEM	0.010 (0.028)	-0.029 (0.042)	0.048 (0.034)	0.061 (0.054)	0.004 (0.037)	0.024 (0.051)	-0.009 (0.032)	0.009 (0.048)	0.004 (0.039)
Intercept	0.55	0.56	0.53	0.57	0.52	0.60	0.52	0.56	0.56
<i>Panel D: Difference in the share of STEM- vs. HSS-related job tasks that enjoy</i>									
STEM	0.187*** (0.046)	0.192*** (0.065)	0.181*** (0.059)	0.203*** (0.069)	0.208*** (0.055)	0.127* (0.075)	0.172*** (0.053)	0.237*** (0.081)	0.194*** (0.056)
Intercept	-0.19	-0.17	-0.21	-0.14	-0.21	0.03	-0.27	-0.24	-0.18
<i>Panel E: Share of O*NET-recommended jobs that are STEM-related</i>									
STEM	0.105*** (0.032)	0.159*** (0.052)	0.050 (0.034)	0.094* (0.051)	0.176*** (0.062)	0.024 (0.073)	0.181*** (0.047)	0.191** (0.089)	0.077* (0.039)
Intercept	0.14	0.13	0.14	0.13	0.12	0.23	0.08	0.03	0.16
<i>Panel F: Share of O*NET-recommended jobs that are HSS-related</i>									
STEM	-0.097** (0.048)	-0.111 (0.073)	-0.084 (0.057)	-0.040 (0.068)	-0.190*** (0.060)	-0.048 (0.086)	-0.113** (0.055)	-0.144* (0.082)	-0.100 (0.060)
Intercept	0.43	0.44	0.42	0.32	0.50	0.29	0.51	0.51	0.40
<i>Panel G: Share of O*NET-recommended jobs that are Other</i>									
STEM	-0.007 (0.044)	-0.048 (0.062)	0.034 (0.055)	-0.053 (0.072)	0.014 (0.056)	0.023 (0.082)	-0.068 (0.055)	-0.047 (0.082)	0.023 (0.056)
Intercept	0.43	0.42	0.44	0.55	0.37	0.48	0.41	0.46	0.44
Student-cutoffs	1,199	575	624	540	550	373	728	618	581

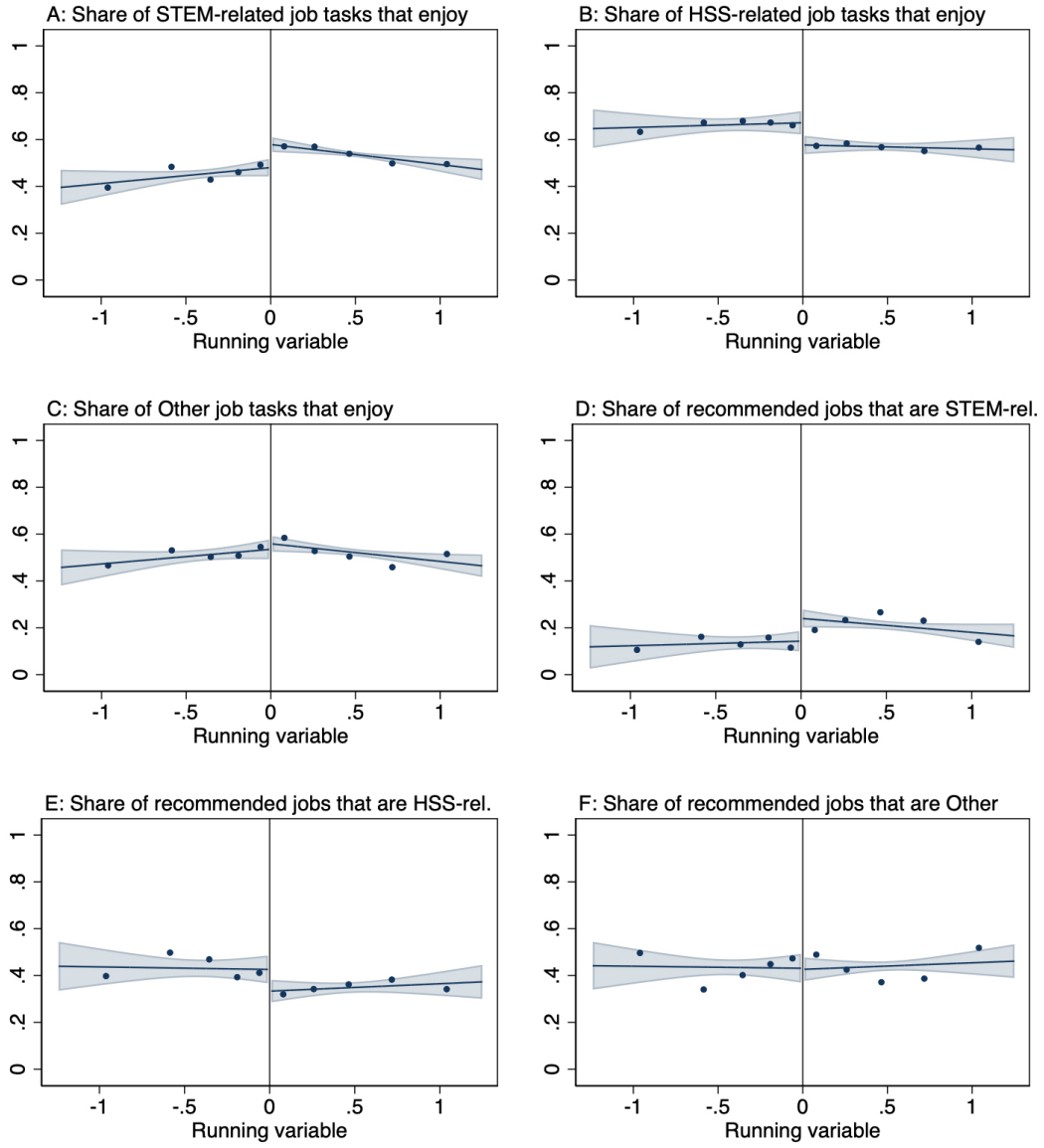
The table shows heterogeneity for the effects in Table 9. The outcome in Panel D is the difference between the outcomes in Panels A and B. The sample is the same for all outcomes, so it is presented in a single row at the bottom of the table. See the notes to Table A6 for definitions of the column headings.

Table A47: Robustness for effects on O*NET outcomes

	Main	Adding covariates	Optimal bandwidths		Alternative fixed bandwidths				Global quartic polynomial
			By cutoff	Mean	0.75	1.00	1.50	1.75	
<i>Panel A: Share of STEM-related job tasks that enjoy</i>									
STEM	0.096*** (0.024)	0.089*** (0.023)	0.095*** (0.026)	0.095*** (0.024)	0.110*** (0.028)	0.096*** (0.026)	0.085*** (0.023)	0.087*** (0.023)	0.086*** (0.026)
<i>Panel B: Share of HSS-related job tasks that enjoy</i>									
STEM	-0.091** (0.041)	-0.081** (0.038)	-0.089** (0.042)	-0.097** (0.042)	-0.081 (0.050)	-0.087* (0.046)	-0.089** (0.039)	-0.078** (0.038)	-0.083* (0.043)
<i>Panel C: Share of Other job tasks that enjoy</i>									
STEM	0.010 (0.028)	0.001 (0.027)	0.012 (0.031)	0.006 (0.030)	0.051 (0.033)	0.029 (0.031)	0.004 (0.027)	-0.004 (0.026)	0.019 (0.031)
<i>Panel D: Share of O*NET-recommended jobs that are STEM-related</i>									
STEM	0.105*** (0.032)	0.103*** (0.033)	0.112*** (0.033)	0.106*** (0.033)	0.077* (0.042)	0.085** (0.036)	0.096*** (0.032)	0.096*** (0.031)	0.086** (0.035)
<i>Panel E: Share of O*NET-recommended jobs that are HSS-related</i>									
STEM	-0.097** (0.048)	-0.087* (0.045)	-0.096* (0.049)	-0.097* (0.050)	-0.075 (0.056)	-0.088* (0.050)	-0.092* (0.048)	-0.083* (0.045)	-0.082 (0.051)
<i>Panel F: Share of O*NET-recommended jobs that are Other</i>									
STEM	-0.007 (0.044)	-0.016 (0.042)	-0.018 (0.045)	-0.007 (0.044)	-0.002 (0.052)	0.003 (0.046)	-0.004 (0.042)	-0.014 (0.040)	-0.003 (0.047)
Student-cutoffs	1,199	1,199	1,140	1,167	883	1,068	1,271	1,318	1,432

The table shows robustness for the effects in Table 9. Specifications match those in Table A18. The one exception is that the added covariates in “Adding covariates” now also include mother’s and father’s years of schooling. The sample is the same for all outcomes, so it is presented in a single row at the bottom of the table. Results for specifications that use alternative fixed effects are in Appendix A14.

Figure A14: RD plots of effects on O*NET outcomes



The figure displays RD plots for effects on the outcomes in Table 9. See the notes to Figure A3 for details on RD plots.

Table A48: Heterogeneity for effects on choices and beliefs regarding representative jobs

	All	Prefer		Stronger in		Gender		Achievement	
		STEM	HSS	Math	Language	Male	Female	Low	High
<i>Panel A: Would choose to be a technology or engineering worker</i>									
STEM	0.146*** (0.047)	0.222*** (0.066)	0.071 (0.056)	0.081 (0.089)	0.238*** (0.066)	0.212** (0.096)	0.085 (0.061)	0.235*** (0.082)	0.130* (0.066)
Intercept	0.20	0.18	0.23	0.29	0.12	0.23	0.22	0.12	0.21
<i>Panel B: Difference in beliefs about pay</i>									
STEM	-0.006 (0.097)	0.029 (0.123)	-0.041 (0.125)	-0.073 (0.209)	0.082 (0.100)	-0.227 (0.217)	0.100 (0.110)	0.215 (0.155)	-0.046 (0.163)
Intercept	1.00	1.08	0.92	0.99	0.96	1.18	0.93	0.67	1.10
<i>Panel C: Difference in beliefs about enjoying the work content</i>									
STEM	0.736*** (0.211)	0.663** (0.323)	0.808*** (0.246)	0.665** (0.299)	0.959*** (0.259)	0.756* (0.386)	0.536** (0.233)	0.642* (0.354)	0.826*** (0.309)
Intercept	-0.64	-0.44	-0.85	-0.27	-0.82	-0.05	-0.77	-0.50	-0.70
<i>Panel D: Difference in beliefs about coworkers and work conditions</i>									
STEM	0.251 (0.187)	0.265 (0.298)	0.237 (0.190)	-0.183 (0.290)	0.624*** (0.214)	0.262 (0.346)	0.101 (0.227)	0.659** (0.328)	0.204 (0.281)
Intercept	-0.32	-0.16	-0.48	0.15	-0.67	0.10	-0.39	-0.68	-0.24
<i>Panel E: Difference in beliefs about own preparation</i>									
STEM	0.896*** (0.232)	0.931** (0.360)	0.861*** (0.255)	0.831** (0.393)	1.23*** (0.286)	1.05*** (0.383)	0.639** (0.277)	1.01*** (0.367)	1.07*** (0.317)
Intercept	-0.73	-0.54	-0.92	-0.33	-1.12	-0.17	-0.97	-0.85	-0.78
<i>Panel F: Difference in beliefs about approval of friends and family</i>									
STEM	0.304** (0.120)	0.221 (0.161)	0.387** (0.153)	0.540** (0.232)	0.341** (0.139)	0.410 (0.310)	0.174 (0.127)	0.004 (0.210)	0.559*** (0.182)
Intercept	0.05	0.18	-0.09	0.03	0.04	0.26	-0.03	0.29	-0.04
Student-cutoffs	1,199	575	624	540	550	373	728	618	581

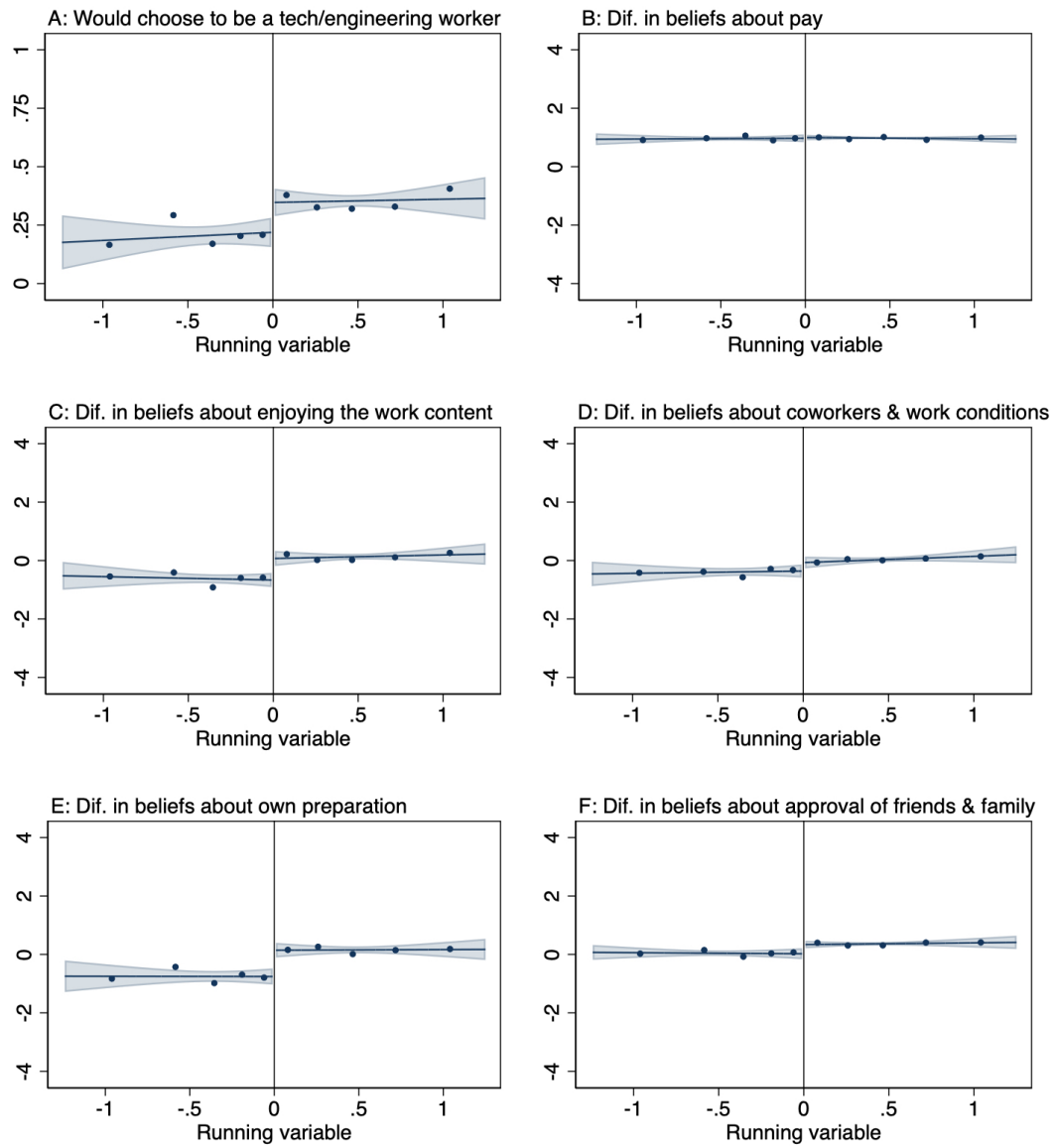
The table shows heterogeneity for effects on outcomes related to those in Table 10. “Difference in beliefs” is the difference between a student’s beliefs about being a technology or engineering worker and the average of their beliefs about being a humanities teacher or a person who runs a small business. The sample is the same for all outcomes, so it is presented in a single row at the bottom of the table. See the notes to Table A6 for definitions of the column headings.

Table A49: Robustness for effects on choices and beliefs regarding representative jobs

	Main	Adding covariates	Optimal bandwidths		Alternative fixed bandwidths				Global quartic polynomial
			By cutoff	Mean	0.75	1.00	1.50	1.75	
<i>Panel A: Would choose to be a technology or engineering worker</i>									
STEM	0.146*** (0.047)	0.139*** (0.046)	0.155*** (0.047)	0.147*** (0.048)	0.193*** (0.056)	0.149*** (0.051)	0.139*** (0.046)	0.114** (0.046)	0.135*** (0.050)
<i>Panel B: Difference in beliefs about pay</i>									
STEM	-0.006 (0.097)	-0.000 (0.098)	0.002 (0.097)	0.009 (0.098)	0.048 (0.124)	0.023 (0.110)	0.010 (0.094)	0.007 (0.087)	0.079 (0.101)
<i>Panel C: Difference in beliefs about enjoying the work content</i>									
STEM	0.736*** (0.211)	0.684*** (0.194)	0.767*** (0.218)	0.745*** (0.213)	0.869*** (0.240)	0.752*** (0.226)	0.648*** (0.205)	0.551*** (0.197)	0.641*** (0.217)
<i>Panel D: Difference in beliefs about coworkers and work conditions</i>									
STEM	0.251 (0.187)	0.216 (0.175)	0.248 (0.190)	0.264 (0.189)	0.257 (0.217)	0.197 (0.196)	0.225 (0.178)	0.212 (0.169)	0.229 (0.188)
<i>Panel E: Difference in beliefs about own preparation</i>									
STEM	0.896*** (0.232)	0.847*** (0.211)	0.902*** (0.227)	0.899*** (0.233)	1.03*** (0.257)	0.822*** (0.241)	0.840*** (0.223)	0.811*** (0.212)	0.864*** (0.227)
<i>Panel F: Difference in beliefs about approval of friends and family</i>									
STEM	0.304** (0.120)	0.298** (0.121)	0.327*** (0.125)	0.304** (0.121)	0.341** (0.142)	0.276** (0.133)	0.249** (0.116)	0.229** (0.111)	0.256** (0.124)
Student-cutoffs	1,199	1,199	1,147	1,167	883	1,068	1,271	1,318	1,432

The table shows robustness for effects on the outcomes in Table A48. Specifications match those in Table A18. The one exception is that the added covariates in “Adding covariates” now also include mother’s and father’s years of schooling. The sample is the same for all outcomes, so it is presented in a single row at the bottom of the table. Results for specifications that use alternative fixed effects are in Appendix A14.

Figure A15: RD plots of effects on choices and beliefs regarding representative jobs



The figure displays RD plots for effects on the outcomes in Table A48. See the notes to Figure A3 for details on RD plots.

A11 The factors that students weigh in their decisions

This appendix provides additional analysis of how students make choices regarding college fields of study and careers.

In the first follow-up survey, we gave students a list of reasons that could explain their choice of college field of study, and we asked them to select yes (rather than no) if a reason was a factor they considered. In the second follow-up survey, we did the same but for what they expect will drive their choice of job at age 30. We convert the responses into indicator variables and run separate RD regressions for each of them. We use the RD regressions to (a) predict the share selecting yes at the RD threshold, both overall and for students assigned to STEM or HSS, and (b) estimate the ATE of STEM assignment, β , on the share selecting yes.

The results are in Tables A50 (for college field of study) and A51 (for job at age 30). The questions for college field of study were not asked to students who had never attended college by the time of the first follow-up survey. Table A50 excludes these students; we find that results are qualitatively similar if we instead include the dropped students by giving them a value of zero (“no”) for all outcomes.

The tables offer a few insights. First, almost all students care about whether an option matches their interests or abilities. The shares at the RD threshold who indicated these attributes were important for the choice of field of study are 96% and 97%, respectively. The shares who expect to care about these attributes for their future job choice are 95% and 97%. Second, considerably fewer students report being influenced by switching costs or by teachers and peers. 38% of observations at the RD threshold indicated that part of why they chose their college field of study was because it matched what they studied in high school and they believed that switching to a different area would be difficult. 46% report being influenced by encouragement from their high school teachers, while 22% chose a field because it is the same as what their friends chose. Similarly, 43% of observations at the RD threshold expect to choose a job because it is related to what they studied in high school and changing would be difficult. Meanwhile, 42% expect their job choice to be influenced by what their high school teachers encouraged them to do. Third, students care about a number of other attributes of a field of study or job, such as family approval, location, prestige, earnings, and coworkers and working conditions. Finally, being assigned to STEM vs. HSS generally has a modest effect on the attributes that students care about. A few exceptions are that STEM assignment makes students more likely to say that switching costs constrained their field choice and more likely to report that their job choice will be influenced by family approval, encouragement from high school teachers, work content, and earnings.

Table A50: The explanations students selected for why they chose their college field of study

Explanation	Share at the RD threshold selecting yes			The effect of STEM assignment	
	All	Assigned to		Coefficient estimate	Standard error
		STEM	HSS		
The subject matched my abilities	0.97	0.97	0.96	0.01	0.03
The subject matched my interests	0.96	0.96	0.96	-0.01	0.03
The subject would lead to a job with high earnings	0.92	0.94	0.89	0.05	0.04
The subject would lead to a job that I would be happy with	0.91	0.91	0.92	-0.01	0.04
My parents wanted me to study the subject	0.77	0.77	0.77	0.00	0.05
In high school, I learned about career paths related to the subject	0.58	0.57	0.58	0.00	0.06
My high school teachers encouraged me to study the subject	0.46	0.51	0.42	0.09	0.07
The subject matched what I studied in high school, and it was hard to change	0.38	0.47	0.29	0.18***	0.06
I thought the subject would be easy	0.25	0.24	0.27	-0.03	0.06
The subject is the same as what my friends chose	0.22	0.19	0.25	-0.05	0.05

The table provides information on the factors that students considered when choosing their college field of study. Each row is for a different RD regression. Outcomes are from the first follow-up survey and are indicators equal to 1 if a student selected yes rather than no when asked if a listed explanation was one of the factors they considered. The sample contains 965 student-cutoff observations; it is the RD sample for the first follow-up survey, but excluding students who had never attended college by the time of this survey. “Share at the RD threshold selecting yes” is the predicted value of an outcome at the RD threshold, either overall (“All”) or on the STEM or HSS sides of the threshold (“Assigned to STEM” and “Assigned to HSS”). “The effect of STEM assignment” is the coefficient estimate and standard error for the ATE of STEM assignment, β . “Assigned to HSS” is the same as the “Intercept” term in Panel A of Table 2. “Assigned to STEM” is the sum of “Assigned to HSS” and the β estimate. “All” is the simple average of “Assigned to STEM” and “Assigned to HSS”. Other details are as in Panel A of Table 2.

Table A51: The explanations that students expect will drive their choice of job at age 30

Explanation	Share at the RD threshold selecting yes			The effect of STEM assignment	
	All	Assigned to		Coefficient estimate	Standard error
		STEM	HSS		
The job has good coworkers and working conditions	0.98	0.97	0.98	-0.01	0.01
The job provides a high income	0.97	0.99	0.95	0.04**	0.02
The job matches my abilities and educational background	0.97	0.97	0.96	0.00	0.02
The content of the work involved in the job matches my interests	0.95	0.98	0.93	0.05*	0.02
The job would let me live close to where my friends and family are	0.86	0.87	0.86	0.02	0.05
The job is prestigious	0.78	0.78	0.78	0.00	0.05
My family would approve of me doing the job	0.67	0.76	0.58	0.18***	0.05
The job is easy	0.48	0.53	0.43	0.10	0.07
The job is related to what I studied in high school, and it is hard to change	0.43	0.45	0.42	0.03	0.06
My high school teachers encouraged me to do the job	0.42	0.48	0.36	0.12*	0.07

The table provides information on the factors that students expect to be important for determining which job they will choose at age 30. Each row is for a different RD regression. Outcomes are from the second follow-up survey and are indicators equal to 1 if a student selected yes rather than no when asked if a listed explanation is likely to be one of the factors that they will consider. The sample is the RD sample for the second follow-up survey (1,199 student-cutoff observations). Other details are the same as in Table A50.

A12 Additional results for wellbeing and school satisfaction

Table A52: Effects on components of the wellbeing index

	Index	Components		
		Overall, am happy	Belong at school	Not depressed
<i>Panel A: End of high school</i>				
STEM	0.133*** (0.051)	0.146** (0.072)	0.199** (0.077)	0.066 (0.055)
Intercept	-0.14	3.66	2.58	2.67
Std. dev.	0.74	1.05	1.12	0.78
Student-cutoffs	3,987	3,987	3,987	3,987
<i>Panel B: 1.5 years after high school</i>				
STEM	-0.033 (0.068)	-0.018 (0.095)	-0.131 (0.107)	0.027 (0.082)
Intercept	0.66	4.21	3.80	3.28
Std. dev.	0.56	0.77	0.90	0.63
Student-cutoffs	1,199	1,198	1,196	1,199

The table shows effects on components of the wellbeing index used in Table 11. The outcomes in Panel A (B) are from the end-of-high-school (second follow-up) survey. “Overall, am happy” and “Belong at school” are responses from questions where students were asked how much they agree with a statement on a scale of 1-5. The respective statements are: (i) “Overall, I am happy” and (ii) “I feel like I belong at school” (Panel A) or “I feel like I belong in my school, workplace, or community” (Panel B). “Not depressed” is on a scale of 1-4. It is the reverse of a student’s average score on the PHQ-4 Depression and Anxiety Questionnaire (Kroenke et al. 2009). In each panel, the wellbeing index for a student is the average of standardized versions of the student’s values of the components. The average ignores any components that are missing for the student. To enhance comparability, the components are always standardized using the mean and standard deviation from the end-of-high-school survey, calculated over students with non-missing end-of-high-school values for each component. Other details are as in Panel A of Table 2.

Table A53: Heterogeneity for effects on wellbeing and high school satisfaction at the end of high school

	All	Prefer		Stronger in		Gender		Achievement	
		STEM	HSS	Math	Language	Male	Female	Low	High
<i>Panel A: Wellbeing</i>									
STEM	0.133*** (0.051)	0.111 (0.077)	0.155*** (0.053)	0.192** (0.077)	0.081 (0.077)	0.055 (0.088)	0.142** (0.068)	0.094 (0.085)	0.114 (0.077)
Intercept	-0.14	-0.14	-0.13	-0.15	-0.09	-0.00	-0.20	-0.06	-0.16
<i>Panel B: Liked the high school experience</i>									
STEM	0.113* (0.068)	0.171* (0.094)	0.054 (0.084)	0.112 (0.113)	0.164* (0.084)	0.096 (0.127)	0.112 (0.083)	0.260** (0.111)	0.013 (0.098)
Intercept	3.69	3.56	3.82	3.62	3.73	3.61	3.75	3.60	3.74
<i>Panel C: Liked the high school curriculum</i>									
STEM	0.002 (0.066)	-0.002 (0.087)	0.007 (0.083)	-0.001 (0.110)	-0.010 (0.091)	0.037 (0.133)	-0.006 (0.071)	0.107 (0.107)	-0.150 (0.108)
Intercept	3.32	3.26	3.38	3.25	3.42	3.13	3.41	3.37	3.32
<i>Panel D: Liked the high school peers</i>									
STEM	0.233*** (0.072)	0.309*** (0.091)	0.156* (0.094)	0.139 (0.118)	0.356*** (0.106)	0.160 (0.121)	0.235*** (0.090)	0.352*** (0.133)	0.169 (0.109)
Intercept	3.53	3.47	3.60	3.59	3.42	3.75	3.43	3.44	3.58
<i>Panel E: Liked the high school teachers</i>									
STEM	-0.022 (0.068)	-0.082 (0.086)	0.037 (0.088)	-0.109 (0.111)	0.059 (0.086)	0.026 (0.139)	-0.025 (0.074)	0.094 (0.114)	-0.175* (0.096)
Intercept	3.60	3.56	3.64	3.52	3.64	3.48	3.65	3.60	3.63
<i>Panel F: The high school curriculum was a good fit for my abilities</i>									
STEM	0.001 (0.053)	0.037 (0.069)	-0.035 (0.064)	-0.060 (0.087)	0.074 (0.076)	-0.019 (0.108)	-0.017 (0.062)	0.202*** (0.069)	-0.089 (0.088)
Intercept	3.39	3.33	3.45	3.40	3.36	3.32	3.45	3.28	3.42
Student-cutoffs	3,987	1,925	2,062	1,921	1,985	1,396	2,510	2,093	1,894

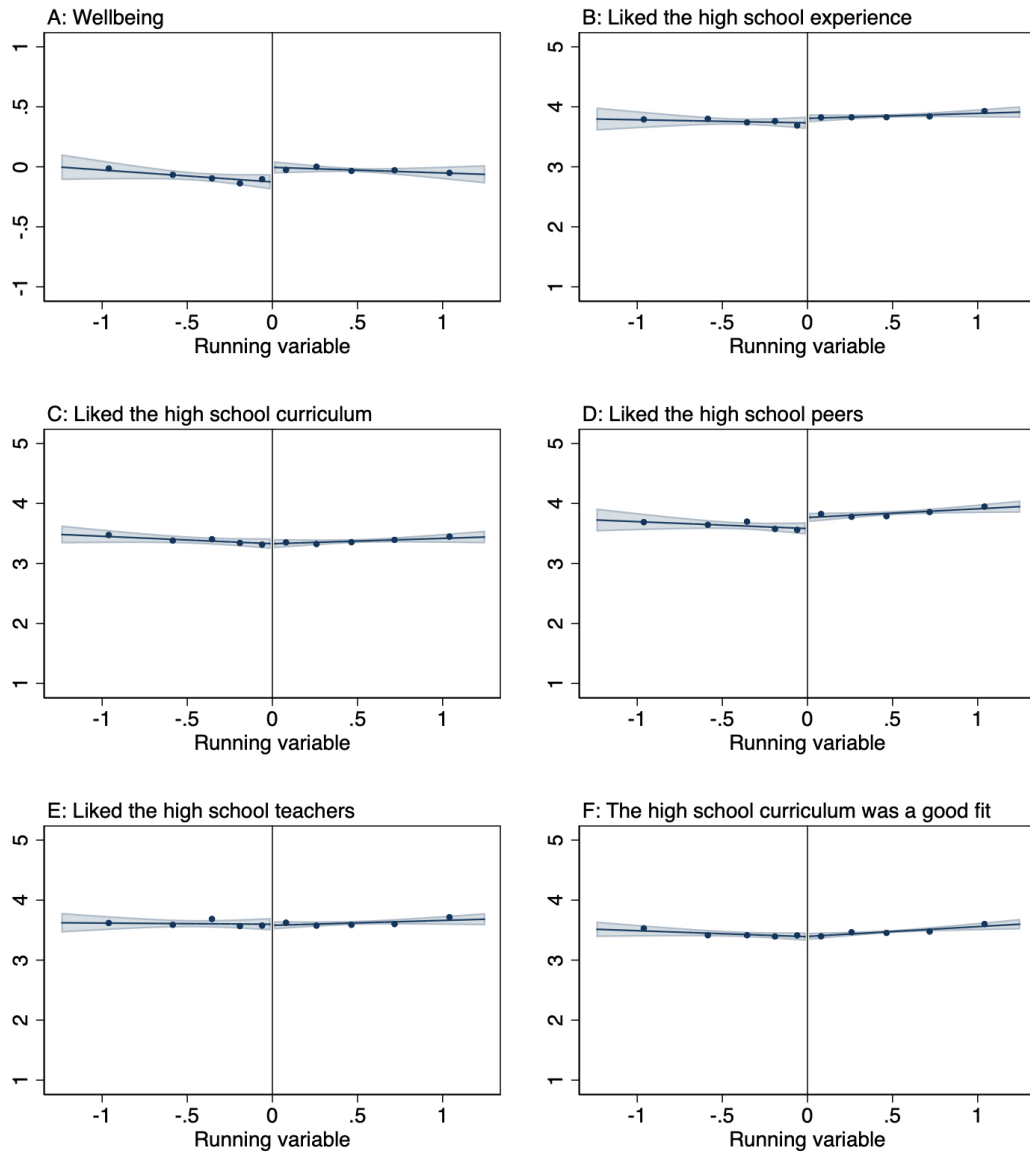
The table shows heterogeneity for the effects in Panel A of Table 11. The sample is the same for all outcomes, so it is presented in a single row at the bottom of the table. See the notes to Table A6 for definitions of the column headings.

Table A54: Robustness for effects on wellbeing and high school satisfaction
at the end of high school

	Main	Adding covariates	Optimal bandwidths		Alternative fixed bandwidths				Global quartic polynomial
			By cutoff	Mean	0.75	1.00	1.50	1.75	
<i>Panel A: Wellbeing</i>									
STEM	0.133*** (0.051)	0.124** (0.049)	0.078 (0.050)	0.137*** (0.052)	0.122** (0.056)	0.144*** (0.051)	0.143*** (0.047)	0.128*** (0.044)	0.127*** (0.046)
<i>Panel B: Liked the high school experience</i>									
STEM	0.113* (0.068)	0.122* (0.067)	0.084 (0.069)	0.105 (0.070)	0.133* (0.077)	0.113 (0.070)	0.109* (0.066)	0.114* (0.062)	0.096 (0.064)
<i>Panel C: Liked the high school curriculum</i>									
STEM	0.002 (0.066)	0.019 (0.065)	0.028 (0.068)	0.032 (0.068)	0.049 (0.076)	0.027 (0.068)	-0.019 (0.063)	-0.017 (0.061)	-0.051 (0.065)
<i>Panel D: Liked the high school peers</i>									
STEM	0.233*** (0.072)	0.214*** (0.071)	0.235*** (0.070)	0.246*** (0.072)	0.246*** (0.078)	0.262*** (0.072)	0.222*** (0.072)	0.228*** (0.069)	0.185** (0.072)
<i>Panel E: Liked the high school teachers</i>									
STEM	-0.022 (0.068)	-0.015 (0.068)	0.016 (0.073)	0.007 (0.071)	0.040 (0.084)	-0.000 (0.072)	-0.011 (0.066)	-0.023 (0.065)	-0.054 (0.067)
<i>Panel F: The high school curriculum was a good fit for my abilities</i>									
STEM	0.001 (0.053)	0.012 (0.052)	-0.017 (0.057)	-0.004 (0.057)	-0.019 (0.065)	0.005 (0.058)	-0.006 (0.051)	0.016 (0.049)	-0.018 (0.052)
Student-cutoffs	3,987	3,987	3,364	3,631	2,993	3,573	4,249	4,415	4,822

The table shows robustness for the effects in Panel A of Table 11. Specifications match those in Table A18. The one exception is that the added covariates in “Adding covariates” now also include mother’s and father’s years of schooling. The sample is the same for all outcomes, so it is presented in a single row at the bottom of the table. Results for specifications that use alternative fixed effects are in Appendix A14.

Figure A16: RD plots of effects on wellbeing and high school satisfaction at the end of high school



The figure displays RD plots for effects on the outcomes in Panel A of Table 11. See the notes to Figure A3 for details on RD plots.

Table A55: Heterogeneity for effects on wellbeing and high school satisfaction
1-1.5 years after high school

	All	Prefer		Stronger in		Gender		Achievement	
		STEM	HSS	Math	Language	Male	Female	Low	High
<i>Panel A: Wellbeing</i>									
STEM	-0.033 (0.068)	0.012 (0.091)	-0.077 (0.087)	-0.056 (0.107)	0.045 (0.103)	0.034 (0.143)	-0.044 (0.088)	0.102 (0.124)	-0.100 (0.107)
Intercept	0.66	0.60	0.72	0.68	0.61	0.58	0.72	0.56	0.67
Student-cutoffs	1,199	575	624	540	550	373	728	618	581
<i>Panel B: Liked the high school experience</i>									
STEM	0.217* (0.110)	0.335** (0.136)	0.099 (0.156)	0.376** (0.172)	0.161 (0.136)	0.709*** (0.246)	-0.088 (0.134)	0.120 (0.225)	0.381*** (0.126)
Intercept	3.90	3.77	4.03	3.85	3.90	3.54	4.02	3.98	3.72
Student-cutoffs	1,159	589	570	546	520	325	746	547	612
<i>Panel C: Liked the high school curriculum</i>									
STEM	-0.004 (0.137)	0.043 (0.159)	-0.050 (0.208)	0.140 (0.219)	0.010 (0.179)	0.187 (0.309)	-0.212 (0.161)	0.015 (0.277)	0.037 (0.168)
Intercept	3.13	3.02	3.25	3.03	3.13	2.91	3.28	3.19	3.03
Student-cutoffs	1,159	589	570	546	520	325	746	547	612
<i>Panel D: Liked the high school peers</i>									
STEM	0.287** (0.138)	0.543*** (0.171)	0.031 (0.194)	0.462** (0.210)	0.220 (0.218)	0.567** (0.260)	0.064 (0.184)	0.091 (0.235)	0.422* (0.217)
Intercept	3.82	3.66	3.97	3.76	3.80	3.79	3.87	3.99	3.68
Student-cutoffs	1,159	589	570	546	520	325	746	547	612
<i>Panel E: Liked the high school teachers</i>									
STEM	0.128 (0.130)	0.132 (0.166)	0.123 (0.188)	-0.026 (0.194)	0.254 (0.182)	0.492* (0.272)	-0.038 (0.141)	0.145 (0.283)	0.078 (0.144)
Intercept	3.75	3.62	3.89	3.83	3.66	3.27	3.88	3.72	3.81
Student-cutoffs	1,159	589	570	546	520	325	746	547	612
<i>Panel F: The high school curriculum was a good fit for my abilities</i>									
STEM	-0.123 (0.118)	0.199 (0.143)	-0.446*** (0.164)	-0.085 (0.182)	-0.151 (0.161)	0.213 (0.250)	-0.364** (0.140)	0.152 (0.246)	-0.292* (0.157)
Intercept	3.49	3.36	3.63	3.41	3.51	3.21	3.65	3.29	3.57
Student-cutoffs	1,159	589	570	546	520	325	746	547	612

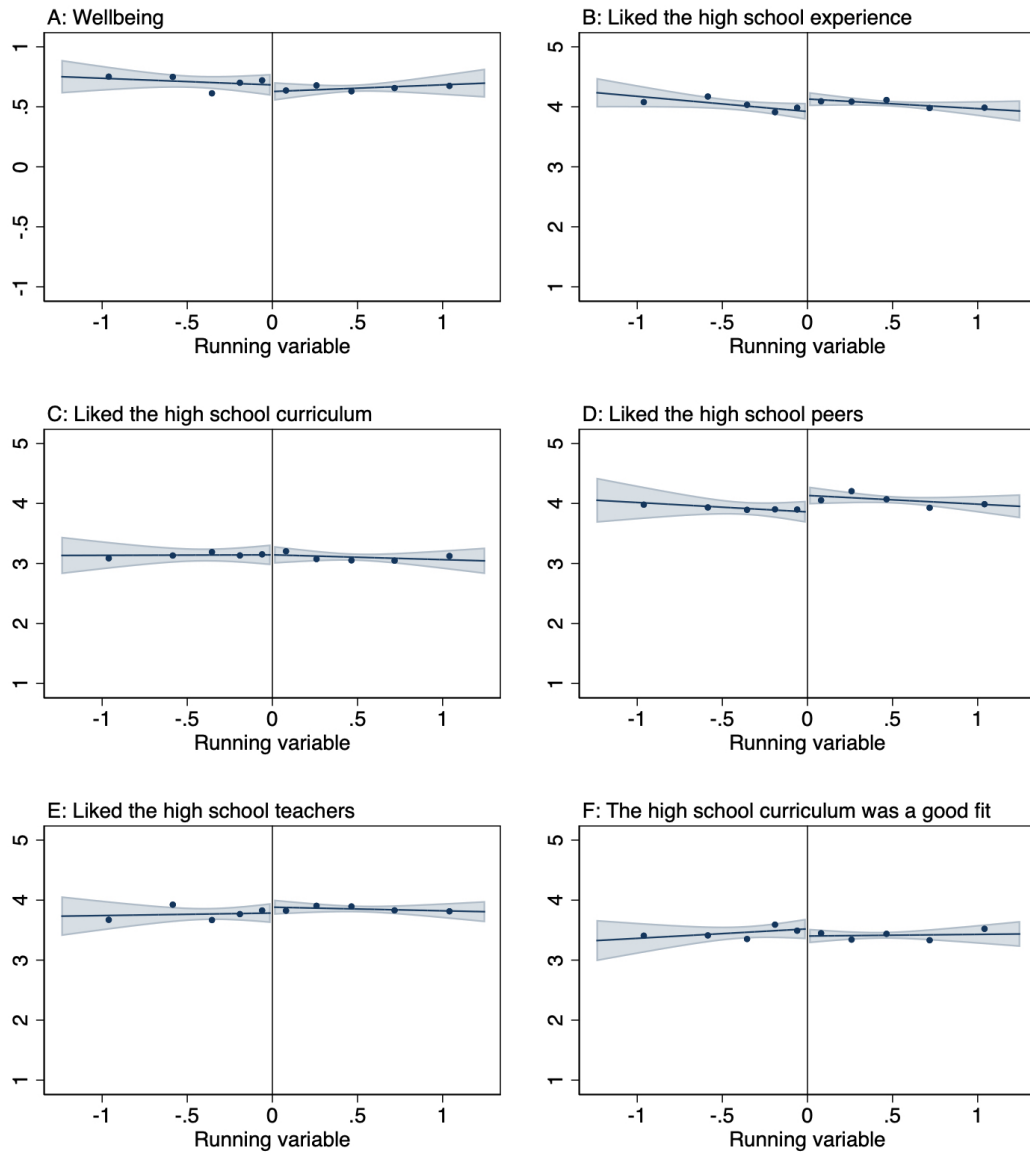
The table shows heterogeneity for the effects in Panel B of Table 11. See the notes to Table A6 for definitions of the column headings.

Table A56: Robustness for effects on wellbeing and high school satisfaction
1-1.5 years after high school

	Main	Adding covariates	Optimal bandwidths		Alternative fixed bandwidths				Global quartic polynomial
			By cutoff	Mean	0.75	1.00	1.50	1.75	
<i>Panel A: Wellbeing</i>									
STEM	-0.033 (0.068)	-0.033 (0.066)	-0.038 (0.068)	-0.032 (0.068)	-0.055 (0.073)	-0.053 (0.070)	-0.025 (0.066)	-0.036 (0.063)	-0.045 (0.070)
Student-cutoffs	1,199	1,199	1,158	1,177	883	1,068	1,271	1,318	1,432
<i>Panel B: Liked the high school experience</i>									
STEM	0.217* (0.110)	0.206* (0.108)	0.143 (0.110)	0.210* (0.113)	0.278** (0.132)	0.197 (0.120)	0.175* (0.106)	0.141 (0.105)	0.206* (0.118)
Student-cutoffs	1,159	1,159	1,086	1,134	886	1,041	1,263	1,319	1,389
<i>Panel C: Liked the high school curriculum</i>									
STEM	-0.004 (0.137)	0.003 (0.138)	-0.030 (0.144)	-0.001 (0.137)	0.037 (0.157)	-0.053 (0.147)	-0.022 (0.134)	-0.039 (0.129)	0.023 (0.141)
Student-cutoffs	1,159	1,159	1,104	1,140	886	1,041	1,263	1,319	1,389
<i>Panel D: Liked the high school peers</i>									
STEM	0.287** (0.138)	0.284** (0.137)	0.246* (0.145)	0.288** (0.141)	0.366** (0.168)	0.283* (0.150)	0.232* (0.133)	0.211* (0.127)	0.226 (0.152)
Student-cutoffs	1,159	1,159	1,113	1,140	886	1,041	1,263	1,319	1,389
<i>Panel E: Liked the high school teachers</i>									
STEM	0.128 (0.130)	0.136 (0.132)	0.080 (0.131)	0.139 (0.132)	0.158 (0.143)	0.128 (0.137)	0.067 (0.126)	0.077 (0.119)	0.109 (0.134)
Student-cutoffs	1,159	1,159	1,095	1,134	886	1,041	1,263	1,319	1,389
<i>Panel F: The high school curriculum was a good fit for my abilities</i>									
STEM	-0.123 (0.118)	-0.114 (0.116)	-0.179 (0.122)	-0.101 (0.119)	-0.055 (0.135)	-0.058 (0.129)	-0.143 (0.114)	-0.126 (0.112)	-0.096 (0.122)
Student-cutoffs	1,159	1,159	1,098	1,140	886	1,041	1,263	1,319	1,389

The table shows robustness for the effects in Panel B of Table 11. Specifications match those in Table A18. The one exception is that the added covariates in “Adding covariates” now also include mother’s and father’s years of schooling. Results for specifications that use alternative fixed effects are in Appendix A14.

Figure A17: RD plots of effects on wellbeing and high school satisfaction
1-1.5 years after high school



The figure displays RD plots for effects on the outcomes in Panel B of Table 11. See the notes to Figure A3 for details on RD plots.

Table A57: Effects on wellbeing and high school satisfaction
for students from the follow-up surveys

	Wellbeing	Liked the high school:				H.s. curriculum was a good fit for my abilities
		Experience	Curriculum	Peers	Teachers	
<i>Panel A: End of high school</i>						
STEM	0.051 (0.078)	0.127 (0.125)	0.185 (0.117)	0.242** (0.117)	0.228* (0.131)	0.275*** (0.100)
Intercept	-0.10	3.63	3.17	3.47	3.39	3.22
Std. dev.	0.72	0.99	1.00	1.09	1.02	0.85
<i>Panel B: 1-1.5 years after high school</i>						
STEM	-0.033 (0.068)	0.217* (0.110)	-0.004 (0.137)	0.287** (0.138)	0.128 (0.130)	-0.123 (0.118)
Intercept	0.66	3.90	3.13	3.82	3.75	3.49
Std. dev.	0.56	0.89	1.06	1.08	0.93	1.02
<i>Panel C: Change versus the end of high school</i>						
STEM	-0.084 (0.088)	0.090 (0.118)	-0.189 (0.138)	0.045 (0.140)	-0.100 (0.134)	-0.398*** (0.124)
Intercept	0.76	0.27	-0.04	0.34	0.36	0.27
Std. dev.	0.80	1.06	1.25	1.15	1.10	1.21
Student-cutoffs	1,199	1,159	1,159	1,159	1,159	1,159

The table is analogous to Table 11 but for restricted samples. The sample in the “Wellbeing” column (remaining columns) is restricted to students from the second (first) follow-up survey. The outcomes in Panel C are the difference between those in Panels B and A. The number of observations is the same across panels, so it is displayed in a single row at the bottom of the table.

Table A58: Heterogeneity for effects on regret over high school application choices one year after high school

	All	Prefer		Stronger in		Gender		Achievement	
		STEM	HSS	Math	Language	Male	Female	Low	High
<i>Panel A: Am happy with the high school application choices that I made</i>									
STEM	0.049 (0.129)	0.282 (0.189)	-0.185 (0.155)	0.215 (0.219)	-0.172 (0.191)	-0.049 (0.189)	0.112 (0.156)	-0.452** (0.183)	0.265 (0.172)
Intercept	4.20	4.14	4.25	3.99	4.31	4.27	4.16	4.48	4.06
<i>Panel B: If could do over, would make no change</i>									
STEM	0.045 (0.042)	0.081 (0.054)	0.009 (0.057)	0.063 (0.076)	0.122** (0.060)	0.066 (0.100)	0.042 (0.047)	-0.015 (0.068)	0.093 (0.062)
Intercept	0.80	0.77	0.83	0.78	0.78	0.79	0.80	0.80	0.81
<i>Panel C: If could do over, would rank STEM tracks higher</i>									
STEM	-0.032 (0.035)	-0.042 (0.047)	-0.022 (0.047)	-0.021 (0.062)	-0.062 (0.043)	-0.047 (0.072)	-0.028 (0.042)	0.034 (0.036)	-0.053 (0.051)
Intercept	0.10	0.11	0.10	0.12	0.09	0.13	0.11	0.06	0.10
<i>Panel D: If could do over, would rank HSS tracks higher</i>									
STEM	-0.018 (0.032)	-0.037 (0.040)	0.001 (0.047)	-0.079 (0.051)	-0.032 (0.051)	-0.049 (0.074)	-0.007 (0.035)	-0.003 (0.057)	-0.052 (0.042)
Intercept	0.08	0.09	0.06	0.11	0.10	0.09	0.07	0.09	0.08
<i>Panel E: If could do over, would rank Other tracks higher</i>									
STEM	0.005 (0.013)	-0.002 (0.015)	0.012 (0.018)	0.037* (0.020)	-0.028 (0.025)	0.031* (0.016)	-0.007 (0.022)	-0.016 (0.045)	0.012 (0.009)
Intercept	0.02	0.03	0.01	-0.00	0.03	-0.00	0.02	0.05	0.01
Student-cutoffs	1,159	589	570	546	520	325	746	547	612

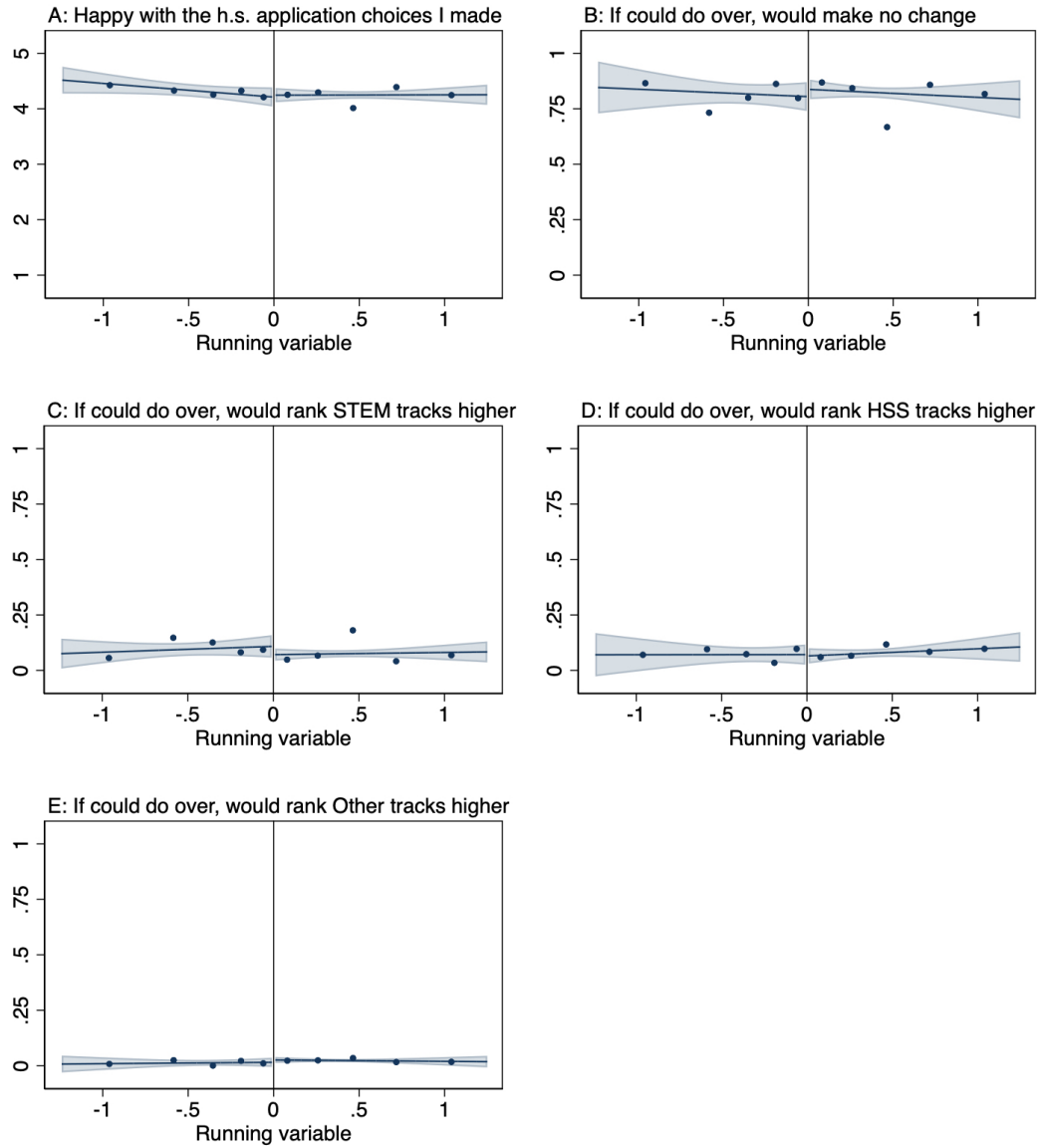
The table shows heterogeneity for the effects in Table 12. The sample is the same for all outcomes, so it is presented in a single row at the bottom of the table. See the notes to Table A6 for definitions of the column headings.

Table A59: Robustness for effects on regret over high school application choices
one year after high school

	Main	Adding covariates	Optimal bandwidths		Alternative fixed bandwidths				Global quartic polynomial
			By cutoff	Mean	0.75	1.00	1.50	1.75	
<i>Panel A: Am happy with the high school application choices that I made</i>									
STEM	0.049 (0.129)	0.046 (0.129)	0.023 (0.133)	0.045 (0.129)	0.115 (0.147)	0.077 (0.133)	0.059 (0.126)	0.043 (0.121)	0.072 (0.130)
<i>Panel B: If could do over, would make no change</i>									
STEM	0.045 (0.042)	0.047 (0.041)	0.022 (0.043)	0.039 (0.042)	0.044 (0.058)	0.025 (0.047)	0.043 (0.041)	0.043 (0.042)	0.066 (0.044)
<i>Panel C: If could do over, would rank STEM tracks higher</i>									
STEM	-0.032 (0.035)	-0.033 (0.034)	-0.028 (0.034)	-0.030 (0.035)	-0.030 (0.045)	-0.025 (0.037)	-0.029 (0.033)	-0.023 (0.033)	-0.047 (0.033)
<i>Panel D: If could do over, would rank HSS tracks higher</i>									
STEM	-0.018 (0.032)	-0.017 (0.032)	-0.015 (0.032)	-0.013 (0.032)	-0.018 (0.038)	-0.006 (0.032)	-0.021 (0.031)	-0.030 (0.031)	-0.022 (0.034)
<i>Panel E: If could do over, would rank Other tracks higher</i>									
STEM	0.005 (0.013)	0.004 (0.013)	0.008 (0.013)	0.005 (0.013)	0.005 (0.015)	0.005 (0.014)	0.007 (0.013)	0.010 (0.013)	0.003 (0.013)
Student-cutoffs	1,159	1,159	1,152	1,154	886	1,041	1,263	1,319	1,389

The table shows robustness for the effects in Table 12. Specifications match those in Table A18. The one exception is that the added covariates in “Adding covariates” now also include mother’s and father’s years of schooling. The sample is the same for all outcomes, so it is presented in a single row at the bottom of the table. Results for specifications that use alternative fixed effects are in Appendix A14.

Figure A18: RD plots of effects on regret over high school application choices one year after high school



The figure displays RD plots for effects on the outcomes in Table 12. See the notes to Figure A3 for details on RD plots.

Table A60: Heterogeneity for effects on college satisfaction one year after high school

	All	Prefer		Stronger in		Gender		Achievement	
		STEM	HSS	Math	Language	Male	Female	Low	High
<i>Panel A: Like the college experience</i>									
STEM	-0.443*** (0.138)	-0.521*** (0.188)	-0.366** (0.166)	-0.746*** (0.192)	-0.213 (0.227)	-0.496** (0.246)	-0.477*** (0.167)	-0.441 (0.320)	-0.413** (0.172)
Intercept	2.20	2.34	2.05	2.52	1.95	2.10	2.30	2.13	2.26
<i>Panel B: Like the college curriculum</i>									
STEM	-0.227* (0.129)	-0.433** (0.165)	-0.021 (0.176)	-0.543** (0.215)	-0.064 (0.178)	-0.224 (0.235)	-0.188 (0.169)	-0.324 (0.303)	-0.205 (0.137)
Intercept	2.25	2.39	2.11	2.72	1.95	2.24	2.23	2.23	2.33
<i>Panel C: Like the college peers</i>									
STEM	-0.398** (0.166)	-0.470** (0.215)	-0.326 (0.219)	-0.515* (0.291)	-0.262 (0.218)	-0.319 (0.301)	-0.453** (0.182)	-0.574 (0.368)	-0.348* (0.203)
Intercept	2.18	2.36	2.00	2.49	1.89	2.05	2.27	2.32	2.16
<i>Panel D: Like the college instructors</i>									
STEM	-0.352*** (0.134)	-0.376** (0.169)	-0.328* (0.192)	-0.353 (0.233)	-0.386* (0.221)	-0.131 (0.259)	-0.443*** (0.158)	-0.303 (0.281)	-0.345** (0.160)
Intercept	2.31	2.44	2.17	2.58	2.16	2.12	2.40	2.11	2.44
<i>Panel E: The college curriculum is a good fit for my abilities</i>									
STEM	0.082 (0.156)	-0.052 (0.205)	0.216 (0.223)	0.200 (0.271)	-0.040 (0.239)	-0.157 (0.342)	0.031 (0.201)	-0.454* (0.240)	0.206 (0.174)
Intercept	3.48	3.75	3.21	3.61	3.37	3.55	3.57	3.59	3.74
<i>Panel F: Am happy with the college application choices that I made</i>									
STEM	0.122 (0.179)	-0.207 (0.267)	0.451** (0.212)	0.368 (0.269)	0.021 (0.273)	0.282 (0.364)	0.022 (0.230)	-0.215 (0.275)	0.198 (0.220)
Intercept	3.72	4.14	3.31	3.79	3.59	3.49	3.86	3.61	4.05
Student-cutoffs	1,159	589	570	546	520	325	746	547	612

The table shows heterogeneity for the effects in Table 13. The sample is the same for all outcomes, so it is presented in a single row at the bottom of the table. See the notes to Table A6 for definitions of the column headings.

Table A61: Effects on college satisfaction one year after high school
with no imputations for missing responses

	Like the college:				College curric. is a good fit for my abilities	Am happy w/ the appl. choices I made
	Experience	Curriculum	Peers	Instructors		
STEM	-0.462*** (0.156)	-0.222* (0.133)	-0.455** (0.178)	-0.386*** (0.142)	0.163 (0.106)	0.226* (0.136)
Intercept	2.40	2.50	2.39	2.54	3.97	4.26
Std. dev.	1.13	1.06	1.22	1.12	0.79	0.92
Student-cutoffs	965	965	965	965	965	965

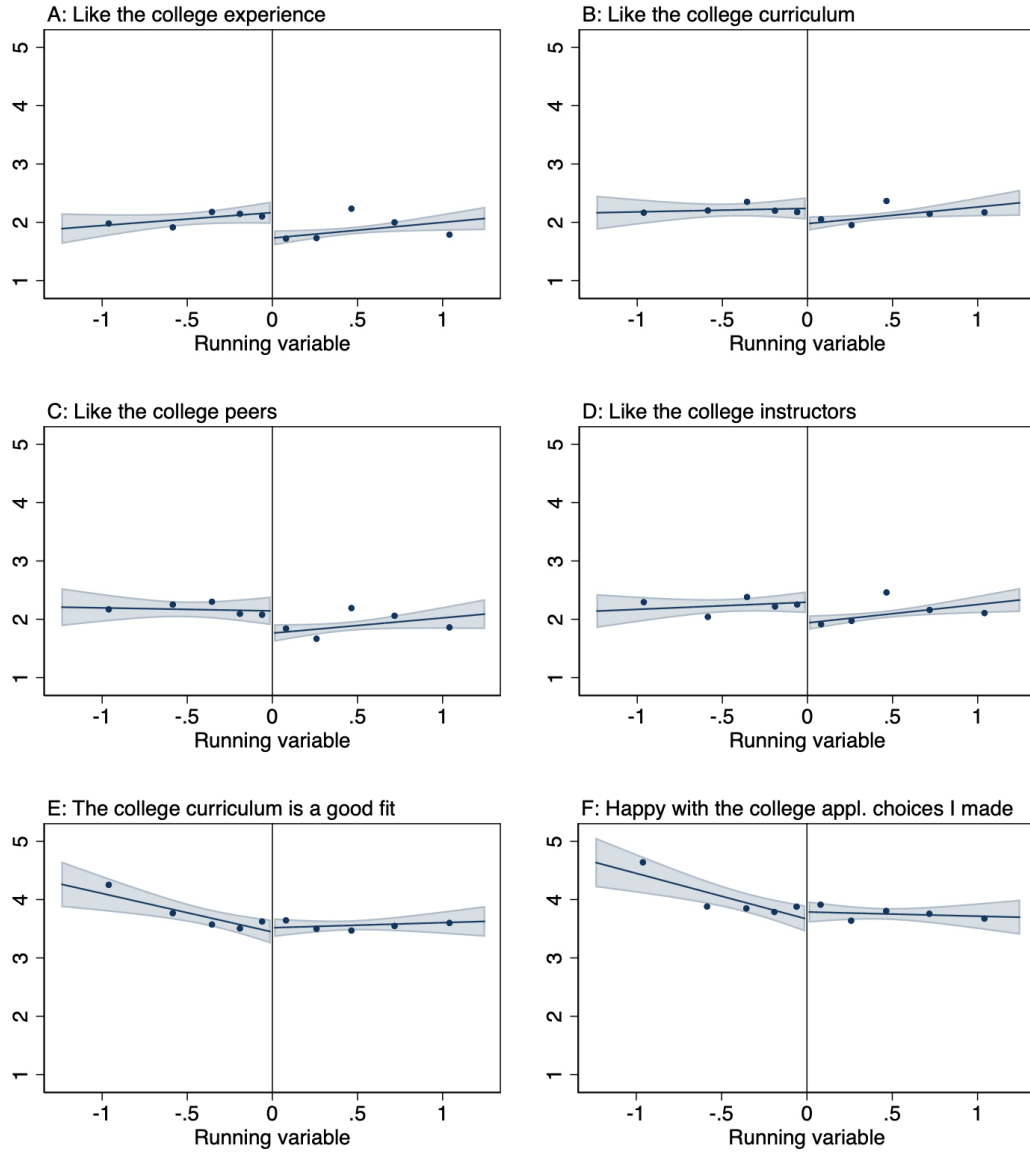
The table is analogous to Table 13, but it does not impute values of 1 for students who had never attended college by the time of the first follow-up survey. Instead, it drops these students.

Table A62: Robustness for effects on college satisfaction one year after high school

	Main	Adding covariates	Optimal bandwidths		Alternative fixed bandwidths				Global quartic polynomial
			By cutoff	Mean	0.75	1.00	1.50	1.75	
Panel A: Like the college experience									
STEM	-0.443*** (0.138)	-0.450*** (0.140)	-0.455*** (0.148)	-0.466*** (0.139)	-0.653*** (0.161)	-0.498*** (0.150)	-0.420*** (0.134)	-0.384*** (0.131)	-0.510*** (0.141)
Panel B: Like the college curriculum									
STEM	-0.227* (0.129)	-0.238* (0.131)	-0.214 (0.134)	-0.237* (0.130)	-0.297** (0.149)	-0.266** (0.130)	-0.211* (0.121)	-0.161 (0.118)	-0.259** (0.123)
Panel C: Like the college peers									
STEM	-0.398** (0.166)	-0.402** (0.169)	-0.422** (0.171)	-0.386** (0.167)	-0.502** (0.196)	-0.382** (0.173)	-0.430*** (0.156)	-0.382** (0.150)	-0.456*** (0.170)
Panel D: Like the college instructors									
STEM	-0.352*** (0.134)	-0.365*** (0.137)	-0.349** (0.139)	-0.349** (0.135)	-0.547*** (0.154)	-0.409*** (0.136)	-0.320** (0.128)	-0.290** (0.124)	-0.443*** (0.133)
Panel E: The college curriculum is a good fit for my abilities									
STEM	0.082 (0.156)	0.079 (0.159)	0.065 (0.162)	0.108 (0.155)	0.067 (0.183)	0.141 (0.159)	0.054 (0.148)	0.097 (0.148)	0.097 (0.162)
Panel F: Am happy with the college application choices that I made									
STEM	0.122 (0.179)	0.125 (0.182)	0.079 (0.184)	0.153 (0.176)	0.090 (0.212)	0.169 (0.183)	0.091 (0.174)	0.097 (0.172)	0.157 (0.185)
Student-cutoffs	1,159	1,159	1,114	1,140	886	1,041	1,263	1,319	1,389

The table shows robustness for the effects in Table 13. Specifications match those in Table A18. The one exception is that the added covariates in “Adding covariates” now also include mother’s and father’s years of schooling. The sample is the same for all outcomes, so it is presented in a single row at the bottom of the table. Results for specifications that use alternative fixed effects are in Appendix A14.

Figure A19: RD plots of effects on college satisfaction one year after high school



The figure displays RD plots for effects on the outcomes in Table 13. See the notes to Figure A3 for details on RD plots.

A13 Additional results for social and civic outcomes

Table A63: Heterogeneity for effects on social and civic outcomes

	All	Prefer		Stronger in		Gender		Achievement	
		STEM	HSS	Math	Language	Male	Female	Low	High
<i>Panel A: Traditionalist expectations</i>									
STEM	0.096*** (0.034)	0.110** (0.051)	0.081** (0.037)	0.126** (0.051)	0.139*** (0.043)	0.116** (0.048)	0.034 (0.033)	0.074 (0.061)	0.091** (0.044)
Intercept	-0.10	-0.12	-0.07	-0.03	-0.16	0.22	-0.29	-0.03	-0.13
Std. dev.	0.49	0.51	0.47	0.51	0.46	0.41	0.41	0.48	0.50
Student-cutoffs	3,987	1,925	2,062	1,921	1,985	1,396	2,510	2,093	1,894
<i>Panel B: Right-wing political views</i>									
STEM	0.042 (0.031)	0.024 (0.047)	0.060* (0.034)	0.071 (0.049)	0.042 (0.045)	0.145*** (0.052)	-0.012 (0.036)	0.044 (0.048)	0.023 (0.043)
Intercept	-0.06	-0.07	-0.05	-0.07	-0.04	-0.05	-0.08	-0.01	-0.08
Std. dev.	0.44	0.44	0.44	0.45	0.42	0.45	0.42	0.42	0.45
Student-cutoffs	3,987	1,925	2,062	1,921	1,985	1,396	2,510	2,093	1,894
<i>Panel C: Trust</i>									
STEM	0.016 (0.050)	0.019 (0.065)	0.013 (0.065)	-0.031 (0.076)	0.008 (0.075)	-0.012 (0.085)	0.028 (0.068)	-0.021 (0.077)	-0.023 (0.080)
Intercept	-0.03	0.01	-0.08	0.03	-0.05	0.05	-0.10	0.04	-0.04
Std. dev.	0.78	0.76	0.79	0.77	0.79	0.77	0.78	0.79	0.77
Student-cutoffs	3,987	1,925	2,062	1,921	1,985	1,396	2,510	2,093	1,894
<i>Panel D: Turn out to vote in the Fall 2024 elections</i>									
STEM	0.027 (0.036)	0.036 (0.053)	0.018 (0.041)	0.009 (0.077)	0.019 (0.037)	-0.005 (0.077)	0.049 (0.043)	-0.010 (0.046)	0.056 (0.051)
Intercept	0.87	0.84	0.90	0.88	0.90	0.89	0.87	0.92	0.84
Std. dev.	0.26	0.26	0.26	0.27	0.23	0.28	0.24	0.26	0.26
Student-cutoffs	1,199	575	624	540	550	373	728	618	581
<i>Panel E: Good friends</i>									
STEM	-0.005 (0.185)	0.143 (0.273)	-0.154 (0.200)	0.303 (0.312)	-0.052 (0.249)	-0.066 (0.321)	-0.041 (0.236)	0.128 (0.263)	-0.304 (0.249)
Intercept	6.07	6.13	6.02	6.06	5.87	6.61	5.78	5.92	6.32
Std. dev.	2.76	2.73	2.77	2.75	2.75	2.69	2.78	2.76	2.75
Student-cutoffs	3,987	1,925	2,062	1,921	1,985	1,396	2,510	2,093	1,894
<i>Panel F: Very close friends</i>									
STEM	-0.041 (0.096)	-0.000 (0.127)	-0.082 (0.119)	-0.101 (0.149)	-0.049 (0.138)	0.004 (0.171)	-0.063 (0.114)	0.042 (0.162)	-0.161 (0.137)
Intercept	3.22	3.28	3.15	3.26	3.18	3.42	3.07	3.05	3.38
Std. dev.	1.50	1.49	1.51	1.49	1.51	1.51	1.48	1.51	1.48
Student-cutoffs	3,987	1,925	2,062	1,921	1,985	1,396	2,510	2,093	1,894
<i>Panel G: Good female vs. male friends</i>									
STEM	-0.764*** (0.191)	-0.868*** (0.299)	-0.660*** (0.202)	-0.983*** (0.299)	-0.812*** (0.250)	-0.706** (0.281)	-0.491*** (0.188)	-0.362 (0.293)	-1.04*** (0.261)
Intercept	0.62	0.60	0.64	0.34	0.91	-1.03	1.50	0.12	0.97
Std. dev.	2.76	2.86	2.65	2.87	2.63	2.48	2.30	2.71	2.80
Student-cutoffs	3,987	1,925	2,062	1,921	1,985	1,396	2,510	2,093	1,894
<i>Panel H: Verbal development</i>									
STEM	-0.279*** (0.033)	-0.272*** (0.048)	-0.285*** (0.039)	-0.289*** (0.046)	-0.308*** (0.051)	-0.260*** (0.052)	-0.273*** (0.043)	-0.255*** (0.057)	-0.304*** (0.047)
Intercept	0.18	0.18	0.18	0.07	0.27	-0.01	0.29	0.15	0.21
Std. dev.	0.49	0.47	0.51	0.49	0.48	0.48	0.47	0.50	0.48
Student-cutoffs	3,987	1,925	2,062	1,921	1,985	1,396	2,510	2,093	1,894
<i>Panel I: Empathy</i>									
STEM	-0.116** (0.054)	-0.051 (0.076)	-0.181*** (0.066)	-0.158* (0.091)	-0.103 (0.084)	-0.145 (0.093)	-0.082 (0.069)	-0.053 (0.109)	-0.128* (0.069)
Intercept	0.06	0.02	0.10	0.00	0.12	-0.04	0.10	0.03	0.02
Std. dev.	0.89	0.87	0.91	0.89	0.89	0.92	0.86	0.92	0.86
Student-cutoffs	3,987	1,925	2,062	1,921	1,985	1,396	2,510	2,093	1,894

The table shows heterogeneity for effects on the main outcomes in Section 6.5. See the notes to Table A6 for definitions of the column headings.

Table A64: Effects on expectations about one's household wealth decile at age 30

Decile	Decile groups				
	≤ 4	5 – 6	7 – 8	9 – 10	
<i>Panel A: All students</i>					
STEM	0.133 (0.086)	-0.003 (0.006)	0.021 (0.025)	-0.089*** (0.030)	0.071*** (0.023)
Intercept	7.40	0.01	0.17	0.68	0.14
Std. dev.	1.33	0.08	0.37	0.45	0.38
Student-cutoffs	3,987	3,987	3,987	3,987	3,987
<i>Panel B: Male students</i>					
STEM	0.030 (0.147)	-0.008 (0.014)	0.055 (0.041)	-0.073 (0.055)	0.026 (0.041)
Intercept	7.51	0.02	0.14	0.66	0.18
Std. dev.	1.34	0.10	0.36	0.45	0.38
Student-cutoffs	1,396	1,396	1,396	1,396	1,396
<i>Panel C: Female students</i>					
STEM	0.172 (0.114)	0.001 (0.007)	0.007 (0.034)	-0.108*** (0.037)	0.100*** (0.032)
Intercept	7.33	0.00	0.20	0.69	0.11
Std. dev.	1.32	0.08	0.38	0.45	0.38
Student-cutoffs	2,510	2,510	2,510	2,510	2,510

The table presents results for the ATE of STEM assignment, β , for outcomes capturing how wealthy a student expects their household to be when the student is 30 years old. The outcomes are based on a question where students were asked to predict their household wealth decile. We use this question to create a variable equal to the raw response (“Decile”) as well as indicator variables that equal one when a response is in a given category (“Decile groups”). The question was asked in both the end-of-high-school and second follow-up surveys. The outcomes in the table use data from both surveys. As in Table A32, they are equal to the student fixed effects from a panel regression of a given derived variable on student and survey fixed effects, centered at the overall mean of the variable for students with non-missing values in both surveys. Other details match those in Panel A of Table 2.

Table A65: Effects on the amount of education that students expect to obtain

Years	Degree					
	High school	Foreman's	Bachelor's	Master's/Professional	Doctoral	
Panel A: All students						
STEM	-0.375* (0.209)	-0.008 (0.021)	0.002 (0.012)	0.079* (0.043)	0.013 (0.058)	-0.085 (0.053)
Intercept	17.16	0.05	0.02	0.15	0.53	0.26
Std. dev.	1.89	0.20	0.12	0.38	0.49	0.40
Student-cutoffs	1,199	1,199	1,199	1,199	1,199	1,199
Panel B: Male students						
STEM	-0.896** (0.388)	-0.028 (0.034)	0.049** (0.024)	-0.030 (0.088)	0.325*** (0.108)	-0.316*** (0.094)
Intercept	17.35	0.06	0.00	0.23	0.34	0.37
Std. dev.	1.86	0.20	0.15	0.41	0.50	0.37
Student-cutoffs	373	373	373	373	373	373
Panel C: Female students						
STEM	0.083 (0.275)	-0.008 (0.023)	-0.013 (0.017)	0.095* (0.054)	-0.139** (0.068)	0.065 (0.068)
Intercept	17.06	0.05	0.02	0.12	0.62	0.20
Std. dev.	1.87	0.18	0.11	0.36	0.49	0.41
Student-cutoffs	728	728	728	728	728	728

The table presents results for the ATE of STEM assignment, β , for outcomes related to how much education a student expects to obtain. Outcomes are from the second follow-up survey and are based on a question where students were asked to select their expected final degree from the options shown in the table. The “Degree” outcomes are indicator variables equal to one if a student selected the given degree. “Years” is the number of years of schooling associated with a student’s selected degree: 12 for High school, 14 for Foreman’s, 15 for Bachelor’s, 17 for Master’s/Professional, and 20 for Doctoral. Other details match those in Panel A of Table 2.

Table A66: Effects on components of the trust index

	Index	Components	
		People can be trusted	Government workers can be trusted
<i>Panel A: All students</i>			
STEM	0.016 (0.050)	0.018 (0.062)	0.014 (0.067)
Intercept	-0.03	2.34	2.35
Std. dev.	0.78	0.99	0.95
Student-cutoffs	3,987	3,987	3,987
<i>Panel B: Male students</i>			
STEM	-0.012 (0.085)	-0.025 (0.111)	-0.006 (0.105)
Intercept	0.05	2.46	2.41
Std. dev.	0.77	0.97	0.95
Student-cutoffs	1,396	1,396	1,396
<i>Panel C: Female students</i>			
STEM	0.028 (0.068)	0.040 (0.089)	0.021 (0.086)
Intercept	-0.10	2.25	2.30
Std. dev.	0.78	1.00	0.95
Student-cutoffs	2,510	2,510	2,510

The table shows effects on components of the trust index used in Table 16. The components are based on questions where students were asked how much they agree with a statement on a scale of 1-5. The statements are: “Generally speaking, people can be trusted” and “Generally speaking, government workers can be trusted”. The questions were asked in both the end-of-high-school and second follow-up surveys. The components and index use data from both surveys; they are calculated using the procedure described in the notes to Table A32. Other details match those in Panel A of Table 2.

Table A67: Effects on components of the index of verbal development

Index	Components											
	Beliefs about own high school humanities abilities	Beliefs about own high school soc. studies abilities	Pref. for high school humanities subjects	Pref. for high school soc. studies subjects	Beliefs about own college reading & writing abilities	Pref. for reading & writing- heavy college subjects	Romanian language & literature bacc. score	In books & movies, prefer characters to plots	Hours spent reading on a typical weekday	Would like to write books or plays	Would like to write scripts for movies/TV	
Panel A: All students												
STEM	-0.279*** (0.033)	-0.281*** (0.059)	-0.620*** (0.071)	-0.405*** (0.102)	-0.697*** (0.135)	-0.396** (0.154)	-0.525*** (0.164)	-0.315*** (0.073)	0.071 (0.075)	-0.284*** (0.093)	-0.191*** (0.073)	-0.084 (0.068)
Intercept	0.18	4.14	4.04	4.13	4.08	3.23	3.24	7.92	2.63	2.04	0.56	0.53
Std. dev.	0.49	0.86	1.10	1.04	1.21	1.32	1.31	1.28	1.18	1.31	0.50	0.50
Student-cutoffs	3,987	3,987	3,987	1,159	1,159	1,159	1,159	3,951	3,987	3,987	1,199	1,199
Panel B: Male students												
STEM	-0.260*** (0.052)	-0.279*** (0.107)	-0.560*** (0.131)	-0.550** (0.232)	-0.293 (0.293)	-0.452 (0.324)	-0.413 (0.352)	-0.338*** (0.111)	0.036 (0.133)	-0.494*** (0.149)	-0.068 (0.118)	0.046 (0.110)
Intercept	-0.01	3.96	4.01	3.90	3.79	2.98	2.85	7.54	2.61	1.77	0.33	0.32
Std. dev.	0.48	0.91	1.16	1.16	1.20	1.22	1.22	1.30	1.16	1.25	0.46	0.48
Student-cutoffs	1,396	1,396	1,396	325	325	325	325	1,384	1,396	1,396	373	373
Panel C: Female students												
STEM	-0.273*** (0.043)	-0.252*** (0.074)	-0.655*** (0.086)	-0.300** (0.140)	-0.778*** (0.162)	-0.512*** (0.194)	-0.659*** (0.191)	-0.276*** (0.082)	0.056 (0.103)	-0.145 (0.123)	-0.222*** (0.083)	-0.091 (0.080)
Intercept	0.29	4.26	4.04	4.25	4.07	3.43	3.46	8.18	2.68	2.23	0.65	0.60
Std. dev.	0.47	0.80	1.08	0.94	1.22	1.33	1.32	1.21	1.19	1.31	0.50	0.50
Student-cutoffs	2,510	2,510	2,510	746	746	746	746	2,488	2,510	2,510	728	728

The table shows effects on components of the index of verbal development used in Table 16. “Beliefs about own high school humanities abilities” combines the “Humanities subjects” outcomes in Panels A and B of Table 7. It is equal to the student fixed effects from a panel regression of the responses for the relevant question from the end-of-high-school and first follow-up surveys on student and survey fixed effects, centered at the mean response for students with non-missing values in both surveys. “Beliefs about own high school soc. studies abilities” is the same, but for the “Social studies subjects” outcomes in Panels A and B of Table 7. “Pref. for high school humanities subjects” (“Pref. for high school soc. studies subjects”) is the “Humanities subjects” (“Social studies subjects”) outcome in Panel C of Table 7. “Beliefs about own college reading & writing abilities” (“Pref. for reading & writing-heavy college subjects”) is the “Subjects that involve reading and writing” outcome in Panel A (B) of Table 8. “Romanian language & literature bacc. score” is the outcome in Panel A of Table A43. “In books & movies, prefer characters to plots” is the negative of the outcome in Panel B of the same table. “Hours spent reading on a typical weekday” combines the Panel F outcomes in Tables A27 and A29. Similar to before, it equals the student fixed effects from a panel regression of the responses for the relevant question from the end-of-high-school and second follow-up surveys on student and survey fixed effects, centered at the mean response for students with non-missing values in both surveys. “Would like to write books or plays” and “Would like to write scripts for movies/TV” are from the O*NET Interest Profiler included in the second follow-up survey; as in Table A45, they are indicator variables equal to one if a student reports that they would enjoy doing the given activity as a job task. Finally, the verbal development index is calculated using the procedure described in the notes to Table A32. Other details match those in Panel A of Table 2

Table A68: Effects on components of the empathy index

	Index	Components	
		Notice when people are sad or angry even when they hide it	Would enjoy helping people with personal/ emotional problems
<i>Panel A: All students</i>			
STEM	-0.116** (0.054)	-0.092* (0.051)	-0.108** (0.044)
Intercept	0.06	4.35	0.90
Std. dev.	0.89	0.83	0.37
Student-cutoffs	3,987	3,987	1,199
<i>Panel B: Male students</i>			
STEM	-0.145 (0.093)	-0.128 (0.086)	-0.127 (0.099)
Intercept	-0.04	4.28	0.86
Std. dev.	0.92	0.86	0.43
Student-cutoffs	1,396	1,396	373
<i>Panel C: Female students</i>			
STEM	-0.082 (0.069)	-0.056 (0.065)	-0.094* (0.049)
Intercept	0.10	4.39	0.93
Std. dev.	0.86	0.80	0.32
Student-cutoffs	2,510	2,510	728

The table shows effects on components of the empathy index used in Table 16. The first component is based on a question where students were asked how much they agree (on a scale of 1-5) with the statement: “I notice when people are sad or angry even when they try to hide it”. This question was asked in both the end-of-high-school and second follow-up surveys. The second component is from the O*NET Interest Profiler included in the second follow-up survey; as in Table A45, it is an indicator variable equal to one if a student reports that they would enjoy doing the listed activity as a job task. The first component and the overall empathy index are calculated using the procedure described in the notes to Table A32. Other details match those in Panel A of Table 2.

Table A69: Effects on social and civic outcomes using only the end-of-high-school survey

	Traditionalist expectations	Right-wing political views	Trust	Friends			Verbal development	Empathy
				Good	Very close	Good female vs. male		
Panel A: All students								
STEM	0.094*** (0.033)	0.056* (0.031)	0.025 (0.049)	0.054 (0.197)	-0.087 (0.100)	-0.852*** (0.198)	-0.213*** (0.033)	-0.099* (0.057)
Intercept	-0.07	-0.08	-0.07	6.72	3.18	0.69	0.20	0.09
Std. dev.	0.50	0.44	0.81	2.94	1.61	2.94	0.49	0.93
Student-cutoffs	3,987	3,987	3,987	3,987	3,987	3,987	3,987	3,987
Panel B: Male students								
STEM	0.125*** (0.048)	0.144*** (0.052)	-0.024 (0.085)	0.071 (0.326)	-0.061 (0.196)	-0.945*** (0.321)	-0.249*** (0.054)	-0.137 (0.107)
Intercept	0.23	-0.06	0.05	7.23	3.41	-0.79	0.07	0.00
Std. dev.	0.43	0.45	0.79	2.86	1.63	2.72	0.48	0.97
Student-cutoffs	1,396	1,396	1,396	1,396	1,396	1,396	1,396	1,396
Panel C: Female students								
STEM	0.025 (0.033)	0.009 (0.035)	0.050 (0.069)	-0.041 (0.260)	-0.086 (0.122)	-0.523*** (0.200)	-0.184*** (0.042)	-0.065 (0.075)
Intercept	-0.26	-0.10	-0.15	6.44	3.01	1.49	0.30	0.13
Std. dev.	0.42	0.43	0.81	2.97	1.59	2.51	0.46	0.90
Student-cutoffs	2,510	2,510	2,510	2,510	2,510	2,510	2,510	2,510

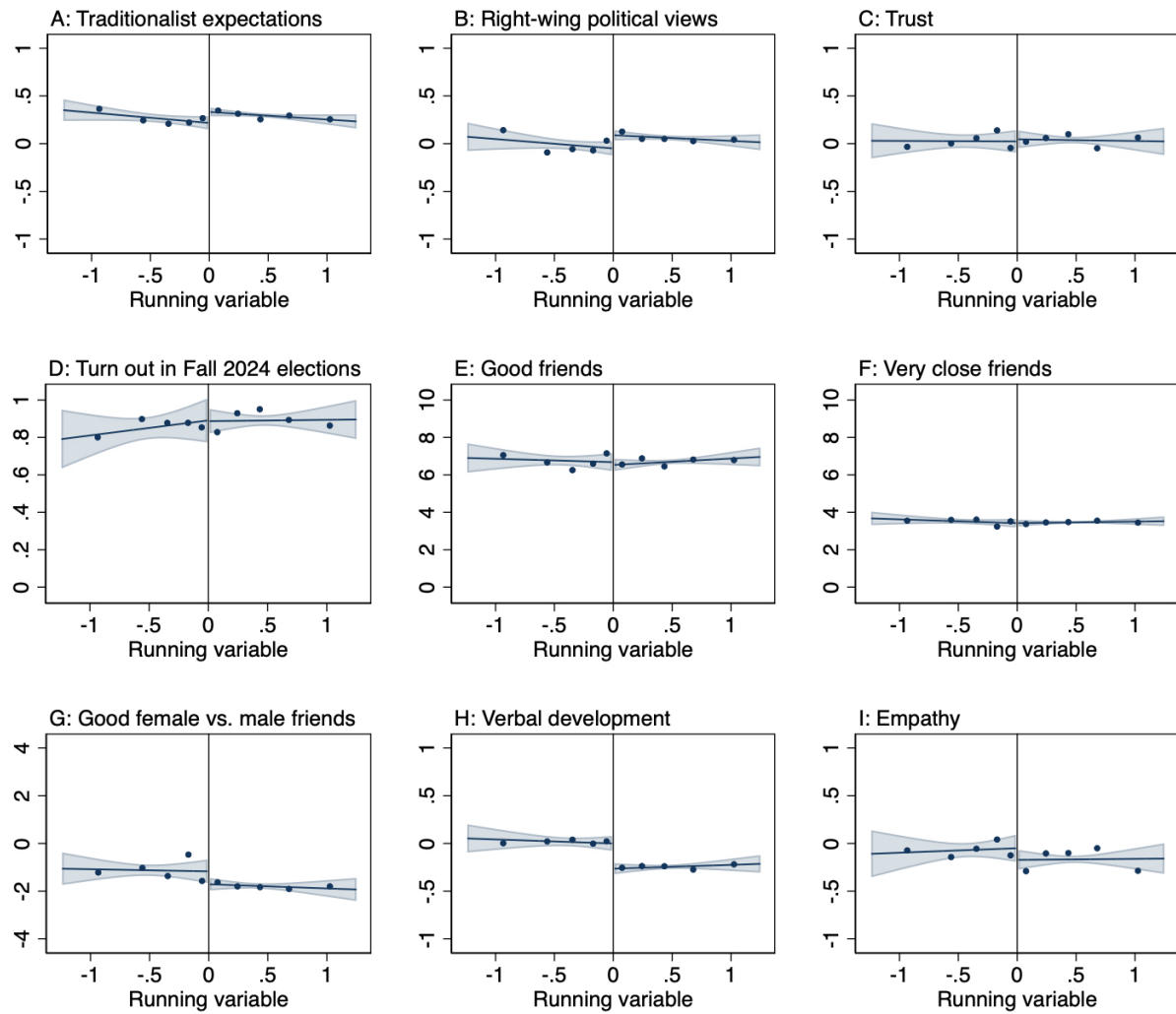
The table shows effects on versions of the main outcomes in Section 6.5 that use data only from the end-of-high-school survey. All indices are calculated using the procedure described in the notes to Table A52. Other details follow Panel A of Table 2.

Table A70: Robustness for effects on social and civic outcomes: male students

	Main	Adding covariates	Optimal bandwidths		Alternative fixed bandwidths				Global quartic polynomial
			By cutoff	Mean	0.75	1.00	1.50	1.75	
<i>Panel A: Traditionalist expectations</i>									
STEM	0.116** (0.048)	0.117** (0.048)	0.118** (0.052)	0.120** (0.049)	0.116** (0.057)	0.111** (0.050)	0.110** (0.046)	0.097** (0.045)	0.117** (0.046)
Student-cutoffs	1,396	1,396	1,334	1,366	1,073	1,252	1,470	1,526	1,629
<i>Panel B: Right-wing political views</i>									
STEM	0.145*** (0.052)	0.150*** (0.052)	0.129** (0.056)	0.144*** (0.051)	0.094 (0.064)	0.152*** (0.056)	0.140*** (0.050)	0.150*** (0.048)	0.148*** (0.049)
Student-cutoffs	1,396	1,396	1,325	1,366	1,073	1,252	1,470	1,526	1,629
<i>Panel C: Trust</i>									
STEM	-0.012 (0.085)	-0.013 (0.083)	-0.009 (0.086)	-0.005 (0.087)	-0.006 (0.096)	-0.041 (0.095)	-0.021 (0.081)	-0.039 (0.082)	-0.014 (0.089)
Student-cutoffs	1,396	1,396	1,345	1,371	1,073	1,252	1,470	1,526	1,629
<i>Panel D: Turn out to vote in the Fall 2024 elections</i>									
STEM	-0.005 (0.077)	-0.010 (0.079)	-0.005 (0.077)	-0.005 (0.077)	0.015 (0.088)	-0.013 (0.080)	-0.001 (0.076)	0.001 (0.076)	-0.010 (0.075)
Student-cutoffs	373	373	373	373	272	334	392	399	426
<i>Panel E: Good friends</i>									
STEM	-0.066 (0.321)	-0.064 (0.317)	-0.152 (0.320)	-0.011 (0.324)	-0.554 (0.384)	-0.119 (0.352)	0.052 (0.304)	0.064 (0.294)	-0.000 (0.297)
Student-cutoffs	1,396	1,396	1,344	1,371	1,073	1,252	1,470	1,526	1,629
<i>Panel F: Very close friends</i>									
STEM	0.004 (0.171)	0.005 (0.172)	0.019 (0.178)	-0.018 (0.174)	-0.113 (0.211)	-0.064 (0.180)	-0.089 (0.167)	-0.039 (0.165)	-0.061 (0.171)
Student-cutoffs	1,396	1,396	1,320	1,358	1,073	1,252	1,470	1,526	1,629
<i>Panel G: Good female vs. male friends</i>									
STEM	-0.706** (0.281)	-0.722** (0.280)	-0.739** (0.293)	-0.700** (0.283)	-0.611* (0.344)	-0.760*** (0.290)	-0.677** (0.271)	-0.538** (0.267)	-0.515* (0.268)
Student-cutoffs	1,396	1,396	1,339	1,371	1,073	1,252	1,470	1,526	1,629
<i>Panel H: Verbal development</i>									
STEM	-0.260*** (0.052)	-0.263*** (0.049)	-0.277*** (0.056)	-0.263*** (0.053)	-0.302*** (0.068)	-0.269*** (0.057)	-0.267*** (0.050)	-0.246*** (0.049)	-0.253*** (0.050)
Student-cutoffs	1,396	1,396	1,327	1,359	1,073	1,252	1,470	1,526	1,629
<i>Panel I: Empathy</i>									
STEM	-0.145 (0.093)	-0.145 (0.092)	-0.156 (0.100)	-0.144 (0.096)	-0.251** (0.121)	-0.186* (0.103)	-0.174* (0.090)	-0.140 (0.086)	-0.152 (0.094)
Student-cutoffs	1,396	1,396	1,316	1,358	1,073	1,252	1,470	1,526	1,629

The table shows robustness for the effects in Table A63 for male students. Specifications match those in Table A18. The one exception is that the added covariates in “Adding covariates” now also include mother’s and father’s years of schooling. Results for specifications that use alternative fixed effects are in Appendix A14.

Figure A20: RD plots of effects on social and civic outcomes: male students



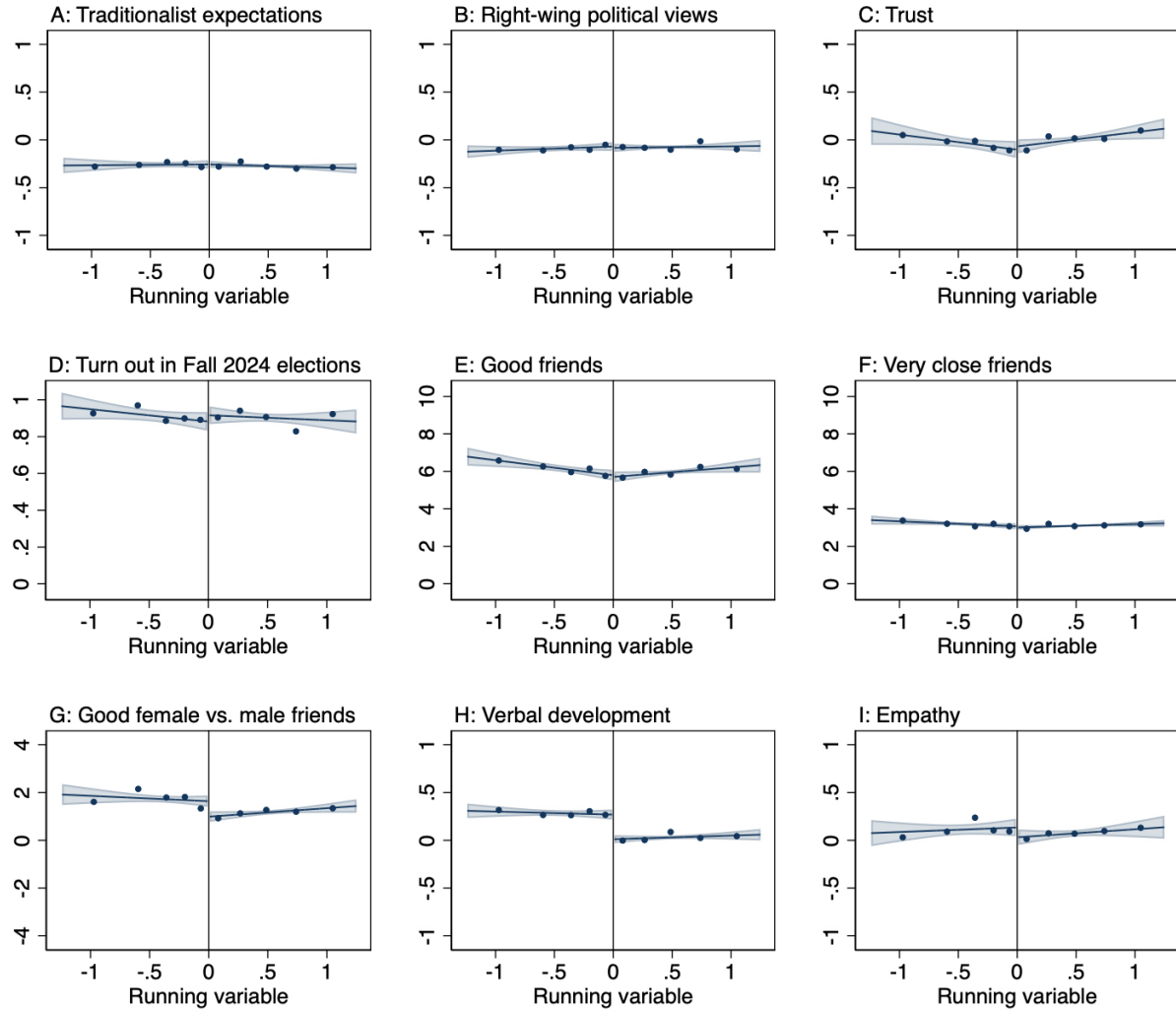
The figure displays RD plots for effects on the outcomes in Table A63. See the notes to Figure A3 for details on RD plots.

Table A71: Robustness for effects on social and civic outcomes: female students

	Main	Adding covariates	Optimal bandwidths		Alternative fixed bandwidths				Global quartic polynomial
			By cutoff	Mean	0.75	1.00	1.50	1.75	
<i>Panel A: Traditionalist expectations</i>									
STEM	0.034 (0.033)	0.032 (0.032)	0.034 (0.035)	0.033 (0.036)	0.052 (0.040)	0.038 (0.037)	0.043 (0.031)	0.025 (0.032)	0.007 (0.034)
Student-cutoffs	2,510	2,510	2,143	2,295	1,846	2,241	2,696	2,807	3,101
<i>Panel B: Right-wing political views</i>									
STEM	-0.012 (0.036)	-0.012 (0.036)	-0.030 (0.040)	-0.009 (0.039)	-0.025 (0.045)	-0.010 (0.040)	-0.007 (0.034)	-0.016 (0.033)	-0.014 (0.034)
Student-cutoffs	2,510	2,510	2,125	2,295	1,846	2,241	2,696	2,807	3,101
<i>Panel C: Trust</i>									
STEM	0.028 (0.068)	0.034 (0.067)	0.034 (0.067)	0.039 (0.070)	-0.012 (0.081)	0.017 (0.073)	0.029 (0.066)	0.015 (0.064)	0.021 (0.063)
Student-cutoffs	2,510	2,510	2,174	2,342	1,846	2,241	2,696	2,807	3,101
<i>Panel D: Turn out to vote in the Fall 2024 elections</i>									
STEM	0.049 (0.043)	0.050 (0.043)	0.045 (0.046)	0.053 (0.044)	0.055 (0.056)	0.070 (0.050)	0.040 (0.040)	0.046 (0.039)	0.037 (0.046)
Student-cutoffs	728	728	702	717	526	638	779	819	901
<i>Panel E: Good friends</i>									
STEM	-0.041 (0.236)	-0.023 (0.234)	-0.060 (0.248)	-0.040 (0.248)	0.105 (0.275)	0.034 (0.257)	-0.082 (0.229)	-0.016 (0.220)	-0.005 (0.224)
Student-cutoffs	2,510	2,510	2,134	2,309	1,846	2,241	2,696	2,807	3,101
<i>Panel F: Very close friends</i>									
STEM	-0.063 (0.114)	-0.058 (0.113)	-0.165 (0.128)	-0.042 (0.124)	-0.035 (0.144)	-0.013 (0.127)	-0.072 (0.110)	-0.057 (0.106)	-0.015 (0.113)
Student-cutoffs	2,510	2,510	2,150	2,309	1,846	2,241	2,696	2,807	3,101
<i>Panel G: Good female vs. male friends</i>									
STEM	-0.491*** (0.188)	-0.496*** (0.188)	-0.525*** (0.197)	-0.543*** (0.201)	-0.234 (0.229)	-0.474** (0.209)	-0.435** (0.183)	-0.451*** (0.173)	-0.499*** (0.187)
Student-cutoffs	2,510	2,510	2,160	2,309	1,846	2,241	2,696	2,807	3,101
<i>Panel H: Verbal development</i>									
STEM	-0.273*** (0.043)	-0.271*** (0.043)	-0.298*** (0.041)	-0.275*** (0.044)	-0.292*** (0.051)	-0.270*** (0.046)	-0.258*** (0.041)	-0.259*** (0.039)	-0.254*** (0.040)
Student-cutoffs	2,510	2,510	2,192	2,342	1,846	2,241	2,696	2,807	3,101
<i>Panel I: Empathy</i>									
STEM	-0.082 (0.069)	-0.083 (0.069)	-0.061 (0.076)	-0.071 (0.077)	-0.055 (0.088)	-0.080 (0.078)	-0.074 (0.067)	-0.055 (0.065)	-0.066 (0.069)
Student-cutoffs	2,510	2,510	2,119	2,278	1,846	2,241	2,696	2,807	3,101

The table shows robustness for the effects in Table A63 for female students. Specifications match those in Table A18. The one exception is that the added covariates in “Adding covariates” now also include mother’s and father’s years of schooling. Results for specifications that use alternative fixed effects are in Appendix A14.

Figure A21: RD plots of effects on social and civic outcomes: female students



The figure displays RD plots for effects on the outcomes in Table A63. See the notes to Figure A3 for details on RD plots.

Table A72: Effects on changes in social and civic outcomes after high school

	Traditionalist expectations	Trust	Friends		
			Good	Very close	Good female vs. male
<i>Panel A: End of high school</i>					
STEM	-0.039 (0.055)	0.119 (0.086)	0.437 (0.393)	-0.212 (0.195)	-0.733** (0.339)
Intercept	0.04	-0.18	6.81	3.30	0.60
Std. dev.	0.48	0.78	2.86	1.57	3.00
<i>Panel B: 1.5 years after high school</i>					
STEM	-0.036 (0.056)	0.107 (0.089)	0.107 (0.351)	0.146 (0.188)	-0.452 (0.323)
Intercept	-0.08	0.28	5.64	3.25	0.36
Std. dev.	0.46	0.73	2.80	1.51	2.58
<i>Panel C: Change in the 1.5 years after high school</i>					
STEM	0.003 (0.053)	-0.011 (0.123)	-0.330 (0.442)	0.358 (0.229)	0.280 (0.392)
Intercept	-0.12	0.46	-1.17	-0.04	-0.23
Std. dev.	0.45	0.96	3.48	2.00	2.86
Student-cutoffs	1,199	1,199	1,199	1,199	1,199

The table shows effects on changes in social and civic outcomes in the time after high school. The outcomes included are those from Section 6.5 that can be calculated in the same manner in both the end-of-high-school survey and a follow-up survey—i.e., those that are entirely based on questions that are repeated in multiple surveys. For all such outcomes, the relevant follow-up survey is the second; as such, the change is always in the 1.5 years after high school. To match Panel B, the sample in Panel A is restricted to students who appear in the second follow-up survey. Thus, the sample is the same for all panels, and the number of observations is listed in a single row at the bottom of the table. In Panels A and B, the “Traditionalist expectations” and “Trust” indices are calculated using the procedure described in the notes to Table A52. The outcomes in Panel C are the difference between those in Panels B and A. Other details follow Panel A of Table 2.

Table A73: Female-male transition score gaps
in the tracks of students at STEM vs. HSS cutoffs

	Transition score			
	Overall score	Math exam score	Language exam score	Grades 5-8 GPA
<i>Panel A: 2015-2017 and 2019 cohorts</i>				
Gap on the STEM side	0.07	-0.20	0.28	0.17
Gap on the HSS side	0.08	-0.21	0.31	0.18
Difference: STEM v. HSS	-0.010*** (0.004)	0.008 (0.008)	-0.027*** (0.007)	-0.014*** (0.004)
Student-cutoffs	73,432	73,432	73,432	73,432
<i>Panel B: Second follow-up survey</i>				
Gap on the STEM side	0.07	-0.23	0.29	0.19
Gap on the HSS side	0.10	-0.16	0.29	0.22
Difference: STEM v. HSS	-0.030* (0.017)	-0.061* (0.035)	-0.002 (0.033)	-0.024 (0.021)
Student-cutoffs	1,199	1,199	1,199	1,199

The table summarizes gender gaps in incoming achievement for classmates of students at STEM vs. HSS cutoffs. The results are obtained using our main RD specification. The outcomes are the difference in average values of a given transition score variable for the female and male students in a student's assigned track. "Gap on the STEM side" ("Gap on the HSS side") is the predicted value of the gap on the STEM (HSS) side of the RD threshold. "Difference: STEM vs. HSS" is the coefficient estimate and standard error for the ATE of STEM assignment, β . "Gap on the HSS side" is the same as the "Intercept" term in Panel A of Table 2. "Gap on the STEM side" is the sum of "Gap on the HSS side" and the β estimate. The table uses the administrative data. Panel A is for the entire 2015-2017 and 2019 cohorts. Panel B restricts to students in the second follow-up survey, as these are the students of whom we asked whether boys are naturally better than girls in math. Other details follow Panel A of Table 2.

Table A74: Female-male gaps in baccalaureate performance
in the tracks of students at STEM vs. HSS cutoffs

	Baccalaureate performance					
	Take the exam	Exam score	Math exam score	History exam score	Romanian exam score	Elective exam score
<i>Panel A: 2015-2017 and 2019 cohorts</i>						
Gap on the STEM side	-0.00	0.27	0.06	-	0.53	0.22
Gap on the HSS side	0.00	0.34	-	0.19	0.60	0.23
Difference: STEM vs. HSS	-0.003 (0.002)	-0.076*** (0.011)	- -	- -	-0.069*** (0.013)	-0.012 (0.015)
Student-cutoffs	71,793	71,773	41,076	30,476	71,793	71,774
<i>Panel B: Second follow-up survey</i>						
Gap on the STEM side	0.01	0.34	0.09	-	0.65	0.27
Gap on the HSS side	0.01	0.42	-	0.22	0.72	0.30
Difference: STEM v. HSS	-0.000 (0.008)	-0.078 (0.053)	- -	- -	-0.071 (0.053)	-0.022 (0.068)
Student-cutoffs	1,199	1,199	676	434	1,199	1,199

The table is analogous to Table A73 but for gaps in baccalaureate performance. Students in HSS (STEM) tracks do not take the math (history) component of the baccalaureate exam. Hence, for these outcomes, we can only estimate gaps on one side of a cutoff.

A14 Robustness to alternative fixed effects

This appendix explores the robustness of our results to alternative ways of specifying the fixed effects in Equation (1). Our main specification, used elsewhere in the paper, includes fixed effects for the interaction of cutoff type, cutoff score, and admission year. In this appendix, we estimate β under three alternative sets of fixed effects: (i) the interaction of cutoff type and admission year, (ii) the interaction of cutoff type, cutoff score, admission year, and the identity of the track (i.e., school-curriculum combination) above the cutoff, and (iii) the interaction of cutoff type, cutoff score, admission year, and the identities of the tracks above and below the cutoff. The first of the alternative approaches is a coarse specification that tries to minimally adjust the data. The other two adjust for finer groupings than in our main approach. For instance, (ii) adjusts for the fact that multiple tracks can have the same cutoff score, and (iii) additionally adjusts for the fact that students who prefer the same track may have different fallbacks.

Results from the different approaches are in Tables A75-A77. As in Appendix A5, each row in the tables corresponds to a different outcome, and rows are included for all the outcomes that are in Tables A10-A12. The values under “Main approach” match those from elsewhere in the paper; values under the “Alternative f.e.” headings are based on the alternative specifications. In these headings, the columns titled “In C.I.” indicate whether the β estimate from the given specification falls within the 95% confidence interval for β from the main approach.

The results show that our findings do not depend on the precise way in which we specify fixed effects. Instead, β estimates are highly similar across all four fixed effect specifications. In each of the first two alternative specifications, there is only one outcome (out of 89) for which the β estimate falls outside the main-approach 95% confidence interval. In the third alternative specification, there are only seven. Moreover, even for these outcomes, the magnitudes of the differences in estimates are always modest. We prefer our main fixed effect specification because it acknowledges that the treatment of interest is at the level of a curricular category (i.e., STEM or HSS), not a track. However, it is apparent that our results would hardly change if we were to adjust for track identities.

Table A75: Robustness to alternative fixed effects

Outcome	Main approach			Alternative f.e. 1			Alternative f.e. 2			Alternative f.e. 3					
	Est.	s.e.	N	Est.	s.e.	In C.I.	N	Est.	s.e.	In C.I.	N	Est.	s.e.	In C.I.	N
Panel A: High school enrollment and graduation															
Years of enrollment	0.00	0.01	55,221	0.00	0.01	Y	55,424	0.00	0.01	Y	51,983	0.00	0.01	Y	45,216
Years of STEM enrollment	3.08	0.02	55,221	3.10	0.02	Y	55,424	3.07	0.02	Y	51,983	3.06	0.03	Y	45,216
Years of HSS enrollment	-3.09	0.02	55,221	-3.10	0.02	Y	55,424	-3.09	0.02	Y	51,983	-3.09	0.03	Y	45,216
Years of Other enrollment	0.01	0.01	55,221	0.01	0.01	Y	55,424	0.02	0.01	Y	51,983	0.02	0.01	Y	45,216
Graduate	0.00	0.00	55,221	0.00	0.00	Y	55,424	0.00	0.00	Y	51,983	-0.01	0.00	Y	45,216
Graduate in STEM	0.69	0.01	55,221	0.70	0.01	Y	55,424	0.69	0.01	Y	51,983	0.69	0.01	Y	45,216
Graduate in HSS	-0.70	0.01	55,221	-0.71	0.01	Y	55,424	-0.70	0.01	Y	51,983	-0.70	0.01	Y	45,216
Graduate in Other	0.01	0.00	55,221	0.01	0.00	Y	55,424	0.01	0.00	Y	51,983	0.01	0.00	Y	45,216
Panel B: College outcomes															
College plans at the end of high school: any college	0.01	0.02	3,987	0.01	0.02	Y	4,056	0.01	0.02	Y	3,651	0.03	0.02	Y	3,127
College plans at the end of high school: STEM	0.23	0.04	3,987	0.24	0.04	Y	4,056	0.25	0.04	Y	3,651	0.27	0.04	Y	3,127
College plans at the end of high school: HLSS	-0.22	0.03	3,987	-0.22	0.03	Y	4,056	-0.24	0.04	Y	3,651	-0.22	0.04	Y	3,127
College enrollment one year after high school: any college	-0.04	0.03	3,327	-0.05	0.03	Y	3,392	-0.04	0.03	Y	3,024	-0.03	0.03	Y	2,549
College enrollment one year after high school: STEM	0.25	0.04	3,327	0.23	0.04	Y	3,392	0.26	0.04	Y	3,024	0.22	0.05	Y	2,549
College enrollment one year after high school: HLSS	-0.26	0.03	3,327	-0.26	0.03	Y	3,392	-0.25	0.03	Y	3,024	-0.21	0.03	Y	2,549
Panel C: Desired careers one year after high school															
Technology or engineering	0.25	0.05	1,159	0.23	0.05	Y	1,235	0.25	0.06	Y	962	0.25	0.07	Y	731
Medicine	-0.04	0.04	1,159	-0.02	0.04	Y	1,235	-0.06	0.05	Y	962	-0.03	0.06	Y	731
Business or economics	-0.04	0.06	1,159	-0.04	0.05	Y	1,235	0.01	0.07	Y	962	-0.01	0.08	Y	731
Art, education, law, or social services	-0.14	0.05	1,159	-0.14	0.04	Y	1,235	-0.18	0.05	Y	962	-0.15	0.07	Y	731
Other/unsure	-0.03	0.04	1,159	-0.03	0.04	Y	1,235	-0.02	0.05	Y	962	-0.07	0.06	Y	731
Panel D: Baccalaureate performance															
Take the exam	-0.01	0.00	73,470	-0.01	0.00	Y	73,740	-0.02	0.00	Y	69,082	-0.02	0.00	N	60,117
Pass the exam	-0.04	0.00	73,470	-0.03	0.00	Y	73,740	-0.05	0.00	N	69,082	-0.05	0.01	N	60,117
Exam score	-0.35	0.01	69,938	-0.30	0.02	N	70,206	-0.38	0.02	Y	65,606	-0.38	0.02	Y	56,849
Pass in STEM	0.62	0.01	73,470	0.63	0.01	Y	73,740	0.61	0.01	Y	69,082	0.60	0.01	N	60,117
Pass in HSS	-0.70	0.01	73,470	-0.70	0.01	Y	73,740	-0.69	0.01	Y	69,082	-0.68	0.01	N	60,117
Pass in Other	0.04	0.00	73,470	0.04	0.00	Y	73,740	0.04	0.00	Y	69,082	0.03	0.00	Y	60,117
Panel E: Confidence in and preferences for high school subjects															
Beliefs about own high school STEM abilities at the end of high school	0.87	0.08	3,987	0.90	0.08	Y	4,056	0.88	0.09	Y	3,651	0.77	0.10	Y	3,127
Beliefs about own high school HSS abilities at the end of high school	-0.40	0.06	3,987	-0.38	0.05	Y	4,056	-0.38	0.06	Y	3,651	-0.37	0.08	Y	3,127
Beliefs about own high school STEM abilities one year after high school	0.74	0.12	1,159	0.84	0.11	Y	1,235	0.71	0.15	Y	962	0.54	0.17	Y	731
Beliefs about own high school HSS abilities one year after high school	-0.51	0.09	1,159	-0.55	0.08	Y	1,235	-0.56	0.11	Y	962	-0.52	0.14	Y	731
Preferences for STEM high school subjects one year after high school	1.01	0.15	1,159	1.12	0.13	Y	1,235	1.03	0.18	Y	962	0.80	0.22	Y	731
Preferences for HSS high school subjects one year after high school	-0.55	0.08	1,159	-0.58	0.08	Y	1,235	-0.56	0.10	Y	962	-0.48	0.14	Y	731

The table presents results for alternative versions of Equation (1). “Main approach” is the version used elsewhere in the paper, which includes fixed effects for the interaction of cutoff type, cutoff score, and admission year. “Alternative f.e. 1” replaces these with fixed effects for the interaction of cutoff type and admission year. “Alternative f.e. 2” uses the interaction of cutoff type, cutoff score, admission year, and the identities of the tracks above and below the cutoff. “Est.” is the coefficient estimate for β , and “s.e.” is the standard error. “In C.I.” indicates whether a coefficient estimate is in the main-approach 95% confidence interval for β . “N” is the number of student-cutoff observations. Observations get dropped when there is only one for a given value of the fixed effects. Rows correspond to different outcomes and are included for all the outcomes in Tables A10-A12.

Table A76: Robustness to alternative fixed effects, cont.

Outcome	Main approach			Alternative f.e. 1			Alternative f.e. 2			Alternative f.e. 3					
	Est.	s.e.	N	Est.	s.e.	In C.I.	N	Est.	s.e.	In C.I.	N	Est.	s.e.	In C.I.	N
Panel F: Confidence in and preferences for college subjects															
Beliefs about own college STEM abilities one year after high school	0.66	0.15	1,159	0.70	0.13	Y	1,235	0.66	0.19	Y	962	0.46	0.23	Y	731
Beliefs about own college non-STEM abilities one year after high school	-0.34	0.14	1,159	-0.31	0.13	Y	1,235	-0.42	0.16	Y	962	-0.63	0.22	N	731
Preferences for STEM college subjects one year after high school	0.78	0.15	1,159	0.82	0.14	Y	1,235	0.78	0.19	Y	962	0.61	0.24	Y	731
Preferences for non-STEM college subjects one year after high school	-0.45	0.15	1,159	-0.33	0.13	Y	1,235	-0.49	0.17	Y	962	-0.60	0.22	Y	731
Panel G: O*NET outcomes 1.5 years after high school															
Share of STEM-related job tasks that enjoy	0.10	0.02	1,199	0.08	0.02	Y	1,265	0.12	0.03	Y	966	0.15	0.04	N	752
Share of HSS-related job tasks that enjoy	-0.09	0.04	1,199	-0.11	0.04	Y	1,265	-0.08	0.05	Y	966	-0.09	0.06	Y	752
Share of Other job tasks that enjoy	0.01	0.03	1,199	0.02	0.03	Y	1,265	0.04	0.03	Y	966	0.05	0.05	Y	752
Dif. in the share of STEM- vs. HSS-related job tasks that enjoy	0.19	0.05	1,199	0.19	0.04	Y	1,265	0.20	0.05	Y	966	0.24	0.07	Y	752
Share of O*NET-recommended jobs that are STEM-related	0.10	0.03	1,199	0.07	0.03	Y	1,265	0.11	0.04	Y	966	0.12	0.04	Y	752
Share of O*NET-recommended jobs that are HSS-related	-0.10	0.05	1,199	-0.09	0.04	Y	1,265	-0.07	0.05	Y	966	-0.07	0.07	Y	752
Share of O*NET-recommended jobs that are Other	-0.01	0.04	1,199	0.02	0.04	Y	1,265	-0.04	0.05	Y	966	-0.05	0.07	Y	752
Panel H: Choices and beliefs regarding representative jobs 1.5 years after high school															
Would choose to be a technology or engineering worker	0.15	0.05	1,199	0.12	0.04	Y	1,265	0.13	0.05	Y	966	0.20	0.07	Y	752
Difference in beliefs about pay	-0.01	0.10	1,199	0.01	0.09	Y	1,265	-0.03	0.11	Y	966	0.04	0.13	Y	752
Difference in beliefs about enjoying the work content	0.74	0.21	1,199	0.71	0.20	Y	1,265	0.77	0.22	Y	966	0.97	0.32	Y	752
Difference in beliefs about coworkers and work conditions	0.25	0.19	1,199	0.23	0.17	Y	1,265	0.29	0.21	Y	966	0.37	0.28	Y	752
Difference in beliefs about own preparation	0.90	0.23	1,199	0.84	0.21	Y	1,265	0.96	0.25	Y	966	1.08	0.36	Y	752
Difference in beliefs about approval of friends and family	0.30	0.12	1,199	0.30	0.11	Y	1,265	0.34	0.13	Y	966	0.32	0.17	Y	752
Panel I: Wellbeing, high school satisfaction, and high school curricular fit at the end of high school															
Wellbeing	0.13	0.05	3,987	0.13	0.05	Y	4,056	0.14	0.06	Y	3,651	0.12	0.06	Y	3,127
Liked the high school experience	0.11	0.07	3,987	0.14	0.07	Y	4,056	0.16	0.07	Y	3,651	0.20	0.08	Y	3,127
Liked the high school curriculum	0.00	0.07	3,987	0.00	0.06	Y	4,056	0.06	0.07	Y	3,651	0.09	0.09	Y	3,127
Liked the high school peers	0.23	0.07	3,987	0.26	0.07	Y	4,056	0.23	0.08	Y	3,651	0.22	0.10	Y	3,127
Liked the high school teachers	-0.02	0.07	3,987	-0.03	0.07	Y	4,056	0.01	0.07	Y	3,651	0.01	0.08	Y	3,127
The high school curriculum was a good fit for my abilities	0.00	0.05	3,987	-0.01	0.05	Y	4,056	0.00	0.06	Y	3,651	0.05	0.07	Y	3,127
Panel J: Wellbeing, high school satisfaction, and high school curricular fit 1-1.5 years after high school															
Wellbeing	-0.03	0.07	1,199	-0.05	0.06	Y	1,265	-0.04	0.08	Y	966	-0.03	0.11	Y	752
Liked the high school experience	0.22	0.11	1,159	0.29	0.10	Y	1,235	0.16	0.13	Y	962	0.11	0.16	Y	731
Liked the high school curriculum	0.00	0.14	1,159	0.09	0.12	Y	1,235	-0.10	0.15	Y	962	-0.15	0.19	Y	731
Liked the high school peers	0.29	0.14	1,159	0.34	0.12	Y	1,235	0.28	0.16	Y	962	0.24	0.19	Y	731
Liked the high school teachers	0.13	0.13	1,159	0.14	0.11	Y	1,235	0.16	0.15	Y	962	0.08	0.19	Y	731
The high school curriculum was a good fit for my abilities	-0.12	0.12	1,159	-0.12	0.11	Y	1,235	-0.34	0.14	Y	962	-0.23	0.18	Y	731

The table is a continuation of Table A75.

Table A77: Robustness to alternative fixed effects, cont.

Outcome	Main approach			Alternative f.e. 1			Alternative f.e. 2			Alternative f.e. 3					
	Est.	s.e.	N	Est.	s.e.	In C.I.	N	Est.	s.e.	In C.I.	N	Est.	s.e.	In C.I.	N
Panel K: Regret over high school application choices one year after high school															
Am happy with the high school application choices that I made	0.05	0.13	1,159	0.10	0.11	Y	1,235	-0.01	0.13	Y	962	-0.04	0.17	Y	731
If could do over, would make no change	0.05	0.04	1,159	0.07	0.04	Y	1,235	0.04	0.05	Y	962	0.00	0.06	Y	731
If could do over, would rank STEM tracks higher	-0.03	0.03	1,159	-0.04	0.03	Y	1,235	-0.01	0.04	Y	962	-0.03	0.05	Y	731
If could do over, would rank HSS tracks higher	-0.02	0.03	1,159	-0.04	0.03	Y	1,235	-0.03	0.04	Y	962	0.02	0.05	Y	731
If could do over, would rank Other tracks higher	0.00	0.01	1,159	0.01	0.01	Y	1,235	0.00	0.02	Y	962	0.01	0.02	Y	731
Panel L: College satisfaction, curricular fit, and regret one year after high school															
Like the college experience	-0.44	0.14	1,159	-0.35	0.12	Y	1,235	-0.38	0.16	Y	962	-0.42	0.21	Y	731
Like the college curriculum	-0.23	0.13	1,159	-0.19	0.11	Y	1,235	-0.21	0.15	Y	962	-0.19	0.20	Y	731
Like the college peers	-0.40	0.17	1,159	-0.34	0.13	Y	1,235	-0.39	0.19	Y	962	-0.38	0.24	Y	731
Like the college instructors	-0.35	0.13	1,159	-0.34	0.12	Y	1,235	-0.38	0.17	Y	962	-0.37	0.20	Y	731
The college curriculum is a good fit for my abilities	0.08	0.16	1,159	0.01	0.14	Y	1,235	0.06	0.19	Y	962	-0.03	0.24	Y	731
Am happy with the college application choices that I made	0.12	0.18	1,159	0.04	0.16	Y	1,235	0.14	0.21	Y	962	0.18	0.28	Y	731
Panel M: Social and civic outcomes: male students															
Traditionalist expectations	0.12	0.05	1,396	0.10	0.04	Y	1,464	0.14	0.05	Y	1,181	0.14	0.07	Y	912
Right-wing political views	0.14	0.05	1,396	0.10	0.05	Y	1,464	0.17	0.06	Y	1,181	0.15	0.07	Y	912
Trust	-0.01	0.08	1,396	-0.02	0.08	Y	1,464	0.00	0.10	Y	1,181	0.19	0.13	N	912
Turn out to vote in the Fall 2024 elections	0.00	0.08	373	0.02	0.06	Y	456	-0.01	0.10	Y	261	0.02	0.14	Y	170
Good friends	-0.07	0.32	1,396	-0.07	0.30	Y	1,464	-0.29	0.36	Y	1,181	0.00	0.46	Y	912
Very close friends	0.00	0.17	1,396	0.03	0.15	Y	1,464	-0.04	0.18	Y	1,181	0.02	0.22	Y	912
Good female vs. male friends	-0.71	0.28	1,396	-0.70	0.26	Y	1,464	-0.58	0.30	Y	1,181	-0.69	0.41	Y	912
Verbal development	-0.26	0.05	1,396	-0.24	0.05	Y	1,464	-0.27	0.06	Y	1,181	-0.30	0.07	Y	912
Empathy	-0.15	0.09	1,396	-0.13	0.09	Y	1,464	-0.13	0.11	Y	1,181	-0.27	0.13	Y	912
Panel N: Social and civic outcomes: female students															
Traditionalist expectations	0.03	0.03	2,510	0.03	0.03	Y	2,592	0.03	0.04	Y	2,219	0.05	0.05	Y	1,823
Right-wing political views	-0.01	0.04	2,510	0.01	0.04	Y	2,592	-0.03	0.04	Y	2,219	-0.02	0.06	Y	1,823
Trust	0.03	0.07	2,510	0.03	0.07	Y	2,592	-0.06	0.08	Y	2,219	0.01	0.09	Y	1,823
Turn out to vote in the Fall 2024 elections	0.05	0.04	728	0.01	0.03	Y	809	0.04	0.04	Y	553	0.08	0.06	Y	425
Good friends	-0.04	0.24	2,510	0.01	0.23	Y	2,592	-0.23	0.26	Y	2,219	0.17	0.32	Y	1,823
Very close friends	-0.06	0.11	2,510	0.03	0.12	Y	2,592	-0.26	0.13	Y	2,219	-0.04	0.15	Y	1,823
Good female vs. male friends	-0.49	0.19	2,510	-0.49	0.19	Y	2,592	-0.52	0.21	Y	2,219	-0.46	0.26	Y	1,823
Verbal development	-0.27	0.04	2,510	-0.24	0.04	Y	2,592	-0.28	0.05	Y	2,219	-0.28	0.05	Y	1,823
Empathy	-0.08	0.07	2,510	-0.06	0.07	Y	2,592	-0.12	0.08	Y	2,219	-0.02	0.09	Y	1,823

The table is a continuation of Table A75.